

A. Isochronous Applications §

A.1 Introduction §

The design goal of isochronous mechanisms in PCI Express is to ensure that isochronous traffic receives its allocated bandwidth over a relevant time period while also preventing starvation of other non-isochronous traffic.

Furthermore, there may exist data traffic that requires a level of service falling in between what is required for bulk data traffic and isochronous data traffic. This type of traffic can be supported through the use of Port arbitration within Switches, the use of TC labels [1:7], and optional additional VC resources. Policies for assignment of TC labels and VC resources that are not isochronous-focused are outside the scope of the PCI Express specification.

Two paradigms of PCI Express communication are supported by the PCI Express isochronous mechanisms: Endpoint-to-Root-Complex communication model and peer-to-peer (Endpoint-to-Endpoint) communication model. In the Endpoint-to-Root-Complex communication model, the primary isochronous traffic is memory read and write requests to the Root Complex and read completions from the Root Complex. § Figure A-1 shows an example of a simple system with both communication models. In the figure, devices A, B, called Requesters, are PCI Express Endpoints capable of issuing isochronous request transactions, while device C and Root Complex, called Completers, are capable of being the targets of isochronous request transactions. An Endpoint-to-Root-Complex communication is established between device A and the Root Complex, and a peer-to-peer communication is established between device B and device C. In the rest of this section, Requester and Completer will be used to make reference to PCI Express elements involved in transactions. The specific aspects of each communication model will be called out explicitly.

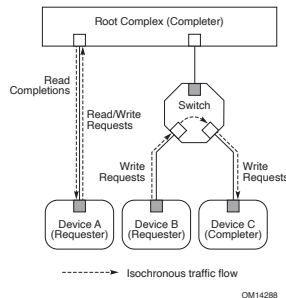


Figure A-1 An Example Showing Endpoint-to-Root-Complex and Peer-to-Peer Communication Models §

Guaranteed bandwidth and deterministic latency require end-to-end configuration of fabric resources. If isochronous traffic is intermixed with non-isochronous traffic, it may not be possible to provide any guarantees/determinism as required by the application usage model. It is recommended that system software configure and assign fabric resources such that traffic intermix either does not occur or is such that the application usage model guarantees can be met. This can be accomplished by assigning dedicated VC resources and corresponding TC labels to the isochronous traffic flow(s) on a given path within the fabric.

Note that there may be one or more isochronous traffic flows per VC/TC label and it is up to system software to insure that the aggregation of these flows does not exceed the requisite bandwidth and latency requirements.

It is also possible for a fabric to support multiple isochronous traffic flows separated across multiple VC (a given flow cannot span multiple VC/TC labels).

In general, as long as the device can meet the isochronous bandwidth and latency requirements, there is nothing to preclude a single VC device from supporting isochronous traffic if multiple TC labels are supported to delineate such traffic from non-isochronous traffic within the fabric.

A.2 Isochronous Contract and Contract Parameters §

In order to support isochronous data transfer with guaranteed bandwidth and deterministic latency, an isochronous contract must be established between a Requester/Completer pair and the PCI Express fabric. This contract must enforce both resource reservation and traffic regulation. Without such a contract, two basic problems, over-subscription and congestion, may occur as illustrated in § Figure A-2. When interconnect bandwidth resources are over-subscribed, the increased latency may cause failure of isochronous service and starvation of non-isochronous services. Traffic congestion occurs when flow control credits are not returned possibly due to a higher than expected/provisioned packet injection rate. This may cause excessive service latencies for both isochronous traffic and non-isochronous traffic.

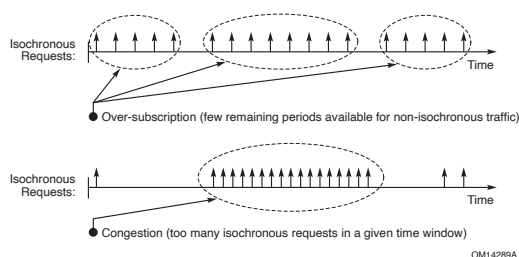


Figure A-2 Two Basic Bandwidth Resourcing Problems: Over-Subscription and Congestion §

The isochronous transfer mechanism in this specification addresses these problems with traffic regulation including admission control and service discipline. Under a software managed admission control, a Requester must not issue isochronous transactions unless the required isochronous bandwidth and resource have been allocated. Specifically, the isochronous bandwidth is given by the following formula:

$$BW = \frac{N * Y}{T}$$

Equation A-1 Isochronous Bandwidth §

The formula defines allocated bandwidth (BW) as a function of specified number (N) of transactions of a specified payload size (Y) within a specified time period (T). Another important parameter in the isochronous contract is latency. Based on the contract, isochronous transactions are completed within a specified latency (L). Once a Requester/Completer pair is admitted for isochronous communication, the bandwidth and latency are guaranteed to the Requester (a PCI Express Endpoint) by the Completer (Root Complex for Endpoint-to-Root-Complex communication and another PCI Express Endpoint for peer-to-peer communication) and by the PCI Express fabric components (Switches).

Specific service disciplines must be implemented by isochronous-capable PCI Express components. The service disciplines are imposed to PCI Express Switches and Completers in such a manner that the service of isochronous requests is subject to a specific service interval (t). This mechanism is used to provide the method of controlling when an isochronous packet injected by a Requester is serviced. Consequently, isochronous traffic is policed in such manner that only packets that can be injected into the fabric in compliance with the isochronous contract are allowed to make immediate progress and start being serviced by the PCI Express fabric. A non-compliant Requester that tries to inject more isochronous transactions than what was being allowed by the contract is prevented from doing so by the

flow-control mechanism thereby allowing compliant Requesters to correctly operate independent of non-compliant Requesters.

In the Endpoint-to-Root-Complex model, since the aggregated isochronous traffic is eventually limited by the host memory subsystem's bandwidth capabilities, isochronous read requests, and write requests (and Messages) are budgeted together. A Requester may divide the isochronous bandwidth between read requests and write requests as appropriate.

A.2.1 Isochronous Time Period and Isochronous Virtual Timeslot §

The PCI Express isochronous time period (T) is uniformly divided into units of virtual timeslots (t). To provide precise isochronous bandwidth distribution only one isochronous request packet is allowed per virtual timeslot. The virtual timeslot supported by a PCI Express component is reported through the Reference Clock field in the Virtual Channel Capability structure or the Multi-Function Virtual Channel Capability structure. When Reference Clock = 00b, duration of a virtual timeslot t is 100 ns. Duration of isochronous time period T depends on the number of phases of the supported time-based WRR Port arbitration table size. When the time-based WRR Port Arbitration Table size equals to 128, there are 128 virtual timeslots (t) in an isochronous time period (i.e., $T = 12.8 \mu\text{s}$).

Note that isochronous period T as well as virtual timeslots t do not need to be aligned and synchronized among different PCI Express isochronous devices, i.e., the notion of $\{T, t\}$ is local to each individual isochronous device.

A.2.2 Isochronous Payload Size §

The payload size (Y) for isochronous transactions must not exceed Tx_MPS_Limit (see § Section 2.2.2). After configuration, the Tx_MPS_Limit is set and fixed for each path that supports isochronous service with a value required to meet isochronous latency. The fixed Tx_MPS_Limit value is used for isochronous bandwidth budgeting regardless of the actual size of data payload associated with isochronous transactions. For isochronous bandwidth budgeting, we have

$$Y = \text{Tx_MPS_Limit}$$

Equation A-2 Isochronous Payload Size §

A transaction with partial writes is treated as a normally accounted transaction. A Completer must account for partial writes as part of bandwidth assignment (for worst case servicing time).

A.2.3 Isochronous Bandwidth Allocation §

Given T , t and Y , the maximum virtual timeslots within a time period is

$$N_{\max} = \frac{T}{t}$$

Equation A-3 $N_{\max}$ §

and the maximum specifiable isochronous bandwidth is

$$BW_{\max} = \frac{Y}{t}$$

Equation A-4 $BW_{\max}$ §

The granularity with which isochronous bandwidth can be allocated is defined as:

$$BW_{\text{granularity}} = \frac{Y}{T}$$

Equation A-5 $BW_{\text{granularity}}$ §

Given T and t at 12.8 μs and 100 ns, respectively, $N_{\max}$ is 128. As shown in § Table A-1, $BW_{\max}$ and $BW_{\text{granularity}}$ are functions of the isochronous payload size Y .

Table A-1 Isochronous Bandwidth Ranges and Granularities §

Y (bytes)	128	256	512	1024
$BW_{\max}$ (MB/s)	1280	2560	5120	10240
$BW_{\text{granularity}}$ (MB/s)	10	20	40	80

Similar to bandwidth budgeting, isochronous service disciplines including arbitration schemes are based on counting requests (not the sizes of those requests). Therefore, assigning isochronous bandwidth BW_{link} to a PCI Express Link is equivalent to assigning N_{link} virtual timeslots per isochronous period, where N_{link} is given by

$$N_{\text{link}} = \frac{BW_{\text{link}}}{BW_{\text{granularity}}}$$

Equation A-6 N_{link} §

A Switch Port serving as an Egress Port (or an RCRB serving as a “virtual” Egress Port) for an isochronous traffic, the $N_{\max}$ virtual timeslots within T are represented by the time-based WRR Port Arbitration Table in the PCI Express Virtual Channel Capability structure detailed in § Section 7.9.1. The table consists of $N_{\max}$ entries. An entry in the table represents one virtual timeslot in the isochronous time period. When a table entry is given a value of PN, it means that the timeslot is assigned to an Ingress Port (in respect to the isochronous traffic targeting the Egress Port) designated by a Port Number of PN. Therefore, N_{link} virtual timeslots are assigned to the Ingress Port when there are N_{link} entries in the table with value of PN. The Egress Port may admit one isochronous request transaction from the Ingress Port for further service only when the table entry reached by the Egress Port's isochronous time ticker (that increments by 1 every t time and wraps around when reaching T) is set to PN. Even if there are outstanding isochronous requests ready in the Ingress Port, they will not be served until next round of time-based WRR arbitration. In this manner, the time-based Port Arbitration Table serves for both isochronous bandwidth assignment and isochronous traffic regulation.

For an Endpoint serving as a Requester or a Completer, isochronous bandwidth allocation is accomplished through negotiation between system software and device driver, which is outside of the scope of this specification.

A.2.4 Isochronous Transaction Latency §

Transaction latency is composed of the latency through the PCI Express fabric and the latency contributed by the Completer. Isochronous transaction latency is defined for each transaction and measured in units of virtual timeslot t .

- The *read latency* is defined as the round-trip latency. This is the delay from the time when the device submits a memory read request packet to its Transaction Layer (Transmit side) to the time when the corresponding read completion arrives at the device's Transaction Layer (Receive side).
- The *write latency* is defined as the delay from the time when the Requester posts a memory write request to its PCI Express Transaction Layer (Transmit side) to the time when the data write becomes globally visible within the memory subsystem of the Completer. A write to memory reaches the point of global visibility when all agents accessing that memory address get the updated data.

When the upper bound and the lower bound of isochronous transaction latency are provided, the size of isochronous data buffers in a Requester can be determined. For most of common platforms, the minimum isochronous transaction latency is much smaller than the maximum. As a conservative measure, the minimum isochronous transaction latency is assumed to be zero; only guidelines on measuring the maximum isochronous transaction latency are provided here.

For a Requester, the maximum isochronous (read or write) transaction latency (L) can be accounted as the following:

$$L = L_{Fabric} + L_{Completer}$$

Equation A-7 Max Isochronous Transaction Latency §

where L_{Fabric} is the maximum latency of the PCI Express fabric and $L_{Completer}$ is the maximum latency of the Completer.

L_{Fabric} which applies to both read and write transactions, depends on the topology, latency across each PCI Express Link, and the arbitration point in the path between the Requester to the Completer. The latency on a PCI Express Link depends on pipeline delays, width and operational frequency of the Link, transmission of electrical signals across the medium, wake up latency from low power states, and delays caused by Data Link Layer Retry.

A restriction on the PCI Express topology may be imposed for each targeted platform in order to provide a practically meaningful guideline for L_{Fabric} . The values of L_{Fabric} should be reasonable and serve as practical upper limits under normal operating conditions.

The value of $L_{Completer}$ depends on the memory technology, memory configuration, and the arbitration policies in the Completer that comprehend PCI Express isochronous traffic. The target value for $L_{Completer}$ should provide enough headroom to allow for implementation tradeoffs.

Definitions of read and write transaction latencies for a Completer are different:

- Read transaction latency for the Completer is defined as the delay from the time a memory read transaction is available at the Receiver end of a PCI Express Port in the Completer to the time the corresponding read completion transaction is posted to the transmission end of the PCI Express Port.
- Write transaction latency is defined as the delay from the time a memory write transaction is available at the Receiver end of a PCI Express Port in the Completer to the time that the transmitted data is globally visible.

All of the isochronous transaction latencies defined above are based on the assumption that the Requester injects isochronous transactions uniformly. According to an isochronous contract of $\{N, T, t\}$, the uniform traffic injection is defined such that up to N transactions are evenly distributed over the isochronous period T based on a ticker granularity

of virtual timeslot t . For a Requester with non-uniform isochronous transaction injection, the Requester is responsible of accounting for any additional delay due to the deviation of its injection pattern from a uniform injection pattern.

A.2.5 An Example Illustrating Isochronous Parameters §

§ Figure A-3 illustrates the key isochronous parameters using a simplified example with $T = 20t$ and $L = 22t$. A Requester has reserved isochronous bandwidth of four transactions per T . The device shares the allocated isochronous bandwidths for both read requests and write requests. As shown, during one isochronous time period, the Requester issues two read requests and two write requests. All requests are completed within the designated transaction latency L . Also shown in the figure, there is no time dependency between the service time of write requests and the arrival time of read completions.

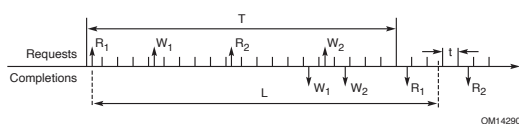


Figure A-3 A Simplified Example Illustrating PCI Express Isochronous Parameters §

A.3 Isochronous Transaction Rules §

Isochronous transactions follow the same rules as described in § Chapter 2. . In order to assist the Completer to meet latency requirements, the following additional rules further illustrate and clarify the proper behavior of isochronous transactions:

- The value in the Length field of read requests must never exceed Tx_MPS_Limit.

A.4 Transaction Ordering §

In general, isochronous transactions follow the ordering rules described in § Section 2.4 . The following ordering rule further illustrates and clarifies the proper behavior of isochronous transactions:

- There are no ordering guarantees between any isochronous and non-isochronous transactions because the traffic has been segregated into distinct VC resources.
- Isochronous write requests are serviced on any PCI Express Link in strictly the same order as isochronous write requests are posted.
- Switches must allow isochronous posted requests to pass isochronous read completions.

A.5 Isochronous Data Coherency §

Cache coherency for isochronous transactions is an operating system software and Root Complex hardware issue. PCI Express provides the necessary mechanism to control Root Complex behavior in terms of enforcing hardware cache coherency on a per transaction basis.

For platforms where snoop latency in a Root Complex is either unbounded or can be excessively large, in order to meet tight maximum isochronous transaction latency $L_{Completer}$, or more precisely $L_{Root_Complex}$, all isochronous transactions should have the No Snoop Attribute bit set.

A Root Complex must report the Root Complex's capability to the system software by setting the Reject Snoop Transactions field in the VC Resource Capability register (for any VC resource capable of supporting isochronous traffic) in its RCRB. Based on whether or not a Root Complex is capable of providing hardware enforced cache coherency for isochronous traffic while still meeting isochronous latency target, system software can then inform the device driver of Endpoints to set or unset the No Snoop Attribute bit for isochronous transactions.

Note that cache coherency considerations for isochronous traffic do not apply to peer-to-peer communication.

A.6 Flow Control §

Completers and PCI Express fabric components should implement proper sizing of buffers such that under normal operating conditions, no backpressure due to flow control should be applied to isochronous traffic injected uniformly by a Requester. For Requesters that are compliant to the isochronous contract, but have bursty injection behavior, Switches and Completers may apply flow control backpressure as long as the admitted isochronous traffic is uniform and compliant to the isochronous contract. Under abnormal conditions when isochronous traffic jitter becomes significant or when isochronous traffic is oversubscribed due to excessive Data Link Layer Retry, flow control provides a natural mechanism to ensure functional correctness.

A.7 Considerations for Bandwidth Allocation §

A.7.1 Isochronous Bandwidth of PCI Express Links §

Isochronous bandwidth budgeting for PCI Express Links can be derived based on Link parameters such as isochronous payload size and the speed and width of the Link.

Isochronous bandwidth allocation for a PCI Express Link should be limited to certain percentage of the maximum effective Link bandwidth in order to leave sufficient bandwidth for non-isochronous traffic and to account for temporary Link bandwidth reduction due to retries. Link utilization is counted based on the actual cycles consumed on the physical PCI Express Link. The maximum number of virtual slots allowed per Link (N_{link}) depends on the isochronous packet payload size and the speed and width of the Link.

As isochronous bandwidth allocation on a PCI Express Link is based on number of requests N_{link} per isochronous period. There is no distinction between read requests and write requests in budgeting isochronous bandwidth on a PCI Express Link.

A.7.2 Isochronous Bandwidth of Endpoints §

For peer-to-peer communication, the device driver is responsible for reporting to system software if the device is capable of being a Completer for isochronous transactions. In addition, the driver must report the device's isochronous bandwidth capability. The specifics of the report mechanism are outside the scope of this specification.

A.7.3 Isochronous Bandwidth of Switches §

Allocation of isochronous bandwidth for a Switch must consider the capacity and utilization of PCI Express Links associated with the Ingress Port and the Egress Port of the Switch that connect the Requester and the Completer, respectively. The lowest common denominator of the two determines if a requested isochronous bandwidth can be supported.

A.7.4 Isochronous Bandwidth of Root Complex §

Isochronous bandwidth of Root Complex is reported to the software through its RCRB structure. Specifically, the Maximum Time Slots field of the VC Resource Capability register in VC Extended Capability structure indicates the total isochronous bandwidth shared by the Root Ports associated with the RCRB. Details of the platform budgeting for available isochronous bandwidth within a Root Complex are outside of the scope of this specification.

A.8 Considerations for PCI Express Components §

A.8.1 An Endpoint as a Requester §

Before an Endpoint as a Requester can start issuing isochronous request transactions, the following configuration steps must be performed by software:

- Configuration of at least one VC resource capable of supporting isochronous communication and assignment of at least one TC label.
- Enablement of this VC resource.

When the Requester uniformly injects isochronous requests, the Receive Port, either a Switch Port or a Root Port, should issue Flow Control credits back promptly such that no backpressure should be applied to the associated VC. This type of Requester may size its buffer based on the PCI Express fabric latency L_{Fabric} plus the Completer's latency $L_{Completer}$.

When isochronous transactions are injected non-uniformly, either some transactions experience longer PCI Express fabric delay or the Requester gets back-pressured on the associated VC. This type of Requester must size its buffer to account for the deviation of its injection pattern from uniformity.

A.8.2 An Endpoint as a Completer §

An Endpoint may serve as a Completer for isochronous peer-to-peer communication. Before an Endpoint starts serving isochronous transactions, system software must identify/configure a VC resource capable of supporting isochronous traffic and assigned a corresponding TC label.

An Endpoint Completer must observe the maximum isochronous transaction latency ($L_{Completer}$). An Endpoint Completer does not have to regulate isochronous request traffic if attached to a Switch since Switches implement traffic regulation. However, an Endpoint Completer must size its internal buffer such that no backpressure should be applied to the corresponding VC.

A.8.3 Switches §

A Switch may have multiple ports capable of supporting isochronous transactions. Before a Switch starts serving isochronous transactions for a Port, the software must perform the following configuration steps:

- Configuration/enablement of at least one VC resource capable of supporting isochronous communication.
- Configuration of the Port as an Ingress Port:

- Configuration (or reconfiguration if the associated VC of the Egress Port is already enabled) of the time-based WRR Port Arbitration Table of the targeting Egress Port to include N_{link} entries set to the Ingress Port's Port Number. Here N_{link} is the isochronous allocation for the Ingress Port.
- Enabling the targeting Egress Port to load newly programmed Port Arbitration Table.
- Configuration of the Port as an Egress Port:
 - Configuration of each VC's Port Arbitration Table with number of entries set according to the assigned isochronous bandwidth for all Ingress Ports.
 - Select proper VC Arbitration, e.g., as strict-priority based VC Arbitration.
 - If required, configuration of the Port's VC Arbitration Table with large weights assigned accordingly to each associated VC.

Each VC associated with isochronous traffic may be served as the highest priority in arbitrating for the shared PCI Express Link resource at an Egress Port. This is comprehended by a Switch's internal arbitration scheme.

In addition, a Switch Port may use “just in time” scheduling mechanism to reduce VC arbitration latency. Instead of pipelining non-isochronous Transport Layer packets to the Data Link Layer of the Egress Port in a manner that Data Link Layer transmit buffer becomes saturated, the Switch Port may hold off scheduling of a new non-isochronous packet to the Data Link Layer as long as it is possible without incurring unnecessary Link idle time.

When a VC configured to support isochronous traffic is enabled for a Switch Port (ingress) that is connected to a Requester, the Switch must enforce proper traffic regulation to ensure that isochronous traffic from the Port conforms to this specification. With such enforcement, normal isochronous transactions from compliant Requesters will not be impacted by ill behavior of any non-compliant Requester.

The above isochronous traffic regulation mechanism only applies to request transactions but not to completion transactions. When Endpoint-to-Root-Complex and peer-to-peer communications co-exist in a Switch, an Egress Port may mix isochronous write requests and read completions in the same direction. In the case of contention, the Egress Port must allow write requests to pass read completions to ensure the Switch meets latency requirement for isochronous requests.

A.8.4 Root Complex §

A Root Complex may have multiple Root Ports capable of supporting isochronous transactions. Before a Root Complex starts serving isochronous transactions for a Root Port, the Port must be configured by software to enable VC to support isochronous traffic using the following configuration steps:

- Configuration of at least one VC resource capable of supporting isochronous communication and assignment of at least one TC label.
- Configuration of the Root Port as an Ingress Port:
 - Configuration (or reconfiguration if the associated VC in RCRB is already enabled) of the time-based WRR Port Arbitration Table of the targeting RCRB to include N_{link} entries set to the Ingress Port's Port Number. Here N_{link} is the isochronous allocation for the Port.
 - Enabling the targeting RCRB to load newly programmed Port Arbitration Table.
- Configuration of the Root Port as an Egress Port:
 - If supported, configuration of the Root Port's VC Arbitration Table with large weights assigned to the associated VC.
 - If the Root Complex supports peer-to-peer traffic between Root Ports, configuration of the Root Port's Port Arbitration Table number of entries is set according to the assigned isochronous bandwidth for all Ingress Ports.

A Root Complex must observe the maximum isochronous transaction latency ($L_{Completer}$ or more precisely $L_{Root_Complex}$) that applies to all the Root Ports in the Root Complex. How a Root Complex schedules memory cycles for PCI Express isochronous transactions and other memory transactions is outside of the scope of this specification as long as $L_{Root_Complex}$ is met for PCI Express isochronous transactions.

When a VC is enabled to support isochronous traffic for a Root Port, the Root Complex must enforce proper traffic regulation to ensure that isochronous traffic from the Root Port conforms to this specification. With such enforcement, normal isochronous transactions from compliant Requesters will not be impacted by ill behavior of any non-compliant Requesters. Isochronous traffic regulation is implemented using the time-based Port Arbitration Table in RCRB.

Root Complex may perform the following operations for invalid isochronous transactions:

- Return partial completions for read requests with the value in the Length field exceeding Tx_MPS_Limit.

B. Symbol Encoding §

§ Table B-1 shows the byte-to-Symbol encodings for data characters. § Table B-2 shows the Symbol encodings for the Special Symbols used for TLP/DLLP Framing and for interface management. RD- and RD+ refer to the Running Disparity of the Symbol sequence on a per-Lane basis.

Table B-1 8b/10b Data Symbol Codes §

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD- abcdei fghj (binary)	Current RD+ abcdei fghj (binary)
D0.0	00	000 00000	100111 0100	011000 1011
D1.0	01	000 00001	011101 0100	100010 1011
D2.0	02	000 00010	101101 0100	010010 1011
D3.0	03	000 00011	110001 1011	110001 0100
D4.0	04	000 00100	110101 0100	001010 1011
D5.0	05	000 00101	101001 1011	101001 0100
D6.0	06	000 00110	011001 1011	011001 0100
D7.0	07	000 00111	111000 1011	000111 0100
D8.0	08	000 01000	111001 0100	000110 1011
D9.0	09	000 01001	100101 1011	100101 0100
D10.0	0A	000 01010	010101 1011	010101 0100
D11.0	0B	000 01011	110100 1011	110100 0100
D12.0	0C	000 01100	001101 1011	001101 0100
D13.0	0D	000 01101	101100 1011	101100 0100
D14.0	0E	000 01110	011100 1011	011100 0100
D15.0	0F	000 01111	010111 0100	101000 1011
D16.0	10	000 10000	011011 0100	100100 1011
D17.0	11	000 10001	100011 1011	100011 0100
D18.0	12	000 10010	010011 1011	010011 0100
D19.0	13	000 10011	110010 1011	110010 0100
D20.0	14	000 10100	001011 1011	001011 0100
D21.0	15	000 10101	101010 1011	101010 0100
D22.0	16	000 10110	011010 1011	011010 0100
D23.0	17	000 10111	111010 0100	000101 1011

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD- abcdei fghj (binary)	Current RD+ abcdei fghj (binary)
D24.0	18	000 11000	110011 0100	001100 1011
D25.0	19	000 11001	100110 1011	100110 0100
D26.0	1A	000 11010	010110 1011	010110 0100
D27.0	1B	000 11011	110110 0100	001001 1011
D28.0	1C	000 11100	001110 1011	001110 0100
D29.0	1D	000 11101	101110 0100	010001 1011
D30.0	1E	000 11110	011110 0100	100001 1011
D31.0	1F	000 11111	101011 0100	010100 1011
D0.1	20	001 00000	100111 1001	011000 1001
D1.1	21	001 00001	011101 1001	100010 1001
D2.1	22	001 00010	101101 1001	010010 1001
D3.1	23	001 00011	110001 1001	110001 1001
D4.1	24	001 00100	110101 1001	001010 1001
D5.1	25	001 00101	101001 1001	101001 1001
D6.1	26	001 00110	011001 1001	011001 1001
D7.1	27	001 00111	111000 1001	000111 1001
D8.1	28	001 01000	111001 1001	000110 1001
D9.1	29	001 01001	100101 1001	100101 1001
D10.1	2A	001 01010	010101 1001	010101 1001
D11.1	2B	001 01011	110100 1001	110100 1001
D12.1	2C	001 01100	001101 1001	001101 1001
D13.1	2D	001 01101	101100 1001	101100 1001
D14.1	2E	001 01110	011100 1001	011100 1001
D15.1	2F	001 01111	010111 1001	101000 1001
D16.1	30	001 10000	011011 1001	100100 1001
D17.1	31	001 10001	100011 1001	100011 1001
D18.1	32	001 10010	010011 1001	010011 1001
D19.1	33	001 10011	110010 1001	110010 1001
D20.1	34	001 10100	001011 1001	001011 1001
D21.1	35	001 10101	101010 1001	101010 1001

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD-abcdei fghj (binary)	Current RD+abcdei fghj (binary)
D22.1	36	001 10110	011010 1001	011010 1001
D23.1	37	001 10111	111010 1001	000101 1001
D24.1	38	001 11000	110011 1001	001100 1001
D25.1	39	001 11001	100110 1001	100110 1001
D26.1	3A	001 11010	010110 1001	010110 1001
D27.1	3B	001 11011	110110 1001	001001 1001
D28.1	3C	001 11100	001110 1001	001110 1001
D29.1	3D	001 11101	101110 1001	010001 1001
D30.1	3E	001 11110	011110 1001	100001 1001
D31.1	3F	001 11111	101011 1001	010100 1001
D0.2	40	010 00000	100111 0101	011000 0101
D1.2	41	010 00001	011101 0101	100010 0101
D2.2	42	010 00010	101101 0101	010010 0101
D3.2	43	010 00011	110001 0101	110001 0101
D4.2	44	010 00100	110101 0101	001010 0101
D5.2	45	010 00101	101001 0101	101001 0101
D6.2	46	010 00110	011001 0101	011001 0101
D7.2	47	010 00111	111000 0101	000111 0101
D8.2	48	010 01000	111001 0101	000110 0101
D9.2	49	010 01001	100101 0101	100101 0101
D10.2	4A	010 01010	010101 0101	010101 0101
D11.2	4B	010 01011	110100 0101	110100 0101
D12.2	4C	010 01100	001101 0101	001101 0101
D13.2	4D	010 01101	101100 0101	101100 0101
D14.2	4E	010 01110	011100 0101	011100 0101
D15.2	4F	010 01111	010111 0101	101000 0101
D16.2	50	010 10000	011011 0101	100100 0101
D17.2	51	010 10001	100011 0101	100011 0101
D18.2	52	010 10010	010011 0101	010011 0101
D19.2	53	010 10011	110010 0101	110010 0101

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD- abcdei fghj (binary)	Current RD+ abcdei fghj (binary)
D20.2	54	010 10100	001011 0101	001011 0101
D21.2	55	010 10101	101010 0101	101010 0101
D22.2	56	010 10110	011010 0101	011010 0101
D23.2	57	010 10111	111010 0101	000101 0101
D24.2	58	010 11000	110011 0101	001100 0101
D25.2	59	010 11001	100110 0101	100110 0101
D26.2	5A	010 11010	010110 0101	010110 0101
D27.2	5B	010 11011	110110 0101	001001 0101
D28.2	5C	010 11100	001110 0101	001110 0101
D29.2	5D	010 11101	101110 0101	010001 0101
D30.2	5E	010 11110	011110 0101	100001 0101
D31.2	5F	010 11111	101011 0101	010100 0101
D0.3	60	011 00000	100111 0011	011000 1100
D1.3	61	011 00001	011101 0011	100010 1100
D2.3	62	011 00010	101101 0011	010010 1100
D3.3	63	011 00011	110001 1100	110001 0011
D4.3	64	011 00100	110101 0011	001010 1100
D5.3	65	011 00101	101001 1100	101001 0011
D6.3	66	011 00110	011001 1100	011001 0011
D7.3	67	011 00111	111000 1100	000111 0011
D8.3	68	011 01000	111001 0011	000110 1100
D9.3	69	011 01001	100101 1100	100101 0011
D10.3	6A	011 01010	010101 1100	010101 0011
D11.3	6B	011 01011	110100 1100	110100 0011
D12.3	6C	011 01100	001101 1100	001101 0011
D13.3	6D	011 01101	101100 1100	101100 0011
D14.3	6E	011 01110	011100 1100	011100 0011
D15.3	6F	011 01111	010111 0011	101000 1100
D16.3	70	011 10000	011011 0011	100100 1100
D17.3	71	011 10001	100011 1100	100011 0011

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD-abcdei fghj (binary)	Current RD+abcdei fghj (binary)
D18.3	72	011 10010	010011 1100	010011 0011
D19.3	73	011 10011	110010 1100	110010 0011
D20.3	74	011 10100	001011 1100	001011 0011
D21.3	75	011 10101	101010 1100	101010 0011
D22.3	76	011 10110	011010 1100	011010 0011
D23.3	77	011 10111	111010 0011	000101 1100
D24.3	78	011 11000	110011 0011	001100 1100
D25.3	79	011 11001	100110 1100	100110 0011
D26.3	7A	011 11010	010110 1100	010110 0011
D27.3	7B	011 11011	110110 0011	001001 1100
D28.3	7C	011 11100	001110 1100	001110 0011
D29.3	7D	011 11101	101110 0011	010001 1100
D30.3	7E	011 11110	011110 0011	100001 1100
D31.3	7F	011 11111	101011 0011	010100 1100
D0.4	80	100 00000	100111 0010	011000 1101
D1.4	81	100 00001	011101 0010	100010 1101
D2.4	82	100 00010	101101 0010	010010 1101
D3.4	83	100 00011	110001 1101	110001 0010
D4.4	84	100 00100	110101 0010	001010 1101
D5.4	85	100 00101	101001 1101	101001 0010
D6.4	86	100 00110	011001 1101	011001 0010
D7.4	87	100 00111	111000 1101	000111 0010
D8.4	88	100 01000	111001 0010	000110 1101
D9.4	89	100 01001	100101 1101	100101 0010
D10.4	8A	100 01010	010101 1101	010101 0010
D11.4	8B	100 01011	110100 1101	110100 0010
D12.4	8C	100 01100	001101 1101	001101 0010
D13.4	8D	100 01101	101100 1101	101100 0010
D14.4	8E	100 01110	011100 1101	011100 0010
D15.4	8F	100 01111	010111 0010	101000 1101

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD- abcdei fghj (binary)	Current RD+ abcdei fghj (binary)
D16.4	90	100 10000	011011 0010	100100 1101
D17.4	91	100 10001	100011 1101	100011 0010
D18.4	92	100 10010	010011 1101	010011 0010
D19.4	93	100 10011	110010 1101	110010 0010
D20.4	94	100 10100	001011 1101	001011 0010
D21.4	95	100 10101	101010 1101	101010 0010
D22.4	96	100 10110	011010 1101	011010 0010
D23.4	97	100 10111	111010 0010	000101 1101
D24.4	98	100 11000	110011 0010	001100 1101
D25.4	99	100 11001	100110 1101	100110 0010
D26.4	9A	100 11010	010110 1101	010110 0010
D27.4	9B	100 11011	110110 0010	001001 1101
D28.4	9C	100 11100	001110 1101	001110 0010
D29.4	9D	100 11101	101110 0010	010001 1101
D30.4	9E	100 11110	011110 0010	100001 1101
D31.4	9F	100 11111	101011 0010	010100 1101
D0.5	A0	101 00000	100111 1010	011000 1010
D1.5	A1	101 00001	011101 1010	100010 1010
D2.5	A2	101 00010	101101 1010	010010 1010
D3.5	A3	101 00011	110001 1010	110001 1010
D4.5	A4	101 00100	110101 1010	001010 1010
D5.5	A5	101 00101	101001 1010	101001 1010
D6.5	A6	101 00110	011001 1010	011001 1010
D7.5	A7	101 00111	111000 1010	000111 1010
D8.5	A8	101 01000	111001 1010	000110 1010
D9.5	A9	101 01001	100101 1010	100101 1010
D10.5	AA	101 01010	010101 1010	010101 1010
D11.5	AB	101 01011	110100 1010	110100 1010
D12.5	AC	101 01100	001101 1010	001101 1010
D13.5	AD	101 01101	101100 1010	101100 1010

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD- abcdei fghj (binary)	Current RD+ abcdei fghj (binary)
D14.5	AE	101 01110	011100 1010	011100 1010
D15.5	AF	101 01111	010111 1010	101000 1010
D16.5	B0	101 10000	011011 1010	100100 1010
D17.5	B1	101 10001	100011 1010	100011 1010
D18.5	B2	101 10010	010011 1010	010011 1010
D19.5	B3	101 10011	110010 1010	110010 1010
D20.5	B4	101 10100	001011 1010	001011 1010
D21.5	B5	101 10101	101010 1010	101010 1010
D22.5	B6	101 10110	011010 1010	011010 1010
D23.5	B7	101 10111	111010 1010	000101 1010
D24.5	B8	101 11000	110011 1010	001100 1010
D25.5	B9	101 11001	100110 1010	100110 1010
D26.5	BA	101 11010	010110 1010	010110 1010
D27.5	BB	101 11011	110110 1010	001001 1010
D28.5	BC	101 11100	001110 1010	001110 1010
D29.5	BD	101 11101	101110 1010	010001 1010
D30.5	BE	101 11110	011110 1010	100001 1010
D31.5	BF	101 11111	101011 1010	010100 1010
D0.6	C0	110 00000	100111 0110	011000 0110
D1.6	C1	110 00001	011101 0110	100010 0110
D2.6	C2	110 00010	101101 0110	010010 0110
D3.6	C3	110 00011	110001 0110	110001 0110
D4.6	C4	110 00100	110101 0110	001010 0110
D5.6	C5	110 00101	101001 0110	101001 0110
D6.6	C6	110 00110	011001 0110	011001 0110
D7.6	C7	110 00111	111000 0110	000111 0110
D8.6	C8	110 01000	111001 0110	000110 0110
D9.6	C9	110 01001	100101 0110	100101 0110
D10.6	CA	110 01010	010101 0110	010101 0110
D11.6	CB	110 01011	110100 0110	110100 0110

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD- abcdei fghj (binary)	Current RD+ abcdei fghj (binary)
D12.6	CC	110 01100	001101 0110	001101 0110
D13.6	CD	110 01101	101100 0110	101100 0110
D14.6	CE	110 01110	011100 0110	011100 0110
D15.6	CF	110 01111	010111 0110	101000 0110
D16.6	D0	110 10000	011011 0110	100100 0110
D17.6	D1	110 10001	100011 0110	100011 0110
D18.6	D2	110 10010	010011 0110	010011 0110
D19.6	D3	110 10011	110010 0110	110010 0110
D20.6	D4	110 10100	001011 0110	001011 0110
D21.6	D5	110 10101	101010 0110	101010 0110
D22.6	D6	110 10110	011010 0110	011010 0110
D23.6	D7	110 10111	111010 0110	000101 0110
D24.6	D8	110 11000	110011 0110	001100 0110
D25.6	D9	110 11001	100110 0110	100110 0110
D26.6	DA	110 11010	010110 0110	010110 0110
D27.6	DB	110 11011	110110 0110	001001 0110
D28.6	DC	110 11100	001110 0110	001110 0110
D29.6	DD	110 11101	101110 0110	010001 0110
D30.6	DE	110 11110	011110 0110	100001 0110
D31.6	DF	110 11111	101011 0110	010100 0110
D0.7	E0	111 00000	100111 0001	011000 1110
D1.7	E1	111 00001	011101 0001	100010 1110
D2.7	E2	111 00010	101101 0001	010010 1110
D3.7	E3	111 00011	110001 1110	110001 0001
D4.7	E4	111 00100	110101 0001	001010 1110
D5.7	E5	111 00101	101001 1110	101001 0001
D6.7	E6	111 00110	011001 1110	011001 0001
D7.7	E7	111 00111	111000 1110	000111 0001
D8.7	E8	111 01000	111001 0001	000110 1110
D9.7	E9	111 01001	100101 1110	100101 0001

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD-abcdei fghj (binary)	Current RD+abcdei fghj (binary)
D10.7	EA	111 01010	010101 1110	010101 0001
D11.7	EB	111 01011	110100 1110	110100 1000
D12.7	EC	111 01100	001101 1110	001101 0001
D13.7	ED	111 01101	101100 1110	101100 1000
D14.7	EE	111 01110	011100 1110	011100 1000
D15.7	EF	111 01111	010111 0001	101000 1110
D16.7	F0	111 10000	011011 0001	100100 1110
D17.7	F1	111 10001	100011 0111	100011 0001
D18.7	F2	111 10010	010011 0111	010011 0001
D19.7	F3	111 10011	110010 1110	110010 0001
D20.7	F4	111 10100	001011 0111	001011 0001
D21.7	F5	111 10101	101010 1110	101010 0001
D22.7	F6	111 10110	011010 1110	011010 0001
D23.7	F7	111 10111	111010 0001	000101 1110
D24.7	F8	111 11000	110011 0001	001100 1110
D25.7	F9	111 11001	100110 1110	100110 0001
D26.7	FA	111 11010	010110 1110	010110 0001
D27.7	FB	111 11011	110110 0001	001001 1110
D28.7	FC	111 11100	001110 1110	001110 0001
D29.7	FD	111 11101	101110 0001	010001 1110
D30.7	FE	111 11110	011110 0001	100001 1110
D31.7	FF	111 11111	101011 0001	010100 1110

Table B-2 8b/10b Special Character Symbol Codes ^{\$}

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD-abcdei fghj (binary)	Current RD+abcdei fghj (binary)
<u>K28.0</u>	1C	000 11100	001111 0100	110000 1011
<u>K28.1</u>	3C	001 11100	001111 1001	110000 0110
<u>K28.2</u>	5C	010 11100	001111 0101	110000 1010
<u>K28.3</u>	7C	011 11100	001111 0011	110000 1100
<u>K28.4</u>	9C	100 11100	001111 0010	110000 1101

Data Byte Name	Data Byte Value (hex)	Bits HGF EDCBA (binary)	Current RD- abcdei fghj (binary)	Current RD+ abcdei fghj (binary)
<u>K28.5</u>	BC	101 11100	001111 1010	110000 0101
<u>K28.6</u>	DC	110 11100	001111 0110	110000 1001
<u>K28.7</u>	FC	111 11100	001111 1000	110000 0111
<u>K23.7</u>	F7	111 10111	111010 1000	000101 0111
<u>K27.7</u>	FB	111 11011	110110 1000	001001 0111
<u>K29.7</u>	FD	111 11101	101110 1000	010001 0111
<u>K30.7</u>	FE	111 11110	011110 1000	100001 0111

C. Physical Layer Appendix §

C.1 8b/10b Data Scrambling Example §

The following subroutines encode and decode an 8-bit value contained in “inbyte” with the LFSR. This is presented as one example only; there are many ways to obtain the proper output. This example demonstrates how to advance the LFSR eight times in one operation and how to XOR the data in one operation. Many other implementations are possible but they must all produce the same output as that shown here.

The following algorithm uses the “C” programming language conventions, where “<<” and “>>” represent the shift left and shift right operators, “>” is the compare greater than operator, and “^” is the exclusive or operator, and “&” is the logical “AND” operator.

```
/*
  this routine implements the serial descrambling algorithm in parallel form
  for the LSFR polynomial:  $x^{16}+x^5+x^4+x^3+1$ 
  this advances the LSFR 8 bits every time it is called
  this requires fewer than 25 xor gates to implement (with a static register)
  The XOR required to advance 8 bits/clock is:
bit 0  1  2  3  4  5  6  7  8  9 10 11 12 13 14 15
      8  9 10 11 12 13 14 15 0  1  2  3  4  5  6  7
          8  9 10 11 12 13 14 15
              8  9 10 11 12 13 14 15
                  8  9 10 11 12 13 14 15
  The serial data is just the reverse of the upper byte:
bit 0  1  2  3  4  5  6  7
    15 14 13 12 11 10 9  8
*/
```

```

int scramble_byte(int inbyte)
{
    static int scrambit[16];
    static int bit[16];
    static int bit_out[16];
    static unsigned short lfsr = 0xffff; // 16 bit short for polynomial
    int i, outbyte;

    if (inbyte == COMMA)           // if this is a comma
    {
        lfsr = 0xffff;           // reset the LFSR
        return (COMMA);          // and return the same data
    }

    if (inbyte == SKIP)           // don't advance or encode on skip
        return (SKIP);

    for (i=0; i<16;i++)           // convert LFSR to bit array for legibility
        bit[i] = (lfsr >> i) & 1;

        for (i=0; i<8; i++)       // convert byte to be scrambled for legibility
            scrambit[i] = (inbyte >> i) & 1;

        // apply the xor to the data
    if (! (inbyte & 0x100) &&      // if not a KCODE, scramble the data
        ! (TrainingSequence == TRUE)) // and if not in the middle of
    {                               // a training sequence
        scrambit[0] ^= bit[15];
        scrambit[1] ^= bit[14];
        scrambit[2] ^= bit[13];
        scrambit[3] ^= bit[12];
        scrambit[4] ^= bit[11];
        scrambit[5] ^= bit[10];
        scrambit[6] ^= bit[9];
        scrambit[7] ^= bit[8];
    }

    // Now advance the LFSR 8 serial clocks
    bit_out[ 0] = bit[ 8];
    bit_out[ 1] = bit[ 9];
    bit_out[ 2] = bit[10];
    bit_out[ 3] = bit[11] ^ bit[ 8];
    bit_out[ 4] = bit[12] ^ bit[ 9] ^ bit[ 8];
    bit_out[ 5] = bit[13] ^ bit[10] ^ bit[ 9] ^ bit[ 8];
    bit_out[ 6] = bit[14] ^ bit[11] ^ bit[10] ^ bit[ 9];
    bit_out[ 7] = bit[15] ^ bit[12] ^ bit[11] ^ bit[10];
    bit_out[ 8] = bit[ 0] ^ bit[13] ^ bit[12] ^ bit[11];
    bit_out[ 9] = bit[ 1] ^ bit[14] ^ bit[13] ^ bit[12];
    bit_out[10] = bit[ 2] ^ bit[15] ^ bit[14] ^ bit[13];
    bit_out[11] = bit[ 3] ^ bit[15] ^ bit[14];
    bit_out[12] = bit[ 4] ^ bit[15];
    bit_out[13] = bit[ 5];
    bit_out[14] = bit[ 6];
    bit_out[15] = bit[ 7];

    lfsr = 0;
    for (i=0; i <16; i++) // convert the LFSR back to an integer
        lfsr += (bit_out[i] << i);

```

```
    outbyte = 0;
    for (i= 0; i<8; i++)      // convert data back to an integer
        outbyte += (scrambit[i] << i);

    return outbyte;
}
```

```

/* NOTE THAT THE DESCramBLE ROUTINE IS IDENTICAL TO THE SCramBLE ROUTINE
   this routine implements the serial descrambling algorithm in parallel form
   this advances the lfsr 8 bits every time it is called
   this uses fewer than 25 xor gates to implement (with a static register)
   The XOR tree is the same as the scrambling routine
*/

int unscramble_byte(int inbyte)
{
    static int descrambit[8];
    static int bit[16];
    static int bit_out[16];
    static unsigned short lfsr = 0xffff; // 16 bit short for polynomial
    int outbyte, i;

    if (inbyte == COMMA)    // if this is a comma
    {
        lfsr = 0xffff;      // reset the LFSR
        return (COMMA);     // and return the same data
    }
    if (inbyte == SKIP)     // don't advance or encode on skip
        return (SKIP);

    for (i=0; i<16;i++)    // convert the LFSR to bit array for legibility
        bit[i] = (lfsr >> i) & 1;

    for (i=0; i<8; i++)    // convert byte to be de-scrambled for legibility
        descrambit[i] = (inbyte >> i) & 1;

    // apply the xor to the data
    if (! (inbyte & 0x100) && // if not a KCODE, scramble the data
        ! (TrainingSequence == TRUE)) // and if not in the middle of
    {                               // a training sequence
        descrambit[0] ^= bit[15];
        descrambit[1] ^= bit[14];
        descrambit[2] ^= bit[13];
        descrambit[3] ^= bit[12];
        descrambit[4] ^= bit[11];
        descrambit[5] ^= bit[10];
        descrambit[6] ^= bit[9];
        descrambit[7] ^= bit[8];
    }

    // Now advance the LFSR 8 serial clocks
    bit_out[ 0] = bit[ 8];
    bit_out[ 1] = bit[ 9];
    bit_out[ 2] = bit[10];
    bit_out[ 3] = bit[11] ^ bit[ 8];
    bit_out[ 4] = bit[12] ^ bit[ 9] ^ bit[ 8];
    bit_out[ 5] = bit[13] ^ bit[10] ^ bit[ 9] ^ bit[ 8];
    bit_out[ 6] = bit[14] ^ bit[11] ^ bit[10] ^ bit[ 9];
    bit_out[ 7] = bit[15] ^ bit[12] ^ bit[11] ^ bit[10];
    bit_out[ 8] = bit[ 0] ^ bit[13] ^ bit[12] ^ bit[11];
    bit_out[ 9] = bit[ 1] ^ bit[14] ^ bit[13] ^ bit[12];
    bit_out[10] = bit[ 2] ^ bit[15] ^ bit[14] ^ bit[13];
    bit_out[11] = bit[ 3] ^ bit[15] ^ bit[14];
    bit_out[12] = bit[ 4] ^ bit[15];
    bit_out[13] = bit[ 5];
    bit_out[14] = bit[ 6];
    bit_out[15] = bit[ 7];

    lfsr = 0;
    for (i=0; i <16; i++)    // convert the LFSR back to an integer
        lfsr += (bit_out[i] << i);

    outbyte = 0;

    for (i= 0; i<8; i++)    // convert data back to an integer
        outbyte += (descrambit[i] << i);

    return outbyte;
}

```

The initial 16-bit values of the LFSR for the first 128 LFSR advances following a reset are listed below:

	0, 8	1, 9	2, A	3, B	4, C	5, D	6, E	7, F
00	FFFF	E817	0328	284B	4DE8	E755	404F	4140
08	4E79	761E	1466	6574	7DBD	B6E5	FDA6	B165
10	7D09	02E5	E572	673D	34CF	CB54	4743	4DEF
18	E055	40E0	EE40	54BE	B334	2C7B	7D0C	07E5
20	E5AF	BA3D	248A	8DC4	D995	85A1	BD5D	4425
28	2BA4	A2A3	B8D2	CBF8	EB43	5763	6E7F	773E
30	345F	5B54	5853	5F18	14B7	B474	6CD4	DC4C
38	5C7C	70FC	F6F0	E6E6	F376	603B	3260	64C2
40	CB84	9743	5CBF	B3FC	E47B	6E04	0C3E	3F2C
48	29D7	D1D1	C069	7BC0	CB73	6043	4A60	6FFA
50	F207	1102	01A9	A939	2351	566B	6646	4FF6
58	F927	3081	85B0	AC5D	478C	82EF	F3F2	E43B
60	2E04	027E	7E72	79AE	A501	1A7D	7F2A	2197
68	9019	0610	1096	9590	8FCD	D0E7	F650	46E6
70	E8D6	C228	3AB2	B70A	129F	9CE2	FC3C	2B5C
78	5AA3	AF6A	70C7	CDF0	E3D5	C0AB	B9C0	D9C1

An 8-bit value of 0 repeatedly encoded with the LFSR after reset produces the following consecutive 8-bit values:

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00	FF	17	C0	14	B2	E7	02	82	72	6E	28	A6	BE	6D	BF	8D
10	BE	40	A7	E6	2C	D3	E2	B2	07	02	77	2A	CD	34	BE	E0
20	A7	5D	24	B1	9B	A1	BD	22	D4	45	1D	D3	D7	EA	76	EE
30	2C	DA	1A	FA	28	2D	36	3B	3A	0E	6F	67	CF	06	4C	26
40	D3	E9	3A	CD	27	76	30	FC	94	8B	03	DE	D3	06	52	F6
50	4F	88	80	95	C4	6A	66	F2	9F	0C	A1	35	E2	41	CF	27
60	74	40	7E	9E	A5	58	FE	84	09	60	08	A9	F1	0B	6F	62
70	17	43	5C	ED	48	39	3F	D4	5A	F5	0E	B3	C7	03	9D	9B
80	8B	0D	8E	5C	33	98	77	AE	2D	AC	0B	3E	DA	0B	42	7A
90	7C	D1	CF	A8	1C	12	EE	41	C2	3F	38	7A	0D	69	F4	01
A0	DA	31	72	C5	A0	D7	93	0E	DC	AF	A4	55	E7	F0	72	16
B0	68	D5	38	84	DD	00	CD	18	9E	CA	30	59	4C	75	1B	77

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
C0	31	C5	ED	CF	91	64	6E	3D	FE	E8	29	04	CF	6C	FC	C4
D0	0B	5E	DA	62	BA	5B	AB	DF	59	B7	7D	37	5E	E3	1A	C6
E0	88	14	F5	4F	8B	C8	56	CB	D3	10	42	63	04	8A	B4	F7
F0	84	01	A0	01	83	49	67	EE	3E	2A	8B	A4	76	AF	14	D5
100	4F	AC	60	B6	79	D6	62	B7	43	E7	E5	2A	40	2C	6E	7A
110	56	61	63	20	6A	97	4A	38	05	E5	DD	68	0D	78	4C	53
120	8B	D6	86	57	B2	AA	1A	80	18	DC	BA	FC	03	A3	4B	30

At 2.5 GT/s, scrambling produces the power spectrum (in the 10-bit domain) shown in § Figure C-1.

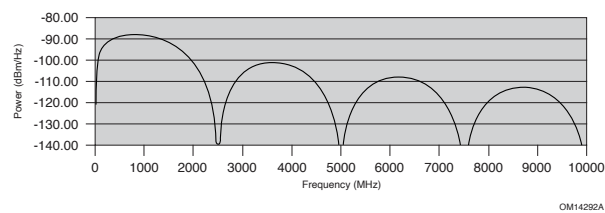


Figure C-1 Scrambling Spectrum at 2.5 GT/s for Data Value of 0 §

C.2 128b/130b Data Scrambling Example §

The following subroutines illustrate how calculate the next value of the 128b/130b scrambling LFSR and how to scramble (or descramble) an 8-bit value, both given the current value of the LFSR.

This is presented as one example only; there are many ways to obtain the proper output. This example demonstrates how to advance the LFSR eight times in one operation and how to XOR the data in one operation. Many other implementations are possible but they must all produce the same output as that shown here.

The following algorithm uses the “C” programming language conventions, where “<<” and “>>” represent the shift left and shift right operators, “^” is the exclusive or operator, and “|=” is the assignment by bitwise or operator.

```

#include <stdio.h>

// "Set Bit" – Sets bit number "bit" of "var" to the value "val". Bit "bit" of "var" must start cleared.
#define SB(var,bit,val) var |= (val & 1) << bit

// "Get Bit" – Returns the value of bit number "bit" of "var".
#define GB(var,bit) ((var >> bit) & 1)

// Function to advance the LFSR value by 8 bits, given the current LFSR value
unsigned long int calc_next_lfsr(unsigned long int lfsr) {
    unsigned long int next_lfsr = 0;
    SB(next_lfsr,22, GB(lfsr,14) ^ GB(lfsr,16) ^ GB(lfsr,18) ^ GB(lfsr,20) ^ GB(lfsr,21) ^ GB(lfsr,22));
    SB(next_lfsr,21, GB(lfsr,13) ^ GB(lfsr,15) ^ GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,20) ^ GB(lfsr,21));
    SB(next_lfsr,20, GB(lfsr,12) ^ GB(lfsr,19) ^ GB(lfsr,21));
    SB(next_lfsr,19, GB(lfsr,11) ^ GB(lfsr,18) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(next_lfsr,18, GB(lfsr,10) ^ GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,21));
    SB(next_lfsr,17, GB(lfsr,9) ^ GB(lfsr,16) ^ GB(lfsr,18) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(next_lfsr,16, GB(lfsr,8) ^ GB(lfsr,15) ^ GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,21) ^ GB(lfsr,22));
    SB(next_lfsr,15, GB(lfsr,7) ^ GB(lfsr,22));
    SB(next_lfsr,14, GB(lfsr,6) ^ GB(lfsr,21));
    SB(next_lfsr,13, GB(lfsr,5) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(next_lfsr,12, GB(lfsr,4) ^ GB(lfsr,19) ^ GB(lfsr,21) ^ GB(lfsr,22));
    SB(next_lfsr,11, GB(lfsr,3) ^ GB(lfsr,18) ^ GB(lfsr,20) ^ GB(lfsr,21) ^ GB(lfsr,22));
    SB(next_lfsr,10, GB(lfsr,2) ^ GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,20) ^ GB(lfsr,21) ^ GB(lfsr,22));
    SB(next_lfsr,9, GB(lfsr,1) ^ GB(lfsr,16) ^ GB(lfsr,18) ^ GB(lfsr,19) ^ GB(lfsr,20) ^ GB(lfsr,21));
    SB(next_lfsr,8, GB(lfsr,0) ^ GB(lfsr,15) ^ GB(lfsr,17) ^ GB(lfsr,18) ^ GB(lfsr,19) ^ GB(lfsr,20));
    SB(next_lfsr,7, GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,20) ^ GB(lfsr,21));
    SB(next_lfsr,6, GB(lfsr,16) ^ GB(lfsr,18) ^ GB(lfsr,19) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(next_lfsr,5, GB(lfsr,15) ^ GB(lfsr,17) ^ GB(lfsr,18) ^ GB(lfsr,19) ^ GB(lfsr,21) ^ GB(lfsr,22));
    SB(next_lfsr,4, GB(lfsr,17));
    SB(next_lfsr,3, GB(lfsr,16));
    SB(next_lfsr,2, GB(lfsr,15) ^ GB(lfsr,22));
    SB(next_lfsr,1, GB(lfsr,16) ^ GB(lfsr,18) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(next_lfsr,0, GB(lfsr,15) ^ GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,21) ^ GB(lfsr,22));
    return(next_lfsr);
}

// Function to scramble a byte, given the current LFSR value
unsigned char scramble_data(unsigned long lfsr, unsigned char data_in) {
    unsigned char data_out = 0;
    SB(data_out,7, GB(data_in,7) ^ GB(lfsr,15) ^ GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,21) ^ GB(lfsr,22));
    SB(data_out,6, GB(data_in,6) ^ GB(lfsr,16) ^ GB(lfsr,18) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(data_out,5, GB(data_in,5) ^ GB(lfsr,17) ^ GB(lfsr,19) ^ GB(lfsr,21));
    SB(data_out,4, GB(data_in,4) ^ GB(lfsr,18) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(data_out,3, GB(data_in,3) ^ GB(lfsr,19) ^ GB(lfsr,21));
    SB(data_out,2, GB(data_in,2) ^ GB(lfsr,20) ^ GB(lfsr,22));
    SB(data_out,1, GB(data_in,1) ^ GB(lfsr,21));
    SB(data_out,0, GB(data_in,0) ^ GB(lfsr,22));
    return(data_out);
}

// Example of LFSR and scrambled data "0" sequence for Lane 0
main() {
    unsigned long lfsr = 0x1DBFBC; // Lane 0 reset LFSR value
    unsigned char unscrambled_data = 0x00;
    int i;
    printf("Iteration LFSR Next LFSR Scrambled Data\n");
    printf("- - - -\n");
    for (i = 0; i < 128; i++) {
        unsigned char scrambled_data = scramble_data(lfsr,unscrambled_data);
        unsigned long next_lfsr = calc_next_lfsr(lfsr);
        printf("%3d %06X %06X %02X\n", i, lfsr, next_lfsr, scrambled_data);
        lfsr = next_lfsr;
    }
}

```

The initial 23-bit values of the LFSR for the first 128 LFSR advances for an 8-bit quantity following a reset for Lane 0 are listed below (ordered left to right, top to bottom):

	0, 8	1, 9	2, A	3, B	4, C	5, D	6, E	7, F
00	1DBFBC	498C2E	1186E9	0FC5AD	7CB75D	3D8DA2	0ECC8F	379717
08	046BDF	21D462	423FCE	5177D6	1447EF	67CBA0	794F7A	6CA2AF

	0, 8	1, 9	2, A	3, B	4, C	5, D	6, E	7, F
10	425060	3ED9D6	3DBF74	3C1A8F	7AE2E0	073E71	137D99	70B139
18	44F521	559720	1FBBA8	689D9F	376B02	597FFA	297C0E	7E450A
20	4BDE36	669B58	6BB530	78C3F9	0322C0	45C7FB	254F6A	323D89
28	07FDD2	71DFBC	68726B	799E27	1CFE8A	4AB864	42CB12	04AAF3
30	41F947	51F808	3A98CA	36A816	796895	4B4DAF	54037D	68E5C7
38	4F3302	60A51F	3AFCE3	528116	248131	1F65E6	17D2BA	34987E
40	6C0524	44DA45	7AF320	16FE71	5A5134	60B5F5	2A16E3	73AFF1
48	3D3ADA	18B5AA	4B9306	2BAB58	2D179E	5FDE68	46E1F8	644B91
50	3F78A4	7FCE1B	23CC59	7F017F	4DA97C	7EDF8B	705E13	0ADE04
58	6F1775	318CBE	70CC0C	39C021	0944ED	33F8AB	21DCBD	4AE0CE
60	1A6112	1B2F92	17ADD8	4BFA7E	42D358	1CE0F3	54C164	0BFDE2
68	0EF33F	082717	1200E1	4FCB73	39D53A	1C5FED	4ADE41	24EE12
70	7046E6	122B04	642E73	5A9AA4	0A24D0	34C250	362B24	5B5BB0
78	2833BF	73F640	648BDA	5E3281	490B97	373ECC	0CB1FA	6FE7A6

An 8-bit value of 0x00 repeatedly encoded with the LFSR after reset for Lane 0 produces the following consecutive 8-bit values (ordered left to right, top to bottom):

	00	01	02	03	04	05	06	07	08	09	0A	0B	0C	0D	0E	0F
00	6C	BD	94	98	53	C6	D8	CE	50	6A	75	C1	04	4F	C3	07
10	75	26	C6	06	A3	B0	B4	AB	05	11	CC	57	4E	69	42	73
20	1D	0F	B7	03	E0	45	BA	5E	30	EB	D7	43	2C	5D	F5	D0
30	15	41	76	8E	C3	9D	D1	57	CD	FF	76	A1	7A	4C	64	2E
40	87	05	A3	24	89	FF	A2	4B	46	7C	1D	62	12	19	A5	2F
50	E6	B3	CA	33	ED	F3	2B	88	67	3E	AB	96	E8	9E	6A	5D
60	5C	1C	64	1D	F5	2C	51	C8	D8	A8	F4	4D	96	AC	5D	7A
70	2B	F4	2F	09	08	2E	0E	C9	02	4B	AF	D9	3D	4E	78	E7

D. Request Dependencies §

This specification does not specify the rules governing the creation of resource dependencies between TLPs using different Traffic Classes. Dependencies between packets in different Traffic Classes can create potential deadlock if devices make different assumptions about what is allowed and what is not. Dependencies can be created when a packet is forwarded (transmitted verbatim) or translated (transmitted with modification) from an input Port to an output Port.

Resource dependencies are created when received packets are forwarded/translated on the same or a different Link. Due to the fact that the forwarding/translating device has finite buffer resources this behavior creates a dependency between the ability to receive a packet and the ability to transmit a packet (in potentially a different VC or sub-channel).

The following notation is used to create a framework to enumerate the possibilities:

$X(m) \rightarrow Y(n)$

This means:

- a request in sub-channel $X(TC = m)$ is forwarded/translated in sub-channel $Y(TC = n)$.
- n and m are between 0-7.
- X and Y are either **P** (Posted Request), **N** (Non-Posted Request), or **C** (Completion).

The list of possible dependencies is:

$P(m) \rightarrow P(n)$

$P(m) \rightarrow N(n)$

$P(m) \rightarrow C(n)$

$N(m) \rightarrow P(n)$

$N(m) \rightarrow N(n)$

$N(m) \rightarrow C(n)$

$C(m) \rightarrow P(n)$

$C(m) \rightarrow N(n)$

$C(m) \rightarrow C(n)$

For a given system, each of these dependencies needs to be classified as legal or illegal for each of the following cases:

- Root Port forwarding to own Link.
- Root Port forwarding to different Root Port's Link.
- Endpoint or Bridge forwarding to own Link.

A Switch is not allowed to modify the TLPs that flow through it, but must ensure complete independence of resources assigned to separate VCs. System software must comprehend the system dependency rules when configuring TC/VC mapping throughout the system.

One possible legal mapping is:

	RC (Same Port)	RC (Different Port)	Endpoint
$P(m) \rightarrow P(n)$	$m \leq n$	$m \leq n$	$m < n$
$P(m) \rightarrow N(n)$	$m < n$	$m < n$	$m < n$
$P(m) \rightarrow C(n)$	illegal	illegal	illegal
$N(m) \rightarrow P(n)$	$m < n$	$m < n$	$m < n$

	RC (Same Port)	RC (Different Port)	Endpoint
$N(m) \rightarrow N(n)$	$m \leq n$	$m \leq n$	$m < n$
$N(m) \rightarrow C(n)$	$m = n$	$m = n$	$m = n$
$C(m) \rightarrow P(n)$	illegal	illegal	illegal
$C(m) \rightarrow N(n)$	illegal	illegal	illegal
$C(m) \rightarrow C(n)$	$m \geq n$	$m \geq n$	$m > n$

Note that this discussion only deals with avoiding the deadlock caused by the creation of resource dependencies. It does not deal with the additional livelock issues (or lack of forward progress) caused by the system's Virtual Channel arbitration policies.

Some of these potential dependencies are illegal or unreachable:

- $P(m) \rightarrow P(n), N(m) \rightarrow N(n)$
 - $m = n$ - This case is illegal and will lead to deadlock, except when a Request is being forwarded from one Port to another of a Switch or Root Complex.
 - $m \neq n$ - See discussion below.
- $P(m) \rightarrow N(n)$
 - $m = n$ - This case is illegal and will lead to deadlock.
 - $m \neq n$ - See discussion below.
- $N(m) \rightarrow P(n)$ - See discussion below.
- $P(m) \rightarrow C(n)$ - This case is illegal and will lead to deadlock.
- $N(m) \rightarrow C(n)$
 - $m = n$ - This case occurs during the normal servicing of a non-posted request by either a root complex or an endpoint.
 - $m \neq n$ - This case is unreachable and should never happen. Completions always use the same TC as the corresponding request.
- $C(m) \rightarrow P(n), C(m) \rightarrow N(n)$ - These cases are unreachable and should never happen due to the fact that completion buffers must be preallocated.
- $C(m) \rightarrow C(n)$
 - $m = n$ - This case is illegal and will lead to deadlock, except when a Completion is being forwarded from one Port to another of a Switch or Root Complex.
 - $m \neq n$ - This case will occur if $N(n) \rightarrow N(m)$ dependencies are allowed.

Other potential dependencies may be legal when comprehended as part of a specific usage model. For example, these cases:

$P(m) \rightarrow P(n), N(m) \rightarrow N(n), P(m) \rightarrow N(n), N(m) \rightarrow P(n)$ where $m \neq n$

might exist where a device services incoming requests by issuing modified versions of those requests using a different Traffic Class (for example, remapping the first requests address and generating the new request with the resulting address). In these cases, suitable rules must be applied to prevent circular dependencies that would lead to deadlock or livelock.

Examples of devices that may find the above mappings useful:

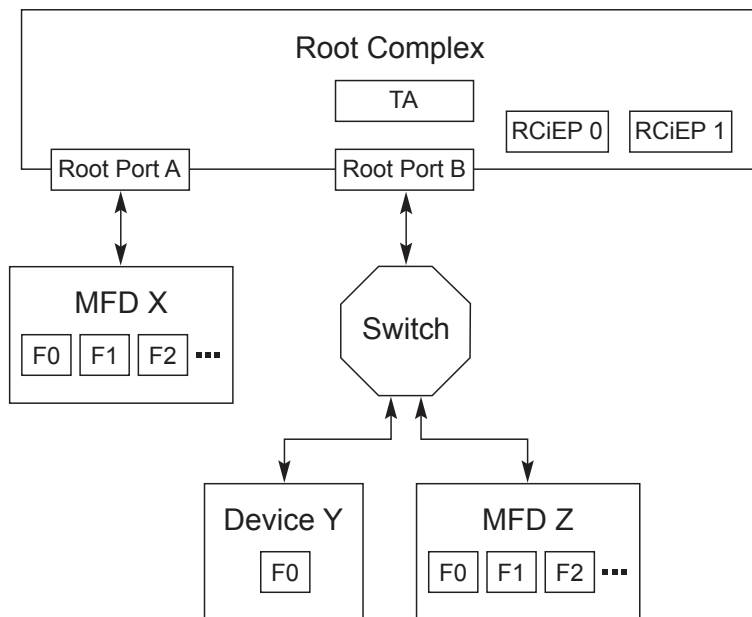
- Bridges to complex protocols that require state to be save/restored to/from host memory, i.e., PCI Express to Infiniband bridges.
- Messaging engines that must do address translation based upon page tables stored in host memory.
- UMA graphics devices that store their frame buffer in host memory.

E. ID-Based Ordering Usage §

E.1 Introduction §

ID-Based Ordering (IDO) is a mechanism that permits certain ordering restrictions to be relaxed as a means to improve performance. IDO permits certain TLPs to pass other TLPs in cases where otherwise such passing would be forbidden. The passing permitted by IDO is not required for proper operation (e.g., deadlock avoidance); it is only a means of improving performance.

For discussing IDO, it's useful to introduce the concept of a “TLP stream”, which is a set of TLPs that all have the same originator.²²³ For several important cases where TLP passing is normally forbidden, IDO permits such passing to occur if the TLPs belong to different TLP streams.



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Figure E-1 Reference Topology for IDO Use §

§ Figure E-1 shows a reference topology. The reference topology is not intended to discourage the use of IDO with other topologies, but rather to provide specific examples for discussion.

Devices X and Z are Multi-Function Devices (MFDs); device Y is a Single-Function Device. The RCiEPs are Root Complex Integrated Endpoint Functions, and might or might not be part of the same Device. We will assume that one or more Functions are using the Translation Agent (TA) in the Root Complex (RC).

Referring to the ordering table and descriptions in § Section 2.4.1, having the IDO bit set in a Posted Request, Non-Posted Request, or Completion TLP permits that TLP to pass a Posted Request TLP if the two TLPs belong to different TLP streams. In the following examples, DMAR and DMAW stand for Direct Memory Access Read and Write; PIOR and PIOW stand for Programmed I/O Read and Write.

223. That is, the Requester IDs of Requests and the Completer IDs of Completions are all the same.

E.2 Potential Benefits with IDO Use §

Here are some example potential benefits that are envisioned with IDO use. Generally, IDO provides the most benefit when multiple TLP streams share a common Link and that Link becomes congested, either due to high utilization or due to temporary lack of Flow Control (FC) credit.

E.2.1 Benefits for MFD/RP Direct Connect §

Here are some examples in the context of traffic between MFD X and the RC in § Figure E-1.

- Posted Request passing another Posted Request: when a DMAW from F0 is stalled due to an TA miss, if IDO is set for a DMAW from F1, it is permitted within the RC for this DMAW to pass the stalled DMAW from F0.
- Non-Posted Request passing a Posted Request: when a DMAW from F0 is stalled due to an TA miss, if IDO is set for a DMAR Request from F1, it is permitted within the RC for this DMAR Request to pass the stalled DMAW from F0.
- Completion passing a Posted Request: when a DMAW from F0 is stalled due to an TA miss, if IDO is set for a PIOR Completion from F1, it is permitted within the RC for this PIOR Completion to pass the stalled DMAW from F0.

E.2.2 Benefits for Switched Environments §

Here are some examples in the context of traffic within the Switch in § Figure E-1.

- Non-Posted Request passing a Posted Request: when a DMAW from Device Y is stalled within the Switch due to a lack of FC credit from Root Port B, if IDO is set for a DMAR Request from MFD Z, it is permitted within the Switch for this DMAR Request to pass the stalled DMAW from Device Y. The same also holds for a DMAR Request from one Function in MFD Z passing a stalled DMAW from a different Function in MFD Z.
- Completion passing a Posted Request: when a DMAW from Device Y is stalled within the Switch due to a lack of FC credit from Root Port B, if IDO is set for a PIOR Completion from MFD Z, it is permitted within the Switch for this PIOR Completion to pass the stalled DMAW from Device Y. The same also holds for a PIOR Completion from one Function in MFD Z passing a stalled DMAW from a different Function in MFD Z.
- Posted Request passing another Posted Request: within a Switch, there is little or no envisioned benefit from having a DMAW from one TLP stream passing a DMAW from a different TLP stream. However, it is not prohibited for Switches to implement such passing as permitted by IDO.

E.2.3 Benefits for Integrated Endpoints §

Here are some examples for the Root Complex Integrated Endpoints (RCiEPs) in § Figure E-1. The benefits are basically the same as for the MFD/RP Direct Connect case.

- Posted Request passing another Posted Request: when a DMAW from RCiEP 0 is stalled due to an TA miss, if IDO is set for a DMAW from RCiEP 1, it is permitted for this DMAW to pass the stalled DMAW from RCiEP 0.
- Non-Posted Request passing a Posted Request: when a DMAW from RCiEP 0 is stalled due to an TA miss, if IDO is set for a DMAR Request from RCiEP 1, it is permitted for this DMAR Request to pass the stalled DMAW from RCiEP 0.

- Completion passing a Posted Request: when a DMAW from RCiEP 0 is stalled due to an TA miss, if IDO is set for a PIOR Completion from RCiEP 1, it is permitted for this PIOR Completion to pass the stalled DMAW from RCiEP 0.

E.2.4 IDO Use in Conjunction with RO §

IDO and RO²²⁴ are orthogonal. Certain instances of passing; for example, a Posted Request passing another Posted Request, might be permitted by IDO, RO, or both at the same time. While IDO and RO have significant overlap for some cases, it is strongly recommended that both be used whenever safely possible. RO permits certain TLP passing within the same TLP stream, which is never permitted by IDO. For traffic in different TLP streams, IDO permits control traffic to pass any other traffic, and generally it is not safe to Set RO with control traffic.

E.3 When to Use IDO §

With Endpoint Functions²²⁵, it is safe to Set IDO in all applicable TLPs originated by the Endpoint when the Endpoint is directly communicating with only one other entity, most commonly the RC. For the RC case, “directly communicating” specifically includes DMA traffic, PIO traffic, and interrupt traffic; communicating with RCiEPs or communicating using P2P Root Port traffic constitutes communicating with multiple entities.

With a Root Port, there are no envisioned high-benefit use models where it is safe to Set IDO in all applicable TLPs that it originates. Use models where a Root Port Sets IDO in a subset of the applicable TLPs it originates are outside the scope of this specification.

E.4 When Not to Use IDO §

E.4.1 When Not to Use IDO with Endpoints §

With Endpoint Functions, it is not always safe to Set IDO in applicable TLPs it originates if the Endpoint directly communicates with multiple entities. It may be safe to Set IDO in some TLPs and not others, but such use models are outside the scope of this specification.

For example, in § Figure E-1 if Device Y and MFD Z are communicating with P2P traffic and also communicating via host memory, it is not always safe for them to Set IDO in the TLPs they originate. As an example failure case, let's assume that Device Y does a DMAW (to host memory) followed by a P2P Write to MFD Z. Upon observing the P2P Write, let's assume that MFD Z then does a DMAW to the same location earlier targeted by the DMAW from Device Y. Normal ordering rules would guarantee that the DMAW from Device Y would be observed by host memory before the DMAW from MFD Z. However, if IDO is set in the DMAW from MFD Z, the RC would be permitted to have the second DMAW pass the first, causing a different end result in host memory contents.

Synchronization techniques like performing zero-length Reads might be used to avoid such communication failures when IDO is used, but specific use models are outside the scope of this specification.

224. In this Appendix, “RO” is an abbreviation for the Relaxed Ordering Attribute field.

225. Endpoint Functions include PCI Express Endpoints, Legacy PCI Express Endpoints, and Root Complex Integrated Endpoints.

E.4.2 When Not to Use IDO with Root Ports §

With Root Ports, it is not always safe to Set IDO in applicable TLPs it originates if Endpoint Functions in the hierarchy do any P2P traffic. It may be safe to Set IDO in some TLPs and not others, but such use models are outside the scope of this specification.

As an example, in § Figure E-1 if Device Y and MFD Z are communicating with P2P traffic and also communicating with host software, it is not always safe for Root Port B to Set IDO in the TLPs it originates. For example, let's assume that Device Y does a P2P Write to MFD Z followed by a DMAW (to host memory). Upon observing the DMAW, let's assume that the host does a PIO to MFD Z. Normal ordering rules would guarantee that the P2P Write from Device Y would be observed by MFD Z before the PIO from the host. However, if IDO is set in the PIO from the host, the Switch would be permitted to have the PIO pass the P2P Write, ultimately having the two Writes arrive at MFD Z out of order.

IMPLEMENTATION NOTE: REQUESTER AND COMPLETER IDS FOR RC-ORIGINATED TLPs §

With RC implementations where the Requester ID in a PIO Request does not match the Completer ID in a DMAR Completion, this enables another potential communication failure case if IDO is Set in the Completion. For this case, if a PIO is followed by a DMAR Completion with IDO Set, a Switch below the Root Port could permit the DMAR Completion to pass the PIO, violating the normal ordering rule that a non-RO Read Completion must not pass Posted Requests. The PIO and DMAR Completion would appear to belong to different TLP streams, though logically they belong to the same TLP stream. Special caution is advised in setting IDO with TLPs originating from such RCs.

E.5 Software Control of IDO Use §

E.5.1 Software Control of Endpoint IDO Use §

By default, Endpoints are not enabled to Set IDO in any TLPs they originate.

IMPLEMENTATION NOTE: THE “SIMPLE” POLICY FOR IDO USE §

It is envisioned that Endpoints designed primarily to communicate directly with only one other entity (e.g., the RC) may find a “simple” policy for setting IDO to be adequate. Here's the envisioned “simple” policy. If the IDO Request Enable bit is Set, the Endpoint Sets IDO in all applicable Request TLPs that it originates. If the IDO Completion Enable bit is Set, the Endpoint Sets IDO in all Completion TLPs that it originates.

It is envisioned that a software driver associated with each Endpoint will determine when it is safe for the Endpoint to set IDO in applicable TLPs it originates. A driver should be able to determine if the Endpoint is communicating with multiple other entities, and should know the Endpoint's capabilities as far as setting IDO with all applicable TLPs when enabled, versus setting IDO selectively. If a driver determines that it is safe to enable the setting of IDO, the driver can set the IDO Request Enable and/or IDO Completion Enable bits either indirectly via OS services or directly, subject to OS policy.

If an Endpoint is designed for communication models where it is not safe to utilize the “simple” policy for IDO use, the Endpoint can implement more complex policies for determining when the Endpoint sets the IDO bit. Such implementations might utilize device-specific controls that are managed by the device driver. Such policies and device-specific control mechanisms are outside the scope of this specification.

E.5.2 Software Control of Root Port IDO Use §

Since there are no envisioned high-benefit “simple” use models for Root Ports setting the IDO bit with TLPs they originate, and there are known communication failure cases if Root Ports set the IDO bit with all applicable TLPs they originate, it is anticipated that Root Ports will rarely be enabled to set IDO in TLPs they originate. Such use models and policies for Root Ports setting IDO are outside the scope of this specification.

F. Message Code Usage §

§ Table F-1 contains a list of currently defined PCI Express Message Codes. Message codes are defined in this specification and in other specifications. This table will be updated as Messages are defined in other specifications but due to document release schedules, this table might not contain recently defined Messages.

Table F-1 Message Code Usage §

Message Code	Routing r[2:0]	Type	Description
0000 0000	011	Msg	Unlock, see § Section 2.2.8.4
0000 0001	010	MsgD	Invalidate Request Message, see § Section 10.3.1
0000 0010	010	Msg	Invalidate Completion Message, see § Section 10.3.2
0000 0100	000	Msg	Page Request Message, see § Section 10.4.1
0000 0101	010	Msg	PRG Response Message, see § Section 10.4.2
0001 0000	100	Msg	Latency Tolerance Reporting (LTR) Message, see § Section 2.2.8.8
0001 0010	100	Msg	Optimized Buffer Flush/Fill (OBFF) Message, see § Section 2.2.8.9
0001 0100	100	Msg	PM_Active_State_Nak, see § Section 2.2.8.2
0001 1000	000	Msg	PM_PME, see § Section 2.2.8.2
0001 1001	011	Msg	PME_Turn_Off, see § Section 2.2.8.2
0001 1011	101	Msg	PME_TO_Ack, see § Section 2.2.8.2
0010 0000	100	Msg	Assert_INTA, see § Section 2.2.8.1
0010 0001	100	Msg	Assert_INTB, see § Section 2.2.8.1
0010 0010	100	Msg	Assert_INTC, see § Section 2.2.8.1
0010 0011	100	Msg	Assert_INTD, see § Section 2.2.8.1
0010 0100	100	Msg	Deassert_INTA, see § Section 2.2.8.1
0010 0101	100	Msg	Deassert_INTB, see § Section 2.2.8.1
0010 0110	100	Msg	Deassert_INTC, see § Section 2.2.8.1
0010 0111	100	Msg	Deassert_INTD, see § Section 2.2.8.1
0011 0000	000	Msg	ERR_COR, see § Section 2.2.8.3
0011 0001	000	Msg	ERR_NONFATAL, see § Section 2.2.8.3
0011 0011	000	Msg	ERR_FATAL, see § Section 2.2.8.3
0100 0000	100	Msg	Ignored Message, see § Section 2.2.8.7
0100 0001	100	Msg	Ignored Message, see § Section 2.2.8.7
0100 0011	100	Msg	Ignored Message, see § Section 2.2.8.7

Message Code	Routing r[2:0]	Type	Description
0100 0100	100	Msg	Ignored Message, see § Section 2.2.8.7
0100 0101	100	Msg	Ignored Message, see § Section 2.2.8.7
0100 0111	100	Msg	Ignored Message, see § Section 2.2.8.7
0100 1000	100	Msg	Ignored Message, see § Section 2.2.8.7
0101 0000	100	MsgD	Set_Slot_Power_Limit, see § Section 2.2.8.5
0101 0010	100	Msg	PTM Request, see § Section 2.2.8.10
0101 0011	100	Msg/MsgD	PTM Response/PTM ResponseD, see § Section 2.2.8.10
0111 1110	000, 010, 011, or 100	Msg/MsgD	Vendor-Defined Type 0, see § Section 2.2.8.6
0111 1111	000, 010, 011, or 100	Msg/MsgD	Vendor-Defined Type 1, see § Section 2.2.8.6
0101 0100	010 / 100	Msg	IDE Sync (see § Section 2.2.8.11)
0101 0101	010 / 100	Msg	IDE Fail (see § Section 2.2.8.11)

§ Table F-2 contains a list of currently defined Subtype codes for PCI-SIG-Defined VDMs (see § Section 2.2.8.6.1). Subtype codes are defined in this specification and in other specifications. This table will be updated as Subtype codes are defined in other specifications but due to document release schedules, this table might not contain recently defined Subtypes.

Table F-2 PCI-SIG-Defined VDM Subtype Usage §

Subtype	Routing r[2:0]	Type	Description
0000 0000	010 or 011	MsgD	Deprecated; formerly used for LN Message (Lightweight Notification)
0000 0001	011	MsgD	Hierarchy ID Message, See § Section 2.2.8.6.4 and § Section 6.25
0000 1000	100	Msg	Device Readiness Status, see § Section 2.2.8.6.2
0000 1001	000	Msg	Function Readiness Status, see § Section 2.2.8.6.3

G. Protocol Multiplexing §

The Protocol Multiplexing mechanism provides a standard mechanism to transport non-PCI Express protocols across a PCI Express Link. The mechanism supports the multiplexing of PMUX Packets and TLPs onto a single PCI Express Link.

An example system topology using Protocol Multiplexing is shown in § Figure G-1. In this example, the Link may operate in two modes:

- PCI Express Link. Protocol Multiplexing is disabled.
- PMUX Link. Protocol Multiplexing is enabled. Both TLPs and PMUX Packets are used in a coordinated fashion. PMUX Packets may be used to support additional protocols efficiently.

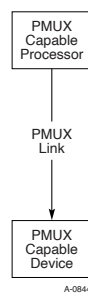


Figure G-1 Device and Processor Connected Using a PMUX Link §

A PMUX Link is shown in § Figure G-2. Arbitration and encapsulation occurs between the transmit queues and the Link. Demultiplexing and decapsulation occurs between the Link and the various receive queues. Packets are sent from transmit queues to the corresponding receive queues. Packets are identified as either PMUX Packets or TLPs.

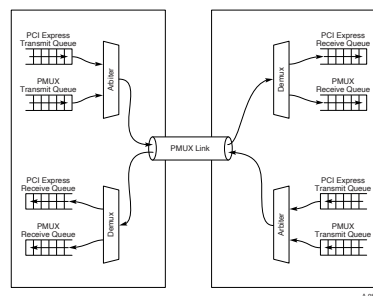


Figure G-2 PMUX Link §

Important attributes of the Protocol Multiplexing mechanism are:

- Protocol Multiplexing support is optional normative.
- Protocol Multiplexing is only supported in Non-Flit Mode. Protocol Multiplexing is not supported in Flit Mode.
- Protocol Multiplexing has no impact on PCI Express components that do not support it.
- Protocol Multiplexing has no impact on PCI Express TLPs and DLLPs, even when it is enabled.

- A Link may be used for both TLPs and PMUX Packets at the same time.
- Protocol Multiplexing does not consume or interfere with PCI Express resources (sequence numbers, credits, etc.). PMUX Packets use distinct resources associated with the specific multiplexed protocol.
- Protocol Multiplexing is disabled by default and is enabled by software. PMUX Packets must not be sent unless enabled by software. PMUX Packets received at Ports that support Protocol Multiplexing are ignored until Protocol Multiplexing is enabled by software.
- Protocol Multiplexing is selectable on a per-Link basis. Protocol Multiplexing may be used on any collection of Links in a system.
- A PMUX Link may support up to four simultaneously active PMUX Channels. Software configures the protocol used on each PMUX Channel.
- PMUX Packets contain an LCRC. This is used to provide data resiliency in a similar fashion as PCI Express TLPs.
- PMUX Packets do not use the ACK/NAK mechanism of PCI Express. Multiplexed protocol specific acknowledgement mechanisms can be used to provide reliable delivery when needed.
- PMUX Packets do not contain a TLP Sequence Number. Instead, they contain a 12 bit PMUX Packet Metadata field that is available for multiplexed protocol specific use.
- PMUX Packet transmitters must contain some arbitration/QoS mechanism for scheduling sending of PMUX Packets, TLPs and DLLPs; however, the mechanism used is outside the scope of this specification.
- The Protocol Multiplexing mechanism does not define any addressing or routing mechanism for PMUX Packets.

PMUX Packets are similar to PCI Express TLPs. The PMUX Packet Flow Through is shown in § Figure G-3. The PCI Express Packet flow through the layers is shown in § Figure 1-5. Changes from PCI Express Packet Flow Through are:

- PMUX Packets use a protocol specific PMUX Protocol Layer instead of the PCI Express Transaction Layer.
- PMUX Packets use a simplified Data Link Layer. The packet integrity portion of the Data Link Layer is mostly unchanged (LCRC computation uses a different seed value). The reliability and flow control aspects of the Data Link Layer are removed (the TLP Sequence Number field is repurposed as PMUX Packet Metadata).
- The Physical Layer is slightly modified to provide a mechanism to identify PMUX Packets.

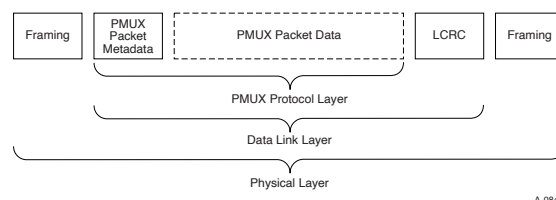


Figure G-3 PMUX Packet Flow Through the Layers §

G.1 Protocol Multiplexing Interactions with PCI Express §

§ Table G-1 and § Table G-2 describe interactions between Protocol Multiplexing and PCI Express. § Table G-1 describes how PCI Express attributes affect Protocol Multiplexing. § Table G-2 describes how Protocol Multiplexing features affect PCI Express attributes.

Table G-1 PCI Express Attribute Impact on Protocol Multiplexing §

PCI Express Attribute	Impact on Protocol Multiplexing
Link Speed	All PMUX Channels are disabled when the Current Link Speed corresponds to a speed that is not supported by Protocol Multiplexing (see § Section G.5.4). A PMUX Channel may be disabled when the Current Link Speed corresponds to a speed that is not supported by the associated protocol.²²⁶ Link speed can change either explicitly due to a change in the Target Link Speed field or automatically due to an autonomous Link speed change (see § Section 6.11). The PMUX Protocol Layer is permitted to influence the mechanism used by a component to determine when it requests an autonomous Link speed change. In addition, setting Hardware Autonomous Speed Disable at each end of the Link will prevent certain autonomous Link speed changes (see § Section 7.5.3.18). The PMUX Protocol Layer may be notified of the change.
Link Width	A PMUX Channel may be disabled when the Link Width corresponds to a width that is not supported by the associated protocol.²²⁷ The PMUX Protocol Layer is permitted to influence the mechanism used by a component to determine when it requests a Link Width change. The PMUX Protocol Layer may be notified of the change.
FLR initiated	All PMUX Channels on a Link are disabled if an FLR is directed to the Upstream Port's Function 0. No PMUX Channels are affected if an FLR is directed to any other Function.
DL_Down	All PMUX Channels on a Link are disabled.
Hot Reset	All PMUX Channels on a Link are disabled.
PERST#	All PMUX Channels on a Link are disabled.
TLP Replay	No effect on Protocol Multiplexing.
DLLP Lost	No effect on Protocol Multiplexing.
TLP Prefix	No effect on Protocol Multiplexing.
Locked Transactions	No effect on Protocol Multiplexing (including no effect on any protocol specific forwarding of PMUX Packets).
AtomicOp Transactions	No effect on Protocol Multiplexing.
Multicast Transactions	No effect on Protocol Multiplexing.
Access Control Services (ACS)	No effect on Protocol Multiplexing.
Alternative Routing ID-Interpretation (ARI)	No effect on Protocol Multiplexing.
TPH Requester	No effect on Protocol Multiplexing.
Virtual Channels	No effect on Protocol Multiplexing. PCI Express Links remain capable of supporting a full complement of VCs.

226. The mechanism software uses to determine what Link Speeds are supported by a protocol is outside the scope of this specification.

227. The mechanism software uses to determine what Link Widths are supported by a protocol is outside the scope of this specification.

PCI Express Attribute	Impact on Protocol Multiplexing
Internal Error	Corrected or Uncorrectable Internal Errors in the PMUX Protocol Layer may be reported as PCI Express Internal Errors.
<u>L0s</u> Link Power State	Protocol Multiplexing tracks the Link state. The PMUX Protocol Layer may request the Link transition back to L0.
<u>L1</u> Link Power State	Protocol Multiplexing tracks the Link state. The PMUX Protocol Layer may request the Link transition back to L0.
Disabled LTSSM State	Disabling a Link also disables all PMUX Channels on the Link.
Loopback LTSSM State	Entering Loopback state disables all PMUX Channels on the Link.
Recovery LTSSM State	No effect on Protocol Multiplexing. The PMUX Protocol Layer may be notified.
Receiver or Framing Error	The error is reported to the PMUX Protocol Layer to indicate that data might have been lost. This can be used to initiate protocol specific error recovery mechanisms. The Error is reported to software using PCI Express Mechanisms.
Lane Reversal	No effect on Protocol Multiplexing. Support for Lane Reversal remains optional.
Polarity Inversion	No effect on Protocol Multiplexing.
Crosslink	No effect on Protocol Multiplexing. Support for Crosslink remains optional. If supported, the PMUX Protocol Layer may be notified of the outcome of the Crosslink Upstream / Downstream negotiation.
Lane assignment rules	Placement and frequency rules for STP Symbols and STP Tokens are not changed (see § Section 4.2.1.2 and § Section 4.2.2.3.2). These rules apply identically to PCI Express TLPs and PMUX Packets.
PCI Power Management Power State	All PMUX Channels on a Link are disabled if the Upstream Port's Function 0 is sent to non-D0 state.
Dynamic Power Allocation (DPA)	No effect on Protocol Multiplexing. The PMUX Protocol Layer is notified of the change and may participate in the power reduction. PCI Express power management includes any power used by the PMUX Protocol Layer.
PCI Power Management Power Consumed / Power Dissipated / Aux_Current	Power required by the PMUX Protocol Layer is included in the PCI structures.
Power Budgeting	Power required by the PMUX Protocol Layer is included in the PCI Express structures.
Slot Power Limit	Slot Power Limit includes power available to PMUX Protocol Layer.
ASPM L0s Entry Condition	The definition of Idle is extended to include: - No pending PMUX Packets to transmit over the Link. - For PMUX Channels that use protocol specific Flow Control, no credits are available to send PMUX Packets in that PMUX Channel.
ASPM L1 Entry Condition	A Link may not enter <u>L1</u> if PMUX Packets are pending or scheduled to be transmitted.
ASPM L0s/L1 Exit Conditions	A Link may be directed to exit L0s or L1 if a component needs to transmit a PMUX Packet. Routing of PMUX Packets through routing elements is outside the scope of this specification; the associated <u>L0s/L1</u> exit rules are also unspecified.

PCI Express Attribute	Impact on Protocol Multiplexing
Bus Renumbering	No effect on Protocol Multiplexing.
Hot Plug	No direct effect on Protocol Multiplexing. Note Hot Plug events indirectly affect Data Link State which, in turn affects Protocol Multiplexing.
TLP Sequence Number	No effect on Protocol Multiplexing. PMUX Packets do not consume TLP Sequence Numbers.
PCI Express Flow Control	No effect on Protocol Multiplexing. PMUX Packets do not consume PCI Express Flow Control credits. Flow Control Update DLLPs must be sent as required by PCI Express.
Error Reporting	No direct effect on Protocol Multiplexing. The PMUX Protocol Layer may be notified when an error is signaled or when an error message is received.
LCRC Errors in TLPs	No effect on Protocol Multiplexing.
Nullified TLPs	No effect on Protocol Multiplexing.
VC Arbitration	No effect on Protocol Multiplexing. Arbitration within PCI Express is unaffected by Protocol Multiplexing.
Port Arbitration	No effect on Protocol Multiplexing. Arbitration within PCI Express is unaffected by Protocol Multiplexing.
Electrical Idle Inference	PMUX Packets count as TLPs for the purpose of inferring Electrical Idle.
MR-IOV	Protocol Multiplexing may co-exist with MR-IOV. PMUX Packets are not part of any MR-IOV Virtual Hierarchy. Protocol Multiplexing is controlled using configuration space in the Management VH(s).

Table G-2 PMUX Attribute Impact on PCI Express §

PMUX Attribute	Impact on PCI Express
PMUX Protocol Error	No effect on PCI Express.
LCRC Errors in PMUX Packets	No effect on PCI Express. PMUX Packets with LCRC errors are discarded without triggering PCI Express replay. This error is reported to the PMUX Protocol Layer and can be used to initiate protocol specific error recovery and/or error reporting mechanisms.
Link Unreliability	The PMUX Protocol Layer is permitted to influence the mechanism used by a component to determine if it requests an autonomous link speed change.
Nullified PMUX Packets	No effect on PCI Express. It is protocol specific whether PMUX Packets within a specific PMUX Channel may be nullified. If supported, PMUX Packets are nullified in the same manner as TLPs (e.g., inverting the LCRC and signaling nullification at the Physical Layer). Receiving a nullified PMUX Packet may be reported to the PMUX Protocol Layer.
Electrical Idle Inference	PMUX Packets count as TLPs for the purpose of inferring Electrical Idle.
PMUX Protocol Layer directs LTSSM to enter Recovery	Both PCI Express and the PMUX Protocol Layer are permitted to direct a transition from L0 to Recovery.
PMUX Channel Enabled / Disabled	No effect on PCI Express
PMUX Packet Receiver Buffer Overflow	No effect on PCI Express. This is a protocol problem within the PMUX Channel. The PMUX Transport Layer must continue to accept such packets dispose of them using protocol specific mechanisms.
Received PMUX Packet larger or smaller than supported by the associated protocol	No effect on PCI Express. These are protocol problems within the PMUX Channel. The PMUX Transport Layer must accept such packets and dispose of them using protocol specific mechanisms.

PMUX Attribute	Impact on PCI Express
Received PMUX Packet that contains more than 125 DWORDs of PMUX Packet Data	No effect on PCI Express. This is an invalid PMUX Packet. The PMUX Transport Layer must accept such a packet and dispose of it. A protocol specific mechanism may be used to report the error. Note: This situation only exists for a packet encoded using 8b/10b. The TLP Length field of a packet encoded using 128b/130b cannot contain values that cause this situation.
PMUX Packet Received on disabled PMUX Channel	No effect on PCI Express. No effect on any other PMUX Channel. Receivers must silently ignore such packets regardless of packet length and regardless of whether or not the packet is nullified. PMUX Packets arriving on a disabled PMUX Channel may occur normally when software is in the process of initializing Protocol Multiplexing.
PMUX Packet Received at component that does not support Protocol Multiplexing	Software should not enable PMUX Packets unless both ends of a Link support Protocol Multiplexing. In the 128b/130b encoding, receiving a PMUX Packet by a component that does not support Protocol Multiplexing is a Framing Error (see § Section 4.2.2.3.1). In the 8b/10b encoding, the PMUX Packet LCRC is computed differently than the TLP LCRC. Receivers that do not support Protocol Multiplexing will interpret PMUX Packets as TLPs with LCRC errors and will not process them.
Large PMUX Packets when PCI Express Max_Payload_Size is small	Under certain conditions, it is possible for a large PMUX Packet to trigger a premature PCI Express replay. For example, this can occur when the time needed to transmit a PMUX Packet is larger than the REPLAY_TIMER (see § Section 3.6.2.1). To avoid this issue, implementations are permitted to not advance (hold) their REPLAY_TIMER during the reception of PMUX Packets. Note: The PCI Express REPLAY_TIMER mechanism has adequate headroom for most cases. This issue exists when (1) Max_Payload_Size is 000b, (2) PMUX Packets are larger than about 80 DWORDs, and (3) the REPLAY_TIMER is at the low end of the -0%/+100% tolerance.

G.2 PMUX Packets §

A PMUX Packet contains the information shown in § Figure G-4.

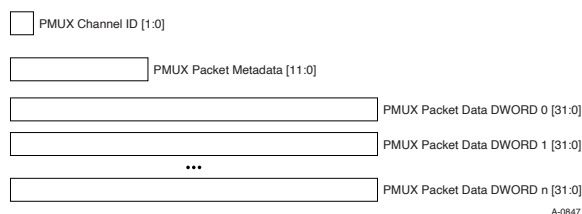


Figure G-4 PMUX Packet §

PMUX Channel ID is a 2 bit field that identifies which protocol is associated with a PMUX Packet. PMUX Channel ID values are between 0 and 3 (inclusive).

PMUX Packet Metadata is a 12 bit field that provides information about the PMUX Packet. Definition of this field is protocol specific and is outside the scope of this specification.

A PMUX Packet consists of between 0 and 125 DWORDs of PMUX Packet Data. Layout and usage of these DWORDs is protocol specific and is outside the scope of this specification. A PMUX Packet need not have any PMUX Packet Data and may consist only of PMUX Channel ID and PMUX Packet Metadata.

G.3 PMUX Packet Layout §

There are two layouts defined for PMUX Packets. One layout is used for 2.5 and 5.0 GT/s data rates and another layout is used for 8.0 GT/s and higher data rates. These layouts are discussed in the following sections.

G.3.1 PMUX Packet Layout for 8b/10b Encoding §

§ Figure G-5 and § Table G-3 show the layout of PMUX Packets when using 8b/10b encoding. For reference, the 8b/10b encoding of a TLP is also shown (see § Section 4.2.1.2 for the official definition).

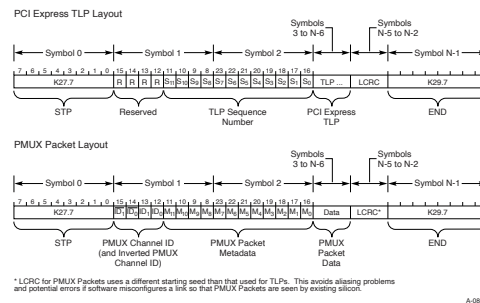


Figure G-5 TLP and PMUX Packet Framing (8b/10b Encoding) §

Table G-3 PMUX Packet Layout (8b/10b Encoding) §

Symbol	Field	Bit Position(s)	PMUX Packet Usage	TLP Usage
0	Start TLP Indicator	7:0	K27.7	
1	Inverted PMUX Channel ID[1:0]	7:6	Inverted (1s complement) of Symbol 1 bits 6:4	Reserved
	PMUX Channel ID[1:0]	5:4	PMUX Channel ID	Reserved
	PMUX Packet Metadata[11:8]	3:0	PMUX Packet Metadata[11:8]	TLP Sequence Number[11:8]
2	PMUX Packet Metadata[7:0]	7:0	PMUX Packet Metadata[7:0]	TLP Sequence Number[7:0]
3 to N-6	Packet	7:0	PMUX Packet	TLP
N-5 to N-2	LCRC	7:0	PMUX LCRC	PCI Express LCRC
N-1	END	7:0	K29.7	

For PMUX Packets, symbols 1 and 2 contain PMUX Packet Metadata in the same bit positions that TLPs use for TLP Sequence Number.

The PMUX LCRC algorithm is identical to the TLP LCRC algorithm as described in [§ Section 3.6.2](#) with the following modifications:

- The seed value is FB3E E248h (TLP LCRC uses FFFF FFFFh).
- The PMUX Channel ID field in Symbol 1 bits 7:4 is included in the PMUX LCRC in the same manner as the 4 reserved bits in the TLP LCRC.
- The PMUX Packet Metadata field is included in the PMUX LCRC in the same manner as the TLP Sequence Number field is included in the TLP LCRC.

IMPLEMENTATION NOTE: PMUX PACKETS AT RECEIVERS THAT DO NOT SUPPORT PROTOCOL MULTIPLEXING §

The bits used for PMUX Channel ID are reserved [unless](#) Protocol Multiplexing is supported. As such, Receivers that do not support Protocol Multiplexing must ignore the PMUX Channel ID bits. If software misconfigures Protocol Multiplexing, a component that does not support Protocol Multiplexing could receive a PMUX Packet. To prevent that component from misinterpreting such a PMUX Packet as a valid TLP, the LCRC computation is changed for PMUX Packets. The result is that a valid PMUX Packet will never be misinterpreted as a valid TLP. These LCRC “errors” may trigger PCI Express replay and may result in [REPLAY_NUM](#) Rollover correctable errors being reported.

IMPLEMENTATION NOTE: PMUX PACKET LCRC §

The PMUX Channel ID field is covered by the LCRC. As such, when using 8b/10b encoding, receivers must wait until the LCRC is checked to make firm decisions based on the PMUX Channel ID value. The Inverted PMUX Channel ID can be compared against the PMUX Channel ID to make tentative decisions.

Note: The value of the LCRC associated with a given PMUX Packet is independent of the encoding used to transmit the packet.

G.3.2 PMUX Packet Layout at 128b/130b Encoding §

§ Figure G-6 and § Table G-4 show the layout of PMUX Packets when using 128b/130b encoding. For reference, the 128b/130b encoding of a TLP is also shown (see [§ Section 4.2.2.2](#) for the official definition).

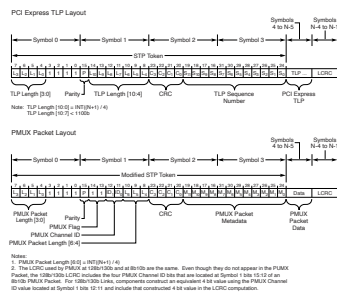


Figure G-6 TLP and PMUX Packet Framing (128b/130b Encoding) §

Table G-4 PMUX Packet Layout (128b/130b Encoding) §

Symbol	Field	Bit Position(s)	PMUX Packet Usage	TLP Usage
0	Start TLP Indicator	3:0	Value of 1111b	
	PMUX Packet Length[3:0]	7:4	Bits [3:0] of the PMUX Packet Length. Bit 0 is the least significant PMUX Packet Length bit.	Bits [3:0] of the TLP Length field. Bit 0 is the least significant TLP Length bit.
1	Frame Parity (P)	7	Even parity of Symbol 0 bits [7:4], Symbol 1 bits [6:0] and Symbol 2 bits [7:4]	
	PMUX Packet Indicator	6:5	Value of 11b	
	PMUX Channel ID[1:0]	4:3	PMUX Channel ID	Bits [10:4] of the TLP Length field. Bit 10 is the most significant TLP Length bit.
	PMUX Packet Length[6:4]	2:0	Bits [6:4] of PMUX Packet Length. Bit 6 is the most significant PMUX Packet Length bit.	
2	PMUX Packet Metadata[11:8]	3:0	PMUX Packet Metadata[11:8]	TLP Sequence Number[11:8]
	Frame CRC (C[3:0])	7:4	CRC of Symbol 0, bits [7:4] and Symbol 1 bits [6:0]	
3	PMUX Packet Metadata[7:0]	7:0	PMUX Packet Metadata[7:0]	TLP Sequence Number[7:0]
4 to N-5	Packet	7:0	PMUX Packet	TLP
N-4 to N-1	LCRC	7:0	PMUX LCRC	PCI Express LCRC

§ Table G-5 describes the encodings of Symbol 1 bits [6:3] in more detail. If these bits contain a value less than 1001b, the packet is a TLP and is processed as described in § Section 4.2.2.²²⁸ If these bits contain 1001b, 1010b, or 1011b, the encoding is reserved for future standardization and is processed as described in § Section 4.2.2.3.3. If these bits contain a value greater than or equal to 1100b, the packet is a PMUX Packet is defined as specified in this appendix.²²⁹

228. The value 1001b supports a maximum TLP Length [10:0] value of 1151 DWORDs (decimal). This will accommodate a TLP consisting of 4096 bytes of payload, 16 bytes of TLP Header, 4 bytes of TLP digest, and 480 bytes of TLP Prefix.

Table G-5 Symbol 1 Bits [6:3] §

Symbol 1 bits [6:3]	Meaning
0xxx or 1000b	Packet is a TLP. Bits [6:3] are TLP Length [10:7].
1001b, 1010b, or 1011b	Encoding reserved for future standardization. Receivers detecting these encodings shall process them as described in § Section 4.2.2.3.3.
1100b	Packet is a PMUX Packet. PMUX Channel ID is 0.
1101b	Packet is a PMUX Packet. PMUX Channel ID is 1.
1110b	Packet is a PMUX Packet. PMUX Channel ID is 2.
1111b	Packet is a PMUX Packet. PMUX Channel ID is 3.

For PMUX Packets, the packet length in DWORDs is contained in PMUX Packet Length [6:0]. Other than being a smaller field, PMUX Packet Length is interpreted in the same manner as TLP Length. Specifically, PMUX Packet Length also includes the framing and PMUX LCRC DWORDs (see § Section 4.2.2.2).

For PMUX Packets, symbols 2 and 3 contain PMUX Packet Metadata in the same bit positions that TLPs use for TLP Sequence Number.

The PMUX LCRC algorithm is identical to the TLP LCRC algorithm as described in § Section 3.6.2 with the following modifications:

- The seed value is FB3E E248h (TLP LCRC uses FFFF FFFFh).
- The PMUX Channel ID field in Symbol 1 bits 4:3 is used to compute a 4 bit value that is included in the PMUX LCRC in the same manner as the 4 reserved bits in the TLP LCRC. This 4 bit value contains the value that would be used, by the 8b/10b encoding, for Symbol 1 bits 7:4. Specifically, the lower 2 bits of this 4 bit value contain the PMUX Channel ID and the upper 2 bits contain the inverse (1s complement) of the PMUX Channel ID.
- The PMUX Packet Metadata field is included in the PMUX LCRC in the same manner as the TLP Sequence Number field is included in the TLP LCRC.

The Frame CRC and Frame Parity fields are computed as shown below. This is the same algorithm computed over the same bit positions as defined in § Section 4.2.2.2.

$$\begin{aligned}
 C[0] &= 1b \wedge PMUX_Channel_ID[0] \\
 &\quad \wedge L[6] \wedge L[4] \wedge L[2] \wedge L[1] \wedge L[0] \\
 C[1] &= 1b \wedge 1b \wedge PMUX_Channel_ID[0] \\
 &\quad \wedge L[5] \wedge L[4] \wedge L[3] \wedge L[2] \\
 C[2] &= 1b \wedge PMUX_Channel_ID[1] \\
 &\quad \wedge L[6] \wedge L[4] \wedge L[3] \wedge L[2] \wedge L[1] \\
 C[3] &= PMUX_Channel_ID[1] \\
 &\quad \wedge PMUX_Channel_ID[0] \\
 &\quad \wedge L[5] \wedge L[3] \wedge L[2] \wedge L[1] \wedge L[0] \\
 P &= 1b \wedge 1b \wedge PMUX_Channel_ID[1] \\
 &\quad \wedge PMUX_Channel_ID[0] \\
 &\quad \wedge L[6] \wedge L[5] \wedge L[4] \wedge L[3] \wedge L[2] \wedge L[1] \wedge L[0] \\
 &\quad \wedge C[3] \wedge C[2] \wedge C[1] \wedge C[0]
 \end{aligned}$$

229. The value 1100b was chosen to simplify distinguishing PMUX Packets from TLPs and from the reserved encodings.

IMPLEMENTATION NOTE:

PMUX CHANNEL ID AND FRAME CRC §

When using 128b/130b encoding, the PMUX Channel ID field is covered by the Frame CRC and Frame Parity fields. As such, receivers may make decisions based on the PMUX Channel ID value as soon as the Frame CRC and Frame Parity is checked and need not wait until the PMUX LCRC is checked.

Note: The PMUX Channel ID is also covered by the LCRC. The value of the LCRC associated with a given PMUX Packet is independent of the encoding used to transmit the packet.

G.4 PMUX Control §

Protocol Multiplexing is disabled by default. Each PMUX Channel must be explicitly enabled by software at each end of the associated Link. Protocol Multiplexing is disabled whenever the link drops (Data Link Layer indicates DL_Down).

A component that supports Protocol Multiplexing indicates such by the presence of the PMUX Extended Capability.

The following rules apply to components that support Protocol Multiplexing:

- PMUX Packets received in a PMUX Channel that is not enabled are silently ignored.
- PMUX Packets may not be transmitted unless the associated PMUX Channel is enabled. A PMUX Channel may also require additional, protocol specific, initialization mechanisms before PMUX Packets may be transmitted.

G.5 PMUX Extended Capability §

§ Figure G-7 shows the PMUX Extended Capability structure. The presence of this capability indicates that the Port supports the optional Protocol Multiplexing mechanism. This capability is optional and may be present in any Downstream Port and in Function 0 of any Upstream Port. It must not be present in non-zero Functions of Upstream Ports or in RCRBs.

The length of the PMUX Extended Capability is determined by the PMUX Protocol Array Size field (see § Section G.5.2).

This capability contains a list of the protocols supported by the Link (the PMUX Protocol Array). It also contains the mechanism software uses to enable and configure PMUX Channels. This capability must be present in both the Upstream and Downstream Ports of a Link in order for Protocol Multiplexing to be successfully enabled.

Software may enable the Upstream and Downstream Ports of a Link in either order. Software may enable multiple PMUX Channels using a single write to the PMUX Control Register.

Behavior is undefined if software enables Protocol Multiplexing in one Port and the other Port of the Link does not support Protocol Multiplexing. Behavior is also undefined if software configures a PMUX Channel inconsistently (the same PMUX Channel in the Ports on each end of a Link configured with incompatible protocols).

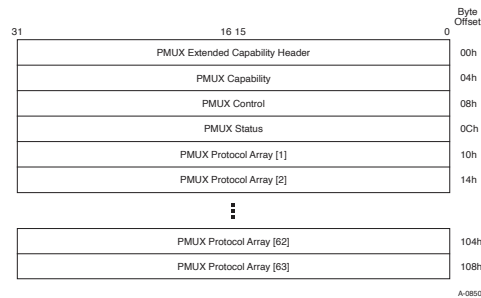


Figure G-7 PMUX Extended Capability §

G.5.1 PMUX Extended Capability Header (Offset 00h) §

§ Figure G-8 details the allocation of fields in the PMUX Extended Capability header; § Table G-6 provides the respective bit definitions.

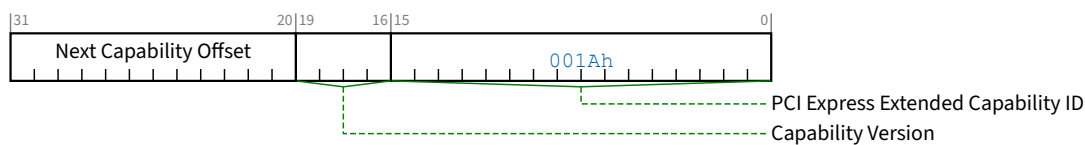


Figure G-8 PMUX Extended Capability Header §

Table G-6 PMUX Extended Capability Header §

Bit Location	Register Description	Attributes
15:0	PCI Express Extended Capability ID - This field is a PCI-SIG defined ID number that indicates the nature and format of the Extended Capability. The Extended Capability ID for the PMUX Extended Capability is 001Ah.	RO
19:16	Capability Version - This field is a PCI-SIG defined version number that indicates the version of the Capability structure present. Must be 1h for this version of the specification.	RO
31:20	Next Capability Offset - This field contains the offset to the next PCI Express Capability structure or 000h if no other items exist in the linked list of capabilities. This offset is relative to the beginning of PCI compatible Configuration Space and thus must always be either 000h (for terminating the list of Capabilities) or greater than 0FFh.	RO

G.5.2 PMUX Capability Register (Offset 04h) §

§ Figure G-9 details the allocation of fields in the PMUX Capability Register. § Table G-7 provides the respective bit definitions.

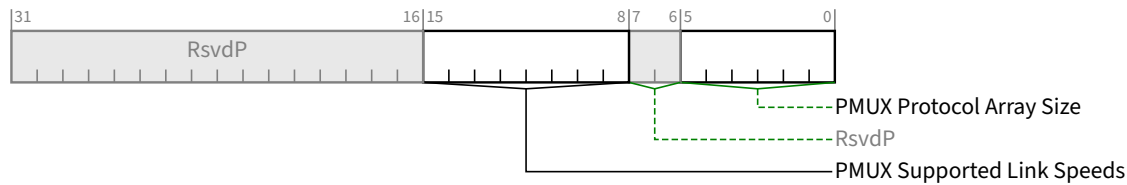


Figure G-9 PMUX Capability Register §

Table G-7 PMUX Capability Register §

Bit Location	Register Description	Attributes
5:0	PMUX Protocol Array Size - Indicates the size of this Function's PMUX Protocol Array. This field may be 0 to indicate that even though no protocols are supported, the Port will ignore all received PMUX Packets.	RO
15:8	PMUX Supported Link Speeds - This field indicates the Link speed(s) where Protocol Multiplexing is supported. Each bit corresponds to a Link speed. If a bit is Set, Protocol Multiplexing is supported at that Link speed. If a bit is Clear, Protocol Multiplexing is not supported at that Link speed. Bit definitions are: Bit 8 2.5 GT/s Bit 9 5.0 GT/s Bit 10 8.0 GT/s Bit 11 16.0 GT/s Bit 12 32.0 GT/s Bits 15:13 RsvdP At least one Link speed must be supported (i.e., the field must be non-zero). A Port may support any combination of Link speeds. For example, this field could contain the value 0000 0100b indicating that Protocol Multiplexing is only supported at 8.0 GT/s. This field must not indicate support for Link speeds that are not supported by the Link (see § Section 7.5.3.18). Note that this field indicates the Link speeds supported by Protocol Multiplexing for the Link. The Link speeds that a particular protocol supports and the mechanism used to report that information are protocol specific.	RO / RsvdP

G.5.3 PMUX Control Register (Offset 08h) §

§ Figure G-10 details the allocation of fields in the PMUX Control Register. § Table G-8 provides the respective bit definitions.

Channel *n* is enabled and available for use by the PMUX Protocol Layer when all of the following are true:

- The Channel *n* Assignment field is non-zero.
- The Channel *n* Assignment field is less than or equal to PMUX Protocol Array Size.
- The Channel *n* Assignment field indicates an implemented entry in the PMUX Protocol Array (see § Section G.5.5).
- All of the PMUX Channel *n* Disabled bits are Clear (see § Section G.5.4).

Otherwise, Channel n is disabled.

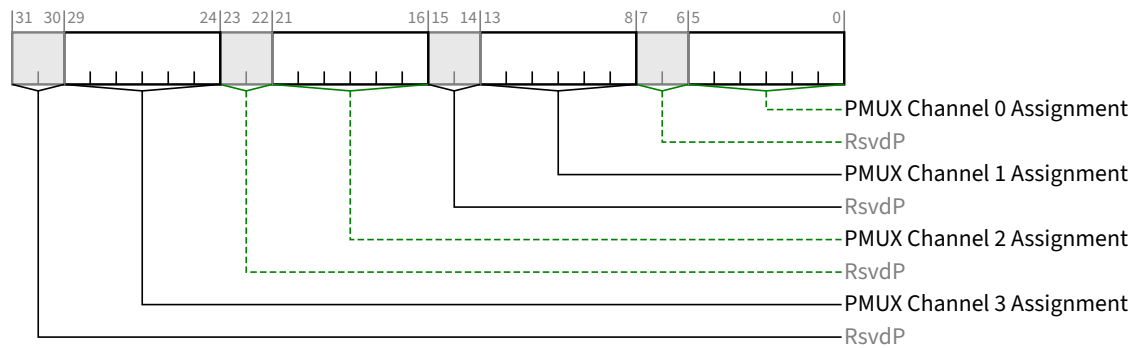


Figure G-10 PMUX Control Register §

Table G-8 PMUX Control Register §

Bit Location	Register Description	Attributes
5:0	PMUX Channel 0 Assignment - This field indicates the protocol assigned to PMUX Channel 0. If the field is 0h, no protocol is assigned. If the field is non-zero, it is the index in the PMUX Protocol Array of the protocol assigned to PMUX Channel 0. If PMUX Protocol Array Size is less than 63 (see § Section G.5.2), unused upper bits of this field may be hardwired to 0b. If PMUX Protocol Array Size is 0, this entire field may be hardwired to 0. This field defaults to 0h.	RW
13:8	PMUX Channel 1 Assignment - This field indicates the protocol assigned to PMUX Channel 1. If the field is 0h, no protocol is assigned. If the field is non-zero, it is the index in the PMUX Protocol Array of the protocol assigned to PMUX Channel 1. If PMUX Protocol Array Size is less than 63 (see § Section G.5.2), unused upper bits of this field may be hardwired to 0b. If PMUX Protocol Array Size is 0, this entire field may be hardwired to 0. This field defaults to 0h.	RW
21:16	PMUX Channel 2 Assignment - This field indicates the protocol assigned to PMUX Channel 2. If the field is 0h, no protocol is assigned. If the field is non-zero, it is the index in the PMUX Protocol Array of the protocol assigned to PMUX Channel 2. If PMUX Protocol Array Size is less than 63 (see § Section G.5.2), unused upper bits of this field may be hardwired to 0b. If PMUX Protocol Array Size is 0, this entire field may be hardwired to 0. This field defaults to 0h.	RW
29:24	PMUX Channel 3 Assignment - This field indicates the protocol assigned to PMUX Channel 3. If the field is 0h, no protocol is assigned. If the field is non-zero, it is the index in the PMUX Protocol Array of the protocol assigned to PMUX Channel 3. If PMUX Protocol Array Size is less than 63 (see § Section G.5.2), unused upper bits of this field may be hardwired to 0b. If PMUX Protocol Array Size is 0, this entire field may be hardwired to 0. This field defaults to 0h.	RW

G.5.4 PMUX Status Register (Offset 0Ch) §

§ Figure G-11 details the allocation of fields in the PMUX Status Register. § Table G-9 provides the respective bit definitions.

Each channel has a set of Disabled bits. When Channel *n* Assignment field is non-zero, the Channel *n* Disabled bits reflect the error status of the channel. The following Disabled bits are defined:

- PMUX Channel *n* Disabled: Link Speed
- PMUX Channel *n* Disabled: Link Width
- PMUX Channel *n* Disabled: Protocol Specific

When there are multiple reasons for disabling a channel, an implementation may choose which reason(s) to report. For example, if a protocol needs bandwidth equivalent to x1 8.0 GT/s, when there is inadequate bandwidth (e.g., the Link is operating at x1 5.0 GT/s, x1 2.5 GT/s, or x2 2.5 GT/s), it could disable the PMUX Channel by indicating any or all of Disabled: Link Width, Disabled: Link Speed, or Disabled: Protocol Specific.

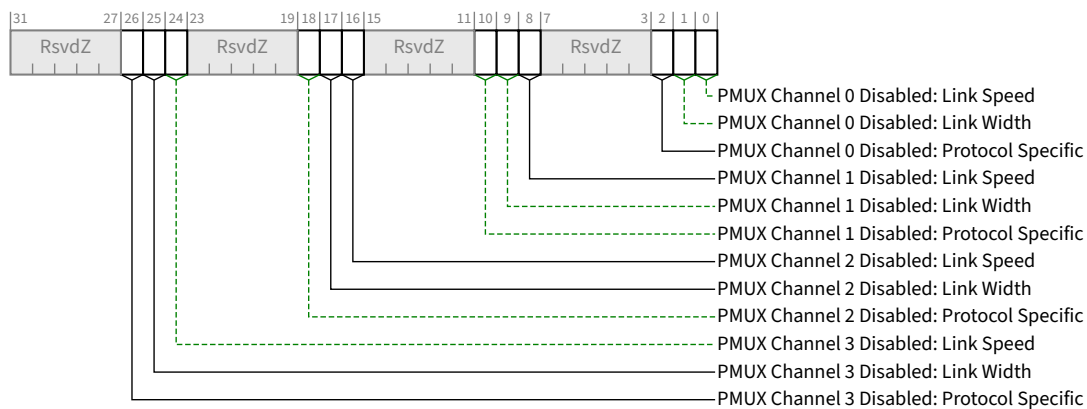


Figure G-11 PMUX Status Register §

Table G-9 PMUX Status Register §

Bit Location	Register Description	Attributes
0	PMUX Channel 0 Disabled: Link Speed - If Set, Channel 0 is disabled because the Current Link Speed (§ Section 7.8.9) is not supported by Protocol Multiplexing or by the protocol assigned to Channel 0. This bit is 0 when no protocol is assigned to Channel 0 (i.e., Channel 0 Control field is 0h).	RO
1	PMUX Channel 0 Disabled: Link Width - If Set, Channel 0 is disabled because the current Link Width is not supported by the protocol assigned to Channel 0. This bit is 0 when no protocol is assigned to Channel 0 (i.e., PMUX Channel 0 Assignment field is 0h).	RO
2	PMUX Channel 0 Disabled: Protocol Specific - If Set, Channel 0 is disabled for protocol specific reasons. This bit is 0 when no protocol is assigned to Channel 0 (i.e., PMUX Channel 0 Assignment field is 0h).	RO
8	PMUX Channel 1 Disabled: Link Speed - If Set, Channel 1 is disabled because the Current Link Speed (§ Section 7.8.9) is not supported by Protocol Multiplexing or by the protocol assigned to Channel 1.	RO

Bit Location	Register Description	Attributes
	This bit is 0 when no protocol is assigned to Channel 1 (i.e., <u>PMUX Channel 1 Assignment</u> field is 0h).	
9	PMUX Channel 1 Disabled: Link Width - If Set, Channel 1 is disabled because the current Link Width is not supported by the protocol assigned to Channel 1. This bit is 0 when no protocol is assigned to Channel 1 (i.e., <u>PMUX Channel 1 Assignment</u> field is 0h).	RO
10	PMUX Channel 1 Disabled: Protocol Specific - If Set, Channel 1 is disabled for protocol specific reasons. This bit is 0 when no protocol is assigned to Channel 1 (i.e., <u>PMUX Channel 1 Assignment</u> field is 0h).	RO
16	PMUX Channel 2 Disabled: Link Speed - If Set, Channel 2 is disabled because the Current Link Speed (§ Section 7.8.9) is not supported by Protocol Multiplexing or by the assigned protocol. This bit is 0 when no protocol is assigned to Channel 2 (i.e., <u>PMUX Channel 2 Assignment</u> field is 0h).	RO
17	PMUX Channel 2 Disabled: Link Width - If Set, Channel 2 is disabled because the current Link Width is not supported by the assigned protocol. This bit is 0 when no protocol is assigned to Channel 2 (i.e., <u>PMUX Channel 2 Assignment</u> field is 0h).	RO
18	PMUX Channel 2 Disabled: Protocol Specific - If Set, Channel 2 is disabled for protocol specific reasons. This bit is 0 when no protocol is assigned to Channel 2 (i.e., <u>PMUX Channel 2 Assignment</u> field is 0h).	RO
24	PMUX Channel 3 Disabled: Link Speed - If Set, Channel 3 is disabled because the Current Link Speed (§ Section 7.8.9) is not supported by Protocol Multiplexing or by the assigned protocol. This bit is 0 when no protocol is assigned to Channel 3 (i.e., <u>PMUX Channel 3 Assignment</u> field is 0h).	RO
25	PMUX Channel 3 Disabled: Link Width - If Set, Channel 3 is disabled because the current Link Width is not supported by the assigned protocol. This bit is 0 when no protocol is assigned to Channel 3 (i.e., <u>PMUX Channel 3 Assignment</u> field is 0h).	RO
26	PMUX Channel 3 Disabled: Protocol Specific - If Set, Channel 3 is disabled for protocol specific reasons. This bit is 0 when no protocol is assigned to Channel 3 (i.e., <u>PMUX Channel 3 Assignment</u> field is 0h).	RO

G.5.5 PMUX Protocol Array (Offsets 10h through 48h) §

The PMUX Protocol Array consists of up to 63 entries. The size of the PMUX Protocol Array is indicated by the PMUX Protocol Array Size field (see § Section G.5.2).

\$ Figure G-12 details the allocation of fields in each PMUX Protocol Array Entry. \$ Table G-10 provides the respective bit definitions.



Figure G-12 PMUX Protocol Array Entry §

Table G-10 PMUX Protocol Array Entry §

Bit Location	Register Description	Attributes
15:0	Protocol ID - In conjunction with Authority ID designates a specific protocol and the mechanism by which that protocol is mapped onto Protocol Multiplexing. This value is assigned by the Vendor associated with the <u>Authority ID</u> field.	RO
31:16	Authority ID - Designates the authority controlling the values used in the <u>Protocol ID</u> field. The Authority ID field contains a Vendor ID as assigned by the PCI-SIG.	RO

The value 0000 0000h indicates an unimplemented PMUX Protocol Array Entry. The PMUX Protocol Array is indexed starting at 1.

PMUX Channel n is enabled and configured to support the protocol associated with PMUX Protocol Array Entry at index m when the PMUX Channel n Assignment field contains the value m (see § Section G.5.3).

Entries in the PMUX Protocol Array with the Authority ID value 1 (0001h) represent protocols that are defined by the PCI-SIG.

Duplicate Entries in the PMUX Protocol Array may be used to represent multiple instances of a particular protocol. This permits software control of the mapping between PMUX Channel ID and a specific instance of a protocol.

IMPLEMENTATION NOTE: MULTIPLE PROTOCOL INSTANCES §

A Link may have a single PMUX Protocol assigned to multiple PMUX Channels. Each PMUX Channel is assigned to a different instance of the protocol. Each instance of a protocol corresponds to an entry in the PMUX Protocol Array.

Consider a Port that supports two instances of protocol X. Two entries in the PMUX Protocol Array would indicate protocol X (indexes A and B for example). To assign instance A to PMUX Channel 0 and instance B to PMUX Channel 2, place the value A in the PMUX Channel 0 Assignment field and the value B in the PMUX Channel 2 Assignment field.

H. Flow Control Update Latency and ACK Update Latency Calculations §

This appendix is informational only and should not be considered normative. Earlier revisions of this specification outlined the method used to calculate the Flow Control Update Latency and Ack Update Latency. This appendix describes the method used to derive the values and preserves the UpdateFactor and AckFactor values.

H.1 Flow Control Update Latency §

Recommended Flow Control Update Latency is described in the Implementation Note in § Section 2.6.1.2 entitled, Non-Flit Mode Flow Control Update Latency.

Flow Control Update Latency tables were simplified in the 4.0 Revision of this specification. At that time, the original tables were moved to this appendix. Note that in the 4.0 Revision of this specification, Tables 2-47 and 2-48 were distinct, but identical tables. The 5.0 Revision contains a single table that applies to 8.0 GT/s or higher.

Original Tables		Simplified Tables	
Base 3.1	Base 4.0 or later	Base 4.0	Base 5.0 or later
Table 2-42	§ Table H-1	Table 2-45	§ Table 2-52
Table 2-43	§ Table H-2	Table 2-46	§ Table 2-53
Table 2-44	§ Table H-3	Table 2-47	§ Table 2-54
n/a	n/a	Table 2-48	

The values in the Tables are measured starting from when the Receiver Transaction Layer makes additional receive buffer space available by processing a received TLP, to when the first Symbol of the corresponding UpdateFC DLLP is transmitted. The values are calculated as a function of the largest TLP payload size and Link width using the formula:

$$\frac{(\text{Rx_MPS_Limit} + \text{TLPOverhead}) \times \text{UpdateFactor}}{\text{LinkWidth}} + \text{InternalDelay}$$

Equation H-1 Max UpdateFC Latency §

where:

Rx_MPS_Limit

The Rx_MPS_Limit value as determined in § Section 2.2.2. For a Multi-Function Device where different Functions have different Rx_MPS_Limit values, using the smallest Rx_MPS_Limit value across all Functions is strongly recommended.

TLPOverhead

Represents the additional TLP components which consume Link bandwidth (TLP Prefix, header, LCRC, framing Symbols) and is treated here as a constant value of 28 Symbols.

UpdateFactor

Represents the number of maximum size TLPs sent during the time between UpdateFC receptions, and is used to balance Link bandwidth efficiency and receive buffer sizes - the value varies according to Rx_MPS_Limit and Link width. § Table H-1, § Table H-2, and § Table H-3 below include the UpdateFactor(UF).

LinkWidth

The operating width of the Link

InternalDelay

Represents the internal processing delays for received TLPs and transmitted DLLPs, and is treated here as a constant value of 19 Symbol Times for 2.5 GT/s mode operation, 70 Symbol Times for 5.0 GT/s mode operation, and 115 Symbol Times for 8.0 GT/s and 16.0 GT/s modes of operation.

*Table H-1 Maximum UpdateFC Transmission Latency Guidelines for 2.5 GT/s Mode
Operation by Link Width and Max Payload (Symbol Times) §*

		Link Operating Width						
		x1	x2	x4	x8	x12	x16	x32
Rx_MPS_Limit (bytes)	128	237 UF = 1.4	128 UF = 1.4	73 UF = 1.4	67 UF = 2.5	58 UF = 3.0	48 UF = 3.0	33 UF = 3.0
	256	416 UF = 1.4	217 UF = 1.4	118 UF = 1.4	107 UF = 2.5	90 UF = 3.0	72 UF = 3.0	45 UF = 3.0
	512	559 UF = 1.0	289 UF = 1.0	154 UF = 1.0	86 UF = 1.0	109 UF = 2.0	86 UF = 2.0	52 UF = 2.0
	1024	1071 UF = 1.0	545 UF = 1.0	282 UF = 1.0	150 UF = 1.0	194 UF = 2.0	150 UF = 2.0	84 UF = 2.0
	2048	2095 UF = 1.0	1057 UF = 1.0	538 UF = 1.0	278 UF = 1.0	365 UF = 2.0	278 UF = 2.0	148 UF = 2.0
	4096	4143 UF = 1.0	2081 UF = 1.0	1050 UF = 1.0	534 UF = 1.0	706 UF = 2.0	534 UF = 2.0	276 UF = 2.0

*Table H-2 Maximum UpdateFC Transmission Latency Guidelines for 5.0 GT/s Mode
Operation by Link Width and Max Payload (Symbol Times) §*

		Link Operating Width						
		x1	x2	x4	x8	x12	x16	x32
Rx_MPS_Limit (bytes)	128	288 UF = 1.4	179 UF = 1.4	124 UF = 1.4	118 UF = 2.5	109 UF = 3.0	99 UF = 3.0	84 UF = 3.0
	256	467 UF = 1.4	268 UF = 1.4	169 UF = 1.4	158 UF = 2.5	141 UF = 3.0	123 UF = 3.0	96 UF = 3.0
	512	610 UF = 1.0	340 UF = 1.0	205 UF = 1.0	137 UF = 1.0	160 UF = 2.0	137 UF = 2.0	103 UF = 2.0
	1024	1122 UF = 1.0	596 UF = 1.0	333 UF = 1.0	201 UF = 1.0	245 UF = 2.0	201 UF = 2.0	135 UF = 2.0
	2048	2146 UF = 1.0	1108 UF = 1.0	589 UF = 1.0	329 UF = 1.0	416 UF = 2.0	329 UF = 2.0	199 UF = 2.0

		Link Operating Width						
		x1	x2	x4	x8	x12	x16	x32
	4096	4194 UF = 1.0	2132 UF = 1.0	1101 UF = 1.0	585 UF = 1.0	757 UF = 2.0	585 UF = 2.0	327 UF = 2.0

Table H-3 Maximum UpdateFC Transmission Latency Guidelines for 8.0 GT/s Operation by Link Width and Max Payload (Symbol Times) §

		Link Operating Width						
		x1	x2	x4	x8	x12	x16	x32
Rx_MPS_Limit (bytes)	128	333 UF = 1.4	224 UF = 1.4	169 UF = 1.4	163 UF = 2.5	154 UF = 3.0	144 UF = 3.0	129 UF = 3.0
	256	512 UF = 1.4	313 UF = 1.4	214 UF = 1.4	203 UF = 2.5	186 UF = 3.0	168 UF = 3.0	141 UF = 3.0
	512	655 UF = 1.0	385 UF = 1.0	250 UF = 1.0	182 UF = 1.0	205 UF = 2.0	182 UF = 2.0	148 UF = 2.0
	1024	1167 UF = 1.0	641 UF = 1.0	378 UF = 1.0	246 UF = 1.0	290 UF = 2.0	246 UF = 2.0	180 UF = 2.0
	2048	2191 UF = 1.0	1153 UF = 1.0	634 UF = 1.0	374 UF = 1.0	461 UF = 2.0	374 UF = 2.0	244 UF = 2.0
	4096	4239 UF = 1.0	2177 UF = 1.0	1146 UF = 1.0	630 UF = 1.0	802 UF = 2.0	630 UF = 2.0	372 UF = 2.0

H.2 Ack Latency §

Ack Latency tables were simplified in the 4.0 Revision of this specification. At that time, the original tables were moved to this appendix. Note that in the 4.0 Revision of this specification, Tables 3-9 and 3-10 were distinct, but identical tables. The 5.0 Revision contains a single table that applies to 8.0 GT/s or higher.

Original Tables		Simplified Tables	
Base 3.1	Base 4.0 or later	Base 4.0	Base 5.0 or later
Table 3-7	§ Table H-5	Table 3-7	§ Table 3-10
Table 3-8	§ Table H-6	Table 3-8	§ Table 3-11
Table 3-9	§ Table H-7	Table 3-9	§ Table 3-12
n/a	n/a	Table 3-10	

The Maximum Ack Latency limits are calculated using the formula:

$$\frac{(\text{Rx_MPS_Limit} + \text{TLPOverhead}) \times \text{UpdateFactor}}{\text{LinkWidth}} + \text{InternalDelay}$$

Equation H-2 Max Ack Latency §

where:

Rx_MPS_Limit

The Rx_MPS_Limit value as determined in § Section 2.2.2 . For a Multi-Function Device where different Functions have different Rx_MPS_Limit values, using the smallest Rx_MPS_Limit value across all Functions is strongly recommended.

TLPOverhead

represents the additional TLP components which consume Link bandwidth (TLP Prefix, header, LCRC, framing Symbols) and is treated here as a constant value of 28 Symbols.

AckFactor

represents the number of maximum size TLPs which can be received before an Ack is sent, and is used to balance Link bandwidth efficiency and retry buffer size - the value varies according to Rx_MPS_Limit and Link width. Table H-4, Table H-5, and Table H-6 below include the AckFactor(AF).

LinkWidth

is the operating width of the Link.

InternalDelay

represents the internal processing delays for received TLPs and transmitted DLLPs, and is treated here as a constant value of 19 Symbol Times for 2.5 GT/s mode operation, 70 Symbol Times for 5.0 GT/s operation, and 115 Symbol Times for 8.0 GT/s and 16.0 GT/s operation.

Table H-5 Maximum Ack Latency Limit and AckFactor for 2.5 GT/s (Symbol Times) §

		Link Operating Width						
		x1	x2	x4	x8	x12	x16	x32
Rx_MPS_Limit (bytes)	128	237 AF = 1.4	128 AF = 1.4	73 AF = 1.4	67 AF = 2.5	58 AF = 3.0	48 AF = 3.0	33 AF = 3.0
	256	416 AF = 1.4	217 AF = 1.4	118 AF = 1.4	107 AF = 2.5	90 AF = 3.0	72 AF = 3.0	45 AF = 3.0
	512	559 AF = 1.0	289 AF = 1.0	154 AF = 1.0	86 AF = 1.0	109 AF = 2.0	86 AF = 2.0	52 AF = 2.0
	1024	1071 AF = 1.0	545 AF = 1.0	282 AF = 1.0	150 AF = 1.0	194 AF = 2.0	150 AF = 2.0	84 AF = 2.0
	2048	2095 AF = 1.0	1057 AF = 1.0	538 AF = 1.0	278 AF = 1.0	365 AF = 2.0	278 AF = 2.0	148 AF = 2.0
	4096	4143 AF = 1.0	2081 AF = 1.0	1050 AF = 1.0	534 AF = 1.0	706 AF = 2.0	534 AF = 2.0	276 AF = 2.0

Table H-6 Maximum Ack Transmission Latency Limit and AckFactor for 5.0 GT/s (Symbol Times) §

		Link Operating Width						
		x1	x2	x4	x8	x12	x16	x32
Rx_MPS_Limit (bytes)	128	288 AF = 1.4	179 AF = 1.4	124 AF = 1.4	118 AF = 2.5	109 AF = 3.0	99 AF = 3.0	84 AF = 3.0
	256	467 AF = 1.4	268 AF = 1.4	169 AF = 1.4	158 AF = 2.5	141 AF = 3.0	123 AF = 3.0	96 AF = 3.0
	512	610 AF = 1.0	340 AF = 1.0	205 AF = 1.0	137 AF = 1.0	160 AF = 2.0	137 AF = 2.0	103 AF = 2.0
	1024	1122 AF = 1.0	596 AF = 1.0	333 AF = 1.0	201 AF = 1.0	245 AF = 2.0	201 AF = 2.0	135 AF = 2.0
	2048	2146 AF = 1.0	1108 AF = 1.0	589 AF = 1.0	329 AF = 1.0	416 AF = 2.0	329 AF = 2.0	199 AF = 2.0
	4096	4194 AF = 1.0	2132 AF = 1.0	1101 AF = 1.0	585 AF = 1.0	757 AF = 2.0	585 AF = 2.0	327 AF = 2.0

Table H-7 Maximum Ack Transmission Latency Limit and AckFactor for 8.0 GT/s (Symbol Times) §

		Link Operating Width						
		x1	x2	x4	x8	x12	x16	x32
Rx_MPS_Limit (bytes)	128	333 AF = 1.4	224 AF = 1.4	169 AF = 1.4	163 AF = 2.5	154 AF = 3.0	144 AF = 3.0	129 AF = 3.0
	256	512 AF = 1.4	313 AF = 1.4	214 AF = 1.4	203 AF = 2.5	186 AF = 3.0	168 AF = 3.0	141 AF = 3.0
	512	655 AF = 1.0	385 AF = 1.0	250 AF = 1.0	182 AF = 1.0	205 AF = 2.0	182 AF = 2.0	148 AF = 2.0
	1024	1167 AF = 1.0	641 AF = 1.0	378 AF = 1.0	246 AF = 1.0	290 AF = 2.0	246 AF = 2.0	180 AF = 2.0
	2048	2191 AF = 1.0	1153 AF = 1.0	634 AF = 1.0	374 AF = 1.0	461 AF = 2.0	374 AF = 2.0	244 AF = 2.0
	4096	4239 AF = 1.0	2177 AF = 1.0	1146 AF = 1.0	630 AF = 1.0	802 AF = 2.0	630 AF = 2.0	372 AF = 2.0

I. Async Hot-Plug Reference Model §

This appendix presents a recommended reference model for async hot-plug. The reference model covers three areas:

- Async hot-plug initial configuration
- Async removal configuration and interrupt handling
- Async hot-add configuration and interrupt handling

There are no normative requirements in this section. The entire reference model is contained in a series of implementation notes. The reference model documents how the various hot-plug mechanisms are envisioned be used to implement a robust asynchronous hot-plug model, but does not mandate that they be used that way.

For brevity and readability, this section uses the following acronyms:

DLL

Data Link Layer

DSP

Downstream Port

FWF

firmware first

HPS

Hot-Plug Surprise

OOB

out-of-band

OS

operating system

PD

presence detect

SFW

system firmware

The reference model covers both the HPS and DPC mechanisms. DPC is the recommended mechanism. The reference model covers FWF for DPC but not for HPS.

The reference model provides a recommended framework for DPC software to support Containment Error Recovery (CER) along with async hot-plug. The reference model does not explicitly cover CER outside of async hot-plug, but certain aspects can be leveraged for DPC support of CER when async hot-plug is not being used.

SFW may use the System Firmware Intermediary (SFI) Capability for async hot-plug, orderly removal hot-plug, or other operations. This reference model does not rely on its functionality nor cover its use.

The reference model refers to various bits or fields outlined in § Section 6.7.2 . For brevity, pointers to the associated registers are not replicated in this section.

IMPLEMENTATION NOTE: IN-BAND PRESENCE DETECT MECHANISM DEPRECATED FOR ASYNC HOT-PLUG §

Due to architectural issues, the in-band (Physical-Layer-based) portion of the PD mechanism is deprecated for use with async hot-plug. One issue is that in-band PD as architected does not detect adapter removal during certain LTSSM states, notably the L1 and Disabled States. Another issue is that when both in-band and OOB PD are being used together, the Presence Detect State bit and its associated interrupt mechanism always reflect the logical OR of the inband and OOB PD states, and with some hot-plug hardware configurations, it is important for software to detect and respond to in-band and OOB PD events independently.

If OOB PD is being used and the associated DSP supports In-Band PD Disable, it is recommended that the In-Band PD Disable bit be Set, and the Presence Detect State bit and its associated interrupt mechanism be used exclusively for OOB PD.

As a substitute for in-band PD with async hot-plug, the reference model uses either the DPC or the DLL Link Active mechanism.

The reference model assumes and covers the configurations listed below. While these cover the bulk of the envisioned use cases, many minor variations are not explicitly covered. For example, there are multiple ways to determine if a slot supports OOB PD, and the reference model does not cover them. As another example, the reference model refers to adding or removing an adapter, but some hot-pluggable modules may include a Switch and multiple Endpoint components.

- Reference model assumptions:
 - DSPs support DLL Link Active Reporting
 - DSPs support In-Band PD Disable
 - Operating systems support both HPS and DPC, using DPC if available
- Reference model covers:
 - Slots that support HPS and/or DPC (SW never enables both at same time)
 - Slots with or without OOB PD
 - RPs with or without RP Extensions for DPC

I.1 Async Hot-Plug Initial Configuration §

IMPLEMENTATION NOTE: ASYNC HOT-PLUG INITIAL CONFIGURATION §

Basic Steps	HPS	DPC
Determine capability control entity	• OS requests, and is granted control of PCIe Native Hot-Plug • If FWF, SFW retains control of AER and DPC • Else OS requests and is granted control of AER and DPC	
OS and SFW determine which async hot-plug mechanism to use; OS/SFW interactions here are outside scope of this specification	• If DPC capability then ◦ If HPS bit not Set, use DPC ◦ Else attempt to Clear HPS bit (§ Section 6.7.4.4) ▪ If successful, use DPC ▪ Else use HPS • Else if HPS bit Set, use HPS • Else async hot-plug cannot be supported by this slot • Configure DSP for selected mechanism	
OS determines if adapter present	• If OOB PD supported, use it to determine if adapter is present ◦ Set In-band PD Disable bit (SFW may have it Set by default) ◦ Read PD State bit • Else allow Link to attempt to train; use DLL Link Active to determine if adapter is present (and at least minimally functional)	
If adapter is present, OS waits for adapter to become ready for configuration	• If using Device Readiness Status (DRS), begin configuration if/after DSP receives a DRS Message • If using Configuration RRS Software Visibility, can attempt first Configuration Read (of Vendor ID field) after 100 ms • Otherwise, must wait at least 1000 ms before attempting first Configuration Read	
OS configures for appropriate case	• If adapter is present, configure system for handling async removal • Else configure system for handling async hot-add	

I.2 Async Removal Configuration and Interrupt Handling §

IMPLEMENTATION NOTE: ASYNC REMOVAL CONFIGURATION §

Basic Steps	HPS	DPC
OS/SFW configure appropriate error handling	- If OS and driver support a non-CER error recovery approach, its policies may influence some of these error settings - Configure the error handling supported by HPS	- If OS and driver support CER, its policies may influence some of these error settings - Enable uncorrectable AER errors to trigger DPC, including <u>Surprise Down</u> - If using RP Extensions for DPC, configure RP PIO error handling - Configure RP Completion Timeout handling per platform and OS policies
OS/SFW configure async removal interrupts	OS enables DLL State Changed interrupt	- If FWF, configure DPC for ERR_COR signaling to enable SFW handling - Else configure DPC for interrupt to enable OS handling
	If OOB PD supported, OS enables PD Changed interrupt	

IMPLEMENTATION NOTE:

ASYNC REMOVAL INTERRUPT HANDLING §

Basic Steps	HPS	DPC
Service routine entry	If PD Changed or DLL State Change, OS is interrupted	- If PD Changed, OS is interrupted - If FWF, OS invokes SFW; preferably via OS Setting DPC Software Trigger bit (if implemented)
		- If DPC triggered - If FWF, DPC sends <u>ERR_COR</u> to signal SFW - Else DPC interrupts OS
Prevent further Link activity	- OS Sets Disable Link; Link goes down and stays down - OS polls DLL Link Active until Clear	DPC automatically keeps Link down
Log errors from DSP's AER/DPC		- If FWF, SFW logs DSP errors - SFW invokes OS and exits
	OS logs and then clears DSP errors	
OS notifies impacted software/driver, which cease accessing the adapter		
OS determines if adapter is still present	- Some FFs with OOB PD automatically power off slot and/or disable switched signals - If OOB PD supported, use it to determine if adapter is physically present - If adapter not determined to be absent, re-enable the Link and slot - As applicable, Clear Disable Link or release DPC - Wait until Link trains or adapter is deemed absent or non-functional - If non-functional, optionally perform a slot reset (if supported) - If still non-functional, optionally power-cycle the slot (if supported)	
If OS determined adapter to be present	- OS waits for adapter to become ready for configuration - OS enumerates and configures the adapter - If DPC and FWF, OS invokes SFW to log adapter AER errors	

Basic Steps	HPS	DPC
	• OS logs adapter AER errors and then clears them • If OS determines the adapter is suitable for continued operation ◦ OS configures for async removal handling ◦ OS resumes driver • Else OS takes OS-specific action	
Else (adapter not present)	• OS ensures DSP is in a clean state ready for a new/different adapter • OS configures for async hot-add handling	

I.3 Async Hot-Add Configuration and Interrupt Handling §

IMPLEMENTATION NOTE: ASYNC HOT-ADD CONFIGURATION §

Basic Steps	HPS	DPC
OS configures for async hot-add handling • Enable Link to train if/when an adapter is inserted ◦ E.g., Clear Disable Link or release DPC if needed • If appropriate for form factor, enable power controller prior to adapter insertion • If OOB PD supported, enable OS interrupt on PD Changed • Else enable OS interrupt on DLL State Changed		

IMPLEMENTATION NOTE:

ASYNC HOT-ADD INTERRUPT HANDLING §

Basic Steps	HPS	DPC
OS is interrupted due to PD Changed or DLL State Changed		
OS waits for the adapter to become ready for configuration		
• If appropriate for form factor, enable power controller now (following adapter insertion) • Wait for DLL Link Active • Wait for adapter to become ready for configuration		
OS configures adapter per standard OS conventions		
OS configures for async removal handling		
OS calls driver to complete adapter configuration and begin normal operation		

J. Alpha Power and Reverse lookup assignment §

The following files are attached to the No Changebar PDF for this specification.

- alpha_powers.vh
 - MD5 = f25de382b6f00fbcefa00a4d25199e85
- ecc_84to86_encoder.sv
 - MD5 = 38956a4b15be17181b2cb098c000ff85
- ecc_250to256_encoder.sv
 - MD5 = 55e41b4727c51715d60b8a08c64d25bb
- ecc_86to84_decoder.sv
 - MD5 = 35518f84edf9361688f065394ceff0cb
- ecc_256to250_decoder.sv
 - MD5 = dbdf0bda8eef74cb9d7af00312a7ea53

J.1 Alpha Powers §

alpha_powers.vh

```
//index to alpha power lookup
localparam logic [7:0] i_to_alpha_power_i[255:0] =
'{
8'h01,
8'h8e,
8'h47,
8'had,
8'hd8,
8'h6c,
8'h36,
8'h1b,
8'h83,
8'hcf,
8'he9,
8'hfa,
8'h7d,
8'hb0,
8'h58,
8'h2c,
8'h16,
8'h0b,
8'h8b,
8'hcb,
8'heb,
8'hfb,
8'hf3,
8'hf7,
8'hf5,
8'hf4,
8'h7a,
8'h3d,
8'h90,
8'h48,
8'h24,
8'h12,
8'h09,
8'h8a,
8'h45,
8'hac,
8'h56,
8'h2b,
8'h9b,
8'hc3,
8'hcf,
8'hf9,
8'hf2,
8'h79,
8'hb2,
8'h59,
8'ha2,
8'h51,
8'ha6,
8'h53,
8'ha7,
8'hdd,
8'he0,
8'h70,
8'h38,
8'h1c,
8'h0e,
8'h07,
8'h8d,
8'hc8,
8'h64,
```

8'h32,
8'h19,
8'h82,
8'h41,
8'hae,
8'h57,
8'ha5,
8'hdc,
8'h6e,
8'h37,
8'h95,
8'hc4,
8'h62,
8'h31,
8'h96,
8'h4b,
8'hab,
8'hdb,
8'he3,
8'hff,
8'hf1,
8'hf6,
8'h7b,
8'hb3,
8'hd7,
8'he5,
8'hfc,
8'h7e,
8'h3f,
8'h91,
8'hc6,
8'h63,
8'hbf,
8'hd1,
8'he6,
8'h73,
8'hb7,
8'hd5,
8'he4,
8'h72,
8'h39,
8'h92,
8'h49,
8'haa,
8'h55,
8'ha4,
8'h52,
8'h29,
8'h9a,
8'h4d,
8'ha8,
8'h54,
8'h2a,
8'h15,
8'h84,
8'h42,
8'h21,
8'h9e,
8'h4f,
8'ha9,
8'hda,
8'h6d,
8'hb8,
8'h5c,
8'h2e,

8'h17,
8'h85,
8'hcc,
8'h66,
8'h33,
8'h97,
8'hc5,
8'hec,
8'h76,
8'h3b,
8'h93,
8'hc7,
8'hed,
8'hf8,
8'h7c,
8'h3e,
8'h1f,
8'h81,
8'hce,
8'h67,
8'hbd,
8'hd0,
8'h68,
8'h34,
8'h1a,
8'h0d,
8'h88,
8'h44,
8'h22,
8'h11,
8'h86,
8'h43,
8'haf,
8'hd9,
8'he2,
8'h71,
8'hb6,
8'h5b,
8'ha3,
8'hdf,
8'he1,
8'hfe,
8'h7f,
8'hb1,
8'hd6,
8'h6b,
8'hbb,
8'hd3,
8'he7,
8'hfd,
8'hf0,
8'h78,
8'h3c,
8'h1e,
8'h0f,
8'h89,
8'hca,
8'h65,
8'hbc,
8'h5e,
8'h2f,
8'h99,
8'hc2,
8'h61,
8'hbe,

8'h5f,
8'ha1,
8'hde,
8'h6f,
8'hb9,
8'hd2,
8'h69,
8'hba,
8'h5d,
8'ha0,
8'h50,
8'h28,
8'h14,
8'h0a,
8'h05,
8'h8c,
8'h46,
8'h23,
8'h9f,
8'hc1,
8'hee,
8'h77,
8'hb5,
8'hd4,
8'h6a,
8'h35,
8'h94,
8'h4a,
8'h25,
8'h9c,
8'h4e,
8'h27,
8'h9d,
8'hc0,
8'h60,
8'h30,
8'h18,
8'h0c,
8'h06,
8'h03,
8'h8f,
8'hc9,
8'hea,
8'h75,
8'hb4,
8'h5a,
8'h2d,
8'h98,
8'h4c,
8'h26,
8'h13,
8'h87,
8'hcd,
8'he8,
8'h74,
8'h3a,
8'h1d,
8'h80,
8'h40,
8'h20,
8'h10,
8'h08,
8'h04,
8'h02,
8'h01

```

};
//Reverse index lookup
localparam logic [7:0] alpha_power_i_to_i[255:0] =
'{
8'haf,
8'h58,
8'h50,
8'ha8,
8'hea,
8'hf4,
8'hd6,
8'h74,
8'he8,
8'had,
8'he7,
8'he6,
8'he9,
8'hd5,
8'hae,
8'h4f,
8'hd7,
8'h2c,
8'h75,
8'h7a,
8'heb,
8'h16,
8'hf5,
8'h0b,
8'h51,
8'ha0,
8'ha9,
8'h9c,
8'hb0,
8'h5f,
8'h59,
8'hcb,
8'h5a,
8'h3e,
8'hcc,
8'hbb,
8'hb1,
8'h86,
8'h60,
8'hfb,
8'haa,
8'h55,
8'h9d,
8'h29,
8'h52,
8'h3b,
8'ha1,
8'h6c,
8'hf6,
8'h6f,
8'h0c,
8'h7f,
8'hec,
8'h49,
8'h17,
8'hc4,
8'h76,
8'ha4,
8'h7b,
8'hb7,
8'hd8,

```


8'h43,
8'h2d,
8'h1f,
8'ha2,
8'h41,
8'h6d,
8'h47,
8'h53,
8'h39,
8'h3c,
8'h84,
8'h9e,
8'h5d,
8'h2a,
8'h14,
8'hab,
8'hd3,
8'h56,
8'hf2,
8'h61,
8'hbe,
8'hfc,
8'hdc,
8'hb2,
8'h97,
8'h87,
8'h90,
8'hcd,
8'hcf,
8'hbc,
8'h95,
8'h5b,
8'hd1,
8'h3f,
8'h37,
8'h2e,
8'h89,
8'h20,
8'h23,
8'hd9,
8'h92,
8'h44,
8'h11,
8'h7c,
8'hb4,
8'hb8,
8'h26,
8'h77,
8'h99,
8'ha5,
8'he3,
8'h18,
8'hfe,
8'hc5,
8'h31,
8'hed,
8'hde,
8'h4a,
8'h67,
8'h0d,
8'h63,
8'h80,
8'h8c,
8'hf7,
8'hc0,

8'h70,
8'h07,
8'h57,
8'ha7,
8'hf3,
8'h73,
8'hac,
8'he5,
8'hd4,
8'h4e,
8'h2b,
8'h79,
8'h15,
8'h0a,
8'h9f,
8'h9b,
8'h5e,
8'hca,
8'h3d,
8'hba,
8'h85,
8'hfa,
8'h54,
8'h28,
8'h3a,
8'h6b,
8'h6e,
8'h7e,
8'h48,
8'hc3,
8'ha3,
8'hb6,
8'h42,
8'h1e,
8'h40,
8'h46,
8'h38,
8'h83,
8'h5c,
8'h13,
8'hd2,
8'hf1,
8'hbd,
8'hdb,
8'h96,
8'h8f,
8'hce,
8'h94,
8'hd0,
8'h36,
8'h88,
8'h22,
8'h91,
8'h10,
8'hb3,
8'h25,
8'h98,
8'he2,
8'hfd,
8'h30,
8'hdd,
8'h66,
8'h62,
8'h8b,
8'hbf,

8'h06,
8'ha6,
8'h72,
8'he4,
8'h4d,
8'h78,
8'h09,
8'h9a,
8'hc9,
8'hb9,
8'hf9,
8'h27,
8'h6a,
8'h7d,
8'hc2,
8'hb5,
8'h1d,
8'h45,
8'h82,
8'h12,
8'hf0,
8'hda,
8'h8e,
8'h93,
8'h35,
8'h21,
8'h0f,
8'h24,
8'he1,
8'h2f,
8'h65,
8'h8a,
8'h05,
8'h71,
8'h4c,
8'h08,
8'hc8,
8'hf8,
8'h69,
8'hc1,
8'h1c,
8'h81,
8'hef,
8'h8d,
8'h34,
8'h0e,
8'he0,
8'h64,
8'h04,
8'h4b,
8'hc7,
8'h68,
8'h1b,
8'hee,
8'h33,
8'hdf,
8'h03,
8'hc6,
8'h1a,
8'h32,
8'h02,
8'h19,
8'h01,
8'h00,

```
8'hff
};
```

J.2 84 Byte to 86 Byte Encoder §

ecc_84to86_encoder.sv

```
`ifndef ECC_84T086_ENCODER__SV
`define ECC_84T086_ENCODER__SV
//Description: This module implements the 84B to 86B ECC group encoder
//as defined in Chapter 4 of the PCI Express Base Specification 6.0.
//Input: 84 bytes of Data
//Output: 86 bytes of data. Check is on byte 84, Parity is on byte 85.
module ecc_84to86_encoder
(
    input logic [7:0] data_in[83:0],
    output logic [7:0] data_out[85:0]
);
logic [7:0] synd_parity;
logic [7:0] synd_check;
`include "alpha_powers.vh"
//Calculate the Check and Parity Bytes.
always_comb begin
    synd_parity      = 8'h00;
    synd_check       = 8'h00;
    for(int i = 0 ; i < 84 ; i++) begin
        synd_parity ^= data_in[i];
        synd_check  ^= ({8{data_in[i][0]}} & i_to_alpha_power_i[84-i]) ^
                        ({8{data_in[i][1]}} & i_to_alpha_power_i[85-i]) ^
                        ({8{data_in[i][2]}} & i_to_alpha_power_i[86-i]) ^
                        ({8{data_in[i][3]}} & i_to_alpha_power_i[87-i]) ^
                        ({8{data_in[i][4]}} & i_to_alpha_power_i[88-i]) ^
                        ({8{data_in[i][5]}} & i_to_alpha_power_i[89-i]) ^
                        ({8{data_in[i][6]}} & i_to_alpha_power_i[90-i]) ^
                        ({8{data_in[i][7]}} & i_to_alpha_power_i[91-i]));
    end
end
//Assign Output
always_comb begin
    for (int i = 0 ; i < 84 ; i++) begin
        data_out[i] = data_in[i];
    end
    data_out[84]    = synd_check;
    data_out[85]    = synd_parity;
end
endmodule
`endif
```

J.3 250 Byte to 256 Byte Encoder example §

`ecc_250to256_encoder.sv`

```

`ifndef ECC_250T0256_ENCODER__SV
`define ECC_250T0256_ENCODER__SV
//Description: This module implements ECC for Flit level blocks.
//It splits the input 250 bytes into the 3 groups defined by PCIe 6.0
//and assigns the ECC bytes based on the byte positions shown
//in Chapter 4 of PCIe 6.0 Specification.
//
//Input      : 250 bytes of Flit without ECC.
//Output     : 256 bytes of Flit with ECC added.
//Child Module : ecc_84to86_encoder
module ecc_250to256_encoder
(
    input logic [7:0] data_in[249:0],
    output logic [7:0] data_out[255:0]
);
logic [7:0] ecc_output0[85:0];
logic [7:0] ecc_output1[85:0];
logic [7:0] ecc_output2[85:0];
logic [7:0] ecc0_input[83:0];
logic [7:0] ecc1_input[83:0];
logic [7:0] ecc2_input[83:0];
//Split input into the 3 ECC groups
//Groups 1 and 2 get an extra byte of 0 padded to byte 83.
always_comb begin
    for (int i = 0 ; i < 83 ; i++) begin
        ecc0_input[i] = data_in[i*3];
        ecc1_input[i] = data_in[(i*3) + 1];
        ecc2_input[i] = data_in[(i*3) + 2];
    end
    ecc0_input[83]    = data_in[249];
    ecc1_input[83]    = 8'h00;
    ecc2_input[83]    = 8'h00;
end
ecc_84to86_encoder enc0 (
    .data_in ( ecc0_input ),
    .data_out ( ecc_output0 )
);
ecc_84to86_encoder enc1 (
    .data_in ( ecc1_input ),
    .data_out ( ecc_output1 )
);
ecc_84to86_encoder enc2 (
    .data_in ( ecc2_input ),
    .data_out ( ecc_output2 )
);
//Assign output
always_comb begin
    //First 250 bytes are assigned directly from input.
    for (int i = 0 ; i < 250 ; i++ ) begin
        data_out[i] = data_in[i];
    end
    //Next Six bytes are assigned using the check and parity
    //from the three ECC groups.
    data_out[250] = ecc_output1[84]; //ECC1[0] = Check of Group 1.
    data_out[251] = ecc_output2[84]; //ECC2[0] = Check of Group 2.
    data_out[252] = ecc_output0[84]; //ECC0[0] = Check of Group 0.
    data_out[253] = ecc_output1[85]; //ECC1[1] = Parity of Group 1.
    data_out[254] = ecc_output2[85]; //ECC2[1] = Parity of Group 2.
    data_out[255] = ecc_output0[85]; //ECC0[1] = Parity of Group 0.
end
endmodule
`endif

```

J.4 86 Byte to 84 Byte Decoder §

ecc_86to84_decoder.sv

```

`ifndef ECC_86T084_DECODER__SV
`define ECC_86T084_DECODER__SV
//Description: This module implements the 86 byte to 84 byte decoder for the
//ECC code defined in PCIe 6.0 Specification.
//
//Input   : 86 bytes of encoded data
//Outputs :
//         84 bytes of decoded data (corrected if applicable)
//         synd_check and synd parity for logging
//         Error magnitude – value of the error term
//         Error byte – location of the error byte, not applicable if
//                     (unc_error == 1)
//         no_error – no error was detected.
//         single_error – single error was detected.
//         unc_error – uncorrectable error (error decoding pointed to
//                     non-existent column).
module ecc_86to84_decoder
(
    input logic [7:0] data_in[85:0],
    output logic [7:0] data_out[83:0],
    output logic [7:0] synd_check,
    output logic [7:0] synd_parity,
    output logic [7:0] error_magnitude,
    output logic [6:0] error_byte,
    output logic no_error,
    output logic single_error,
    output logic unc_error
);
logic synd_parity_0;
logic synd_check_0;
logic [7:0] synd_parity_map_i;
logic [7:0] synd_check_map_i;
logic [8:0] error_byte_result;
logic overflow;
`include "alpha_powers.vh"
logic [7:0] encoded_data[85:0];
ecc_84to86_encoder encoder (
    .data_in ( data_in[83:0] ),
    .data_out ( encoded_data[85:0] )
); // The decoder has to first invoke the encoder function to calculate the syndrome
assign synd_parity      = encoded_data[85] ^ data_in[85];
assign synd_check       = encoded_data[84] ^ data_in[84];
assign synd_parity_0    = (synd_parity == 8'h00);
assign synd_check_0     = (synd_check == 8'h00);
assign synd_parity_map_i = alpha_power_i_to_i[synd_parity];
assign synd_check_map_i  = alpha_power_i_to_i[synd_check];
assign error_byte       = error_byte_result[6:0];
always_comb begin
    no_error      = 1'b0;
    single_error  = 1'b0;
    unc_error     = 1'b0;
    error_byte_result = 9'd0;
    error_magnitude = synd_parity;
    overflow       = 1'b0;
    for (int i = 0 ; i < 84 ; i++) begin
        data_out[i] = data_in[i];
    end
    unique casez ({synd_check_0, synd_parity_0})
    2'b11: begin
        no_error = 1'b1;
    end
    2'b10: begin
        single_error = 1'b1;
        error_byte_result = 9'd85;
    end
    default: begin
        unc_error = 1'b1;
    end
end
endmodule

```

```

end
2'b01: begin
    single_error      = 1'b1;
    error_magnitude   = synd_check;
    error_byte_result = 9'd84;
end
2'b00: begin
    //Overflow term in the equations below is ignored, as those cases are
    //outside the correction capability of the ECC, and
    //we rely on CRC to detect those.
    if (synd_check_map_i < synd_parity_map_i) begin
        {overflow, error_byte_result} = 9'd84 - (9'd255 + {1'b0, synd_check_map_i} - {1'b0, synd_parity_map_i});
    end else begin
        {overflow, error_byte_result} = 9'd84 - ({1'b0, synd_check_map_i} - {1'b0, synd_parity_map_i}) ;
    end
    if (error_byte_result >= 9'd84) begin
        unc_error      = 1'b1;
    end else begin
        single_error    = 1'b1;
        data_out[error_byte_result] ^= synd_parity;
    end
end
end
endcase
end
endmodule
`endif

```

J.5 256 Byte to 250 Byte decoder §

ecc_256to250_decoder.sv

```

`ifndef ECC_256T0250_DECODER__SV
`define ECC_256T0250_DECODER__SV
//Description : This module takes in 256 bytes of the flit, and outputs 250 bytes
//of corrected data. It also gives the metrics of the detected error per ECC group
//defined in PCIe 6.0 Specification.
//Note: it is possible that the ECC code corrupted the data further if more than a
//single byte error happened within an ECC group. The final check is the 8B CRC -
//data should only be consumed if CRC passes post FEC correction.
//Input : 256 bytes of the flit.
//Output : 250 bytes of corrected data
//      synd_check and parity for each ECC group for logging (see signal descriptions)
//      Error metrics for each ECC group (see signal descriptions).
module ecc_256to250_decoder (
    input logic [7:0] data_in[255:0],
    output logic [7:0] data_out[249:0],
    output logic [7:0] synd_check0, //Syndrome Check for ECC Group 0. Flit Error log 1, bits 31:24.
    output logic [7:0] synd_parity0, //Syndrome Parity for ECC Group 0. Flit Error log 1, bits 23:16.
    output logic [7:0] synd_check1, //Syndrome Check for ECC Group 1. Flit Error log 2, bits 15:8.
    output logic [7:0] synd_parity1, //Syndrome Parity for ECC Group 1. Flit Error log 2, bits 7:0.
    output logic [7:0] synd_check2, //Syndrome Check for ECC Group 2. Flit Error log 2, bits 31:24.
    output logic [7:0] synd_parity2, //Syndrome Parity for ECC Group 2. Flit Error log 2, bits 23:16.
    //Error metrics
    //Note that error metrics can be incorrect if more than
    //1 byte per ECC group is corrupted, and CRC check ultimately
    //determines whether the flit is consumed or replayed.
    //
    //If unc_error0 == 0, error_byte0 indicates the error location for ECC Group 0 as the
    //byte position between(including) 0 and 85.
    //If unc_error1 == 0, error_byte1 indicates the error location for ECC Group 1 as the
    //byte position between(including) 0 and 85.
    //If unc_error2 == 0, error_byte2 indicates the error location for ECC Group 2 as the
    //byte position between(including) 0 and 85.
    //
    //ECC Group 0 metrics :
    output logic [7:0] error_magnitude0,
    output logic [6:0] error_byte0,
    output logic no_error0,
    output logic single_error0,
    output logic unc_error0,
    //ECC Group 1 metrics :
    output logic [7:0] error_magnitude1,
    output logic [6:0] error_byte1,
    output logic no_error1,
    output logic single_error1,
    output logic unc_error1,
    //ECC Group 2 metrics :
    output logic [7:0] error_magnitude2,
    output logic [6:0] error_byte2,
    output logic no_error2,
    output logic single_error2,
    output logic unc_error2
);
logic [7:0] dec_data_out0 [83:0];
logic [7:0] dec_data_out1 [83:0];
logic [7:0] dec_data_out2 [83:0];
logic [7:0] dec_data_in0 [85:0];
logic [7:0] dec_data_in1 [85:0];
logic [7:0] dec_data_in2 [85:0];
//Split the received flit into the 3 ECC groups.
always_comb begin
    for (int i=0; i<83; i++) begin
        dec_data_in0[i] = data_in[i*3];
        dec_data_in1[i] = data_in[(i*3) + 1];
        dec_data_in2[i] = data_in[(i*3) + 2];
    end
end

```

```

        data_out[i*3]      = dec_data_out0[i];
        data_out[i*3 + 1] = dec_data_out1[i];
        data_out[i*3 + 2] = dec_data_out2[i];
    end
    dec_data_in0[83]      = data_in[249]; //Blue ECC group 0
    dec_data_in0[84]      = data_in[252];
    dec_data_in0[85]      = data_in[255];
    dec_data_in1[83]      = 8'h00;        //Orange ECC group 1
    dec_data_in1[84]      = data_in[250];
    dec_data_in1[85]      = data_in[253];
    dec_data_in2[83]      = 8'h00;        //Green ECC group 2
    dec_data_in2[84]      = data_in[251];
    dec_data_in2[85]      = data_in[254];
    data_out[249]         = dec_data_out0[83];
end
//Instantiate separate 86to84 decoder for each group.
ecc_86to84_decoder dec0 (
    .data_in ( dec_data_in0 ),
    .data_out ( dec_data_out0 ),
    .synd_check ( synd_check0 ),
    .synd_parity ( synd_parity0 ),
    .error_magnitude ( error_magnitude0 ),
    .error_byte ( error_byte0 ),
    .no_error ( no_error0 ),
    .single_error ( single_error0 ),
    .unc_error ( unc_error0 )
);
ecc_86to84_decoder dec1 (
    .data_in ( dec_data_in1 ),
    .data_out ( dec_data_out1 ),
    .synd_check ( synd_check1 ),
    .synd_parity ( synd_parity1 ),
    .error_magnitude ( error_magnitude1 ),
    .error_byte ( error_byte1 ),
    .no_error ( no_error1 ),
    .single_error ( single_error1 ),
    .unc_error ( unc_error1 )
);
ecc_86to84_decoder dec2 (
    .data_in ( dec_data_in2 ),
    .data_out ( dec_data_out2 ),
    .synd_check ( synd_check2 ),
    .synd_parity ( synd_parity2 ),
    .error_magnitude ( error_magnitude2 ),
    .error_byte ( error_byte2 ),
    .no_error ( no_error2 ),
    .single_error ( single_error2 ),
    .unc_error ( unc_error2 )
);
endmodule
`endif

```


K. MATLAB created generator matrix for CRC code §

The following files are attached to the No Changebar PDF for this specification.

- `gen_matrix.txt`
 - MD5 = e6d5d08c0fc89cf28bcd142bcd9c635
- `gen_matrix.v`
 - MD5 = 44362bcba204ff1548c3fbd61d8efdb
- `pci_sig_8B_crc.sv`
 - MD5 = 04f8a99644d87483155921ce870a09f1

K.1 Generator Matrix output §

`gen_matrix.txt` (8 columns and 1936 rows, values in hex):

```

a7 ad 46 a7 3e 67 9d 2d
c6 c3 23 c6 1f a6 db 83
63 f4 84 63 9a 53 f8 d4
a4 7a 42 a4 4d bc 7c 6a
52 3d 21 52 b3 5e 3e 35
29 8b 85 29 cc 2f 1f 8f
81 d0 d7 81 66 82 9a d2
d5 68 fe d5 33 41 4d 69
14 2c c9 87 5a 45 80 cd
0a 16 f1 d6 2d b7 40 f3
05 0b ed 6b 83 ce 20 ec
97 90 e3 a0 d4 67 10 76
de 48 e4 50 6a a6 08 3b
6f 24 72 28 35 53 04 88
a2 12 39 14 8f bc 02 44
51 09 89 0a d2 5e 01 22
a1 38 40 d7 c4 13 ae e5
c5 1c 20 fe 62 9c 57 e7
f7 0e 10 7f 31 4e be e6
ee 07 08 aa 8d 27 5f 73
77 96 04 55 d3 86 ba ac
ae 4b 02 bf fc 43 5d 56
57 b0 01 ca 7e b4 bb 2b
be 58 95 65 3f 5a c8 80
29 71 eb d5 84 db cd 90
81 ad e0 ff 42 f8 f3 48
d5 c3 70 ea 21 7c ec 24
ff f4 38 75 85 3e 76 12
ea 7a 1c af d7 1f 3b 09
75 3d 0e c2 fe 9a 88 91
af 8b 07 61 7f 4d 44 dd
c2 d0 96 a5 aa b3 22 fb
95 4a 8e 60 c1 81 1b 88
df 25 47 30 f5 d5 98 44
fa 87 b6 18 ef ff 4c 22
7d d6 5b 0c e2 ea 26 11
ab 6b b8 06 71 75 13 9d
c0 a0 5c 03 ad af 9c db
60 50 2e 94 c3 c2 4e f8
30 28 17 4a f4 61 27 7c
b5 ba 1f 3e 0d ae 3b a1
cf 5d 9a 1f 93 57 88 c5
f2 bb 4d 9a dc be 44 f7
79 c8 b3 4d 6e 5f 22 ee
a9 64 cc b3 37 ba 11 77
c1 32 66 cc 8e 5d 9d ae
f5 19 33 66 47 bb db 57
ef 99 8c 33 b6 c8 f8 be
26 df c5 91 b8 dd ea 75
13 fa f7 dd 5c fb 75 af
9c 7d ee fb 2e e8 af c2
4e ab 77 e8 17 74 c2 61
27 c0 ae 74 9e 3a 61 a5
86 60 57 3a 4f 1d a5 c7
43 30 be 1d b2 9b c7 f6
b4 18 5f 9b 59 d8 f6 7b
14 6a 47 73 ed 14 08 89
0a 35 b6 ac e3 0a 04 d1
05 8f 5b 56 e4 05 02 fd
97 d2 b8 2b 72 97 01 eb
de 69 5c 80 39 de 95 e0
6f a1 2e 40 89 6f df 70
a2 c5 17 20 d1 a2 fa 38
51 f7 9e 10 fd 51 7d 1c

```

```

e7 b6 b4 60 95 9b ea e5
e6 5b 5a 30 df d8 75 e7
73 b8 2d 18 fa 6c af e6
ac 5c 83 0c 7d 36 c2 73
56 2e d4 06 ab 1b 61 ac
2b 17 6a 03 c0 98 a5 56
80 9e 35 94 60 4c c7 2b
40 4f 8f 4a 30 26 f6 80
c9 1d 2d ea b9 94 93 4e
f1 9b 83 75 c9 4a dc 27
ed d8 d4 af f1 25 6e 86
e3 6c 6a c2 ed 87 37 43
e4 36 35 61 e3 d6 8e b4
72 1b 8f a5 e4 6b 47 5a
39 98 d2 c7 72 a0 b6 2d
89 4c 69 f6 39 50 5b 83
fb 16 50 5f 35 4d 7b f2
e8 0b 28 ba 8f b3 a8 79
74 90 14 5d d2 cc 54 a9
3a 48 0a bb 69 66 2a c1
1d 24 05 c8 a1 33 15 f5
9b 12 97 64 c5 8c 9f ef
d8 09 de 32 f7 46 da e2
6c 91 6f 19 ee 23 6d 71
b6 35 f4 95 5d c8 d9 9e
5b 8f 7a df bb 64 f9 4f
b8 d2 3d fa c8 32 e9 b2
5c 69 8b 7d 64 19 e1 59
2e a1 d0 ab 32 99 e5 b9
17 c5 68 c0 19 d9 e7 c9
9e f7 34 60 99 f9 e6 f1
4f ee 1a 30 d9 e9 73 ed
fd 8c 47 95 8b fc 02 ce
eb 46 b6 df d0 7e 01 67
e0 23 5b fa 68 3f 95 a6
70 84 b8 7d 34 8a df 53
38 42 5c ab 1a 45 fa bc
1c 21 2e c0 0d b7 7d 5e
0e 85 17 60 93 ce ab 2f
07 d7 9e 30 dc 67 c0 82
84 79 6c 83 46 1c 60 c3
42 a9 36 d4 23 0e 30 f4
21 c1 1b 6a 84 07 18 7a
85 f5 98 35 42 96 0c 3d
d7 ef 4c 8f 21 4b 06 8b
fe e2 26 d2 85 b0 03 d0
7f 71 13 69 d7 58 94 68
aa ad 9c a1 fe 2c 4a 34
f7 4a 40 c8 ee 28 41 a2
ee 25 20 64 77 14 b5 51
77 87 10 32 ae 0a cf bd
ae d6 08 19 57 05 f2 cb
57 6b 04 99 be 97 79 f0
be a0 02 d9 5f de a9 78
5f 50 01 f9 ba 6f c1 3c
ba 28 95 e9 5d a2 f5 1e
91 93 c7 35 47 83 a8 24
dd dc f6 8f b6 d4 54 12
fb 6e 7b d2 5b 6a 2a 09
e8 37 a8 69 b8 35 15 91
74 8e 54 a1 5c 8f 9f dd
3a 47 2a c5 2e d2 da fb
1d b6 15 f7 17 69 6d e8
9b 5b 9f ee 9e a1 a3 74
45 78 cf 91 c3 32 88 2e

```

```

b7 3c f2 dd f4 19 44 17
ce 1e 79 fb 7a 99 22 9e
67 0f a9 e8 3d d9 11 4f
a6 92 c1 74 8b f9 9d b2
53 49 f5 3a d0 e9 db 59
bc b1 ef 1d 68 e1 f8 b9
5e cd e2 9b 34 e5 7c c9
42 ef c9 f9 d2 40 a7 65
21 e2 f1 e9 69 20 c6 a7
85 71 ed e1 a1 10 63 c6
d7 ad e3 e5 c5 08 a4 63
fe c3 e4 e7 f7 04 52 a4
7f f4 72 e6 ee 02 29 52
aa 7a 39 73 77 01 81 29
55 3d 89 ac ae 95 d5 81
a8 da 0d 95 39 83 24 51
54 6d 93 df 89 d4 12 bd
2a a3 dc fa d1 6a 09 cb
15 c4 6e 7d fd 35 91 f0
9f 62 37 ab eb 8f dd 78
da 31 8e c0 e0 d2 fb 3c
6d 8d 47 60 70 69 e8 1e
a3 d3 b6 30 38 a1 74 0f
4c 69 76 af 94 4e 0a cc
26 a1 3b c2 4a 27 05 66
13 c5 88 61 25 86 97 33
9c f7 44 a5 87 43 de 8c
4e ee 22 c7 d6 b4 6f 46
27 77 11 f6 6b 5a a2 23
86 ae 9d 7b a0 2d 51 84
43 57 db a8 50 83 bd 42
d4 03 28 29 2e dd 36 39
6a 94 14 81 17 fb 1b 89
35 4a 0a d5 9e e8 98 d1
8f 25 05 ff 4f 74 4c fd
d2 87 97 ea b2 3a 26 eb
69 d6 de 75 59 1d 13 e0
a1 6b 6f af b9 9b 9c 70
c5 a0 a2 c2 c9 d8 4e 38
ef b7 8b c2 7d 64 1c 4b
e2 ce d0 61 ab 32 0e b0
71 67 68 a5 c0 19 07 58
ad a6 34 c7 60 99 96 2c
c3 53 1a f6 30 d9 4b 16
f4 bc 0d 7b 18 f9 b0 0b
7a 5e 93 a8 0c e9 58 90
3d 2f dc 54 06 e1 2c 48
9a 1f ae 50 f5 3c 03 7b
4d 9a 57 28 ef 1e 94 a8
b3 4d be 14 e2 0f 4a 54
cc b3 5f 0a 71 92 25 2a
66 cc ba 05 ad 49 87 15
33 66 5d 97 c3 b1 d6 9f
8c 33 bb de f4 cd 6b da
46 8c c8 6f 7a f3 a0 6d
cf 94 a2 25 9a 04 3e a0
f2 4a 51 87 4d 02 1f 50
79 25 bd d6 b3 01 9a 28
a9 87 cb 6b cc 95 4d 14
c1 d6 f0 a0 66 df b3 0a
f5 6b 78 50 33 fa cc 05
ef a0 3c 28 8c 7d 66 97
e2 50 1e 14 46 ab 33 de
da c2 cd d4 0f 4d 10 af
6d 61 f3 6a 92 b3 08 c2

```

```

a3 a5 ec 35 49 cc 04 61
c4 c7 76 8f b1 66 02 a5
62 f6 3b d2 cd 33 01 c7
31 7b 88 69 f3 8c 95 f6
8d a8 44 a1 ec 46 df 7b
d3 54 22 c5 76 23 fa a8
d4 34 05 19 f4 b1 31 23
6a 1a 97 99 7a cd 8d 84
35 0d de d9 3d f3 d3 42
8f 93 6f f9 8b ec fc 21
d2 dc a2 e9 d0 76 7e 85
69 6e 51 e1 68 3b 3f d7
a1 37 bd e5 34 88 8a fe
c5 8e cb e7 1a 44 45 7f
d8 9a bb 18 11 63 06 4b
6c 4d c8 0c 9d a4 03 b0
36 b3 64 06 db 52 94 58
1b cc 32 03 f8 29 4a 2c
98 66 19 94 7c 81 25 16
4c 33 99 4a 3e d5 87 0b
26 8c d9 25 1f ff d6 90
13 46 f9 87 9a ea 6b 48
0d 92 1e 86 bc 25 4f f1
93 49 0f 43 5e 87 b2 ed
dc b1 92 b4 2f d6 59 e3
6e cd 49 5a 82 6b b9 e4
37 f3 b1 2d 41 a0 c9 72
8e ec cd 83 b5 50 f1 39
47 76 f3 d4 cf 28 ed 89
b6 3b ec 6a f2 14 e3 d1
3c c0 dc 12 69 7f 9d d3
1e 60 6e 09 a1 aa db fc
0f 30 37 91 c5 55 f8 7e
92 18 8e dd f7 bf 7c 3f
49 0c 47 fb ee ca 3e 8a
b1 06 b6 e8 77 65 1f 45
cd 03 5b 74 ae a7 9a b7
f3 94 b8 3a 57 c6 4d ce
7c e4 70 d5 d7 03 76 04
3e 72 38 ff fe 94 3b 02
1f 39 1c ea 7f 4a 88 01
9a 89 0e 75 aa 25 44 95
4d d1 07 af 55 87 22 df
b3 fd 96 c2 bf d6 11 fa
cc eb 4b 61 ca 6b 9d 7d
66 e0 b0 a5 65 a0 db ab
9e 8b 94 ad b4 4e 7a 87
4f d0 4a c3 5a 27 3d d6
b2 68 25 f4 2d 86 8b 6b
59 34 87 7a 83 43 d0 a0
b9 1a d6 3d d4 b4 68 50
c9 0d 6b 8b 6a 5a 34 28
f1 93 a0 d0 35 2d 1a 14
ed dc 50 68 8f 83 0d 0a
72 25 da 4d 24 52 dd 2f
39 87 6d b3 12 29 fb 82
89 d6 a3 cc 09 81 e8 41
d1 6b c4 66 91 d5 74 b5
fd a0 62 33 dd ff 3a cf
eb 50 31 8c fb ea 1d f2
e0 28 8d 46 e8 75 9b 79
70 14 d3 23 74 af d8 a9
a5 47 7f a4 fc 16 ea ef
c7 b6 aa 52 7e 0b 75 e2
f6 5b 55 29 3f 90 af 71

```

```

7b b8 bf 81 8a 48 c2 ad
a8 5c ca d5 45 24 61 c3
54 2e 65 ff b7 12 a5 f4
2a 17 a7 ea ce 09 c7 7a
15 9e c6 75 67 91 f6 3d
7f c5 1d c4 4d b0 d8 1f
aa f7 9b 62 b3 58 6c 9a
55 ee d8 31 cc 2c 36 4d
bf 77 6c 8d 66 16 1b b3
ca ae 36 d3 33 0b 98 cc
65 57 1b fc 8c 90 4c 66
a7 be 98 7e 46 48 26 33
c6 5f 4c 3f 23 24 13 8c
eb 5e ac 63 52 23 b6 3c
e0 2f 56 a4 29 84 5b 1e
70 82 2b 52 81 42 b8 0f
38 41 80 29 d5 21 5c 92
1c b5 40 81 ff 85 2e 49
0e cf 20 d5 ea d7 17 b1
07 f2 10 ff 75 fe 9e cd
96 79 08 ea af 7f 4f f3
5a b3 8a 56 7e b9 6b f4
2d cc 45 2b 3f c9 a0 7a
83 66 b7 80 8a f1 50 3d
d4 33 ce 40 45 ed 28 8b
6a 8c 67 20 b7 e3 14 d0
35 46 a6 10 ce e4 0a 68
8f 23 53 08 67 72 05 34
d2 84 bc 04 a6 39 97 1a
02 de c1 cf 3e ad f7 0e
01 6f f5 f2 1f c3 ee 07
95 a2 ef 79 9a f4 77 96
df 51 e2 a9 4d 7a ae 4b
fa bd 71 c1 b3 3d 57 b0
7d cb ad f5 cc 8b be 58
ab f0 c3 ef 66 d0 5f 2c
c0 78 f4 e2 33 68 ba 16
5f 11 18 bf cb 75 94 d2
ba 9d 0c ca f0 af 4a 69
5d db 06 65 78 c2 25 a1
bb f8 03 a7 3c 61 87 c5
c8 7c 94 c6 1e a5 d6 f7
64 3e 4a 63 0f c7 6b ee
32 1f 25 a4 92 f6 a0 77
19 9a 87 52 49 7b 50 ae
5c af 53 86 0d 3c 83 e8
2e c2 bc 43 93 1e d4 74
17 61 5e b4 dc 0f 6a 3a
9e a5 2f 5a 6e 92 35 1d
4f c7 82 2d 37 49 8f 9b
b2 f6 41 83 8e b1 d2 d8
59 7b b5 d4 47 cd 69 6c
b9 a8 cf 6a b6 f3 a1 36
b6 5c 43 14 11 e8 6e 53
5b 2e b4 0a 9d 74 37 bc
b8 17 5a 05 db 3a 8e 5e
5c 9e 2d 97 f8 1d 47 2f
2e 4f 83 de 7c 9b b6 82
17 b2 d4 6f 3e d8 5b 41
9e 59 6a a2 1f 6c b8 b5
4f b9 35 51 9a 36 5c cf
94 3b c6 d9 ab 4b cf ce
4a 88 63 f9 c0 b0 f2 67
25 44 a4 e9 60 58 79 a6
87 22 52 e1 30 2c a9 53

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d6 11 29 e5 18 16 c1 bc
6b 9d 81 e7 0c 0b f5 5e
a0 db d5 e6 06 90 ef 2f
50 f8 ff 73 03 48 e2 82
11 9a 58 81 f4 3b 30 c8
9d 4d 2c d5 7a 88 18 64
db b3 16 ff 3d 44 0c 32
f8 cc 0b ea 8b 22 06 19
7c 66 90 75 d0 11 03 99
3e 33 48 af 68 9d 94 d9
1f 8c 24 c2 34 db 4a f9
9a 46 12 61 1a f8 25 e9
eb 4a 3d 85 45 cd f9 03
e0 25 8b d7 b7 f3 e9 94
70 87 d0 fe ce ec e1 4a
38 d6 68 7f 67 76 e5 25
1c 6b 34 aa a6 3b e7 87
0e a0 1a 55 53 88 e6 d6
07 50 0d bf bc 44 73 6b
96 28 93 ca 5e 22 ac a0
4e 22 6c 41 90 f6 54 f4
27 11 36 b5 48 7b 2a 7a
86 9d 1b cf 24 a8 15 3d
43 db 98 f2 12 54 9f 8b
b4 f8 4c 79 09 2a da d0
5a 7c 26 a9 91 15 6d 68
2d 3e 13 c1 dd 9f a3 34
83 1f 9c f5 fb da c4 1a
1e c9 11 ac f0 01 94 eb
0f f1 9d 56 78 95 4a e0
92 ed db 2b 3c df 25 70
49 e3 f8 80 1e fa 87 38
b1 e4 7c 40 0f 7d d6 1c
cd 72 3e 20 92 ab 6b 0e
f3 39 1f 10 49 c0 a0 07
ec 89 9a 08 b1 60 50 96
97 0d 9d ae 55 db 2c 02
de 93 db 57 bf f8 16 01
6f dc f8 be ca 7c 0b 95
a2 6e 7c 5f 65 3e 90 df
51 37 3e ba a7 1f 48 fa
bd 8e 1f 5d c6 9a 24 7d
cb 47 9a bb 63 4d 12 ab
f0 b6 4d c8 a4 b3 09 c0
73 79 06 2b 31 1b 2b 73
ac a9 03 80 8d 98 80 ac
56 c1 94 40 d3 4c 40 56
2b f5 4a 20 fc 26 20 2b
80 ef 25 10 7e 13 10 80
40 e2 87 08 3f 9c 08 40
20 71 d6 04 8a 4e 04 20
10 ad 6b 02 45 27 02 10
2c f3 e7 64 86 a1 fb 86
16 ec e6 32 43 c5 e8 43
0b 76 73 19 b4 f7 74 b4
90 3b ac 99 5a ee 3a 5a
48 88 56 d9 2d 77 1d 2d
24 44 2b f9 83 ae 9b 83
12 22 80 e9 d4 57 d8 d4
09 11 40 e1 6a be 6c 6a
eb a5 44 9e 44 d9 5f 6e
e0 c7 22 4f 22 f9 ba 37
70 f6 11 b2 11 e9 5d 8e
38 7b 9d 59 9d e1 bb 47
1c a8 db b9 db e5 c8 b6

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0e 54 f8 c9 f8 e7 64 5b
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f7 5c 57 10 21 14 d1 3d
ee 2e be 08 85 0a fd 8b
77 17 5f 04 d7 05 eb d0
ae 9e ba 02 fe 97 e0 68
57 4f 5d 01 7f de 70 34
be b2 bb 95 aa 6f 38 1a
4a 46 7c 95 c7 4c dc 90
25 23 3e df f6 26 6e 48
87 84 1f fa 7b 13 37 24
d6 42 9a 7d a8 9c 8e 12
6b 21 4d ab 54 4e 47 09
a0 85 b3 c0 2a 27 b6 91
50 d7 cc 60 15 86 5b dd
28 fe 66 30 9f 43 b8 fb
53 52 40 d2 86 a6 ef 64
bc 29 20 69 43 53 e2 32
5e 81 10 a1 b4 bc 71 19
2f d5 08 c5 5a 5e ad 99
82 ff 04 f7 2d 2f c3 d9
41 ea 02 ee 83 82 f4 f9
b5 75 01 77 d4 41 7a e9
cf af 95 ae 6a b5 3d e1
64 41 9a b0 a1 36 14 52
32 b5 4d 58 c5 1b 0a 29
19 cf b3 2c f7 98 05 81
99 f2 cc 16 ee 4c 97 d5
d9 79 66 0b 77 26 de ff
f9 a9 33 90 ae 13 6f ea
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e1 f5 46 24 be 4e 51 af
cd 26 92 2d 7f ce 6e d8
f3 13 49 83 aa 67 37 6c
ec 9c b1 d4 55 a6 8e 36
76 4e cd 6a bf 53 47 1b
3b 27 f3 35 ca bc b6 98
88 86 ec 8f 65 5e 5b 4c
44 43 76 d2 a7 2f b8 26
22 b4 3b 69 c6 82 5c 13
e9 22 12 b0 a3 9f f2 7d
e1 11 09 58 c4 da 79 ab
e5 9d 91 2c 62 6d a9 c0
e7 db dd 16 31 a3 c1 60
e6 f8 fb 0b 8d c4 f5 30
73 7c e8 90 d3 62 ef 18
ac 3e 74 48 fc 31 e2 0c
56 1f 3a 24 7e 8d 71 06
a7 dd 8e 26 a4 7f b0 26
c6 fb 47 13 52 aa 58 13
63 e8 b6 9c 29 55 2c 9c
a4 74 5b 4e 81 bf 16 4e
52 3a b8 27 d5 ca 0b 27
29 1d 5c 86 ff 65 90 86
81 9b 2e 43 ea a7 48 43
d5 d8 17 b4 75 c6 24 b4
64 e4 48 1d 42 68 8b cd
32 72 24 9b 21 34 d0 f3
19 39 12 d8 85 1a 68 ec
99 89 09 6c d7 0d 34 76
d9 d1 91 36 fe 93 1a 3b
f9 fd dd 1b 7f dc 0d 88

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e9 eb fb 98 aa 6e 93 44
e1 e0 e8 4c 55 37 dc 22
68 f4 3f ce 21 51 f1 d8
34 7a 8a 67 85 bd ed 6c
1a 3d 45 a6 d7 cb e3 36
0d 8b b7 53 fe f0 e4 1b
93 d0 ce bc 7f 78 72 98
dc 68 67 5e aa 3c 39 4c
6e 34 a6 2f 55 1e 89 26
37 1a 53 82 bf 0f d1 13
03 35 48 d6 67 5a c5 62
94 8f 24 6b a6 2d f7 31
4a d2 12 a0 53 83 ee 8d
25 69 09 50 bc d4 77 d3
87 a1 91 28 5e 6a ae fc
d6 c5 dd 14 2f 35 57 7e
6b f7 fb 0a 82 8f be 3f
a0 ee e8 05 41 d2 5f 8a
61 f0 ff 33 0f 06 b5 bb
a5 78 ea 8c 92 03 cf c8
c7 3c 75 46 49 94 f2 64
f6 1e af 23 b1 4a 79 32
7b 0f c2 84 cd 25 a9 19
a8 92 61 42 f3 87 c1 99
54 49 a5 21 ec d6 f5 d9
2a b1 c7 85 76 6b ef f9
80 a0 6a 7f b0 01 d5 3e
40 50 35 aa 58 95 ff 1f
20 28 8f 55 2c df ea 9a
10 14 d2 bf 16 fa 75 4d
08 0a 69 ca 0b 7d af b3
04 05 a1 65 90 ab c2 cc
02 97 c5 a7 48 c0 61 66
01 de f7 c6 24 60 a5 33
07 c7 39 17 3f b2 a3 2d
96 f6 89 9e 8a 59 c4 83
4b 7b d1 4f 45 b9 62 d4
b0 a8 fd b2 b7 c9 31 6a
58 54 eb 59 ce f1 8d 35
2c 2a e0 b9 67 ed d3 8f
16 15 70 c9 a6 e3 fc d2
0b 9f 38 f1 53 e4 7e 69
ba 0a bb 42 2b 4f e5 34
5d 05 c8 21 80 b2 e7 1a
bb 97 64 85 40 59 e6 0d
c8 de 32 d7 20 b9 73 93
64 6f 19 fe 10 c9 ac dc
32 a2 99 7f 08 f1 56 6e
19 51 d9 aa 04 ed 2b 37
99 bd f9 55 02 e3 80 8e
b9 75 34 98 73 b1 89 74
c9 af 1a 4c ac cd d1 3a
f1 c2 0d 26 56 f3 fd 1d
ed 61 93 13 2b ec eb 9b
e3 a5 dc 9c 80 76 e0 d8
e4 c7 6e 4e 40 3b 70 6c
72 f6 37 27 20 88 38 36
39 7b 8e 86 10 44 1c 1b
92 42 c7 94 d8 1e 1e cf
49 21 f6 4a 6c 0f 0f f2
b1 85 7b 25 36 92 92 79
cd d7 a8 87 1b 49 49 a9
f3 fe 54 d6 98 b1 b1 c1
ec 7f 2a 6b 4c cd cd f5
76 aa 15 a0 26 f3 f3 ef

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3b 55 9f 50 13 ec ec e2
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60 60 b6 2d 90 b6 5a df
30 30 5b 83 48 5b 2d fa
18 18 b8 d4 24 b8 83 7d
0c 0c 5c 6a 12 5c d4 ab
06 06 2e 35 09 2e 6a c0
03 03 17 8f 91 17 35 60
94 94 9e d2 dd 9e 8f 30
a1 29 3f 6a 66 75 d3 ae
c5 81 8a 35 33 af fc 57
f7 d5 45 8f 8c c2 7e be
ee ff b7 d2 46 61 3f 5f
77 ea ce 69 23 a5 8a ba
ae 75 67 a1 84 c7 45 5d
57 af a6 c5 42 f6 b7 bb
be c2 53 f7 21 7b ce c8
38 0e 56 77 e2 a6 86 90
1c 07 2b ae 71 53 43 48
0e 96 80 57 ad bc b4 24
07 4b 40 be c3 5e 5a 12
96 b0 20 5f f4 2f 2d 09
4b 58 10 ba 7a 82 83 91
b0 2c 08 5d 3d 41 d4 dd
58 16 04 bb 8b b5 6a fb
9b e5 90 77 c2 37 2a 8b
d8 e7 48 ae 61 8e 15 d0
6c e6 24 57 a5 47 9f 68
36 73 12 be c7 b6 da 34
1b ac 09 5f f6 5b 6d 1a
98 56 91 ba 7b b8 a3 0d
4c 2b dd 5d a8 5c c4 93
26 80 fb bb 54 2e 62 dc
e0 c2 7b c7 a2 6c 83 c9
70 61 a8 f6 51 36 d4 f1
38 a5 54 7b bd 1b 6a ed
1c c7 2a a8 cb 98 35 e3
0e f6 15 54 f0 4c 8f e4
07 7b 9f 2a 78 26 d2 72
96 a8 da 15 3c 13 69 39
4b 54 6d 9f 1e 9c a1 89
c0 e1 26 a0 d7 11 77 7a
60 e5 13 50 fe 9d ae 3d
30 e7 9c 28 7f db 57 8b
18 e6 4e 14 aa f8 be d0
0c 73 27 0a 55 7c 5f 68
06 ac 86 05 bf 3e ba 34
03 56 43 97 ca 1f 5d 1a
94 2b b4 de 65 9a bb 0d
80 48 c5 b6 30 b6 3c ae
40 24 f7 5b 18 5b 1e 57
20 12 ee b8 0c b8 0f be
10 09 77 5c 06 5c 92 5f
08 91 ae 2e 03 2e 49 ba
04 dd 57 17 94 17 b1 5d
02 fb be 9e 4a 9e cd bb
01 e8 5f 4f 25 4f f3 c8
ef 68 f0 97 88 5b 33 2d
e2 34 78 de 44 b8 8c 83
71 1a 3c 6f 22 5c 46 d4
ad 0d 1e a2 11 2e 23 6a
c3 93 0f 51 9d 17 84 35
f4 dc 92 bd db 9e 42 8f
7a 6e 49 cb f8 4f 21 d2
3d 37 b1 f0 7c b2 85 69

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45 64 fb a5 ca 13 65 7b
b7 32 e8 c7 65 9c a7 a8
ce 19 74 f6 a7 4e c6 54
67 99 3a 7b c6 27 63 2a
a6 d9 1d a8 63 86 a4 15
53 f9 9b 54 a4 43 52 9f
bc e9 d8 2a 52 b4 29 da
5e e1 6c 15 29 5a 81 6d
5e db fd f0 f3 ad f2 65
2f f8 eb 78 ec c3 79 a7
82 7c e0 3c 76 f4 a9 c6
41 3e 70 1e 3b 7a c1 63
b5 1f 38 0f 88 3d f5 a4
cf 9a 1c 92 44 8b ef 52
f2 4d 0e 49 22 d0 e2 29
79 b3 07 b1 11 68 71 81
43 22 e2 6b e6 1b 79 81
b4 11 71 a0 73 98 a9 d5
5a 9d ad 50 ac 4c c1 ff
2d db c3 28 56 26 f5 ea
83 f8 f4 14 2b 13 ef 75
d4 7c 7a 0a 80 9c e2 af
6a 3e 3d 05 40 4e 71 c2
35 1f 8b 97 20 27 ad 61
b0 99 61 74 51 1c 8d 38
58 d9 a5 3a bd 0e d3 1c
2c f9 c7 1d cb 07 fc 0e
16 e9 f6 9b f0 96 7e 07
0b e1 7b d8 78 4b 3f 96
90 e5 a8 6c 3c b0 8a 4b
48 e7 54 36 1e 58 45 b0
24 e6 2a 1b 0f 2c b7 58
f9 42 f4 31 f5 05 21 93
e9 21 7a 8d ef 97 85 dc
e1 85 3d d3 e2 de d7 6e
e5 d7 8b fc 71 6f fe 37
e7 fe d0 7e ad a2 7f 8e
e6 7f 68 3f c3 51 aa 47
73 aa 34 8a f4 bd 55 b6
ac 55 1a 45 7a cb bf 5b
63 41 4d d4 73 10 22 4c
a4 b5 b3 6a ac 08 11 26
52 cf cc 35 56 04 9d 13
29 f2 66 8f 2b 02 db 9c
81 79 33 d2 80 01 f8 4e
d5 a9 8c 69 40 95 7c 27
ff c1 46 a1 20 df 3e 86
ea f5 23 c5 10 fa 1f 43
b0 c2 5a 82 c0 14 b8 ec
58 61 2d 41 60 0a 5c 76
2c a5 83 b5 30 05 2e 3b
16 c7 d4 cf 18 97 17 88
0b f6 6a f2 0c de 9e 44
90 7b 35 79 06 6f 4f 22
48 a8 8f a9 03 a2 b2 11
24 54 d2 c1 94 51 59 9d
a2 79 02 a0 fd 30 f5 93
51 a9 01 50 eb 18 ef dc
bd c1 95 28 e0 0c e2 6e
cb f5 df 14 70 06 71 37
f0 ef fa 0a 38 03 ad 8e
78 e2 7d 05 1c 94 c3 47
3c 71 ab 97 0e 4a f4 b6
1e ad c0 de 07 25 7a 5b
3c 8b b5 b8 f2 43 6c 2b

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1e d0 cf 5c 79 b4 36 80
0f 68 f2 2e a9 5a 1b 40
92 34 79 17 c1 2d 98 20
49 1a a9 9e f5 83 4c 10
b1 0d c1 4f ef d4 26 08
cd 93 f5 b2 e2 6a 13 04
f3 dc ef 59 71 35 9c 02
37 8d da 4e eb f2 8e 04
8e d3 6d 27 e0 79 47 02
47 fc a3 86 70 a9 b6 01
b6 7e c4 43 38 c1 5b 95
5b 3f 62 b4 1c f5 b8 df
b8 8a 31 5a 0e ef 5c fa
5c 45 8d 2d 07 e2 2e 7d
2e b7 d3 83 96 71 17 ab
37 67 24 51 bc 8d 48 8a
8e a6 12 bd 5e d3 24 45
47 53 09 cb 2f fc 12 b7
b6 bc 91 f0 82 7e 09 ce
5b 5e dd 78 41 3f 91 67
b8 2f fb 3c b5 8a dd a6
5c 82 e8 1e cf 45 fb 53
2e 41 74 0f f2 b7 e8 bc
dd 99 3b 06 c3 4b c6 8a
fb d9 88 03 f4 b0 63 45
e8 f9 44 94 7a 58 a4 b7
74 e9 22 4a 3d 2c 52 ce
3a e1 11 25 8b 16 29 67
1d e5 9d 87 d0 0b 81 a6
9b e7 db d6 68 90 d5 53
d8 e6 f8 6b 34 48 ff bc
f2 f1 7b a8 6b 8b dc 17
79 ed a8 54 a0 d0 6e 9e
a9 e3 54 2a 50 68 37 4f
c1 e4 2a 15 28 34 8e b2
f5 72 15 9f 14 1a 47 59
ef 39 9f da 0a 0d b6 b9
e2 89 da 6d 05 93 5b c9
71 d1 6d a3 97 dc b8 f1
d6 4b dc 4c 1b 70 4f c2
6b b0 6e 26 98 38 b2 61
a0 58 37 13 4c 1c 59 a5
50 2c 8e 9c 26 0e b9 c7
28 16 47 4e 13 07 c9 f6
14 0b b6 27 9c 96 f1 7b
0a 90 5b 86 4e 4b ed a8
05 48 b8 43 27 b0 e3 54
26 93 39 76 b6 9f 7d 99
13 dc 89 3b 5b da ab d9
9c 6e d1 88 b8 6d c0 f9
4e 37 fd 44 5c a3 60 e9
27 8e eb 22 2e c4 30 e1
86 47 e0 11 17 62 18 e5
43 b6 70 9d 9e 31 0c e7
b4 5b 38 db 4f 8d 06 e6
58 96 a0 7d af 83 e4 89
2c 4b 50 ab c2 d4 72 d1
16 b0 28 c0 61 6a 39 fd
0b 58 14 60 a5 35 89 eb
90 2c 0a 30 c7 8f d1 e0
48 16 05 18 f6 d2 fd 70
24 0b 97 0c 7b 69 eb 38
12 90 de 06 a8 a1 e0 1c
a6 24 3d 9f 62 a0 10 dc
53 12 8b da 31 50 08 6e

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bc 09 d0 6d 8d 28 04 37
5e 91 68 a3 d3 14 02 8e
2f dd 34 c4 fc 0a 01 47
82 fb 1a 62 7e 05 95 b6
41 e8 0d 31 3f 97 df 5b
b5 74 93 8d 8a de fa b8
48 3f 0f 0e ae 89 3c a4
24 8a 92 07 57 d1 1e 52
12 45 49 96 be fd 0f 29
09 b7 b1 4b 5f eb 92 81
91 ce cd b0 ba e0 49 d5
dd 67 f3 58 5d 70 b1 ff
fb a6 ec 2c bb 38 cd ea
e8 53 76 16 c8 1c f3 75
ab f1 0c 3a 25 c4 41 b6
c0 ed 06 1d 87 62 b5 5b
60 e3 03 9b d6 31 cf b8
30 e4 94 d8 6b 8d f2 5c
18 72 4a 6c a0 d3 79 2e
0c 39 25 36 50 fc a9 17
06 89 87 1b 28 7e c1 9e
03 d1 d6 98 14 3f f5 4f
33 d0 f0 e7 86 e8 3a 77
8c 68 78 e6 43 74 1d ae
46 34 3c 73 b4 3a 9b 57
23 1a 1e ac 5a 1d d8 be
84 0d 0f 56 2d 9b 6c 5f
42 93 92 2b 83 d8 36 ba
21 dc 49 80 d4 6c 1b 5d
85 6e b1 40 6a 36 98 bb
43 c6 08 15 6a 16 24 05
b4 63 04 9f 35 0b 12 97
5a a4 02 da 8f 90 09 de
2d 52 01 6d d2 48 91 6f
83 29 95 a3 69 24 dd a2
d4 81 df c4 a1 12 fb 51
6a d5 fa 62 c5 09 e8 bd
35 ff 7d 31 f7 91 74 cb
54 73 1f f8 5c 41 09 38
2a ac 9a 7c 2e b5 91 1c
15 56 4d 3e 17 cf dd 0e
9f 2b b3 1f 9e f2 fb 07
da 80 cc 9a 4f 79 e8 96
6d 40 66 4d b2 a9 74 4b
a3 20 33 b3 59 c1 3a b0
c4 10 8c cc b9 f5 1d 58
38 2d 1c 17 df 3c 80 66
1c 83 0e 9e fa 1e 40 33
0e d4 07 4f 7d 0f 20 8c
07 6a 96 b2 ab 92 10 46
96 35 4b 59 c0 49 08 23
4b 8f b0 b9 60 b1 04 84
b0 d2 58 c9 30 cd 02 42
58 69 2c f1 18 f3 01 21
b8 af f0 4a 58 31 dc 8b
5c c2 78 25 2c 8d 6e d0
2e 61 3c 87 16 d3 37 68
17 a5 1e d6 0b fc 8e 34
9e c7 0f 6b 90 7e 47 1a
4f f6 92 a0 48 3f b6 0d
b2 7b 49 50 24 8a 5b 93
59 a8 b1 28 12 45 b8 dc
9d ee eb 6a 6b 0a ac a6
db 77 e0 35 a0 05 56 53
f8 ae 70 8f 50 97 2b bc

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7c 57 38 d2 28 de 80 5e
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9a ba 07 c5 05 51 10 41
4d 5d 96 f7 97 bd 08 b5
43 e2 34 c6 35 47 2b 94
b4 71 1a 63 8f b6 80 4a
5a ad 0d a4 d2 5b 40 25
2d c3 93 52 69 b8 20 87
83 f4 dc 29 a1 5c 10 d6
d4 7a 6e 81 c5 2e 08 6b
6a 3d 37 d5 f7 17 04 a0
35 8b 8e ff ee 9e 02 50
70 4f cc a7 0d 4e 98 38
38 b2 66 c6 93 27 4c 1c
1c 59 33 63 dc 86 26 0e
0e b9 8c a4 6e 43 13 07
07 c9 46 52 37 b4 9c 96
96 f1 23 29 8e 5a 4e 4b
4b ed 84 81 47 2d 27 b0
b0 e3 42 d5 b6 83 86 58
4e 81 42 0c 86 d1 67 3d
27 d5 21 06 43 fd a6 8b
86 ff 85 03 b4 eb 53 d0
43 ea d7 94 5a e0 bc 68
b4 75 fe 4a 2d 70 5e 34
5a af 7f 25 83 38 2f 1a
2d c2 aa 87 d4 1c 82 0d
83 61 55 d6 6a 0e 41 93
bd e7 5c ba d7 32 5d eb
cb e6 2e 5d fe 19 bb e0
f0 73 17 bb 7f 99 c8 70
78 ac 9e c8 aa d9 64 38
3c 56 4f 64 55 f9 32 1c
1e 2b b2 32 bf e9 19 0e
0f 80 59 19 ca e1 99 07
92 40 b9 99 65 e5 d9 96
29 a1 60 19 97 e5 9e 40
81 c5 30 99 de e7 4f 20
d5 f7 18 d9 6f e6 b2 10
ff ee 0c f9 a2 73 59 08
ea 77 06 e9 51 ac b9 04
75 ae 03 e1 bd 56 c9 02
af 57 94 e5 cb 2b f1 01
c2 be 4a e7 f0 80 ed 95
45 c1 42 73 ff d2 cb 88
b7 f5 21 ac ea 69 f0 44
ce ef 85 56 75 a1 78 22
67 e2 d7 2b af c5 3c 11
a6 71 fe 80 c2 f7 1e 9d
53 ad 7f 40 61 ee 0f db
bc c3 aa 20 a5 77 92 f8
5e f4 55 10 c7 ae 49 7c
fb 62 2b c5 32 03 01 65
e8 31 80 f7 19 94 95 a7
74 8d 40 ee 99 4a df c6
3a d3 20 77 d9 25 fa 63
1d fc 10 ae f9 87 7d a4
9b 7e 08 57 e9 d6 ab 52
d8 3f 04 be e1 6b c0 29
6c 8a 02 5f e5 a0 60 81
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61 27 37 49 9c 59 27 4f
a5 86 8e b1 4e b9 86 b2
c7 43 47 cd 27 c9 43 59

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57 68 10 ec ef 93 21 3e
be 34 08 76 e2 dc 85 1f
5f 1a 04 3b 71 6e d7 9a
ba 0d 02 88 ad 37 fe 4d
5d 93 01 44 c3 8e 7f b3
bb dc 95 22 f4 47 aa cc
c8 6e df 11 7a b6 55 66
ee 1f 42 cb b0 85 a2 91
77 9a 21 f0 58 d7 51 dd
ae 4d 85 78 2c fe bd fb
57 b3 d7 3c 16 7f cb e8
be cc fe 1e 0b aa f0 74
5f 66 7f 0f 90 55 78 3a
ba 33 aa 92 48 bf 3c 1d
5d 8c 55 49 24 ca 1e 9b
e7 be 59 48 27 c3 94 12
e6 5f b9 24 86 f4 4a 09
73 ba c9 12 43 7a 25 91
ac 5d f1 09 b4 3d 87 dd
56 bb ed 91 5a 8b d6 fb
2b c8 e3 dd 2d d0 6b e8
80 64 e4 fb 83 68 a0 74
40 32 72 e8 d4 34 50 3a
c1 f0 05 58 e1 ea 64 4e
f5 78 97 2c e5 75 32 27
ef 3c de 16 e7 af 19 86
e2 1e 6f 0b e6 c2 99 43
71 0f a2 90 73 61 d9 b4
ad 92 51 48 ac a5 f9 5a
c3 49 bd 24 56 c7 e9 2d
f4 b1 cb 12 2b f6 e1 83
44 03 c3 55 f8 e4 45 c7
22 94 f4 bf 7c 72 b7 f6
11 4a 7a ca 3e 39 ce 7b
9d 25 3d 65 1f 89 67 a8
db 87 8b a7 9a d1 a6 54
f8 d6 d0 c6 4d fd 53 2a
7c 6b 68 63 b3 eb bc 15
3e a0 34 a4 cc e0 5e 9f
ec 8b f3 17 37 cc 03 0c
76 d0 ec 9e 8e 66 94 06
3b 68 76 4f 47 33 4a 03
88 34 3b b2 b6 8c 25 94
44 1a 88 59 5b 46 87 4a
22 0d 44 b9 b8 23 d6 25
11 93 22 c9 5c 84 6b 87
9d dc 11 f1 2e 42 a0 d6
f2 df 52 4e 08 e0 93 c0
79 fa 29 27 04 70 dc 60
a9 7d 81 86 02 38 6e 30
c1 ab d5 43 01 1c 37 18
f5 c0 ff b4 95 0e 8e 0c
ef 60 ea 5a df 07 47 06
e2 30 75 2d fa 96 b6 03
71 18 af 83 7d 4b 5b 94
f8 62 3a 2f 70 3f 98 c2
7c 31 1d 82 38 8a 4c 61
3e 8d 9b 41 1c 45 26 a5
1f d3 d8 b5 0e b7 13 c7
9a fc 6c cf 07 ce 9c f6

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4d 7e 36 f2 96 67 4e 7b
b3 3f 1b 79 4b a6 27 a8
cc 8a 98 a9 b0 53 86 54
96 e7 ad 84 7a e8 3e 25
4b e6 c3 42 3d 74 1f 87
b0 73 f4 21 8b 3a 9a d6
58 ac 7a 85 d0 1d 4d 6b
2c 56 3d d7 68 9b b3 a0
16 2b 8b fe 34 d8 cc 50
0b 80 d0 7f 1a 6c 66 28
90 40 68 aa 0d 36 33 14
4c 21 d2 d1 31 48 41 1a
26 85 69 fd 8d 24 b5 0d
13 d7 a1 eb d3 12 cf 93
9c fe c5 e0 fc 09 f2 dc
4e 7f f7 70 7e 91 79 6e
27 aa ee 38 3f dd a9 37
86 55 77 1c 8a fb c1 8e
43 bf ae 0e 45 e8 f5 47
9c a7 56 8c 28 96 e0 39
4e c6 2b 46 14 4b 70 89
27 63 80 23 0a b0 38 d1
86 a4 40 84 05 58 1c fd
43 52 20 42 97 2c 0e eb
b4 29 10 21 de 16 07 e0
5a 81 08 85 6f 0b 96 70
2d d5 04 d7 a2 90 4b 38
df 37 2c 50 9a 4a f9 fd
fa 8e 16 28 4d 25 e9 eb
7d 47 0b 14 b3 87 e1 e0
ab b6 90 0a cc d6 e5 70
c0 5b 48 05 66 6b e7 38
60 b8 24 97 33 a0 e6 1c
30 5c 12 de 8c 50 73 0e
18 2e 09 6f 46 28 ac 07
dd 36 fa 70 0c 36 31 c5
fb 1b 7d 38 06 1b 8d f7
e8 98 ab 1c 03 98 d3 ee
74 4c c0 0e 94 4c fc 77
3a 26 60 07 4a 26 7e ae
1d 13 30 96 25 13 3f 57
9b 9c 18 4b 87 9c 8a be
d8 4e 0c b0 d6 4e 45 5f
5d 30 0d 67 16 7c 93 17
bb 18 93 a6 0b 3e dc 9e
c8 0c dc 53 90 1f 6e 4f
64 06 6e bc 48 9a 37 b2
32 03 37 5e 24 4d 8e 59
19 94 8e 2f 12 b3 47 b9
99 4a 47 82 09 cc b6 c9
d9 25 b6 41 91 66 5b f1
fc 6a 5c da 62 b9 dc 3a
7e 35 2e 6d 31 c9 6e 1d
3f 8f 17 a3 8d f1 37 9b
8a d2 9e c4 d3 ed 8e d8
45 69 4f 62 fc e3 47 6c
b7 a1 b2 31 7e e4 b6 36
ce c5 59 8d 3f 72 5b 1b
67 f7 b9 d3 8a 39 b8 98
b7 0a dd bf 30 83 d9 aa
ce 05 fb ca 18 d4 f9 55
67 97 e8 65 0c 6a e9 bf
a6 de 74 a7 06 35 e1 ca
53 6f 3a c6 03 8f e5 65
bc a2 1d 63 94 d2 e7 a7

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5e 51 9b a4 4a 69 e6 c6
2f bd d8 52 25 a1 73 63
17 cd 93 2d f3 bd 7b a7
9e f3 dc 83 ec cb a8 c6
4f ec 6e d4 76 f0 54 63
b2 76 37 6a 3b 78 2a a4
59 3b 8e 35 88 3c 15 52
b9 88 47 8f 44 1e 9f 29
c9 44 b6 d2 22 0f da 81
f1 22 5b 69 11 92 6d d5
14 da c3 2a 69 2b 13 5e
0a 6d f4 15 a1 80 9c 2f
05 a3 7a 9f c5 40 4e 82
97 c4 3d da f7 20 27 41
de 62 8b 6d ee 10 86 b5
6f 31 d0 a3 77 08 43 cf
a2 8d 68 c4 ae 04 b4 f2
51 d3 34 62 57 02 5a 79
57 32 ed e4 aa 80 3d e5
be 19 e3 72 55 40 8b e7
5f 99 e4 39 bf 20 d0 e6
ba d9 72 89 ca 10 68 73
5d f9 39 d1 65 08 34 ac
bb e9 89 fd a7 04 1a 56
c8 e1 d1 eb c6 02 0d 2b
64 e5 fd e0 63 01 93 80
2d 67 29 b5 4b cb 8a dd
83 a6 81 cf b0 f0 45 fb
d4 53 d5 f2 58 78 b7 e8
6a bc ff 79 2c 3c ce 74
35 5e ea a9 16 1e 67 3a
8f 2f 75 c1 0b 0f a6 1d
d2 82 af f5 90 92 53 9b
69 41 c2 ef 48 49 bc d8
aa 03 6b 86 1d e9 49 07
55 94 a0 43 9b e1 b1 96
bf 4a 50 b4 d8 e5 cd 4b
ca 25 28 5a 6c e7 f3 b0
65 87 14 2d 36 e6 ec 58
a7 d6 0a 83 1b 73 76 2c
c6 6b 05 d4 98 ac 3b 16
63 a0 97 6a 4c 56 88 0b
14 df b2 0a 98 a1 c6 1e
0a fa 59 05 4c c5 63 0f
05 7d b9 97 26 f7 a4 92
97 ab c9 de 13 ee 52 49
de c0 f1 6f 9c 77 29 b1
6f 60 ed a2 4e ae 81 cd
a2 30 e3 51 27 57 d5 f3
51 18 e4 bd 86 be ff ec
52 43 cd 15 20 55 7d e5
29 b4 f3 9f 10 bf ab e7
81 5a ec da 08 ca c0 e6
d5 2d 76 6d 04 65 60 73
ff 83 3b a3 02 a7 30 ac
ea d4 88 c4 01 c6 18 56
75 6a 44 62 95 63 0c 2b
af 35 22 31 df a4 06 80
a0 a4 a3 c3 61 e5 d8 3b
50 52 c4 f4 a5 e7 6c 88
28 29 62 7a c7 e6 36 44
14 81 31 3d f6 73 1b 22
0a d5 8d 8b 7b ac 98 11
05 ff d3 d0 a8 56 4c 9d
97 ea fc 68 54 2b 26 db

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de 75 7e 34 2a 80 13 f8
60 fa 01 a5 41 ec 5e f9
30 7d 95 c7 b5 76 2f e9
18 ab df f6 cf 3b 82 e1
0c c0 fa 7b f2 88 41 e5
06 60 7d a8 79 44 b5 e7
03 30 ab 54 a9 22 cf e6
94 18 c0 2a c1 11 f2 73
4a 0c 60 15 f5 9d 79 ac
5f 36 02 e4 69 ab da 57
ba 1b 01 72 a1 c0 6d be
5d 98 95 39 c5 60 a3 5f
bb 4c df 89 f7 30 c4 ba
c8 26 fa d1 ee 18 62 5d
64 13 7d fd 77 0c 31 bb
32 9c ab eb ae 06 8d c8
19 4e c0 e0 57 03 d3 64
7b b5 08 24 d3 72 06 e8
a8 cf 04 12 fc 39 03 74
54 f2 02 09 7e 89 94 3a
2a 79 01 91 3f d1 4a 1d
15 a9 95 dd 8a fd 25 9b
9f c1 df fb 45 eb 87 d8
da f5 fa e8 b7 e0 d6 6c
6d ef 7d 74 ce 70 6b 36
b2 c0 c9 d4 5c d2 5e b3
59 60 f1 6a 2e 69 2f cc
b9 30 ed 35 17 a1 82 66
c9 18 e3 8f 9e c5 41 33
f1 0c e4 d2 4f f7 b5 8c
ed 06 72 69 b2 ee cf 46
e3 03 39 a1 59 77 f2 23
e4 94 89 c5 b9 ae 79 84
21 3a 83 bd 5d 54 30 41
85 1d d4 cb bb 2a 18 b5
d7 9b 6a f0 c8 15 0c cf
fe d8 35 78 64 9f 06 f2
7f 6c 8f 3c 32 da 03 79
aa 36 d2 1e 19 6d 94 a9
55 1b 69 0f 99 a3 4a c1
bf 98 a1 92 d9 c4 25 f5
8c 1f c7 eb fd 22 f4 bd
46 9a f6 e0 eb 11 7a cb
23 4d 7b 70 e0 9d 3d f0
84 b3 a8 38 70 db 8b 78
42 cc 54 1c 38 f8 d0 3c
21 66 2a 0e 1c 7c 68 1e
85 33 15 07 0e 3e 34 0f
d7 8c 9f 96 07 1f 1a 92
c3 dc 09 21 63 e2 01 97
f4 6e 91 85 a4 71 95 de
7a 37 dd d7 52 ad df 6f
3d 8e fb fe 29 c3 fa a2
8b 47 e8 7f 81 f4 7d 51
d0 b6 74 aa d5 7a ab bd
68 5b 3a 55 ff 3d c0 cb
34 b8 1d bf ea 8b 60 f0
e9 df 6d 56 96 03 06 15
e1 fa a3 2b 4b 94 03 9f
e5 7d c4 80 b0 4a 94 da
e7 ab 62 40 58 25 4a 6d
e6 c0 31 20 2c 87 25 a3
73 60 8d 10 16 d6 87 c4
ac 30 d3 08 0b 6b d6 62
56 18 fc 04 90 a0 6b 31

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5a a2 68 13 38 8b d8 26
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83 bd 1a 4e 0e 68 36 9c
d4 cb 0d 27 07 34 1b 4e
6a f0 93 86 96 1a 98 27
35 78 dc 43 4b 0d 4c 86
8f 3c 6e b4 b0 93 26 43
d2 1e 37 5a 58 dc 13 b4
13 3c 84 89 0c 1e 25 0e
9c 1e 42 d1 06 0f 87 07
4e 0f 21 fd 03 92 d6 96
27 92 85 eb 94 49 6b 4b
86 49 d7 e0 4a b1 a0 b0
43 b1 fe 70 25 cd 50 58
b4 cd 7f 38 87 f3 28 2c
5a f3 aa 1c d6 ec 14 16
cc 46 e2 fc 06 5a a5 d1
66 23 71 7e 03 2d c7 fd
33 84 ad 3f 94 83 f6 eb
8c 42 c3 8a 4a d4 7b e0
46 21 f4 45 25 6a a8 70
23 85 7a b7 87 35 54 38
84 d7 3d ce d6 8f 2a 1c
42 fe 8b 67 6b d2 15 0e
5c 3a 3d 1c 04 15 b6 14
2e 1d 8b 0e 02 9f 5b 0a
17 9b d0 07 01 da b8 05
9e d8 68 96 95 6d 5c 97
4f 6c 34 4b df a3 2e de
b2 36 1a b0 fa c4 17 6f
59 1b 0d 58 7d 62 9e a2
b9 98 93 2c ab 31 4f 51
23 32 d9 1d 38 dd 92 53
84 19 f9 9b 1c fb 49 bc
42 99 e9 d8 0e e8 b1 5e
21 d9 e1 6c 07 74 cd 2f
85 f9 e5 36 96 3a f3 82
d7 e9 e7 1b 4b 1d ec 41
fe e1 e6 98 b0 9b 76 b5
7f e5 73 4c 58 d8 3b cf
05 95 b0 0f 12 02 7c 6f
97 df 58 92 09 01 3e a2
de fa 2c 49 91 95 1f 51
6f 7d 16 b1 dd df 9a bd
a2 ab 0b cd fb fa 4d cb
51 c0 90 f3 e8 7d b3 f0
bd 60 48 ec 74 ab cc 78
cb 30 24 76 3a c0 66 3c
69 53 74 ee fd 12 3d e6
a1 bc 3a 77 eb 09 8b 73
c5 5e 1d ae e0 91 d0 ac
f7 2f 9b 57 70 dd 68 56
ee 82 d8 be 38 fb 34 2b
77 41 6c 5f 1c e8 1a 80
ae b5 36 ba 0e 74 0d 40
57 cf 1b 5d 07 3a 93 20
71 16 96 df 17 d7 b6 0b
ad 0b 4b fa 9e fe 5b 90
c3 90 b0 7d 4f 7f b8 48
f4 48 58 ab b2 aa 5c 24
7a 24 2c c0 59 55 2e 12
3d 12 16 60 b9 bf 17 09
8b 09 0b 30 c9 ca 9e 91
d0 91 90 18 f1 65 4f dd
c2 b3 c4 c3 2c be 19 54

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61 cc 62 f4 16 5f 99 2a
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c7 33 8d 3d 90 5d f9 9f
f6 8c d3 8b 48 bb e9 da
7b 46 fc d0 24 c8 e1 6d
a8 23 7e 68 12 64 e5 a3
54 84 3f 34 09 32 e7 c4
53 7a 71 cc f9 5a 88 7c
bc 3d ad 66 e9 2d 44 3e
5e 8b c3 33 e1 83 22 1f
2f d0 f4 8c e5 d4 11 9a
82 68 7a 46 e7 6a 9d 4d
41 34 3d 23 e6 35 db b3
b5 1a 8b 84 73 8f f8 cc
cf 0d d0 42 ac d2 7c 66
4c 70 84 cf 5d 51 0c 52
26 38 42 f2 bb bd 06 29
13 1c 21 79 c8 cb 03 81
9c 0e 85 a9 64 f0 94 d5
4e 07 d7 c1 32 78 4a ff
27 96 fe f5 19 3c 25 ea
86 4b 7f ef 99 1e 87 75
43 b0 aa e2 d9 0f d6 af
cd f1 48 e0 31 db a8 39
f3 ed 24 70 8d f8 54 89
ec e3 12 38 d3 7c 2a d1
76 e4 09 1c fc 3e 15 fd
3b 72 91 0e 7e 1f 9f eb
88 39 dd 07 3f 9a da e0
44 89 fb 96 8a 4d 6d 70
22 d1 e8 4b 45 b3 a3 38
3e f8 df fe b6 59 13 7d
1f 7c fa 7f 5b b9 9c ab
9a 3e 7d aa b8 c9 4e c0
4d 1f ab 55 5c f1 27 60
b3 9a c0 bf 2e ed 86 30
cc 4d 60 ca 17 e3 43 18
66 b3 30 65 9e e4 b4 0c
33 cc 18 a7 4f 72 5a 06
c5 37 4b 8b 97 0f 42 d6
f7 8e b0 d0 de 92 21 6b
ee 47 58 68 6f 49 85 a0
77 b6 2c 34 a2 b1 d7 50
ae 5b 16 1a 51 cd fe 28
57 b8 0b 0d bd f3 7f 14
be 5c 90 93 cb ec aa 0a
5f 2e 48 dc f0 76 55 05
aa c6 95 0a d1 ed c2 48
55 63 df 05 fd e3 61 24
bf a4 fa 97 eb e4 a5 12
ca 52 7d de e0 72 c7 09
65 29 ab 6f 70 39 f6 91
a7 81 c0 a2 38 89 7b dd
c6 d5 60 51 1c d1 a8 fb
63 ff 30 bd 0e fd 54 e8
d1 21 3e c6 9c 2a 89 1e
fd 85 1f 63 4e 15 d1 0f
eb d7 9a a4 27 9f fd 92
e0 fe 4d 52 86 da eb 49
70 7f b3 29 43 6d e0 b1
38 aa cc 81 b4 a3 70 cd
1c 55 66 d5 5a c4 38 f3
0e bf 33 ff 2d 62 1c ec
31 42 1f 8c 75 b5 69 ad
8d 21 9a 46 af cf a1 c3

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d3 85 4d 23 c2 f2 c5 f4
fc d7 b3 84 61 79 f7 7a
7e fe cc 42 a5 a9 ee 3d
3f 7f 66 21 c7 c1 77 8b
8a aa 33 85 f6 f5 ae d0
45 55 8c d7 7b ef 57 68
50 f9 b4 67 51 c7 64 d7
28 e9 5a a6 bd f6 32 fe
14 e1 2d 53 cb 7b 19 7f
0a e5 83 bc f0 a8 99 aa
05 e7 d4 5e 78 54 d9 55
97 e6 6a 2f 3c 2a f9 bf
de 73 35 82 1e 15 e9 ca
6f ac 8f 41 0f 9f e1 65
9b 0d 06 33 95 7e 70 e9
d8 93 03 8c df 3f 38 e1
6c dc 94 46 fa 8a 1c e5
36 6e 4a 23 7d 45 0e e7
1b 37 25 84 ab b7 07 e6
98 8e 87 42 c0 ce 96 73
4c 47 d6 21 60 67 4b ac
26 b6 6b 85 30 a6 b0 56
08 54 3f 90 eb 36 e1 c9
04 2a 8a 48 e0 1b e5 f1
02 15 45 24 70 98 e7 ed
01 9f b7 12 38 4c e6 e3
95 da ce 09 1c 26 73 e4
df 6d 67 91 0e 13 ac 72
fa a3 a6 dd 07 9c 56 39
7d c4 53 fb 96 4e 2b 89
06 02 b1 b9 85 bf f7 35
03 01 cd c9 d7 ca ee 8f
94 95 f3 f1 fe 65 77 d2
4a df ec ed 7f a7 ae 69
25 fa 76 e3 aa c6 57 a1
87 7d 3b e4 55 63 be c5
d6 ab 88 72 bf a4 5f f7
6b c0 44 39 ca 52 ba ee
aa ea eb 2d 15 5a b0 5d
55 75 e0 83 9f 2d 58 bb
bf af 70 d4 da 83 2c c8
ca c2 38 6a 6d d4 16 64
65 61 1c 35 a3 6a 0b 32
a7 a5 0e 8f c4 35 90 19
c6 c7 07 d2 62 8f 48 99
63 f6 96 69 31 d2 24 d9
fd 5f 19 02 2b 58 9c 1e
eb ba 99 01 80 2c 4e 0f
e0 5d d9 95 40 16 27 92
70 bb f9 df 20 0b 86 49
38 c8 e9 fa 10 90 43 b1
1c 64 e1 7d 08 48 b4 cd
0e 32 e5 ab 04 24 5a f3
07 19 e7 c0 02 12 2d ec
57 27 fb 23 e2 82 b0 c3
be 86 e8 84 71 41 58 f4
5f 43 74 42 ad b5 2c 7a
ba b4 3a 21 c3 cf 16 3d
5d 5a 1d 85 f4 f2 0b 8b
bb 2d 9b d7 7a 79 90 d0
c8 83 d8 fe 3d a9 48 68
64 d4 6c 7f 8b c1 24 34
38 71 ee fd 49 46 ac dd
1c ad 77 eb b1 23 56 fb
0e c3 ae e0 cd 84 2b e8

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07 f4 57 70 f3 42 80 74
96 7a be 38 ec 21 40 3a
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b0 8b ba 0e 3b d7 10 9b
58 d0 5d 07 88 fe 08 d8
e4 5d 1a dc 22 1d 67 8b
72 bb 0d 6e 11 9b a6 d0
39 c8 93 37 9d d8 53 68
89 64 dc 8e db 6c bc 34
d1 32 6e 47 f8 36 5e 1a
fd 19 37 b6 7c 1b 2f 0d
eb 99 8e 5b 3e 98 82 93
e0 d9 47 b8 1f 4c 41 dc
76 0b b8 09 6a da 2a f5
3b 90 5c 91 35 6d 15 ef
88 48 2e dd 8f a3 9f e2
44 24 17 fb d2 c4 da 71
22 12 9e e8 69 62 6d ad
11 09 4f 74 a1 31 a3 c3
9d 91 b2 3a c5 8d c4 f4
db dd 59 1d f7 d3 62 7a
a2 ae be c3 b8 ce 2f 60
51 57 5f f4 5c 67 82 30
bd be ba 7a 2e a6 41 18
cb 5f 5d 3d 17 53 b5 0c
f0 ba bb 8b 9e bc cf 06
78 5d c8 d0 4f 5e f2 03
3c bb 64 68 b2 2f 79 94
1e c8 32 34 59 82 a9 4a
eb 37 d6 fd 0c 99 9f 2b
e0 8e 6b eb 06 d9 da 80
70 47 a0 e0 03 f9 6d 40
38 b6 50 70 94 e9 a3 20
1c 5b 28 38 4a e1 c4 10
0e b8 14 1c 25 e5 62 08
07 5c 0a 0e 87 e7 31 04
96 2e 05 07 d6 e6 8d 02
33 c9 14 08 c4 90 7c f4
8c f1 0a 04 62 48 3e 7a
46 ed 05 02 31 24 1f 3d
23 e3 97 01 8d 12 9a 8b
84 e4 de 95 d3 09 4d d0
42 72 6f df fc 91 b3 68
21 39 a2 fa 7e dd cc 34
85 89 51 7d 3f fb 66 1a
5a 22 e7 57 12 50 a7 05
2d 11 e6 be 09 28 c6 97
83 9d 73 5f 91 14 63 de
d4 db ac ba dd 0a a4 6f
6a f8 56 5d fb 05 52 a2
35 7c 2b bb e8 97 29 51
8f 3e 80 c8 74 de 81 bd
d2 1f 40 64 3a 6f d5 cb
93 b3 c0 a3 d7 61 06 0e
dc cc 60 c4 fe a5 03 07
6e 66 30 62 7f c7 94 96
37 33 18 31 aa f6 4a 4b
8e 8c 0c 8d 55 7b 25 b0
47 46 06 d3 bf a8 87 58
b6 23 03 fc ca 54 d6 2c
5b 84 94 7e 65 2a 6b 16
e4 af 8e 80 47 1e 38 fc
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39 61 b6 20 5b 92 0e 3f
89 a5 5b 10 b8 49 07 8a

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d1 c7 b8 08 5c b1 96 45
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84 9f e4 6c 69 85 5d f5
42 da 72 36 a1 d7 bb ef
21 6d 39 1b c5 fe c8 e2
85 a3 89 98 f7 7f 64 71
d7 c4 d1 4c ee aa 32 ad
fe 62 fd 26 77 55 19 c3
7f 31 eb 13 ae bf 99 f4
aa 8d e0 9c 57 ca d9 7a
11 c2 af e7 77 15 77 a2
9d 61 c2 e6 ae 9f ae 51
db a5 61 73 57 da 57 bd
f8 c7 a5 ac be 6d be cb
7c f6 c7 56 5f a3 5f f0
3e 7b f6 2b ba c4 ba 78
1f a8 7b 80 5d 62 5d 3c
9a 54 a8 40 bb 31 bb 1e
b3 bd 5b 06 6b 8a 93 03
cc cb b8 03 a0 45 dc 94
66 f0 5c 94 50 b7 6e 4a
33 78 2e 4a 28 ce 37 25
8c 3c 17 25 14 67 8e 87
46 1e 9e 87 0a a6 47 d6
23 0f 4f d6 05 53 b6 6b
84 92 b2 6b 97 bc 5b a0
89 c0 af 5f 36 d8 cd 28
d1 60 c2 ba 1b 6c f3 14
fd 30 61 5d 98 36 ec 0a
eb 18 a5 bb 4c 1b 76 05
e0 0c c7 c8 26 98 3b 97
70 06 f6 64 13 4c 88 de
38 03 7b 32 9c 26 44 6f
1c 94 a8 19 4e 13 22 a2
e0 57 c6 16 66 b5 c6 71
70 be 63 0b 33 cf 63 ad
38 5f a4 90 8c f2 a4 c3
1c ba 52 48 46 79 52 f4
0e 5d 29 24 23 a9 29 7a
07 bb 81 12 84 c1 81 3d
96 c8 d5 09 42 f5 d5 8b
4b 64 ff 91 21 ef ff d0
55 5c f7 64 0e 54 cf 7a
bf 2e ee 32 07 2a f2 3d
ca 17 77 19 96 15 79 8b
65 9e ae 99 4b 9f a9 d0
a7 4f 57 d9 b0 da c1 68
c6 b2 be f9 58 6d f5 34
63 59 5f e9 2c a3 ef 1a
a4 b9 ba e1 16 c4 e2 0d
c2 ad 7e 90 f9 bb 8f 0f
61 c3 3f 48 e9 c8 d2 92
a5 f4 8a 24 e1 64 69 49
c7 7a 45 12 e5 32 a1 b1
f6 3d b7 09 e7 19 c5 cd
7b 8b ce 91 e6 99 f7 f3
a8 d0 67 dd 73 d9 ee ec
54 68 a6 fb ac f9 77 76
4d c0 22 19 fc cc d3 7c
b3 60 11 99 7e 66 fc 3e
cc 30 9d d9 3f 33 7e 1f
66 18 db f9 8a 8c 3f 9a
33 0c f8 e9 45 46 8a 4d

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8c 06 7c e1 b7 23 45 b3
46 03 3e e5 ce 84 b7 cc
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a8 3f 60 94 9f 45 cb 50
54 8a 30 4a da b7 f0 28
2a 45 18 25 6d ce 78 14
15 b7 0c 87 a3 67 3c 0a
9f ce 06 d6 c4 a6 1e 05
da 67 03 6b 62 53 0f 97
6d a6 94 a0 31 bc 92 de
a3 53 4a 50 8d 5e 49 6f
a9 04 77 09 52 a1 0b cc
c1 02 ae 91 29 c5 90 66
f5 01 57 dd 81 f7 48 33
ef 95 be fb d5 ee 24 8c
e2 df 5f e8 ff 77 12 46
71 fa ba 74 ea ae 09 23
ad 7d 5d 3a 75 57 91 84
c3 ab bb 1d af be dd 42
47 7b 14 11 85 20 da a5
b6 a8 0a 9d d7 10 6d c7
5b 54 05 db fe 08 a3 f6
b8 2a 97 f8 7f 04 c4 7b
5c 15 de 7c aa 02 62 a8
2e 9f 6f 3e 55 01 31 54
17 da a2 1f bf 95 8d 2a
9e 6d 51 9a ca df d3 15
c0 e4 9e 3e a6 90 b6 b7
60 72 4f 1f 53 48 5b ce
30 39 b2 9a bc 24 b8 67
18 89 59 4d 5e 12 5c a6
0c d1 b9 b3 2f 09 2e 53
06 fd c9 cc 82 91 17 bc
03 eb f1 66 41 dd 9e 5e
94 e0 ed 33 b5 fb 4f 2f
85 f0 5b c7 b1 77 f1 ae
d7 78 b8 f6 cd ae ed 57
fe 3c 5c 7b f3 57 e3 be
7f 1e 2e a8 ec be e4 5f
aa 0f 17 54 76 5f 72 ba
55 92 9e 2a 3b ba 39 5d
bf 49 4f 15 88 5d 89 bb
ca b1 b2 9f 44 bb d1 c8
ab 15 fa ea b6 f8 61 cb
c0 9f 7d 75 5b 7c a5 f0
60 da ab af b8 3e c7 78
30 6d c0 c2 5c 1f f6 3c
18 a3 60 61 2e 9a 7b 1e
0c c4 30 a5 17 4d a8 0f
06 62 18 c7 9e b3 54 92
03 31 0c f6 4f cc 2a 49
d7 26 20 74 ba c8 47 77
fe 13 10 3a 5d 64 b6 ae
7f 9c 08 1d bb 32 5b 57
aa 4e 04 9b c8 19 b8 be
55 27 02 d8 64 99 5c 5f
bf 86 01 6c 32 d9 2e ba
ca 43 95 36 19 f9 17 5d
65 b4 df 1b 99 e9 9e bb
9e 07 ff 02 3d d6 85 f0
4f 96 ea 01 8b 6b d7 78
b2 4b 75 95 d0 a0 fe 3c
59 b0 af df 68 50 7f 1e
b9 58 c2 fa 34 28 aa 0f
c9 2c 61 7d 1a 14 55 92

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f1 16 a5 ab 0d 0a bf 49
ed 0b c7 c0 93 05 ca b1
fe 4e 75 c4 bc ad aa 2f
7f 27 af 62 5e c3 55 82
aa 86 c2 31 2f f4 bf 41
55 43 61 8d 82 7a ca b5
bf b4 a5 d3 41 3d 65 cf
ca 5a c7 fc b5 8b a7 f2
65 2d f6 7e cf d0 c6 79
a7 83 7b 3f f2 68 63 a9
12 f3 14 e0 42 77 56 78
09 ec 0a 70 21 ae 2b 3c
91 76 05 38 85 57 80 1e
dd 3b 97 1c d7 be 40 0f
fb 88 de 0e fe 5f 20 92
e8 44 6f 07 7f ba 10 49
74 22 a2 96 aa 5d 08 b1
3a 11 51 4b 55 bb 04 cd
d6 be 75 67 5c 68 9e b8
6b 5f af a6 2e 34 4f 5c
a0 ba c2 53 17 1a b2 2e
50 5d 61 bc 9e 0d 59 17
28 bb a5 5e 4f 93 b9 9e
14 c8 c7 2f b2 dc c9 4f
0a 64 f6 82 59 6e f1 b2
05 32 7b 41 b9 37 ed 59
d3 3a 12 31 ae 4e 07 99
fc 1d 09 8d 57 27 96 d9
7e 9b 91 d3 be 86 4b f9
3f d8 dd fc 5f 43 b0 e9
8a 6c fb 7e ba b4 58 e1
45 36 e8 3f 5d 5a 2c e5
b7 1b 74 8a bb 2d 16 e7
ce 98 3a 45 c8 83 0b e6
ab be 3f 3f 77 b9 74 7f
c0 5f 8a 8a ae c9 3a aa
60 ba 45 45 57 f1 1d 55
30 5d b7 b7 be ed 9b bf
18 bb ce ce 5f e3 d8 ca
0c c8 67 67 ba e4 6c 65
06 64 a6 a6 5d 72 36 a7
03 32 53 53 bb 39 1b c6
7c e3 f5 b5 fb dd f3 77
3e e4 ef cf e8 fb ec ae
1f 72 e2 f2 74 e8 76 57
9a 39 71 79 3a 74 3b be
4d 89 ad a9 1d 3a 88 5f
b3 d1 c3 c1 9b 1d 44 ba
cc fd f4 f5 d8 9b 22 5d
66 eb 7a ef 6c d8 11 bb
99 0e f4 81 6a cb 09 87
d9 07 7a d5 35 f0 91 d6
f9 96 3d ff 8f 78 dd 6b
e9 4b 8b ea d2 3c fb a0
e1 b0 d0 75 69 1e e8 50
e5 58 68 af a1 0f 74 28
e7 2c 34 c2 c5 92 3a 14
e6 16 1a 61 f7 49 1d 0a
8a 76 5a ee 38 cd 15 1b
45 3b 2d 77 1c f3 9f 98
b7 88 83 ae 0e ec da 4c
ce 44 d4 57 07 76 6d 26
67 22 6a be 96 3b a3 13
a6 11 35 5f 4b 88 c4 9c
53 9d 8f ba b0 44 62 4e

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bc db d2 5d 58 22 31 27
02 1a 5e 4c 26 ae 22 ca
01 0d 2f 26 13 57 11 65
95 93 82 13 9c be 9d a7
df dc 41 9c 4e 5f db c6
fa 6e b5 4e 27 ba f8 63
7d 37 cf 27 86 5d 7c a4
ab 8e f2 86 43 bb 3e 52
c0 47 79 43 b4 c8 1f 29
9b 8e 9b a7 c8 a0 50 d2
d8 47 d8 c6 64 50 28 69
6c b6 6c 63 32 28 14 a1
36 5b 36 a4 19 14 0a c5
1b b8 1b 52 99 0a 05 f7
98 5c 98 29 d9 05 97 ee
4c 2e 4c 81 f9 97 de 77
26 17 26 d5 e9 de 6f ae
8b c9 ab cd 35 16 da c9
d0 f1 c0 f3 8f 0b 6d f1
68 ed 60 ec d2 90 a3 ed
34 e3 30 76 69 48 c4 e3
1a e4 18 3b a1 24 62 e4
0d 72 0c 88 c5 12 31 72
93 39 06 44 f7 09 8d 39
dc 89 03 22 ee 91 d3 89
68 83 83 94 ce 20 bd a3
34 d4 d4 4a 67 10 cb c4
1a 6a 6a 25 a6 08 f0 62
0d 35 35 87 53 04 78 31
93 8f 8f d6 bc 02 3c 8d
dc d2 d2 6b 5e 01 1e d3
6e 69 69 a0 2f 95 0f fc
37 a1 a1 50 82 df 92 7e
74 89 12 39 16 16 be 62
3a d1 09 89 0b 0b 5f 31
1d fd 91 d1 90 90 ba 8d
9b eb dd fd 48 48 5d d3
d8 e0 fb eb 24 24 bb fc
6c 70 e8 e0 12 12 c8 7e
36 38 74 70 09 09 64 3f
1b 1c 3a 38 91 91 32 8a
a1 d4 59 3e 12 d8 22 b2
c5 6a b9 1f 09 6c 11 59
f7 35 c9 9a 91 36 9d b9
ee 8f f1 4d dd 1b db c9
77 d2 ed b3 fb 98 f8 f1
ae 69 e3 cc e8 4c 7c ed
57 a1 e4 66 74 26 3e e3
be c5 72 33 3a 13 1f e4
c5 68 02 03 4f 57 9a 90
f7 34 01 94 b2 be 4d 48
ee 1a 95 4a 59 5f b3 24
77 0d df 25 b9 ba cc 12
ae 93 fa 87 c9 5d 66 09
57 dc 7d d6 f1 bb 33 91
be 6e ab 6b ed c8 8c dd
5f 37 c0 a0 e3 64 46 fb
f5 8f 1d d2 89 35 84 48
ef d2 9b 69 d1 8f 42 24
e2 69 d8 a1 fd d2 21 12
71 a1 6c c5 eb 69 85 09
ad c5 36 f7 e0 a1 d7 91
c3 f7 1b ee 70 c5 fe dd
f4 ee 98 77 38 f7 7f fb
7a 77 4c ae 1c ee aa e8

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d5 1e 0a d3 3c c4 d8 f6
ff 0f 05 fc 1e 62 6c 7b
ea 92 97 7e 0f 31 36 a8
75 49 de 3f 92 8d 1b 54
af b1 6f 8a 49 d3 98 2a
c2 cd a2 45 b1 fc 4c 15
61 f3 51 b7 cd 7e 26 9f
a5 ec bd ce f3 3f 13 da
27 fd 8f 05 57 cb 9e 22
86 eb d2 97 be f0 4f 11
43 e0 69 de 5f 78 b2 9d
b4 70 a1 6f ba 3c 59 db
5a 38 c5 a2 5d 1e b9 f8
2d 1c f7 51 bb 0f c9 7c
83 0e ee bd c8 92 f1 3e
d4 07 77 cb 64 49 ed 1f
e3 48 2d 49 c8 21 12 e0
e4 24 83 b1 64 85 09 70
72 12 d4 cd 32 d7 91 38
39 09 6a f3 19 fe dd 1c
89 91 35 ec 99 7f fb 0e
d1 dd 8f 76 d9 aa e8 07
fd fb d2 3b f9 55 74 96
eb e8 69 88 e9 bf 3a 4b
1e 0f 81 9e cf 43 89 c1
0f 92 d5 4f f2 b4 d1 f5
92 49 ff b2 79 5a fd ef
49 b1 ea 59 a9 2d eb e2
b1 cd 75 b9 c1 83 e0 71
cd f3 af c9 f5 d4 70 ad
f3 ec c2 f1 ef 6a 38 c3
ec 76 61 ed e2 35 1c f4
51 9d af 91 17 c6 06 02
bd db c2 dd 9e 63 03 01
cb f8 61 fb 4f a4 94 95
f0 7c a5 e8 b2 52 4a df
78 3e c7 74 59 29 25 fa
3c 1f f6 3a b9 81 87 7d
1e 9a 7b 1d c9 d5 d6 ab
0f 4d a8 9b f1 ff 6b c0
2a 7e 0e a0 a7 5d e8 80
15 3f 07 50 c6 bb 74 40
9f 8a 96 28 63 c8 3a 20
da 45 4b 14 a4 64 1d 10
6d b7 b0 0a 52 32 9b 08
a3 ce 58 05 29 19 d8 04
c4 67 2c 97 81 99 6c 02
62 a6 16 de d5 d9 36 01
ce 17 d2 17 12 67 dc 33
67 9e 69 9e 09 a6 6e 8c
a6 4f a1 4f 91 53 37 46
53 b2 c5 b2 dd bc 8e 23
bc 59 f7 59 fb 5e 47 84
5e b9 ee b9 e8 2f b6 42
2f c9 77 c9 74 82 5b 21
82 f1 ae f1 3a 41 b8 85
8c da 01 89 5f ee ce c6
46 6d 95 d1 ba 77 67 63
23 a3 df fd 5d ae a6 a4
84 c4 fa eb bb 57 53 52
42 62 7d e0 c8 be bc 29
21 31 ab 70 64 5f 5e 81
85 8d c0 38 32 ba 2f d5
d7 d3 60 1c 19 5d 82 ff
06 1a 6b 83 af d8 7a 97

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03 0d a0 d4 c2 6c 3d de
94 93 50 6a 61 36 8b 6f
4a dc 28 35 a5 1b d0 a2
25 6e 14 8f c7 98 68 51
87 37 0a d2 f6 4c 34 bd
d6 8e 05 69 7b 26 1a cb
6b 47 97 a1 a8 13 0d f0
b2 30 d1 07 72 d7 12 5d
59 18 fd 96 39 fe 09 bb
b9 0c eb 4b 89 7f 91 c8
c9 06 e0 b0 d1 aa dd 64
f1 03 70 58 fd 55 fb 32
ed 94 38 2c eb bf e8 19
e3 4a 1c 16 e0 ca 74 99
e4 25 0e 0b 70 65 3a d9
d1 22 50 93 58 18 de 41
fd 11 28 dc 2c 0c 6f b5
eb 9d 14 6e 16 06 a2 cf
e0 db 0a 37 0b 03 51 f2
70 f8 05 8e 90 94 bd 79
38 7c 97 47 48 4a cb a9
1c 3e de b6 24 25 f0 c1
0e 1f 6f 5b 12 87 78 f5
32 2c 4a 48 47 e2 36 ad
19 16 25 24 b6 71 1b c3
99 0b 87 12 5b ad 98 f4
d9 90 d6 09 b8 c3 4c 7a
f9 48 6b 91 5c f4 26 3d
e9 24 a0 dd 2e 7a 13 8b
e1 12 50 fb 17 3d 9c d0
e5 09 28 e8 9e 8b 4e 68
6a 14 59 01 53 5b b3 6c
35 0a b9 95 bc b8 cc 36
8f 05 c9 df 5e 5c 66 1b
d2 97 f1 fa 2f 2e 33 98
69 de ed 7d 82 17 8c 4c
a1 6f e3 ab 41 9e 46 26
c5 a2 e4 c0 b5 4f 23 13
f7 51 72 60 cf b2 84 9c
62 83 50 25 0b 9a eb b0
31 d4 28 87 90 4d e0 58
8d 6a 14 d6 48 b3 70 2c
d3 35 0a 6b 24 cc 38 16
fc 8f 05 a0 12 66 1c 0b
7e d2 97 50 09 33 0e 90
3f 69 de 28 91 8c 07 48
8a a1 6f 14 dd 46 96 24
a7 b7 55 2f 79 9c 09 85
c6 ce bf 82 a9 4e 91 d7
63 67 ca 41 c1 27 dd fe
a4 a6 65 b5 f5 86 fb 7f
52 53 a7 cf ef 43 e8 aa
29 bc c6 f2 e2 b4 74 55
81 5e 63 79 71 5a 3a bf
d5 2f a4 a9 ad 2d 1d ca
0e 3f 41 c0 a1 d1 28 cd
07 8a b5 60 c5 fd 14 f3
96 45 cf 30 f7 eb 0a ec
4b b7 f2 18 ee e0 05 76
b0 ce 79 0c 77 70 97 3b
58 67 a9 06 ae 38 de 88
2c a6 c1 03 57 1c 6f 44
16 53 f5 94 be 0e a2 22
c5 27 b3 5b c8 db 76 68
f7 86 cc b8 64 f8 3b 34

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ee 43 66 5c 32 7c 88 1a
77 b4 33 2e 19 3e 44 0d
ae 5a 8c 17 99 1f 22 93
57 2d 46 9e d9 9a 11 dc
be 83 23 4f f9 4d 9d 6e
5f d4 84 b2 e9 b3 db 37
ba 3e 45 55 05 d9 7c 48
5d 1f b7 bf 97 f9 3e 24
bb 9a ce ca de e9 1f 12
c8 4d 67 65 6f e1 9a 09
64 b3 a6 a7 a2 e5 4d 91
32 cc 53 c6 51 e7 b3 dd
19 66 bc 63 bd e6 cc fb
99 33 5e a4 cb 73 66 e8
8d 8b 23 b6 e5 28 f5 74
d3 d0 84 5b e7 14 ef 3a
fc 68 42 b8 e6 0a e2 1d
7e 34 21 5c 73 05 71 9b
3f 1a 85 2e ac 97 ad d8
8a 0d d7 17 56 de c3 6c
45 93 fe 9e 2b 6f f4 36
b7 dc 7f 4f 80 a2 7a 1b
82 50 aa ec 5a a2 85 fe
41 28 55 76 2d 51 d7 7f
b5 14 bf 3b 83 bd fe aa
cf 0a ca 88 d4 cb 7f 55
f2 05 65 44 6a f0 aa bf
79 97 a7 22 35 78 55 ca
a9 de c6 11 8f 3c bf 65
c1 6f 63 9d d2 1e ca a7
76 d7 7d 7c fa 60 f9 ff
3b fe ab 3e 7d 30 e9 ea
88 7f c0 1f ab 18 e1 75
44 aa 60 9a c0 0c e5 af
22 55 30 4d 60 06 e7 c2
11 bf 18 b3 30 03 e6 61
9d ca 0c cc 18 94 73 a5
db 65 06 66 0c 4a ac c7
7e 6b cb 53 02 1d 25 60
3f a0 f0 bc 01 9b 87 30
8a 50 78 5e 95 d8 d6 18
45 28 3c 2f df 6c 6b 0c
b7 14 1e 82 fa 36 a0 06
ce 0a 0f 41 7d 1b 50 03
67 05 92 b5 ab 98 28 94
a6 97 49 cf c0 4c 14 4a
90 e0 c5 f9 cc 9f 84 55
48 70 f7 e9 66 da 42 bf
24 38 ee e1 33 6d 21 ca
12 1c 77 e5 8c a3 85 65
09 0e ae e7 46 c4 d7 a7
91 07 57 e6 23 62 fe c6
dd 96 be 73 84 31 7f 63
fb 4b 5f ac 42 8d aa a4
e3 12 fd cf ec 5f b4 47
e4 09 eb f2 76 ba 5a b6
72 91 e0 79 3b 5d 2d 5b
39 dd 70 a9 88 bb 83 b8
89 fb 38 c1 44 c8 d4 5c
d1 e8 1c f5 22 64 6a 2e
fd 74 0e ef 11 32 35 17
eb 3a 07 e2 9d 19 8f 9e
44 df 07 ba b1 e5 2e c1
22 fa 96 5d cd e7 17 f5
11 7d 4b bb f3 e6 9e ef

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9d ab b0 c8 ec 73 4f e2
db c0 58 64 76 ac b2 71
f8 60 2c 32 3b 56 59 ad
7c 30 16 19 88 2b b9 c3
3e 18 0b 99 44 80 c9 f4
30 4f 1c 5e 36 a7 05 0c
18 b2 0e 2f 1b c6 97 06
0c 59 07 82 98 63 de 03
06 b9 96 41 4c a4 6f 94
03 c9 4b b5 26 52 a2 4a
94 f1 b0 cf 13 29 51 25
4a ed 58 f2 9c 81 bd 87
25 e3 2c 79 4e d5 cb d6
88 92 98 f1 70 ea 88 be
44 49 4c ed 38 75 44 5f
22 b1 26 e3 1c af 22 ba
11 cd 13 e4 0e c2 11 5d
9d f3 9c 72 07 61 9d bb
db ec 4e 39 96 a5 db c8
f8 76 27 89 4b c7 f8 64
7c 3b 86 d1 b0 f6 7c 32
67 08 96 85 67 b1 1d 18
a6 04 4b d7 a6 cd 9b 0c
53 02 b0 fe 53 f3 d8 06
bc 01 58 7f bc ec 6c 03
5e 95 2c aa 5e 76 36 94
2f df 16 55 2f 3b 1b 4a
82 fa 0b bf 82 88 98 25
41 7d 90 ca 41 44 4c 87
d0 92 8e bf ad 04 f3 63
68 49 47 ca c3 02 ec a4
34 b1 b6 65 f4 01 76 52
1a cd 5b a7 7a 95 3b 29
0d f3 b8 c6 3d df 88 81
93 ec 5c 63 8b fa 44 d5
dc 76 2e a4 d0 7d 22 ff
6e 3b 17 52 68 ab 11 ea
57 9a 98 68 68 8e 59 c4
be 4d 4c 34 34 47 b9 62
5f b3 26 1a 1a b6 c9 31
ba cc 13 0d 0d 5b f1 8d
5d 66 9c 93 93 b8 ed d3
bb 33 4e dc dc 5c e3 fc
c8 8c 27 6e 6e 2e e4 7e
64 46 86 37 37 17 72 3f
85 12 a5 77 45 af ab dd
d7 09 c7 ae b7 c2 c0 fb
fe 91 f6 57 ce 61 60 e8
7f dd 7b be 67 a5 30 74
aa fb a8 5f a6 c7 18 3a
55 e8 54 ba 53 f6 0c 1d
bf 74 2a 5d bc 7b 06 9b
ca 3a 15 bb 5e a8 03 d8
49 eb 4a 1e 6e a2 12 cb
b1 e0 25 0f 37 51 09 f0
cd 70 87 92 8e bd 91 78
f3 38 d6 49 47 cb dd 3c
ec 1c 6b b1 b6 f0 fb 1e
76 0e a0 cd 5b 78 e8 0f
3b 07 50 f3 b8 3c 74 92
88 96 28 ec 5c 1e 3a 49
aa dc e2 2f 3d ab 63 df
55 6e 71 82 8b c0 a4 fa
bf 37 ad 41 d0 60 52 7d
ca 8e c3 b5 68 30 29 ab

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65 47 f4 cf 34 18 81 c0
a7 b6 7a f2 1a 0c d5 60
c6 5b 3d 79 0d 06 ff 30
63 b8 8b a9 93 03 ea 18
cb 56 1b 2a da 8b 1e 1e
f0 2b 98 15 6d d0 0f 0f
78 80 4c 9f a3 68 92 92
3c 40 26 da c4 34 49 49
1e 20 13 6d 62 1a b1 b1
0f 10 9c a3 31 0d cd cd
92 08 4e c4 8d 93 f3 f3
49 04 27 62 d3 dc ec ec
31 f0 47 bd 4c 42 b1 20
8d 78 b6 cb 26 21 cd 10
d3 3c 5b f0 13 85 f3 08
fc 1e b8 78 9c d7 ec 04
7e 0f 5c 3c 4e fe 76 02
3f 92 2e 1e 27 7f 3b 01
8a 49 17 0f 86 aa 88 95
45 b1 9e 92 43 55 44 df
e2 a1 85 5e a6 1f e9 d7
71 c5 d7 2f 53 9a e1 fe
ad f7 fe 82 bc 4d e5 7f
c3 ee 7f 41 5e b3 e7 aa
f4 77 aa b5 2f cc e6 55
7a ae 55 cf 82 66 73 bf
3d 57 bf f2 41 33 ac ca
8b be ca 79 b5 8c 56 65
22 cf 68 25 c2 f9 f3 a8
11 f2 34 87 61 e9 ec 54
9d 79 1a d6 a5 e1 76 2a
db a9 0d 6b c7 e5 3b 15
f8 c1 93 a0 f6 e7 88 9f
7c f5 dc 50 7b e6 44 da
3e ef 6e 28 a8 73 22 6d
1f e2 37 14 54 ac 11 a3
2d 4c 76 20 05 22 ca 06
83 26 3b 10 97 11 65 03
d4 13 88 08 de 9d a7 94
6a 9c 44 04 6f db c6 4a
35 4e 22 02 a2 f8 63 25
8f 27 11 01 51 7c a4 87
d2 86 9d 95 bd 3e 52 d6
69 43 db df cb 1f 29 6b
81 5c fe c8 f4 a9 92 07
d5 2e 7f 64 7a c1 49 96
ff 17 aa 32 3d f5 b1 4b
ea 9e 55 19 8b ef cd b0
75 4f bf 99 d0 e2 f3 58
af b2 ca d9 68 71 ec 2c
c2 59 65 f9 34 ad 76 16
61 b9 a7 e9 1a c3 3b 0b

```

K.2 Flit 8 byte LCRC §

pci_sig_8B_crc.sv

```

`ifndef PCI_SIG_8B_CRC__SV
`define PCI_SIG_8B_CRC__SV
//PCI SIG code for Flit Mode 8B CRC generation
//input: 242 bytes of flit data
//output: 250 bytes of flit data + CRC
module pci_sig_8B_crc (
    input logic [7:0] data_in[241:0],
    output logic [7:0] data_out[249:0]
);
always_comb begin
    //Assign data bytes from input to output
    for (int i =0;i<242;i++) begin
        data_out[i] = data_in[i];
    end
    //Assign CRC bytes
    data_out[249] = ({8{data_in[0][0]}} & 8'h61)^
        ({8{data_in[0][1]}} & 8'hc2)^
        ({8{data_in[0][2]}} & 8'haf)^
        ({8{data_in[0][3]}} & 8'h75)^
        ({8{data_in[0][4]}} & 8'hea)^
        ({8{data_in[0][5]}} & 8'hff)^
        ({8{data_in[0][6]}} & 8'hd5)^
        ({8{data_in[0][7]}} & 8'h81)^
        ({8{data_in[1][0]}} & 8'h69)^
        ({8{data_in[1][1]}} & 8'hd2)^
        ({8{data_in[1][2]}} & 8'h8f)^
        ({8{data_in[1][3]}} & 8'h35)^
        ({8{data_in[1][4]}} & 8'h6a)^
        ({8{data_in[1][5]}} & 8'hd4)^
        ({8{data_in[1][6]}} & 8'h83)^
        ({8{data_in[1][7]}} & 8'h2d)^
        ({8{data_in[2][0]}} & 8'h1f)^
        ({8{data_in[2][1]}} & 8'h3e)^
        ({8{data_in[2][2]}} & 8'h7c)^
        ({8{data_in[2][3]}} & 8'hf8)^
        ({8{data_in[2][4]}} & 8'hdb)^
        ({8{data_in[2][5]}} & 8'h9d)^
        ({8{data_in[2][6]}} & 8'h11)^
        ({8{data_in[2][7]}} & 8'h22)^
        ({8{data_in[3][0]}} & 8'h8b)^
        ({8{data_in[3][1]}} & 8'h3d)^
        ({8{data_in[3][2]}} & 8'h7a)^
        ({8{data_in[3][3]}} & 8'hf4)^
        ({8{data_in[3][4]}} & 8'hc3)^
        ({8{data_in[3][5]}} & 8'had)^
        ({8{data_in[3][6]}} & 8'h71)^
        ({8{data_in[3][7]}} & 8'he2)^
        ({8{data_in[4][0]}} & 8'h45)^
        ({8{data_in[4][1]}} & 8'h8a)^
        ({8{data_in[4][2]}} & 8'h3f)^
        ({8{data_in[4][3]}} & 8'h7e)^
        ({8{data_in[4][4]}} & 8'hfc)^
        ({8{data_in[4][5]}} & 8'hd3)^
        ({8{data_in[4][6]}} & 8'h8d)^
        ({8{data_in[4][7]}} & 8'h31)^
        ({8{data_in[5][0]}} & 8'h49)^
        ({8{data_in[5][1]}} & 8'h92)^
        ({8{data_in[5][2]}} & 8'hf)^
        ({8{data_in[5][3]}} & 8'h1e)^
        ({8{data_in[5][4]}} & 8'h3c)^
        ({8{data_in[5][5]}} & 8'h78)^
        ({8{data_in[5][6]}} & 8'hf0)^
        ({8{data_in[5][7]}} & 8'hcb)^
        ({8{data_in[6][0]}} & 8'h63)^

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({8{data_in[6][1]}} & 8'hc6)^
({8{data_in[6][2]}} & 8'ha7)^
({8{data_in[6][3]}} & 8'h65)^
({8{data_in[6][4]}} & 8'hca)^
({8{data_in[6][5]}} & 8'hbf)^
({8{data_in[6][6]}} & 8'h55)^
({8{data_in[6][7]}} & 8'haa)^
({8{data_in[7][0]}} & 8'h88)^
({8{data_in[7][1]}} & 8'h3b)^
({8{data_in[7][2]}} & 8'h76)^
({8{data_in[7][3]}} & 8'hec)^
({8{data_in[7][4]}} & 8'hf3)^
({8{data_in[7][5]}} & 8'hcd)^
({8{data_in[7][6]}} & 8'hb1)^
({8{data_in[7][7]}} & 8'h49)^
({8{data_in[8][0]}} & 8'hca)^
({8{data_in[8][1]}} & 8'hbf)^
({8{data_in[8][2]}} & 8'h55)^
({8{data_in[8][3]}} & 8'haa)^
({8{data_in[8][4]}} & 8'h7f)^
({8{data_in[8][5]}} & 8'hfe)^
({8{data_in[8][6]}} & 8'hd7)^
({8{data_in[8][7]}} & 8'h85)^
({8{data_in[9][0]}} & 8'h64)^
({8{data_in[9][1]}} & 8'hc8)^
({8{data_in[9][2]}} & 8'hbb)^
({8{data_in[9][3]}} & 8'h5d)^
({8{data_in[9][4]}} & 8'hba)^
({8{data_in[9][5]}} & 8'h5f)^
({8{data_in[9][6]}} & 8'hbe)^
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({8{data_in[10][0]}} & 8'h6e)^
({8{data_in[10][1]}} & 8'hdc)^
({8{data_in[10][2]}} & 8'h93)^
({8{data_in[10][3]}} & 8'hd)^
({8{data_in[10][4]}} & 8'h1a)^
({8{data_in[10][5]}} & 8'h34)^
({8{data_in[10][6]}} & 8'h68)^
({8{data_in[10][7]}} & 8'hd0)^
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({8{data_in[11][1]}} & 8'h82)^
({8{data_in[11][2]}} & 8'h2f)^
({8{data_in[11][3]}} & 8'h5e)^
({8{data_in[11][4]}} & 8'hbc)^
({8{data_in[11][5]}} & 8'h53)^
({8{data_in[11][6]}} & 8'ha6)^
({8{data_in[11][7]}} & 8'h67)^
({8{data_in[12][0]}} & 8'h7c)^
({8{data_in[12][1]}} & 8'hf8)^
({8{data_in[12][2]}} & 8'hdb)^
({8{data_in[12][3]}} & 8'h9d)^
({8{data_in[12][4]}} & 8'h11)^
({8{data_in[12][5]}} & 8'h22)^
({8{data_in[12][6]}} & 8'h44)^
({8{data_in[12][7]}} & 8'h88)^
({8{data_in[13][0]}} & 8'h25)^
({8{data_in[13][1]}} & 8'h4a)^
({8{data_in[13][2]}} & 8'h94)^
({8{data_in[13][3]}} & 8'h3)^
({8{data_in[13][4]}} & 8'h6)^
({8{data_in[13][5]}} & 8'hc)^
({8{data_in[13][6]}} & 8'h18)^
({8{data_in[13][7]}} & 8'h30)^
({8{data_in[14][0]}} & 8'h3e)^
({8{data_in[14][1]}} & 8'h7c)^

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({8{data_in[14][2]}} & 8'hf8)^
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({8{data_in[14][4]}} & 8'h9d)^
({8{data_in[14][5]}} & 8'h11)^
({8{data_in[14][6]}} & 8'h22)^
({8{data_in[14][7]}} & 8'h44)^
({8{data_in[15][0]}} & 8'heb)^
({8{data_in[15][1]}} & 8'hfd)^
({8{data_in[15][2]}} & 8'hd1)^
({8{data_in[15][3]}} & 8'h89)^
({8{data_in[15][4]}} & 8'h39)^
({8{data_in[15][5]}} & 8'h72)^
({8{data_in[15][6]}} & 8'he4)^
({8{data_in[15][7]}} & 8'he3)^
({8{data_in[16][0]}} & 8'hfb)^
({8{data_in[16][1]}} & 8'hdd)^
({8{data_in[16][2]}} & 8'h91)^
({8{data_in[16][3]}} & 8'h9)^
({8{data_in[16][4]}} & 8'h12)^
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({8{data_in[17][0]}} & 8'ha6)^
({8{data_in[17][1]}} & 8'h67)^
({8{data_in[17][2]}} & 8'hce)^
({8{data_in[17][3]}} & 8'hb7)^
({8{data_in[17][4]}} & 8'h45)^
({8{data_in[17][5]}} & 8'h8a)^
({8{data_in[17][6]}} & 8'h3f)^
({8{data_in[17][7]}} & 8'h7e)^
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({8{data_in[18][1]}} & 8'h9d)^
({8{data_in[18][2]}} & 8'h11)^
({8{data_in[18][3]}} & 8'h22)^
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({8{data_in[19][7]}} & 8'h82)^
({8{data_in[20][0]}} & 8'hb7)^
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({8{data_in[20][2]}} & 8'h8a)^
({8{data_in[20][3]}} & 8'h3f)^
({8{data_in[20][4]}} & 8'h7e)^
({8{data_in[20][5]}} & 8'hfc)^
({8{data_in[20][6]}} & 8'hd3)^
({8{data_in[20][7]}} & 8'h8d)^
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({8{data_in[21][7]}} & 8'hba)^
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({8{data_in[22][1]}} & 8'hbe)^
({8{data_in[22][2]}} & 8'h57)^

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({8{data_in[23][3]}} & 8'hb0)^
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({8{data_in[23][5]}} & 8'h96)^
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({8{data_in[27][1]}} & 8'he1)^
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({8{data_in[28][6]}} & 8'hfd)^
({8{data_in[28][7]}} & 8'hd1)^
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({8{data_in[29][4]}} & 8'hc9)^
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({8{data_in[32][2]}} & 8'h5e)^
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({8{data_in[43][3]}} & 8'h93)^
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({8{data_in[44][1]}} & 8'h93)^
({8{data_in[44][2]}} & 8'hd)^
({8{data_in[44][3]}} & 8'h1a)^
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({8{data_in[49][6]}} & 8'h3e)^
({8{data_in[49][7]}} & 8'h7c)^
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({8{data_in[58][4]}} & 8'h7f)^
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({8{data_in[59][4]}} & 8'h18)^
({8{data_in[59][5]}} & 8'h30)^
({8{data_in[59][6]}} & 8'h60)^
({8{data_in[59][7]}} & 8'hc0)^
({8{data_in[60][0]}} & 8'h9e)^
({8{data_in[60][1]}} & 8'h17)^
({8{data_in[60][2]}} & 8'h2e)^
({8{data_in[60][3]}} & 8'h5c)^
({8{data_in[60][4]}} & 8'hb8)^
({8{data_in[60][5]}} & 8'h5b)^
({8{data_in[60][6]}} & 8'hb6)^
({8{data_in[60][7]}} & 8'h47)^
({8{data_in[61][0]}} & 8'hc3)^
({8{data_in[61][1]}} & 8'had)^
({8{data_in[61][2]}} & 8'h71)^
({8{data_in[61][3]}} & 8'he2)^
({8{data_in[61][4]}} & 8'hef)^
({8{data_in[61][5]}} & 8'hf5)^
({8{data_in[61][6]}} & 8'hc1)^
({8{data_in[61][7]}} & 8'ha9)^
({8{data_in[62][0]}} & 8'ha3)^
({8{data_in[62][1]}} & 8'h6d)^
({8{data_in[62][2]}} & 8'hda)^
({8{data_in[62][3]}} & 8'h9f)^
({8{data_in[62][4]}} & 8'h15)^
({8{data_in[62][5]}} & 8'h2a)^
({8{data_in[62][6]}} & 8'h54)^
({8{data_in[62][7]}} & 8'ha8)^

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({8{data_in[63][0]}} & 8'h23)^
({8{data_in[63][1]}} & 8'h46)^
({8{data_in[63][2]}} & 8'h8c)^
({8{data_in[63][3]}} & 8'h33)^
({8{data_in[63][4]}} & 8'h66)^
({8{data_in[63][5]}} & 8'hcc)^
({8{data_in[63][6]}} & 8'hb3)^
({8{data_in[63][7]}} & 8'h4d)^
({8{data_in[64][0]}} & 8'h54)^
({8{data_in[64][1]}} & 8'ha8)^
({8{data_in[64][2]}} & 8'h7b)^
({8{data_in[64][3]}} & 8'hf6)^
({8{data_in[64][4]}} & 8'hc7)^
({8{data_in[64][5]}} & 8'ha5)^
({8{data_in[64][6]}} & 8'h61)^
({8{data_in[64][7]}} & 8'hc2)^
({8{data_in[65][0]}} & 8'ha4)^
({8{data_in[65][1]}} & 8'h63)^
({8{data_in[65][2]}} & 8'hc6)^
({8{data_in[65][3]}} & 8'ha7)^
({8{data_in[65][4]}} & 8'h65)^
({8{data_in[65][5]}} & 8'hca)^
({8{data_in[65][6]}} & 8'hbf)^
({8{data_in[65][7]}} & 8'h55)^
({8{data_in[66][0]}} & 8'h4b)^
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({8{data_in[66][2]}} & 8'h7)^
({8{data_in[66][3]}} & 8'he)^
({8{data_in[66][4]}} & 8'h1c)^
({8{data_in[66][5]}} & 8'h38)^
({8{data_in[66][6]}} & 8'h70)^
({8{data_in[66][7]}} & 8'he0)^
({8{data_in[67][0]}} & 8'h1c)^
({8{data_in[67][1]}} & 8'h38)^
({8{data_in[67][2]}} & 8'h70)^
({8{data_in[67][3]}} & 8'he0)^
({8{data_in[67][4]}} & 8'heb)^
({8{data_in[67][5]}} & 8'hfd)^
({8{data_in[67][6]}} & 8'hd1)^
({8{data_in[67][7]}} & 8'h89)^
({8{data_in[68][0]}} & 8'h84)^
({8{data_in[68][1]}} & 8'h23)^
({8{data_in[68][2]}} & 8'h46)^
({8{data_in[68][3]}} & 8'h8c)^
({8{data_in[68][4]}} & 8'h33)^
({8{data_in[68][5]}} & 8'h66)^
({8{data_in[68][6]}} & 8'hcc)^
({8{data_in[68][7]}} & 8'hb3)^
({8{data_in[69][0]}} & 8'h9a)^
({8{data_in[69][1]}} & 8'h1f)^
({8{data_in[69][2]}} & 8'h3e)^
({8{data_in[69][3]}} & 8'h7c)^
({8{data_in[69][4]}} & 8'hf8)^
({8{data_in[69][5]}} & 8'hdb)^
({8{data_in[69][6]}} & 8'h9d)^
({8{data_in[69][7]}} & 8'h11)^
({8{data_in[70][0]}} & 8'haa)^
({8{data_in[70][1]}} & 8'h7f)^
({8{data_in[70][2]}} & 8'hfe)^
({8{data_in[70][3]}} & 8'hd7)^
({8{data_in[70][4]}} & 8'h85)^
({8{data_in[70][5]}} & 8'h21)^
({8{data_in[70][6]}} & 8'h42)^
({8{data_in[70][7]}} & 8'h84)^
({8{data_in[71][0]}} & 8'he0)^

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({8{data_in[71][1]}} & 8'heb)^
({8{data_in[71][2]}} & 8'hfd)^
({8{data_in[71][3]}} & 8'hd1)^
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({8{data_in[71][6]}} & 8'h72)^
({8{data_in[71][7]}} & 8'he4)^
({8{data_in[72][0]}} & 8'h5b)^
({8{data_in[72][1]}} & 8'hb6)^
({8{data_in[72][2]}} & 8'h47)^
({8{data_in[72][3]}} & 8'h8e)^
({8{data_in[72][4]}} & 8'h37)^
({8{data_in[72][5]}} & 8'h6e)^
({8{data_in[72][6]}} & 8'hdc)^
({8{data_in[72][7]}} & 8'h93)^
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({8{data_in[73][1]}} & 8'h8f)^
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({8{data_in[73][3]}} & 8'h6a)^
({8{data_in[73][4]}} & 8'hd4)^
({8{data_in[73][5]}} & 8'h83)^
({8{data_in[73][6]}} & 8'h2d)^
({8{data_in[73][7]}} & 8'h5a)^
({8{data_in[74][0]}} & 8'h85)^
({8{data_in[74][1]}} & 8'h21)^
({8{data_in[74][2]}} & 8'h42)^
({8{data_in[74][3]}} & 8'h84)^
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({8{data_in[74][7]}} & 8'h33)^
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({8{data_in[75][1]}} & 8'h7)^
({8{data_in[75][2]}} & 8'he)^
({8{data_in[75][3]}} & 8'h1c)^
({8{data_in[75][4]}} & 8'h38)^
({8{data_in[75][5]}} & 8'h70)^
({8{data_in[75][6]}} & 8'he0)^
({8{data_in[75][7]}} & 8'heb)^
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({8{data_in[76][1]}} & 8'h3c)^
({8{data_in[76][2]}} & 8'h78)^
({8{data_in[76][3]}} & 8'hf0)^
({8{data_in[76][4]}} & 8'hcb)^
({8{data_in[76][5]}} & 8'hbd)^
({8{data_in[76][6]}} & 8'h51)^
({8{data_in[76][7]}} & 8'ha2)^
({8{data_in[77][0]}} & 8'hdb)^
({8{data_in[77][1]}} & 8'h9d)^
({8{data_in[77][2]}} & 8'h11)^
({8{data_in[77][3]}} & 8'h22)^
({8{data_in[77][4]}} & 8'h44)^
({8{data_in[77][5]}} & 8'h88)^
({8{data_in[77][6]}} & 8'h3b)^
({8{data_in[77][7]}} & 8'h76)^
({8{data_in[78][0]}} & 8'he0)^
({8{data_in[78][1]}} & 8'heb)^
({8{data_in[78][2]}} & 8'hfd)^
({8{data_in[78][3]}} & 8'hd1)^
({8{data_in[78][4]}} & 8'h89)^
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({8{data_in[78][6]}} & 8'h72)^
({8{data_in[78][7]}} & 8'he4)^
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({8{data_in[79][1]}} & 8'hb0)^

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({8{data_in[79][7]}} & 8'h38)^
({8{data_in[80][0]}} & 8'h64)^
({8{data_in[80][1]}} & 8'hc8)^
({8{data_in[80][2]}} & 8'hbb)^
({8{data_in[80][3]}} & 8'h5d)^
({8{data_in[80][4]}} & 8'hba)^
({8{data_in[80][5]}} & 8'h5f)^
({8{data_in[80][6]}} & 8'hbe)^
({8{data_in[80][7]}} & 8'h57)^
({8{data_in[81][0]}} & 8'h7)^
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({8{data_in[81][2]}} & 8'h1c)^
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({8{data_in[81][4]}} & 8'h70)^
({8{data_in[81][5]}} & 8'he0)^
({8{data_in[81][6]}} & 8'heb)^
({8{data_in[81][7]}} & 8'hfd)^
({8{data_in[82][0]}} & 8'h63)^
({8{data_in[82][1]}} & 8'hc6)^
({8{data_in[82][2]}} & 8'ha7)^
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({8{data_in[82][4]}} & 8'hca)^
({8{data_in[82][5]}} & 8'hbf)^
({8{data_in[82][6]}} & 8'h55)^
({8{data_in[82][7]}} & 8'haa)^
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({8{data_in[83][4]}} & 8'h4a)^
({8{data_in[83][5]}} & 8'h94)^
({8{data_in[83][6]}} & 8'h3)^
({8{data_in[83][7]}} & 8'h6)^
({8{data_in[84][0]}} & 8'h7d)^
({8{data_in[84][1]}} & 8'hfa)^
({8{data_in[84][2]}} & 8'hdf)^
({8{data_in[84][3]}} & 8'h95)^
({8{data_in[84][4]}} & 8'h1)^
({8{data_in[84][5]}} & 8'h2)^
({8{data_in[84][6]}} & 8'h4)^
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({8{data_in[85][3]}} & 8'h1b)^
({8{data_in[85][4]}} & 8'h36)^
({8{data_in[85][5]}} & 8'h6c)^
({8{data_in[85][6]}} & 8'hd8)^
({8{data_in[85][7]}} & 8'h9b)^
({8{data_in[86][0]}} & 8'h6f)^
({8{data_in[86][1]}} & 8'hde)^
({8{data_in[86][2]}} & 8'h97)^
({8{data_in[86][3]}} & 8'h5)^
({8{data_in[86][4]}} & 8'ha)^
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({8{data_in[87][2]}} & 8'h3f)^

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({8{data_in[88][2]}} & 8'h38)^
({8{data_in[88][3]}} & 8'h70)^
({8{data_in[88][4]}} & 8'he0)^
({8{data_in[88][5]}} & 8'heb)^
({8{data_in[88][6]}} & 8'hfd)^
({8{data_in[88][7]}} & 8'hd1)^
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({8{data_in[89][7]}} & 8'haa)^
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({8{data_in[90][1]}} & 8'hbe)^
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({8{data_in[90][3]}} & 8'hae)^
({8{data_in[90][4]}} & 8'h77)^
({8{data_in[90][5]}} & 8'hee)^
({8{data_in[90][6]}} & 8'hf7)^
({8{data_in[90][7]}} & 8'hc5)^
({8{data_in[91][0]}} & 8'h33)^
({8{data_in[91][1]}} & 8'h66)^
({8{data_in[91][2]}} & 8'hcc)^
({8{data_in[91][3]}} & 8'hb3)^
({8{data_in[91][4]}} & 8'h4d)^
({8{data_in[91][5]}} & 8'h9a)^
({8{data_in[91][6]}} & 8'h1f)^
({8{data_in[91][7]}} & 8'h3e)^
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({8{data_in[92][2]}} & 8'h88)^
({8{data_in[92][3]}} & 8'h3b)^
({8{data_in[92][4]}} & 8'h76)^
({8{data_in[92][5]}} & 8'hec)^
({8{data_in[92][6]}} & 8'hf3)^
({8{data_in[92][7]}} & 8'hcd)^
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({8{data_in[93][1]}} & 8'h86)^
({8{data_in[93][2]}} & 8'h27)^
({8{data_in[93][3]}} & 8'h4e)^
({8{data_in[93][4]}} & 8'h9c)^
({8{data_in[93][5]}} & 8'h13)^
({8{data_in[93][6]}} & 8'h26)^
({8{data_in[93][7]}} & 8'h4c)^
({8{data_in[94][0]}} & 8'hcf)^
({8{data_in[94][1]}} & 8'hb5)^
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({8{data_in[94][3]}} & 8'h82)^
({8{data_in[94][4]}} & 8'h2f)^
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({8{data_in[96][3]}} & 8'h7a)^
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({8{data_in[96][5]}} & 8'hc3)^
({8{data_in[96][6]}} & 8'had)^
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({8{data_in[97][2]}} & 8'h77)^
({8{data_in[97][3]}} & 8'hee)^
({8{data_in[97][4]}} & 8'hf7)^
({8{data_in[97][5]}} & 8'hc5)^
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({8{data_in[98][2]}} & 8'h51)^
({8{data_in[98][3]}} & 8'ha2)^
({8{data_in[98][4]}} & 8'h6f)^
({8{data_in[98][5]}} & 8'hde)^
({8{data_in[98][6]}} & 8'h97)^
({8{data_in[98][7]}} & 8'h5)^
({8{data_in[99][0]}} & 8'h7f)^
({8{data_in[99][1]}} & 8'hfe)^
({8{data_in[99][2]}} & 8'hd7)^
({8{data_in[99][3]}} & 8'h85)^
({8{data_in[99][4]}} & 8'h21)^
({8{data_in[99][5]}} & 8'h42)^
({8{data_in[99][6]}} & 8'h84)^
({8{data_in[99][7]}} & 8'h23)^
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({8{data_in[100][2]}} & 8'hb2)^
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({8{data_in[100][4]}} & 8'h9e)^
({8{data_in[100][5]}} & 8'h17)^
({8{data_in[100][6]}} & 8'h2e)^
({8{data_in[100][7]}} & 8'h5c)^
({8{data_in[101][0]}} & 8'h42)^
({8{data_in[101][1]}} & 8'h84)^
({8{data_in[101][2]}} & 8'h23)^
({8{data_in[101][3]}} & 8'h46)^
({8{data_in[101][4]}} & 8'h8c)^
({8{data_in[101][5]}} & 8'h33)^
({8{data_in[101][6]}} & 8'h66)^
({8{data_in[101][7]}} & 8'hcc)^
({8{data_in[102][0]}} & 8'h5a)^
({8{data_in[102][1]}} & 8'hb4)^
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({8{data_in[102][3]}} & 8'h86)^
({8{data_in[102][4]}} & 8'h27)^
({8{data_in[102][5]}} & 8'h4e)^
({8{data_in[102][6]}} & 8'h9c)^
({8{data_in[102][7]}} & 8'h13)^
({8{data_in[103][0]}} & 8'hd2)^
({8{data_in[103][1]}} & 8'h8f)^
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({8{data_in[103][4]}} & 8'hd4)^

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({8{data_in[105][4]}} & 8'h3d)^
({8{data_in[105][5]}} & 8'h7a)^
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({8{data_in[106][1]}} & 8'h85)^
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({8{data_in[106][5]}} & 8'h23)^
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({8{data_in[107][1]}} & 8'h55)^
({8{data_in[107][2]}} & 8'haa)^
({8{data_in[107][3]}} & 8'h7f)^
({8{data_in[107][4]}} & 8'hfe)^
({8{data_in[107][5]}} & 8'hd7)^
({8{data_in[107][6]}} & 8'h85)^
({8{data_in[107][7]}} & 8'h21)^
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({8{data_in[109][2]}} & 8'h9f)^
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({8{data_in[109][7]}} & 8'h7b)^
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({8{data_in[110][1]}} & 8'h32)^
({8{data_in[110][2]}} & 8'h64)^
({8{data_in[110][3]}} & 8'hc8)^
({8{data_in[110][4]}} & 8'hbb)^
({8{data_in[110][5]}} & 8'h5d)^
({8{data_in[110][6]}} & 8'hba)^
({8{data_in[110][7]}} & 8'h5f)^
({8{data_in[111][0]}} & 8'h4a)^
({8{data_in[111][1]}} & 8'h94)^
({8{data_in[111][2]}} & 8'h3)^
({8{data_in[111][3]}} & 8'h6)^
({8{data_in[111][4]}} & 8'hc)^
({8{data_in[111][5]}} & 8'h18)^

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({8{data_in[112][1]}} & 8'h97)^
({8{data_in[112][2]}} & 8'h5)^
({8{data_in[112][3]}} & 8'ha)^
({8{data_in[112][4]}} & 8'h14)^
({8{data_in[112][5]}} & 8'h28)^
({8{data_in[112][6]}} & 8'h50)^
({8{data_in[112][7]}} & 8'ha0)^
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({8{data_in[113][2]}} & 8'hea)^
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({8{data_in[114][1]}} & 8'ha2)^
({8{data_in[114][2]}} & 8'h6f)^
({8{data_in[114][3]}} & 8'hde)^
({8{data_in[114][4]}} & 8'h97)^
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({8{data_in[116][4]}} & 8'h6a)^
({8{data_in[116][5]}} & 8'hd4)^
({8{data_in[116][6]}} & 8'h83)^
({8{data_in[116][7]}} & 8'h2d)^
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({8{data_in[117][1]}} & 8'hc8)^
({8{data_in[117][2]}} & 8'hbb)^
({8{data_in[117][3]}} & 8'h5d)^
({8{data_in[117][4]}} & 8'hba)^
({8{data_in[117][5]}} & 8'h5f)^
({8{data_in[117][6]}} & 8'hbe)^
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({8{data_in[118][2]}} & 8'h6f)^
({8{data_in[118][3]}} & 8'hde)^
({8{data_in[118][4]}} & 8'h97)^
({8{data_in[118][5]}} & 8'h5)^
({8{data_in[118][6]}} & 8'ha)^
({8{data_in[118][7]}} & 8'h14)^
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({8{data_in[119][1]}} & 8'hc9)^
({8{data_in[119][2]}} & 8'hb9)^
({8{data_in[119][3]}} & 8'h59)^
({8{data_in[119][4]}} & 8'hb2)^
({8{data_in[119][5]}} & 8'h4f)^
({8{data_in[119][6]}} & 8'h9e)^

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({8{data_in[120][6]}} & 8'hce)^
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({8{data_in[121][1]}} & 8'hce)^
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({8{data_in[121][4]}} & 8'h8a)^
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({8{data_in[122][7]}} & 8'h5d)^
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({8{data_in[123][1]}} & 8'h9b)^
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({8{data_in[123][3]}} & 8'h3a)^
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({8{data_in[123][5]}} & 8'he8)^
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({8{data_in[124][4]}} & 8'hab)^
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({8{data_in[125][6]}} & 8'h4e)^
({8{data_in[125][7]}} & 8'h9c)^
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({8{data_in[126][1]}} & 8'h86)^
({8{data_in[126][2]}} & 8'h27)^
({8{data_in[126][3]}} & 8'h4e)^
({8{data_in[126][4]}} & 8'h9c)^
({8{data_in[126][5]}} & 8'h13)^
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({8{data_in[127][5]}} & 8'hb0)^
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({8{data_in[128][6]}} & 8'h7c)^
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({8{data_in[129][4]}} & 8'hc1)^
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({8{data_in[134][1]}} & 8'hba)^
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({8{data_in[134][3]}} & 8'hbe)^
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({8{data_in[137][1]}} & 8'hd8)^
({8{data_in[137][2]}} & 8'h9b)^
({8{data_in[137][3]}} & 8'h1d)^
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({8{data_in[144][1]}} & 8'h9a)^

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({8{data_in[144][7]}} & 8'h9d)^
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({8{data_in[158][1]}} & 8'h5c)^
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({8{data_in[159][7]}} & 8'h37)^
({8{data_in[160][0]}} & 8'hf3)^
({8{data_in[160][1]}} & 8'hcd)^
({8{data_in[160][2]}} & 8'hb1)^
({8{data_in[160][3]}} & 8'h49)^

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({8{data_in[160][4]}} & 8'h92)^
({8{data_in[160][5]}} & 8'hf)^
({8{data_in[160][6]}} & 8'h1e)^
({8{data_in[160][7]}} & 8'h3c)^
({8{data_in[161][0]}} & 8'h1e)^
({8{data_in[161][1]}} & 8'h3c)^
({8{data_in[161][2]}} & 8'h78)^
({8{data_in[161][3]}} & 8'hf0)^
({8{data_in[161][4]}} & 8'hcb)^
({8{data_in[161][5]}} & 8'hbd)^
({8{data_in[161][6]}} & 8'h51)^
({8{data_in[161][7]}} & 8'ha2)^
({8{data_in[162][0]}} & 8'h24)^
({8{data_in[162][1]}} & 8'h48)^
({8{data_in[162][2]}} & 8'h90)^
({8{data_in[162][3]}} & 8'hb)^
({8{data_in[162][4]}} & 8'h16)^
({8{data_in[162][5]}} & 8'h2c)^
({8{data_in[162][6]}} & 8'h58)^
({8{data_in[162][7]}} & 8'hb0)^
({8{data_in[163][0]}} & 8'hea)^
({8{data_in[163][1]}} & 8'hff)^
({8{data_in[163][2]}} & 8'hd5)^
({8{data_in[163][3]}} & 8'h81)^
({8{data_in[163][4]}} & 8'h29)^
({8{data_in[163][5]}} & 8'h52)^
({8{data_in[163][6]}} & 8'ha4)^
({8{data_in[163][7]}} & 8'h63)^
({8{data_in[164][0]}} & 8'hac)^
({8{data_in[164][1]}} & 8'h73)^
({8{data_in[164][2]}} & 8'he6)^
({8{data_in[164][3]}} & 8'he7)^
({8{data_in[164][4]}} & 8'he5)^
({8{data_in[164][5]}} & 8'he1)^
({8{data_in[164][6]}} & 8'he9)^
({8{data_in[164][7]}} & 8'hf9)^
({8{data_in[165][0]}} & 8'h24)^
({8{data_in[165][1]}} & 8'h48)^
({8{data_in[165][2]}} & 8'h90)^
({8{data_in[165][3]}} & 8'hb)^
({8{data_in[165][4]}} & 8'h16)^
({8{data_in[165][5]}} & 8'h2c)^
({8{data_in[165][6]}} & 8'h58)^
({8{data_in[165][7]}} & 8'hb0)^
({8{data_in[166][0]}} & 8'h35)^
({8{data_in[166][1]}} & 8'h6a)^
({8{data_in[166][2]}} & 8'hd4)^
({8{data_in[166][3]}} & 8'h83)^
({8{data_in[166][4]}} & 8'h2d)^
({8{data_in[166][5]}} & 8'h5a)^
({8{data_in[166][6]}} & 8'hb4)^
({8{data_in[166][7]}} & 8'h43)^
({8{data_in[167][0]}} & 8'h79)^
({8{data_in[167][1]}} & 8'hf2)^
({8{data_in[167][2]}} & 8'hcf)^
({8{data_in[167][3]}} & 8'hb5)^
({8{data_in[167][4]}} & 8'h41)^
({8{data_in[167][5]}} & 8'h82)^
({8{data_in[167][6]}} & 8'h2f)^
({8{data_in[167][7]}} & 8'h5e)^
({8{data_in[168][0]}} & 8'h5e)^
({8{data_in[168][1]}} & 8'hbc)^
({8{data_in[168][2]}} & 8'h53)^
({8{data_in[168][3]}} & 8'ha6)^
({8{data_in[168][4]}} & 8'h67)^

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({8{data_in[168][5]}} & 8'hce)^
({8{data_in[168][6]}} & 8'hb7)^
({8{data_in[168][7]}} & 8'h45)^
({8{data_in[169][0]}} & 8'h3d)^
({8{data_in[169][1]}} & 8'h7a)^
({8{data_in[169][2]}} & 8'hf4)^
({8{data_in[169][3]}} & 8'hc3)^
({8{data_in[169][4]}} & 8'had)^
({8{data_in[169][5]}} & 8'h71)^
({8{data_in[169][6]}} & 8'he2)^
({8{data_in[169][7]}} & 8'hef)^
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({8{data_in[171][1]}} & 8'h3)^
({8{data_in[171][2]}} & 8'h6)^
({8{data_in[171][3]}} & 8'hc)^
({8{data_in[171][4]}} & 8'h18)^
({8{data_in[171][5]}} & 8'h30)^
({8{data_in[171][6]}} & 8'h60)^
({8{data_in[171][7]}} & 8'hc0)^
({8{data_in[172][0]}} & 8'h4b)^
({8{data_in[172][1]}} & 8'h96)^
({8{data_in[172][2]}} & 8'h7)^
({8{data_in[172][3]}} & 8'he)^
({8{data_in[172][4]}} & 8'h1c)^
({8{data_in[172][5]}} & 8'h38)^
({8{data_in[172][6]}} & 8'h70)^
({8{data_in[172][7]}} & 8'he0)^
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({8{data_in[173][1]}} & 8'h4c)^
({8{data_in[173][2]}} & 8'h98)^
({8{data_in[173][3]}} & 8'h1b)^
({8{data_in[173][4]}} & 8'h36)^
({8{data_in[173][5]}} & 8'h6c)^
({8{data_in[173][6]}} & 8'hd8)^
({8{data_in[173][7]}} & 8'h9b)^
({8{data_in[174][0]}} & 8'h58)^
({8{data_in[174][1]}} & 8'hb0)^
({8{data_in[174][2]}} & 8'h4b)^
({8{data_in[174][3]}} & 8'h96)^
({8{data_in[174][4]}} & 8'h7)^
({8{data_in[174][5]}} & 8'he)^
({8{data_in[174][6]}} & 8'h1c)^
({8{data_in[174][7]}} & 8'h38)^
({8{data_in[175][0]}} & 8'hbe)^
({8{data_in[175][1]}} & 8'h57)^
({8{data_in[175][2]}} & 8'hae)^
({8{data_in[175][3]}} & 8'h77)^
({8{data_in[175][4]}} & 8'hee)^
({8{data_in[175][5]}} & 8'hf7)^
({8{data_in[175][6]}} & 8'hc5)^
({8{data_in[175][7]}} & 8'ha1)^
({8{data_in[176][0]}} & 8'h94)^
({8{data_in[176][1]}} & 8'h3)^
({8{data_in[176][2]}} & 8'h6)^
({8{data_in[176][3]}} & 8'hc)^
({8{data_in[176][4]}} & 8'h18)^
({8{data_in[176][5]}} & 8'h30)^

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({8{data_in[177][0]}} & 8'h3b)^
({8{data_in[177][1]}} & 8'h76)^
({8{data_in[177][2]}} & 8'hec)^
({8{data_in[177][3]}} & 8'hf3)^
({8{data_in[177][4]}} & 8'hcd)^
({8{data_in[177][5]}} & 8'hb1)^
({8{data_in[177][6]}} & 8'h49)^
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({8{data_in[178][0]}} & 8'h39)^
({8{data_in[178][1]}} & 8'h72)^
({8{data_in[178][2]}} & 8'he4)^
({8{data_in[178][3]}} & 8'he3)^
({8{data_in[178][4]}} & 8'hed)^
({8{data_in[178][5]}} & 8'hf1)^
({8{data_in[178][6]}} & 8'hc9)^
({8{data_in[178][7]}} & 8'hb9)^
({8{data_in[179][0]}} & 8'h99)^
({8{data_in[179][1]}} & 8'h19)^
({8{data_in[179][2]}} & 8'h32)^
({8{data_in[179][3]}} & 8'h64)^
({8{data_in[179][4]}} & 8'hc8)^
({8{data_in[179][5]}} & 8'hbb)^
({8{data_in[179][6]}} & 8'h5d)^
({8{data_in[179][7]}} & 8'hba)^
({8{data_in[180][0]}} & 8'hb)^
({8{data_in[180][1]}} & 8'h16)^
({8{data_in[180][2]}} & 8'h2c)^
({8{data_in[180][3]}} & 8'h58)^
({8{data_in[180][4]}} & 8'hb0)^
({8{data_in[180][5]}} & 8'h4b)^
({8{data_in[180][6]}} & 8'h96)^
({8{data_in[180][7]}} & 8'h7)^
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({8{data_in[181][1]}} & 8'h2)^
({8{data_in[181][2]}} & 8'h4)^
({8{data_in[181][3]}} & 8'h8)^
({8{data_in[181][4]}} & 8'h10)^
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({8{data_in[182][1]}} & 8'h54)^
({8{data_in[182][2]}} & 8'ha8)^
({8{data_in[182][3]}} & 8'h7b)^
({8{data_in[182][4]}} & 8'hf6)^
({8{data_in[182][5]}} & 8'hc7)^
({8{data_in[182][6]}} & 8'ha5)^
({8{data_in[182][7]}} & 8'h61)^
({8{data_in[183][0]}} & 8'ha0)^
({8{data_in[183][1]}} & 8'h6b)^
({8{data_in[183][2]}} & 8'hd6)^
({8{data_in[183][3]}} & 8'h87)^
({8{data_in[183][4]}} & 8'h25)^
({8{data_in[183][5]}} & 8'h4a)^
({8{data_in[183][6]}} & 8'h94)^
({8{data_in[183][7]}} & 8'h3)^
({8{data_in[184][0]}} & 8'h37)^
({8{data_in[184][1]}} & 8'h6e)^
({8{data_in[184][2]}} & 8'hdc)^
({8{data_in[184][3]}} & 8'h93)^
({8{data_in[184][4]}} & 8'hd)^
({8{data_in[184][5]}} & 8'h1a)^
({8{data_in[184][6]}} & 8'h34)^

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({8{data_in[185][1]}} & 8'he9)^
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({8{data_in[185][3]}} & 8'hd9)^
({8{data_in[185][4]}} & 8'h99)^
({8{data_in[185][5]}} & 8'h19)^
({8{data_in[185][6]}} & 8'h32)^
({8{data_in[185][7]}} & 8'h64)^
({8{data_in[186][0]}} & 8'hd5)^
({8{data_in[186][1]}} & 8'h81)^
({8{data_in[186][2]}} & 8'h29)^
({8{data_in[186][3]}} & 8'h52)^
({8{data_in[186][4]}} & 8'ha4)^
({8{data_in[186][5]}} & 8'h63)^
({8{data_in[186][6]}} & 8'hc6)^
({8{data_in[186][7]}} & 8'ha7)^
({8{data_in[187][0]}} & 8'h56)^
({8{data_in[187][1]}} & 8'hac)^
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({8{data_in[187][4]}} & 8'he7)^
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({8{data_in[187][6]}} & 8'he1)^
({8{data_in[187][7]}} & 8'he9)^
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({8{data_in[188][1]}} & 8'h44)^
({8{data_in[188][2]}} & 8'h88)^
({8{data_in[188][3]}} & 8'h3b)^
({8{data_in[188][4]}} & 8'h76)^
({8{data_in[188][5]}} & 8'hec)^
({8{data_in[188][6]}} & 8'hf3)^
({8{data_in[188][7]}} & 8'hcd)^
({8{data_in[189][0]}} & 8'he1)^
({8{data_in[189][1]}} & 8'he9)^
({8{data_in[189][2]}} & 8'hf9)^
({8{data_in[189][3]}} & 8'hd9)^
({8{data_in[189][4]}} & 8'h99)^
({8{data_in[189][5]}} & 8'h19)^
({8{data_in[189][6]}} & 8'h32)^
({8{data_in[189][7]}} & 8'h64)^
({8{data_in[190][0]}} & 8'hcf)^
({8{data_in[190][1]}} & 8'hb5)^
({8{data_in[190][2]}} & 8'h41)^
({8{data_in[190][3]}} & 8'h82)^
({8{data_in[190][4]}} & 8'h2f)^
({8{data_in[190][5]}} & 8'h5e)^
({8{data_in[190][6]}} & 8'hbc)^
({8{data_in[190][7]}} & 8'h53)^
({8{data_in[191][0]}} & 8'h28)^
({8{data_in[191][1]}} & 8'h50)^
({8{data_in[191][2]}} & 8'ha0)^
({8{data_in[191][3]}} & 8'h6b)^
({8{data_in[191][4]}} & 8'hd6)^
({8{data_in[191][5]}} & 8'h87)^
({8{data_in[191][6]}} & 8'h25)^
({8{data_in[191][7]}} & 8'h4a)^
({8{data_in[192][0]}} & 8'hbe)^
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({8{data_in[192][2]}} & 8'hae)^
({8{data_in[192][3]}} & 8'h77)^
({8{data_in[192][4]}} & 8'hee)^
({8{data_in[192][5]}} & 8'hf7)^
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({8{data_in[192][7]}} & 8'ha1)^

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({8{data_in[193][0]}} & 8'h96)^
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({8{data_in[193][4]}} & 8'h38)^
({8{data_in[193][5]}} & 8'h70)^
({8{data_in[193][6]}} & 8'he0)^
({8{data_in[193][7]}} & 8'heb)^
({8{data_in[194][0]}} & 8'h9)^
({8{data_in[194][1]}} & 8'h12)^
({8{data_in[194][2]}} & 8'h24)^
({8{data_in[194][3]}} & 8'h48)^
({8{data_in[194][4]}} & 8'h90)^
({8{data_in[194][5]}} & 8'hb)^
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({8{data_in[194][7]}} & 8'h2c)^
({8{data_in[195][0]}} & 8'h10)^
({8{data_in[195][1]}} & 8'h20)^
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({8{data_in[195][3]}} & 8'h80)^
({8{data_in[195][4]}} & 8'h2b)^
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({8{data_in[195][6]}} & 8'hac)^
({8{data_in[195][7]}} & 8'h73)^
({8{data_in[196][0]}} & 8'hf0)^
({8{data_in[196][1]}} & 8'hcb)^
({8{data_in[196][2]}} & 8'hbd)^
({8{data_in[196][3]}} & 8'h51)^
({8{data_in[196][4]}} & 8'ha2)^
({8{data_in[196][5]}} & 8'h6f)^
({8{data_in[196][6]}} & 8'hde)^
({8{data_in[196][7]}} & 8'h97)^
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({8{data_in[198][0]}} & 8'h83)^
({8{data_in[198][1]}} & 8'h2d)^
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({8{data_in[198][7]}} & 8'h4e)^
({8{data_in[199][0]}} & 8'h96)^
({8{data_in[199][1]}} & 8'h7)^
({8{data_in[199][2]}} & 8'he)^
({8{data_in[199][3]}} & 8'h1c)^
({8{data_in[199][4]}} & 8'h38)^
({8{data_in[199][5]}} & 8'h70)^
({8{data_in[199][6]}} & 8'he0)^
({8{data_in[199][7]}} & 8'heb)^
({8{data_in[200][0]}} & 8'h9a)^
({8{data_in[200][1]}} & 8'h1f)^
({8{data_in[200][2]}} & 8'h3e)^
({8{data_in[200][3]}} & 8'h7c)^
({8{data_in[200][4]}} & 8'hf8)^
({8{data_in[200][5]}} & 8'hdb)^
({8{data_in[200][6]}} & 8'h9d)^
({8{data_in[200][7]}} & 8'h11)^
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({8{data_in[201][5]}} & 8'h25)^
({8{data_in[201][6]}} & 8'h4a)^
({8{data_in[201][7]}} & 8'h94)^
({8{data_in[202][0]}} & 8'h4f)^
({8{data_in[202][1]}} & 8'h9e)^
({8{data_in[202][2]}} & 8'h17)^
({8{data_in[202][3]}} & 8'h2e)^
({8{data_in[202][4]}} & 8'h5c)^
({8{data_in[202][5]}} & 8'hb8)^
({8{data_in[202][6]}} & 8'h5b)^
({8{data_in[202][7]}} & 8'hb6)^
({8{data_in[203][0]}} & 8'hb9)^
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({8{data_in[203][2]}} & 8'hb2)^
({8{data_in[203][3]}} & 8'h4f)^
({8{data_in[203][4]}} & 8'h9e)^
({8{data_in[203][5]}} & 8'h17)^
({8{data_in[203][6]}} & 8'h2e)^
({8{data_in[203][7]}} & 8'h5c)^
({8{data_in[204][0]}} & 8'h19)^
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({8{data_in[204][5]}} & 8'h5d)^
({8{data_in[204][6]}} & 8'hba)^
({8{data_in[204][7]}} & 8'h5f)^
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({8{data_in[205][1]}} & 8'hab)^
({8{data_in[205][2]}} & 8'h7d)^
({8{data_in[205][3]}} & 8'hfa)^
({8{data_in[205][4]}} & 8'hdf)^
({8{data_in[205][5]}} & 8'h95)^
({8{data_in[205][6]}} & 8'h1)^
({8{data_in[205][7]}} & 8'h2)^
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({8{data_in[206][3]}} & 8'h6a)^
({8{data_in[206][4]}} & 8'hd4)^
({8{data_in[206][5]}} & 8'h83)^
({8{data_in[206][6]}} & 8'h2d)^
({8{data_in[206][7]}} & 8'h5a)^
({8{data_in[207][0]}} & 8'h96)^
({8{data_in[207][1]}} & 8'h7)^
({8{data_in[207][2]}} & 8'he)^
({8{data_in[207][3]}} & 8'h1c)^
({8{data_in[207][4]}} & 8'h38)^
({8{data_in[207][5]}} & 8'h70)^
({8{data_in[207][6]}} & 8'he0)^
({8{data_in[207][7]}} & 8'heb)^
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({8{data_in[208][4]}} & 8'hbf)^
({8{data_in[208][5]}} & 8'h55)^
({8{data_in[208][6]}} & 8'haa)^
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({8{data_in[209][6]}} & 8'hc7)^
({8{data_in[209][7]}} & 8'ha5)^
({8{data_in[210][0]}} & 8'h70)^
({8{data_in[210][1]}} & 8'he0)^
({8{data_in[210][2]}} & 8'heb)^
({8{data_in[210][3]}} & 8'hfd)^
({8{data_in[210][4]}} & 8'hd1)^
({8{data_in[210][5]}} & 8'h89)^
({8{data_in[210][6]}} & 8'h39)^
({8{data_in[210][7]}} & 8'h72)^
({8{data_in[211][0]}} & 8'hed)^
({8{data_in[211][1]}} & 8'hf1)^
({8{data_in[211][2]}} & 8'hc9)^
({8{data_in[211][3]}} & 8'hb9)^
({8{data_in[211][4]}} & 8'h59)^
({8{data_in[211][5]}} & 8'hb2)^
({8{data_in[211][6]}} & 8'h4f)^
({8{data_in[211][7]}} & 8'h9e)^
({8{data_in[212][0]}} & 8'h66)^
({8{data_in[212][1]}} & 8'hcc)^
({8{data_in[212][2]}} & 8'hb3)^
({8{data_in[212][3]}} & 8'h4d)^
({8{data_in[212][4]}} & 8'h9a)^
({8{data_in[212][5]}} & 8'h1f)^
({8{data_in[212][6]}} & 8'h3e)^
({8{data_in[212][7]}} & 8'h7c)^
({8{data_in[213][0]}} & 8'hf3)^
({8{data_in[213][1]}} & 8'hcd)^
({8{data_in[213][2]}} & 8'hb1)^
({8{data_in[213][3]}} & 8'h49)^
({8{data_in[213][4]}} & 8'h92)^
({8{data_in[213][5]}} & 8'hf)^
({8{data_in[213][6]}} & 8'h1e)^
({8{data_in[213][7]}} & 8'h3c)^
({8{data_in[214][0]}} & 8'hb6)^
({8{data_in[214][1]}} & 8'h47)^
({8{data_in[214][2]}} & 8'h8e)^
({8{data_in[214][3]}} & 8'h37)^
({8{data_in[214][4]}} & 8'h6e)^
({8{data_in[214][5]}} & 8'hdc)^
({8{data_in[214][6]}} & 8'h93)^
({8{data_in[214][7]}} & 8'hd)^
({8{data_in[215][0]}} & 8'h13)^
({8{data_in[215][1]}} & 8'h26)^
({8{data_in[215][2]}} & 8'h4c)^
({8{data_in[215][3]}} & 8'h98)^
({8{data_in[215][4]}} & 8'h1b)^
({8{data_in[215][5]}} & 8'h36)^
({8{data_in[215][6]}} & 8'h6c)^
({8{data_in[215][7]}} & 8'hd8)^
({8{data_in[216][0]}} & 8'hc5)^
({8{data_in[216][1]}} & 8'ha1)^
({8{data_in[216][2]}} & 8'h69)^
({8{data_in[216][3]}} & 8'hd2)^
({8{data_in[216][4]}} & 8'h8f)^
({8{data_in[216][5]}} & 8'h35)^
({8{data_in[216][6]}} & 8'h6a)^
({8{data_in[216][7]}} & 8'hd4)^
({8{data_in[217][0]}} & 8'hd3)^
({8{data_in[217][1]}} & 8'h8d)^
({8{data_in[217][2]}} & 8'h31)^

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({8{data_in[217][3]}} & 8'h62)^
({8{data_in[217][4]}} & 8'hc4)^
({8{data_in[217][5]}} & 8'ha3)^
({8{data_in[217][6]}} & 8'h6d)^
({8{data_in[217][7]}} & 8'hda)^
({8{data_in[218][0]}} & 8'he2)^
({8{data_in[218][1]}} & 8'hef)^
({8{data_in[218][2]}} & 8'hf5)^
({8{data_in[218][3]}} & 8'hc1)^
({8{data_in[218][4]}} & 8'ha9)^
({8{data_in[218][5]}} & 8'h79)^
({8{data_in[218][6]}} & 8'hf2)^
({8{data_in[218][7]}} & 8'hcf)^
({8{data_in[219][0]}} & 8'h46)^
({8{data_in[219][1]}} & 8'h8c)^
({8{data_in[219][2]}} & 8'h33)^
({8{data_in[219][3]}} & 8'h66)^
({8{data_in[219][4]}} & 8'hcc)^
({8{data_in[219][5]}} & 8'hb3)^
({8{data_in[219][6]}} & 8'h4d)^
({8{data_in[219][7]}} & 8'h9a)^
({8{data_in[220][0]}} & 8'h3d)^
({8{data_in[220][1]}} & 8'h7a)^
({8{data_in[220][2]}} & 8'hf4)^
({8{data_in[220][3]}} & 8'hc3)^
({8{data_in[220][4]}} & 8'had)^
({8{data_in[220][5]}} & 8'h71)^
({8{data_in[220][6]}} & 8'he2)^
({8{data_in[220][7]}} & 8'hef)^
({8{data_in[221][0]}} & 8'hc5)^
({8{data_in[221][1]}} & 8'ha1)^
({8{data_in[221][2]}} & 8'h69)^
({8{data_in[221][3]}} & 8'hd2)^
({8{data_in[221][4]}} & 8'h8f)^
({8{data_in[221][5]}} & 8'h35)^
({8{data_in[221][6]}} & 8'h6a)^
({8{data_in[221][7]}} & 8'hd4)^
({8{data_in[222][0]}} & 8'h43)^
({8{data_in[222][1]}} & 8'h86)^
({8{data_in[222][2]}} & 8'h27)^
({8{data_in[222][3]}} & 8'h4e)^
({8{data_in[222][4]}} & 8'h9c)^
({8{data_in[222][5]}} & 8'h13)^
({8{data_in[222][6]}} & 8'h26)^
({8{data_in[222][7]}} & 8'h4c)^
({8{data_in[223][0]}} & 8'ha3)^
({8{data_in[223][1]}} & 8'h6d)^
({8{data_in[223][2]}} & 8'hda)^
({8{data_in[223][3]}} & 8'h9f)^
({8{data_in[223][4]}} & 8'h15)^
({8{data_in[223][5]}} & 8'h2a)^
({8{data_in[223][6]}} & 8'h54)^
({8{data_in[223][7]}} & 8'ha8)^
({8{data_in[224][0]}} & 8'h55)^
({8{data_in[224][1]}} & 8'haa)^
({8{data_in[224][2]}} & 8'h7f)^
({8{data_in[224][3]}} & 8'hfe)^
({8{data_in[224][4]}} & 8'hd7)^
({8{data_in[224][5]}} & 8'h85)^
({8{data_in[224][6]}} & 8'h21)^
({8{data_in[224][7]}} & 8'h42)^
({8{data_in[225][0]}} & 8'h5e)^
({8{data_in[225][1]}} & 8'hbc)^
({8{data_in[225][2]}} & 8'h53)^
({8{data_in[225][3]}} & 8'ha6)^

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({8{data_in[225][4]}} & 8'h67)^
({8{data_in[225][5]}} & 8'hce)^
({8{data_in[225][6]}} & 8'hb7)^
({8{data_in[225][7]}} & 8'h45)^
({8{data_in[226][0]}} & 8'h9b)^
({8{data_in[226][1]}} & 8'h1d)^
({8{data_in[226][2]}} & 8'h3a)^
({8{data_in[226][3]}} & 8'h74)^
({8{data_in[226][4]}} & 8'he8)^
({8{data_in[226][5]}} & 8'hfb)^
({8{data_in[226][6]}} & 8'hdd)^
({8{data_in[226][7]}} & 8'h91)^
({8{data_in[227][0]}} & 8'hba)^
({8{data_in[227][1]}} & 8'h5f)^
({8{data_in[227][2]}} & 8'hbe)^
({8{data_in[227][3]}} & 8'h57)^
({8{data_in[227][4]}} & 8'hae)^
({8{data_in[227][5]}} & 8'h77)^
({8{data_in[227][6]}} & 8'hee)^
({8{data_in[227][7]}} & 8'hf7)^
({8{data_in[228][0]}} & 8'haa)^
({8{data_in[228][1]}} & 8'h7f)^
({8{data_in[228][2]}} & 8'hfe)^
({8{data_in[228][3]}} & 8'hd7)^
({8{data_in[228][4]}} & 8'h85)^
({8{data_in[228][5]}} & 8'h21)^
({8{data_in[228][6]}} & 8'h42)^
({8{data_in[228][7]}} & 8'h84)^
({8{data_in[229][0]}} & 8'h7)^
({8{data_in[229][1]}} & 8'he)^
({8{data_in[229][2]}} & 8'h1c)^
({8{data_in[229][3]}} & 8'h38)^
({8{data_in[229][4]}} & 8'h70)^
({8{data_in[229][5]}} & 8'he0)^
({8{data_in[229][6]}} & 8'heb)^
({8{data_in[229][7]}} & 8'hfd)^
({8{data_in[230][0]}} & 8'h4f)^
({8{data_in[230][1]}} & 8'h9e)^
({8{data_in[230][2]}} & 8'h17)^
({8{data_in[230][3]}} & 8'h2e)^
({8{data_in[230][4]}} & 8'h5c)^
({8{data_in[230][5]}} & 8'hb8)^
({8{data_in[230][6]}} & 8'h5b)^
({8{data_in[230][7]}} & 8'hb6)^
({8{data_in[231][0]}} & 8'h6c)^
({8{data_in[231][1]}} & 8'hd8)^
({8{data_in[231][2]}} & 8'h9b)^
({8{data_in[231][3]}} & 8'h1d)^
({8{data_in[231][4]}} & 8'h3a)^
({8{data_in[231][5]}} & 8'h74)^
({8{data_in[231][6]}} & 8'he8)^
({8{data_in[231][7]}} & 8'hfb)^
({8{data_in[232][0]}} & 8'h89)^
({8{data_in[232][1]}} & 8'h39)^
({8{data_in[232][2]}} & 8'h72)^
({8{data_in[232][3]}} & 8'he4)^
({8{data_in[232][4]}} & 8'he3)^
({8{data_in[232][5]}} & 8'hed)^
({8{data_in[232][6]}} & 8'hf1)^
({8{data_in[232][7]}} & 8'hc9)^
({8{data_in[233][0]}} & 8'h40)^
({8{data_in[233][1]}} & 8'h80)^
({8{data_in[233][2]}} & 8'h2b)^
({8{data_in[233][3]}} & 8'h56)^
({8{data_in[233][4]}} & 8'hac)^

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({8{data_in[233][5]}} & 8'h73)^
({8{data_in[233][6]}} & 8'he6)^
({8{data_in[233][7]}} & 8'he7)^
({8{data_in[234][0]}} & 8'h51)^
({8{data_in[234][1]}} & 8'ha2)^
({8{data_in[234][2]}} & 8'h6f)^
({8{data_in[234][3]}} & 8'hde)^
({8{data_in[234][4]}} & 8'h97)^
({8{data_in[234][5]}} & 8'h5)^
({8{data_in[234][6]}} & 8'ha)^
({8{data_in[234][7]}} & 8'h14)^
({8{data_in[235][0]}} & 8'hb4)^
({8{data_in[235][1]}} & 8'h43)^
({8{data_in[235][2]}} & 8'h86)^
({8{data_in[235][3]}} & 8'h27)^
({8{data_in[235][4]}} & 8'h4e)^
({8{data_in[235][5]}} & 8'h9c)^
({8{data_in[235][6]}} & 8'h13)^
({8{data_in[235][7]}} & 8'h26)^
({8{data_in[236][0]}} & 8'hef)^
({8{data_in[236][1]}} & 8'hf5)^
({8{data_in[236][2]}} & 8'hc1)^
({8{data_in[236][3]}} & 8'ha9)^
({8{data_in[236][4]}} & 8'h79)^
({8{data_in[236][5]}} & 8'hf2)^
({8{data_in[236][6]}} & 8'hcf)^
({8{data_in[236][7]}} & 8'hb5)^
({8{data_in[237][0]}} & 8'h30)^
({8{data_in[237][1]}} & 8'h60)^
({8{data_in[237][2]}} & 8'hc0)^
({8{data_in[237][3]}} & 8'hab)^
({8{data_in[237][4]}} & 8'h7d)^
({8{data_in[237][5]}} & 8'hfa)^
({8{data_in[237][6]}} & 8'hdf)^
({8{data_in[237][7]}} & 8'h95)^
({8{data_in[238][0]}} & 8'hc2)^
({8{data_in[238][1]}} & 8'haf)^
({8{data_in[238][2]}} & 8'h75)^
({8{data_in[238][3]}} & 8'hea)^
({8{data_in[238][4]}} & 8'hff)^
({8{data_in[238][5]}} & 8'hd5)^
({8{data_in[238][6]}} & 8'h81)^
({8{data_in[238][7]}} & 8'h29)^
({8{data_in[239][0]}} & 8'hbe)^
({8{data_in[239][1]}} & 8'h57)^
({8{data_in[239][2]}} & 8'hae)^
({8{data_in[239][3]}} & 8'h77)^
({8{data_in[239][4]}} & 8'hee)^
({8{data_in[239][5]}} & 8'hf7)^
({8{data_in[239][6]}} & 8'hc5)^
({8{data_in[239][7]}} & 8'ha1)^
({8{data_in[240][0]}} & 8'h51)^
({8{data_in[240][1]}} & 8'ha2)^
({8{data_in[240][2]}} & 8'h6f)^
({8{data_in[240][3]}} & 8'hde)^
({8{data_in[240][4]}} & 8'h97)^
({8{data_in[240][5]}} & 8'h5)^
({8{data_in[240][6]}} & 8'ha)^
({8{data_in[240][7]}} & 8'h14)^
({8{data_in[241][0]}} & 8'hd5)^
({8{data_in[241][1]}} & 8'h81)^
({8{data_in[241][2]}} & 8'h29)^
({8{data_in[241][3]}} & 8'h52)^
({8{data_in[241][4]}} & 8'ha4)^
({8{data_in[241][5]}} & 8'h63)^

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        ({8{data_in[241][6]}} & 8'hc6)^
        ({8{data_in[241][7]}} & 8'ha7);
data_out[248] = ({8{data_in[0][0]}} & 8'hb9)^
        ({8{data_in[0][1]}} & 8'h59)^
        ({8{data_in[0][2]}} & 8'hb2)^
        ({8{data_in[0][3]}} & 8'h4f)^
        ({8{data_in[0][4]}} & 8'h9e)^
        ({8{data_in[0][5]}} & 8'h17)^
        ({8{data_in[0][6]}} & 8'h2e)^
        ({8{data_in[0][7]}} & 8'h5c)^
        ({8{data_in[1][0]}} & 8'h43)^
        ({8{data_in[1][1]}} & 8'h86)^
        ({8{data_in[1][2]}} & 8'h27)^
        ({8{data_in[1][3]}} & 8'h4e)^
        ({8{data_in[1][4]}} & 8'h9c)^
        ({8{data_in[1][5]}} & 8'h13)^
        ({8{data_in[1][6]}} & 8'h26)^
        ({8{data_in[1][7]}} & 8'h4c)^
        ({8{data_in[2][0]}} & 8'he2)^
        ({8{data_in[2][1]}} & 8'h ef)^
        ({8{data_in[2][2]}} & 8'h f5)^
        ({8{data_in[2][3]}} & 8'h c1)^
        ({8{data_in[2][4]}} & 8'ha9)^
        ({8{data_in[2][5]}} & 8'h79)^
        ({8{data_in[2][6]}} & 8'h f2)^
        ({8{data_in[2][7]}} & 8'h cf)^
        ({8{data_in[3][0]}} & 8'hbe)^
        ({8{data_in[3][1]}} & 8'h57)^
        ({8{data_in[3][2]}} & 8'hae)^
        ({8{data_in[3][3]}} & 8'h77)^
        ({8{data_in[3][4]}} & 8'hee)^
        ({8{data_in[3][5]}} & 8'h f7)^
        ({8{data_in[3][6]}} & 8'h c5)^
        ({8{data_in[3][7]}} & 8'ha1)^
        ({8{data_in[4][0]}} & 8'hb1)^
        ({8{data_in[4][1]}} & 8'h49)^
        ({8{data_in[4][2]}} & 8'h92)^
        ({8{data_in[4][3]}} & 8'h f)^
        ({8{data_in[4][4]}} & 8'h1e)^
        ({8{data_in[4][5]}} & 8'h3c)^
        ({8{data_in[4][6]}} & 8'h78)^
        ({8{data_in[4][7]}} & 8'h f0)^
        ({8{data_in[5][0]}} & 8'h4)^
        ({8{data_in[5][1]}} & 8'h8)^
        ({8{data_in[5][2]}} & 8'h10)^
        ({8{data_in[5][3]}} & 8'h20)^
        ({8{data_in[5][4]}} & 8'h40)^
        ({8{data_in[5][5]}} & 8'h80)^
        ({8{data_in[5][6]}} & 8'h2b)^
        ({8{data_in[5][7]}} & 8'h56)^
        ({8{data_in[6][0]}} & 8'hb8)^
        ({8{data_in[6][1]}} & 8'h5b)^
        ({8{data_in[6][2]}} & 8'hb6)^
        ({8{data_in[6][3]}} & 8'h47)^
        ({8{data_in[6][4]}} & 8'h8e)^
        ({8{data_in[6][5]}} & 8'h37)^
        ({8{data_in[6][6]}} & 8'h6e)^
        ({8{data_in[6][7]}} & 8'hdc)^
        ({8{data_in[7][0]}} & 8'h96)^
        ({8{data_in[7][1]}} & 8'h7)^
        ({8{data_in[7][2]}} & 8'he)^
        ({8{data_in[7][3]}} & 8'h1c)^
        ({8{data_in[7][4]}} & 8'h38)^
        ({8{data_in[7][5]}} & 8'h70)^
        ({8{data_in[7][6]}} & 8'he0)^

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({8{data_in[7][7]}} & 8'heb)^
({8{data_in[8][0]}} & 8'h3a)^
({8{data_in[8][1]}} & 8'h74)^
({8{data_in[8][2]}} & 8'he8)^
({8{data_in[8][3]}} & 8'hfb)^
({8{data_in[8][4]}} & 8'hdd)^
({8{data_in[8][5]}} & 8'h91)^
({8{data_in[8][6]}} & 8'h9)^
({8{data_in[8][7]}} & 8'h12)^
({8{data_in[9][0]}} & 8'h46)^
({8{data_in[9][1]}} & 8'h8c)^
({8{data_in[9][2]}} & 8'h33)^
({8{data_in[9][3]}} & 8'h66)^
({8{data_in[9][4]}} & 8'hcc)^
({8{data_in[9][5]}} & 8'hb3)^
({8{data_in[9][6]}} & 8'h4d)^
({8{data_in[9][7]}} & 8'h9a)^
({8{data_in[10][0]}} & 8'h3b)^
({8{data_in[10][1]}} & 8'h76)^
({8{data_in[10][2]}} & 8'hec)^
({8{data_in[10][3]}} & 8'hf3)^
({8{data_in[10][4]}} & 8'hcd)^
({8{data_in[10][5]}} & 8'hb1)^
({8{data_in[10][6]}} & 8'h49)^
({8{data_in[10][7]}} & 8'h92)^
({8{data_in[11][0]}} & 8'h7d)^
({8{data_in[11][1]}} & 8'hfa)^
({8{data_in[11][2]}} & 8'hdf)^
({8{data_in[11][3]}} & 8'h95)^
({8{data_in[11][4]}} & 8'h1)^
({8{data_in[11][5]}} & 8'h2)^
({8{data_in[11][6]}} & 8'h4)^
({8{data_in[11][7]}} & 8'h8)^
({8{data_in[12][0]}} & 8'h3b)^
({8{data_in[12][1]}} & 8'h76)^
({8{data_in[12][2]}} & 8'hec)^
({8{data_in[12][3]}} & 8'hf3)^
({8{data_in[12][4]}} & 8'hcd)^
({8{data_in[12][5]}} & 8'hb1)^
({8{data_in[12][6]}} & 8'h49)^
({8{data_in[12][7]}} & 8'h92)^
({8{data_in[13][0]}} & 8'he3)^
({8{data_in[13][1]}} & 8'hed)^
({8{data_in[13][2]}} & 8'hf1)^
({8{data_in[13][3]}} & 8'hc9)^
({8{data_in[13][4]}} & 8'hb9)^
({8{data_in[13][5]}} & 8'h59)^
({8{data_in[13][6]}} & 8'hb2)^
({8{data_in[13][7]}} & 8'h4f)^
({8{data_in[14][0]}} & 8'h18)^
({8{data_in[14][1]}} & 8'h30)^
({8{data_in[14][2]}} & 8'h60)^
({8{data_in[14][3]}} & 8'hc0)^
({8{data_in[14][4]}} & 8'hab)^
({8{data_in[14][5]}} & 8'h7d)^
({8{data_in[14][6]}} & 8'hfa)^
({8{data_in[14][7]}} & 8'hdf)^
({8{data_in[15][0]}} & 8'h3a)^
({8{data_in[15][1]}} & 8'h74)^
({8{data_in[15][2]}} & 8'he8)^
({8{data_in[15][3]}} & 8'hfb)^
({8{data_in[15][4]}} & 8'hdd)^
({8{data_in[15][5]}} & 8'h91)^
({8{data_in[15][6]}} & 8'h9)^
({8{data_in[15][7]}} & 8'h12)^

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({8{data_in[16][0]}} & 8'h4b)^
({8{data_in[16][1]}} & 8'h96)^
({8{data_in[16][2]}} & 8'h7)^
({8{data_in[16][3]}} & 8'he)^
({8{data_in[16][4]}} & 8'h1c)^
({8{data_in[16][5]}} & 8'h38)^
({8{data_in[16][6]}} & 8'h70)^
({8{data_in[16][7]}} & 8'he0)^
({8{data_in[17][0]}} & 8'h97)^
({8{data_in[17][1]}} & 8'h5)^
({8{data_in[17][2]}} & 8'ha)^
({8{data_in[17][3]}} & 8'h14)^
({8{data_in[17][4]}} & 8'h28)^
({8{data_in[17][5]}} & 8'h50)^
({8{data_in[17][6]}} & 8'ha0)^
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({8{data_in[18][4]}} & 8'haa)^
({8{data_in[18][5]}} & 8'h7f)^
({8{data_in[18][6]}} & 8'hfe)^
({8{data_in[18][7]}} & 8'hd7)^
({8{data_in[19][0]}} & 8'h6f)^
({8{data_in[19][1]}} & 8'hde)^
({8{data_in[19][2]}} & 8'h97)^
({8{data_in[19][3]}} & 8'h5)^
({8{data_in[19][4]}} & 8'ha)^
({8{data_in[19][5]}} & 8'h14)^
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({8{data_in[19][7]}} & 8'h50)^
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({8{data_in[20][1]}} & 8'h93)^
({8{data_in[20][2]}} & 8'hd)^
({8{data_in[20][3]}} & 8'h1a)^
({8{data_in[20][4]}} & 8'h34)^
({8{data_in[20][5]}} & 8'h68)^
({8{data_in[20][6]}} & 8'hd0)^
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({8{data_in[21][4]}} & 8'h4d)^
({8{data_in[21][5]}} & 8'h9a)^
({8{data_in[21][6]}} & 8'h1f)^
({8{data_in[21][7]}} & 8'h3e)^
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({8{data_in[22][1]}} & 8'h83)^
({8{data_in[22][2]}} & 8'h2d)^
({8{data_in[22][3]}} & 8'h5a)^
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({8{data_in[22][6]}} & 8'h86)^
({8{data_in[22][7]}} & 8'h27)^
({8{data_in[23][0]}} & 8'h53)^
({8{data_in[23][1]}} & 8'ha6)^
({8{data_in[23][2]}} & 8'h67)^
({8{data_in[23][3]}} & 8'hce)^
({8{data_in[23][4]}} & 8'hb7)^
({8{data_in[23][5]}} & 8'h45)^
({8{data_in[23][6]}} & 8'h8a)^
({8{data_in[23][7]}} & 8'h3f)^
({8{data_in[24][0]}} & 8'h2f)^

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({8{data_in[24][5]}} & 8'h67)^
({8{data_in[24][6]}} & 8'hce)^
({8{data_in[24][7]}} & 8'hb7)^
({8{data_in[25][0]}} & 8'ha1)^
({8{data_in[25][1]}} & 8'h69)^
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({8{data_in[25][3]}} & 8'h8f)^
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({8{data_in[25][5]}} & 8'h6a)^
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({8{data_in[25][7]}} & 8'h83)^
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({8{data_in[26][5]}} & 8'h5)^
({8{data_in[26][6]}} & 8'ha)^
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({8{data_in[28][1]}} & 8'h3e)^
({8{data_in[28][2]}} & 8'h7c)^
({8{data_in[28][3]}} & 8'hf8)^
({8{data_in[28][4]}} & 8'hdb)^
({8{data_in[28][5]}} & 8'h9d)^
({8{data_in[28][6]}} & 8'h11)^
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({8{data_in[30][1]}} & 8'h8e)^
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({8{data_in[30][3]}} & 8'h6e)^
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({8{data_in[30][5]}} & 8'h93)^
({8{data_in[30][6]}} & 8'hd)^
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({8{data_in[31][7]}} & 8'hda)^
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({8{data_in[32][1]}} & 8'hc9)^

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({8{data_in[33][6]}} & 8'h3f)^
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({8{data_in[34][5]}} & 8'hf8)^
({8{data_in[34][6]}} & 8'hdb)^
({8{data_in[34][7]}} & 8'h9d)^
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({8{data_in[38][1]}} & 8'hf3)^
({8{data_in[38][2]}} & 8'hcd)^
({8{data_in[38][3]}} & 8'hb1)^
({8{data_in[38][4]}} & 8'h49)^
({8{data_in[38][5]}} & 8'h92)^
({8{data_in[38][6]}} & 8'hf)^
({8{data_in[38][7]}} & 8'h1e)^
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({8{data_in[39][1]}} & 8'hee)^
({8{data_in[39][2]}} & 8'hf7)^
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({8{data_in[39][5]}} & 8'h69)^
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({8{data_in[39][7]}} & 8'h8f)^
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({8{data_in[40][2]}} & 8'hdc)^

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({8{data_in[41][2]}} & 8'h69)^
({8{data_in[41][3]}} & 8'hd2)^
({8{data_in[41][4]}} & 8'h8f)^
({8{data_in[41][5]}} & 8'h35)^
({8{data_in[41][6]}} & 8'h6a)^
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({8{data_in[42][7]}} & 8'h89)^
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({8{data_in[43][1]}} & 8'h69)^
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({8{data_in[43][3]}} & 8'h8f)^
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({8{data_in[43][7]}} & 8'h83)^
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({8{data_in[46][1]}} & 8'h8e)^
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({8{data_in[46][6]}} & 8'hd)^
({8{data_in[46][7]}} & 8'h1a)^
({8{data_in[47][0]}} & 8'hdb)^
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({8{data_in[50][2]}} & 8'hc8)^
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({8{data_in[50][4]}} & 8'h5d)^
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({8{data_in[51][6]}} & 8'h1d)^
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({8{data_in[52][5]}} & 8'hba)^
({8{data_in[52][6]}} & 8'h5f)^
({8{data_in[52][7]}} & 8'hbe)^
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({8{data_in[54][2]}} & 8'h5a)^
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({8{data_in[54][4]}} & 8'h43)^
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({8{data_in[54][6]}} & 8'h27)^
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({8{data_in[55][2]}} & 8'h2c)^
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({8{data_in[62][7]}} & 8'h3f)^
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({8{data_in[65][7]}} & 8'h5c)^
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({8{data_in[66][3]}} & 8'h5d)^
({8{data_in[66][4]}} & 8'hba)^
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({8{data_in[68][2]}} & 8'h1e)^
({8{data_in[68][3]}} & 8'h3c)^
({8{data_in[68][4]}} & 8'h78)^
({8{data_in[68][5]}} & 8'hf0)^
({8{data_in[68][6]}} & 8'hcb)^
({8{data_in[68][7]}} & 8'hbd)^
({8{data_in[69][0]}} & 8'h54)^
({8{data_in[69][1]}} & 8'ha8)^
({8{data_in[69][2]}} & 8'h7b)^
({8{data_in[69][3]}} & 8'hf6)^
({8{data_in[69][4]}} & 8'hc7)^
({8{data_in[69][5]}} & 8'ha5)^
({8{data_in[69][6]}} & 8'h61)^
({8{data_in[69][7]}} & 8'hc2)^
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({8{data_in[70][2]}} & 8'h62)^
({8{data_in[70][3]}} & 8'hc4)^
({8{data_in[70][4]}} & 8'ha3)^
({8{data_in[70][5]}} & 8'h6d)^
({8{data_in[70][6]}} & 8'hda)^
({8{data_in[70][7]}} & 8'h9f)^
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({8{data_in[71][1]}} & 8'h7b)^
({8{data_in[71][2]}} & 8'hf6)^
({8{data_in[71][3]}} & 8'hc7)^
({8{data_in[71][4]}} & 8'ha5)^
({8{data_in[71][5]}} & 8'h61)^
({8{data_in[71][6]}} & 8'hc2)^
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({8{data_in[72][1]}} & 8'h23)^
({8{data_in[72][2]}} & 8'h46)^
({8{data_in[72][3]}} & 8'h8c)^
({8{data_in[72][4]}} & 8'h33)^
({8{data_in[72][5]}} & 8'h66)^
({8{data_in[72][6]}} & 8'hcc)^

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({8{data_in[73][2]}} & 8'h7c)^
({8{data_in[73][3]}} & 8'hf8)^
({8{data_in[73][4]}} & 8'hdb)^
({8{data_in[73][5]}} & 8'h9d)^
({8{data_in[73][6]}} & 8'h11)^
({8{data_in[73][7]}} & 8'h22)^
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({8{data_in[74][4]}} & 8'he3)^
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({8{data_in[75][1]}} & 8'h5c)^
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({8{data_in[75][4]}} & 8'hb6)^
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({8{data_in[78][6]}} & 8'hbb)^
({8{data_in[78][7]}} & 8'h5d)^
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({8{data_in[79][1]}} & 8'h8b)^
({8{data_in[79][2]}} & 8'h3d)^
({8{data_in[79][3]}} & 8'h7a)^
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({8{data_in[80][2]}} & 8'h2d)^
({8{data_in[80][3]}} & 8'h5a)^
({8{data_in[80][4]}} & 8'hb4)^
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({8{data_in[80][7]}} & 8'h27)^

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({8{data_in[83][4]}} & 8'hdf)^
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({8{data_in[83][6]}} & 8'h1)^
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({8{data_in[84][0]}} & 8'hc4)^
({8{data_in[84][1]}} & 8'ha3)^
({8{data_in[84][2]}} & 8'h6d)^
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({8{data_in[85][4]}} & 8'h6e)^
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({8{data_in[87][0]}} & 8'h55)^
({8{data_in[87][1]}} & 8'haa)^
({8{data_in[87][2]}} & 8'h7f)^
({8{data_in[87][3]}} & 8'hfe)^
({8{data_in[87][4]}} & 8'hd7)^
({8{data_in[87][5]}} & 8'h85)^
({8{data_in[87][6]}} & 8'h21)^
({8{data_in[87][7]}} & 8'h42)^
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({8{data_in[88][1]}} & 8'h55)^
({8{data_in[88][2]}} & 8'haa)^
({8{data_in[88][3]}} & 8'h7f)^
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({8{data_in[88][5]}} & 8'hd7)^
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({8{data_in[90][1]}} & 8'h5c)^
({8{data_in[90][2]}} & 8'hb8)^
({8{data_in[90][3]}} & 8'h5b)^
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({8{data_in[90][6]}} & 8'h8e)^
({8{data_in[90][7]}} & 8'h37)^
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({8{data_in[91][1]}} & 8'hb3)^
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({8{data_in[91][4]}} & 8'h1f)^
({8{data_in[91][5]}} & 8'h3e)^
({8{data_in[91][6]}} & 8'h7c)^
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({8{data_in[93][3]}} & 8'h7)^
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({8{data_in[96][1]}} & 8'h9)^
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({8{data_in[98][3]}} & 8'hab)^
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({8{data_in[99][6]}} & 8'h19)^
({8{data_in[99][7]}} & 8'h32)^
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({8{data_in[100][1]}} & 8'h1b)^
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({8{data_in[100][3]}} & 8'h6c)^
({8{data_in[100][4]}} & 8'hd8)^
({8{data_in[100][5]}} & 8'h9b)^
({8{data_in[100][6]}} & 8'h1d)^
({8{data_in[100][7]}} & 8'h3a)^
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({8{data_in[104][7]}} & 8'hdf)^
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({8{data_in[105][7]}} & 8'hdc)^
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({8{data_in[106][1]}} & 8'h33)^
({8{data_in[106][2]}} & 8'h66)^
({8{data_in[106][3]}} & 8'hcc)^
({8{data_in[106][4]}} & 8'hb3)^
({8{data_in[106][5]}} & 8'h4d)^
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({8{data_in[107][7]}} & 8'h3a)^
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({8{data_in[111][3]}} & 8'h60)^
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({8{data_in[111][5]}} & 8'hab)^
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({8{data_in[114][4]}} & 8'hab)^
({8{data_in[114][5]}} & 8'h7d)^
({8{data_in[114][6]}} & 8'hfa)^
({8{data_in[114][7]}} & 8'hdf)^
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({8{data_in[118][1]}} & 8'h8d)^
({8{data_in[118][2]}} & 8'h31)^
({8{data_in[118][3]}} & 8'h62)^
({8{data_in[118][4]}} & 8'hc4)^
({8{data_in[118][5]}} & 8'ha3)^
({8{data_in[118][6]}} & 8'h6d)^
({8{data_in[118][7]}} & 8'hda)^
({8{data_in[119][0]}} & 8'h22)^
({8{data_in[119][1]}} & 8'h44)^
({8{data_in[119][2]}} & 8'h88)^
({8{data_in[119][3]}} & 8'h3b)^
({8{data_in[119][4]}} & 8'h76)^
({8{data_in[119][5]}} & 8'hec)^
({8{data_in[119][6]}} & 8'hf3)^
({8{data_in[119][7]}} & 8'hcd)^
({8{data_in[120][0]}} & 8'hbd)^
({8{data_in[120][1]}} & 8'h51)^
({8{data_in[120][2]}} & 8'ha2)^
({8{data_in[120][3]}} & 8'h6f)^
({8{data_in[120][4]}} & 8'hde)^
({8{data_in[120][5]}} & 8'h97)^
({8{data_in[120][6]}} & 8'h5)^
({8{data_in[120][7]}} & 8'ha)^
({8{data_in[121][0]}} & 8'hf7)^
({8{data_in[121][1]}} & 8'hc5)^
({8{data_in[121][2]}} & 8'ha1)^
({8{data_in[121][3]}} & 8'h69)^
({8{data_in[121][4]}} & 8'hd2)^

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({8{data_in[121][5]}} & 8'h8f)^
({8{data_in[121][6]}} & 8'h35)^
({8{data_in[121][7]}} & 8'h6a)^
({8{data_in[122][0]}} & 8'h25)^
({8{data_in[122][1]}} & 8'h4a)^
({8{data_in[122][2]}} & 8'h94)^
({8{data_in[122][3]}} & 8'h3)^
({8{data_in[122][4]}} & 8'h6)^
({8{data_in[122][5]}} & 8'hc)^
({8{data_in[122][6]}} & 8'h18)^
({8{data_in[122][7]}} & 8'h30)^
({8{data_in[123][0]}} & 8'h4e)^
({8{data_in[123][1]}} & 8'h9c)^
({8{data_in[123][2]}} & 8'h13)^
({8{data_in[123][3]}} & 8'h26)^
({8{data_in[123][4]}} & 8'h4c)^
({8{data_in[123][5]}} & 8'h98)^
({8{data_in[123][6]}} & 8'h1b)^
({8{data_in[123][7]}} & 8'h36)^
({8{data_in[124][0]}} & 8'h2e)^
({8{data_in[124][1]}} & 8'h5c)^
({8{data_in[124][2]}} & 8'hb8)^
({8{data_in[124][3]}} & 8'h5b)^
({8{data_in[124][4]}} & 8'hb6)^
({8{data_in[124][5]}} & 8'h47)^
({8{data_in[124][6]}} & 8'h8e)^
({8{data_in[124][7]}} & 8'h37)^
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({8{data_in[125][1]}} & 8'h81)^
({8{data_in[125][2]}} & 8'h29)^
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({8{data_in[125][4]}} & 8'ha4)^
({8{data_in[125][5]}} & 8'h63)^
({8{data_in[125][6]}} & 8'hc6)^
({8{data_in[125][7]}} & 8'ha7)^
({8{data_in[126][0]}} & 8'hbf)^
({8{data_in[126][1]}} & 8'h55)^
({8{data_in[126][2]}} & 8'haa)^
({8{data_in[126][3]}} & 8'h7f)^
({8{data_in[126][4]}} & 8'hfe)^
({8{data_in[126][5]}} & 8'hd7)^
({8{data_in[126][6]}} & 8'h85)^
({8{data_in[126][7]}} & 8'h21)^
({8{data_in[127][0]}} & 8'h40)^
({8{data_in[127][1]}} & 8'h80)^
({8{data_in[127][2]}} & 8'h2b)^
({8{data_in[127][3]}} & 8'h56)^
({8{data_in[127][4]}} & 8'hac)^
({8{data_in[127][5]}} & 8'h73)^
({8{data_in[127][6]}} & 8'he6)^
({8{data_in[127][7]}} & 8'he7)^
({8{data_in[128][0]}} & 8'h8a)^
({8{data_in[128][1]}} & 8'h3f)^
({8{data_in[128][2]}} & 8'h7e)^
({8{data_in[128][3]}} & 8'hfc)^
({8{data_in[128][4]}} & 8'hd3)^
({8{data_in[128][5]}} & 8'h8d)^
({8{data_in[128][6]}} & 8'h31)^
({8{data_in[128][7]}} & 8'h62)^
({8{data_in[129][0]}} & 8'h18)^
({8{data_in[129][1]}} & 8'h30)^
({8{data_in[129][2]}} & 8'h60)^
({8{data_in[129][3]}} & 8'hc0)^
({8{data_in[129][4]}} & 8'hab)^
({8{data_in[129][5]}} & 8'h7d)^

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({8{data_in[129][6]}} & 8'hfa)^
({8{data_in[129][7]}} & 8'hdf)^
({8{data_in[130][0]}} & 8'hdc)^
({8{data_in[130][1]}} & 8'h93)^
({8{data_in[130][2]}} & 8'hd)^
({8{data_in[130][3]}} & 8'h1a)^
({8{data_in[130][4]}} & 8'h34)^
({8{data_in[130][5]}} & 8'h68)^
({8{data_in[130][6]}} & 8'hd0)^
({8{data_in[130][7]}} & 8'h8b)^
({8{data_in[131][0]}} & 8'ha0)^
({8{data_in[131][1]}} & 8'h6b)^
({8{data_in[131][2]}} & 8'hd6)^
({8{data_in[131][3]}} & 8'h87)^
({8{data_in[131][4]}} & 8'h25)^
({8{data_in[131][5]}} & 8'h4a)^
({8{data_in[131][6]}} & 8'h94)^
({8{data_in[131][7]}} & 8'h3)^
({8{data_in[132][0]}} & 8'hb1)^
({8{data_in[132][1]}} & 8'h49)^
({8{data_in[132][2]}} & 8'h92)^
({8{data_in[132][3]}} & 8'hf)^
({8{data_in[132][4]}} & 8'h1e)^
({8{data_in[132][5]}} & 8'h3c)^
({8{data_in[132][6]}} & 8'h78)^
({8{data_in[132][7]}} & 8'hf0)^
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({8{data_in[133][1]}} & 8'h64)^
({8{data_in[133][2]}} & 8'hc8)^
({8{data_in[133][3]}} & 8'hbb)^
({8{data_in[133][4]}} & 8'h5d)^
({8{data_in[133][5]}} & 8'hba)^
({8{data_in[133][6]}} & 8'h5f)^
({8{data_in[133][7]}} & 8'hbe)^
({8{data_in[134][0]}} & 8'h8c)^
({8{data_in[134][1]}} & 8'h33)^
({8{data_in[134][2]}} & 8'h66)^
({8{data_in[134][3]}} & 8'hcc)^
({8{data_in[134][4]}} & 8'hb3)^
({8{data_in[134][5]}} & 8'h4d)^
({8{data_in[134][6]}} & 8'h9a)^
({8{data_in[134][7]}} & 8'h1f)^
({8{data_in[135][0]}} & 8'h6e)^
({8{data_in[135][1]}} & 8'hdc)^
({8{data_in[135][2]}} & 8'h93)^
({8{data_in[135][3]}} & 8'hd)^
({8{data_in[135][4]}} & 8'h1a)^
({8{data_in[135][5]}} & 8'h34)^
({8{data_in[135][6]}} & 8'h68)^
({8{data_in[135][7]}} & 8'hd0)^
({8{data_in[136][0]}} & 8'h83)^
({8{data_in[136][1]}} & 8'h2d)^
({8{data_in[136][2]}} & 8'h5a)^
({8{data_in[136][3]}} & 8'hb4)^
({8{data_in[136][4]}} & 8'h43)^
({8{data_in[136][5]}} & 8'h86)^
({8{data_in[136][6]}} & 8'h27)^
({8{data_in[136][7]}} & 8'h4e)^
({8{data_in[137][0]}} & 8'h8a)^
({8{data_in[137][1]}} & 8'h3f)^
({8{data_in[137][2]}} & 8'h7e)^
({8{data_in[137][3]}} & 8'hfc)^
({8{data_in[137][4]}} & 8'hd3)^
({8{data_in[137][5]}} & 8'h8d)^
({8{data_in[137][6]}} & 8'h31)^

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({8{data_in[137][7]}} & 8'h62)^
({8{data_in[138][0]}} & 8'hf4)^
({8{data_in[138][1]}} & 8'hc3)^
({8{data_in[138][2]}} & 8'had)^
({8{data_in[138][3]}} & 8'h71)^
({8{data_in[138][4]}} & 8'he2)^
({8{data_in[138][5]}} & 8'hef)^
({8{data_in[138][6]}} & 8'hf5)^
({8{data_in[138][7]}} & 8'hc1)^
({8{data_in[139][0]}} & 8'hbe)^
({8{data_in[139][1]}} & 8'h57)^
({8{data_in[139][2]}} & 8'hae)^
({8{data_in[139][3]}} & 8'h77)^
({8{data_in[139][4]}} & 8'hee)^
({8{data_in[139][5]}} & 8'hf7)^
({8{data_in[139][6]}} & 8'hc5)^
({8{data_in[139][7]}} & 8'ha1)^
({8{data_in[140][0]}} & 8'h40)^
({8{data_in[140][1]}} & 8'h80)^
({8{data_in[140][2]}} & 8'h2b)^
({8{data_in[140][3]}} & 8'h56)^
({8{data_in[140][4]}} & 8'hac)^
({8{data_in[140][5]}} & 8'h73)^
({8{data_in[140][6]}} & 8'he6)^
({8{data_in[140][7]}} & 8'he7)^
({8{data_in[141][0]}} & 8'h61)^
({8{data_in[141][1]}} & 8'hc2)^
({8{data_in[141][2]}} & 8'haf)^
({8{data_in[141][3]}} & 8'h75)^
({8{data_in[141][4]}} & 8'hea)^
({8{data_in[141][5]}} & 8'hff)^
({8{data_in[141][6]}} & 8'hd5)^
({8{data_in[141][7]}} & 8'h81)^
({8{data_in[142][0]}} & 8'he3)^
({8{data_in[142][1]}} & 8'hed)^
({8{data_in[142][2]}} & 8'hf1)^
({8{data_in[142][3]}} & 8'hc9)^
({8{data_in[142][4]}} & 8'hb9)^
({8{data_in[142][5]}} & 8'h59)^
({8{data_in[142][6]}} & 8'hb2)^
({8{data_in[142][7]}} & 8'h4f)^
({8{data_in[143][0]}} & 8'h8b)^
({8{data_in[143][1]}} & 8'h3d)^
({8{data_in[143][2]}} & 8'h7a)^
({8{data_in[143][3]}} & 8'hf4)^
({8{data_in[143][4]}} & 8'hc3)^
({8{data_in[143][5]}} & 8'had)^
({8{data_in[143][6]}} & 8'h71)^
({8{data_in[143][7]}} & 8'he2)^
({8{data_in[144][0]}} & 8'h5d)^
({8{data_in[144][1]}} & 8'hba)^
({8{data_in[144][2]}} & 8'h5f)^
({8{data_in[144][3]}} & 8'hbe)^
({8{data_in[144][4]}} & 8'h57)^
({8{data_in[144][5]}} & 8'hae)^
({8{data_in[144][6]}} & 8'h77)^
({8{data_in[144][7]}} & 8'hee)^
({8{data_in[145][0]}} & 8'ha8)^
({8{data_in[145][1]}} & 8'h7b)^
({8{data_in[145][2]}} & 8'hf6)^
({8{data_in[145][3]}} & 8'hc7)^
({8{data_in[145][4]}} & 8'ha5)^
({8{data_in[145][5]}} & 8'h61)^
({8{data_in[145][6]}} & 8'hc2)^
({8{data_in[145][7]}} & 8'haf)^

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({8{data_in[146][0]}} & 8'h69)^
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({8{data_in[146][2]}} & 8'h8f)^
({8{data_in[146][3]}} & 8'h35)^
({8{data_in[146][4]}} & 8'h6a)^
({8{data_in[146][5]}} & 8'hd4)^
({8{data_in[146][6]}} & 8'h83)^
({8{data_in[146][7]}} & 8'h2d)^
({8{data_in[147][0]}} & 8'h10)^
({8{data_in[147][1]}} & 8'h20)^
({8{data_in[147][2]}} & 8'h40)^
({8{data_in[147][3]}} & 8'h80)^
({8{data_in[147][4]}} & 8'h2b)^
({8{data_in[147][5]}} & 8'h56)^
({8{data_in[147][6]}} & 8'hac)^
({8{data_in[147][7]}} & 8'h73)^
({8{data_in[148][0]}} & 8'hff)^
({8{data_in[148][1]}} & 8'hd5)^
({8{data_in[148][2]}} & 8'h81)^
({8{data_in[148][3]}} & 8'h29)^
({8{data_in[148][4]}} & 8'h52)^
({8{data_in[148][5]}} & 8'ha4)^
({8{data_in[148][6]}} & 8'h63)^
({8{data_in[148][7]}} & 8'hc6)^
({8{data_in[149][0]}} & 8'h6e)^
({8{data_in[149][1]}} & 8'hdc)^
({8{data_in[149][2]}} & 8'h93)^
({8{data_in[149][3]}} & 8'hd)^
({8{data_in[149][4]}} & 8'h1a)^
({8{data_in[149][5]}} & 8'h34)^
({8{data_in[149][6]}} & 8'h68)^
({8{data_in[149][7]}} & 8'hd0)^
({8{data_in[150][0]}} & 8'hd1)^
({8{data_in[150][1]}} & 8'h89)^
({8{data_in[150][2]}} & 8'h39)^
({8{data_in[150][3]}} & 8'h72)^
({8{data_in[150][4]}} & 8'he4)^
({8{data_in[150][5]}} & 8'he3)^
({8{data_in[150][6]}} & 8'hed)^
({8{data_in[150][7]}} & 8'hf1)^
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({8{data_in[151][1]}} & 8'ha6)^
({8{data_in[151][2]}} & 8'h67)^
({8{data_in[151][3]}} & 8'hce)^
({8{data_in[151][4]}} & 8'hb7)^
({8{data_in[151][5]}} & 8'h45)^
({8{data_in[151][6]}} & 8'h8a)^
({8{data_in[151][7]}} & 8'h3f)^
({8{data_in[152][0]}} & 8'h74)^
({8{data_in[152][1]}} & 8'he8)^
({8{data_in[152][2]}} & 8'hfb)^
({8{data_in[152][3]}} & 8'hdd)^
({8{data_in[152][4]}} & 8'h91)^
({8{data_in[152][5]}} & 8'h9)^
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({8{data_in[152][7]}} & 8'h24)^
({8{data_in[153][0]}} & 8'h90)^
({8{data_in[153][1]}} & 8'hb)^
({8{data_in[153][2]}} & 8'h16)^
({8{data_in[153][3]}} & 8'h2c)^
({8{data_in[153][4]}} & 8'h58)^
({8{data_in[153][5]}} & 8'hb0)^
({8{data_in[153][6]}} & 8'h4b)^
({8{data_in[153][7]}} & 8'h96)^
({8{data_in[154][0]}} & 8'h5b)^

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({8{data_in[154][1]}} & 8'hb6)^
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({8{data_in[154][3]}} & 8'h8e)^
({8{data_in[154][4]}} & 8'h37)^
({8{data_in[154][5]}} & 8'h6e)^
({8{data_in[154][6]}} & 8'hdc)^
({8{data_in[154][7]}} & 8'h93)^
({8{data_in[155][0]}} & 8'h48)^
({8{data_in[155][1]}} & 8'h90)^
({8{data_in[155][2]}} & 8'hb)^
({8{data_in[155][3]}} & 8'h16)^
({8{data_in[155][4]}} & 8'h2c)^
({8{data_in[155][5]}} & 8'h58)^
({8{data_in[155][6]}} & 8'hb0)^
({8{data_in[155][7]}} & 8'h4b)^
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({8{data_in[156][1]}} & 8'h89)^
({8{data_in[156][2]}} & 8'h39)^
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({8{data_in[156][5]}} & 8'he3)^
({8{data_in[156][6]}} & 8'hed)^
({8{data_in[156][7]}} & 8'hf1)^
({8{data_in[157][0]}} & 8'he6)^
({8{data_in[157][1]}} & 8'he7)^
({8{data_in[157][2]}} & 8'he5)^
({8{data_in[157][3]}} & 8'he1)^
({8{data_in[157][4]}} & 8'he9)^
({8{data_in[157][5]}} & 8'hf9)^
({8{data_in[157][6]}} & 8'hd9)^
({8{data_in[157][7]}} & 8'h99)^
({8{data_in[158][0]}} & 8'h41)^
({8{data_in[158][1]}} & 8'h82)^
({8{data_in[158][2]}} & 8'h2f)^
({8{data_in[158][3]}} & 8'h5e)^
({8{data_in[158][4]}} & 8'hbc)^
({8{data_in[158][5]}} & 8'h53)^
({8{data_in[158][6]}} & 8'ha6)^
({8{data_in[158][7]}} & 8'h67)^
({8{data_in[159][0]}} & 8'hb7)^
({8{data_in[159][1]}} & 8'h45)^
({8{data_in[159][2]}} & 8'h8a)^
({8{data_in[159][3]}} & 8'h3f)^
({8{data_in[159][4]}} & 8'h7e)^
({8{data_in[159][5]}} & 8'hfc)^
({8{data_in[159][6]}} & 8'hd3)^
({8{data_in[159][7]}} & 8'h8d)^
({8{data_in[160][0]}} & 8'hdc)^
({8{data_in[160][1]}} & 8'h93)^
({8{data_in[160][2]}} & 8'hd)^
({8{data_in[160][3]}} & 8'h1a)^
({8{data_in[160][4]}} & 8'h34)^
({8{data_in[160][5]}} & 8'h68)^
({8{data_in[160][6]}} & 8'hd0)^
({8{data_in[160][7]}} & 8'h8b)^
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({8{data_in[161][1]}} & 8'h71)^
({8{data_in[161][2]}} & 8'he2)^
({8{data_in[161][3]}} & 8'hef)^
({8{data_in[161][4]}} & 8'hf5)^
({8{data_in[161][5]}} & 8'hc1)^
({8{data_in[161][6]}} & 8'ha9)^
({8{data_in[161][7]}} & 8'h79)^
({8{data_in[162][0]}} & 8'h54)^
({8{data_in[162][1]}} & 8'ha8)^

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({8{data_in[162][5]}} & 8'ha5)^
({8{data_in[162][6]}} & 8'h61)^
({8{data_in[162][7]}} & 8'hc2)^
({8{data_in[163][0]}} & 8'hf5)^
({8{data_in[163][1]}} & 8'hc1)^
({8{data_in[163][2]}} & 8'ha9)^
({8{data_in[163][3]}} & 8'h79)^
({8{data_in[163][4]}} & 8'hf2)^
({8{data_in[163][5]}} & 8'hcf)^
({8{data_in[163][6]}} & 8'hb5)^
({8{data_in[163][7]}} & 8'h41)^
({8{data_in[164][0]}} & 8'h55)^
({8{data_in[164][1]}} & 8'haa)^
({8{data_in[164][2]}} & 8'h7f)^
({8{data_in[164][3]}} & 8'hfe)^
({8{data_in[164][4]}} & 8'hd7)^
({8{data_in[164][5]}} & 8'h85)^
({8{data_in[164][6]}} & 8'h21)^
({8{data_in[164][7]}} & 8'h42)^
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({8{data_in[165][1]}} & 8'he7)^
({8{data_in[165][2]}} & 8'he5)^
({8{data_in[165][3]}} & 8'he1)^
({8{data_in[165][4]}} & 8'he9)^
({8{data_in[165][5]}} & 8'hf9)^
({8{data_in[165][6]}} & 8'hd9)^
({8{data_in[165][7]}} & 8'h99)^
({8{data_in[166][0]}} & 8'h1f)^
({8{data_in[166][1]}} & 8'h3e)^
({8{data_in[166][2]}} & 8'h7c)^
({8{data_in[166][3]}} & 8'hf8)^
({8{data_in[166][4]}} & 8'hdb)^
({8{data_in[166][5]}} & 8'h9d)^
({8{data_in[166][6]}} & 8'h11)^
({8{data_in[166][7]}} & 8'h22)^
({8{data_in[167][0]}} & 8'hb3)^
({8{data_in[167][1]}} & 8'h4d)^
({8{data_in[167][2]}} & 8'h9a)^
({8{data_in[167][3]}} & 8'h1f)^
({8{data_in[167][4]}} & 8'h3e)^
({8{data_in[167][5]}} & 8'h7c)^
({8{data_in[167][6]}} & 8'hf8)^
({8{data_in[167][7]}} & 8'hdb)^
({8{data_in[168][0]}} & 8'he1)^
({8{data_in[168][1]}} & 8'he9)^
({8{data_in[168][2]}} & 8'hf9)^
({8{data_in[168][3]}} & 8'hd9)^
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({8{data_in[168][5]}} & 8'h19)^
({8{data_in[168][6]}} & 8'h32)^
({8{data_in[168][7]}} & 8'h64)^
({8{data_in[169][0]}} & 8'h37)^
({8{data_in[169][1]}} & 8'h6e)^
({8{data_in[169][2]}} & 8'hdc)^
({8{data_in[169][3]}} & 8'h93)^
({8{data_in[169][4]}} & 8'hd)^
({8{data_in[169][5]}} & 8'h1a)^
({8{data_in[169][6]}} & 8'h34)^
({8{data_in[169][7]}} & 8'h68)^
({8{data_in[170][0]}} & 8'he8)^
({8{data_in[170][1]}} & 8'hfb)^
({8{data_in[170][2]}} & 8'hdd)^

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({8{data_in[170][3]}} & 8'h91)^
({8{data_in[170][4]}} & 8'h9)^
({8{data_in[170][5]}} & 8'h12)^
({8{data_in[170][6]}} & 8'h24)^
({8{data_in[170][7]}} & 8'h48)^
({8{data_in[171][0]}} & 8'h2b)^
({8{data_in[171][1]}} & 8'h56)^
({8{data_in[171][2]}} & 8'hac)^
({8{data_in[171][3]}} & 8'h73)^
({8{data_in[171][4]}} & 8'he6)^
({8{data_in[171][5]}} & 8'he7)^
({8{data_in[171][6]}} & 8'he5)^
({8{data_in[171][7]}} & 8'he1)^
({8{data_in[172][0]}} & 8'h54)^
({8{data_in[172][1]}} & 8'ha8)^
({8{data_in[172][2]}} & 8'h7b)^
({8{data_in[172][3]}} & 8'hf6)^
({8{data_in[172][4]}} & 8'hc7)^
({8{data_in[172][5]}} & 8'ha5)^
({8{data_in[172][6]}} & 8'h61)^
({8{data_in[172][7]}} & 8'hc2)^
({8{data_in[173][0]}} & 8'h80)^
({8{data_in[173][1]}} & 8'h2b)^
({8{data_in[173][2]}} & 8'h56)^
({8{data_in[173][3]}} & 8'hac)^
({8{data_in[173][4]}} & 8'h73)^
({8{data_in[173][5]}} & 8'he6)^
({8{data_in[173][6]}} & 8'he7)^
({8{data_in[173][7]}} & 8'he5)^
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({8{data_in[174][1]}} & 8'h2c)^
({8{data_in[174][2]}} & 8'h58)^
({8{data_in[174][3]}} & 8'hb0)^
({8{data_in[174][4]}} & 8'h4b)^
({8{data_in[174][5]}} & 8'h96)^
({8{data_in[174][6]}} & 8'h7)^
({8{data_in[174][7]}} & 8'he)^
({8{data_in[175][0]}} & 8'hc2)^
({8{data_in[175][1]}} & 8'haf)^
({8{data_in[175][2]}} & 8'h75)^
({8{data_in[175][3]}} & 8'hea)^
({8{data_in[175][4]}} & 8'hff)^
({8{data_in[175][5]}} & 8'hd5)^
({8{data_in[175][6]}} & 8'h81)^
({8{data_in[175][7]}} & 8'h29)^
({8{data_in[176][0]}} & 8'h94)^
({8{data_in[176][1]}} & 8'h3)^
({8{data_in[176][2]}} & 8'h6)^
({8{data_in[176][3]}} & 8'hc)^
({8{data_in[176][4]}} & 8'h18)^
({8{data_in[176][5]}} & 8'h30)^
({8{data_in[176][6]}} & 8'h60)^
({8{data_in[176][7]}} & 8'hc0)^
({8{data_in[177][0]}} & 8'h55)^
({8{data_in[177][1]}} & 8'haa)^
({8{data_in[177][2]}} & 8'h7f)^
({8{data_in[177][3]}} & 8'hfe)^
({8{data_in[177][4]}} & 8'hd7)^
({8{data_in[177][5]}} & 8'h85)^
({8{data_in[177][6]}} & 8'h21)^
({8{data_in[177][7]}} & 8'h42)^
({8{data_in[178][0]}} & 8'h7b)^
({8{data_in[178][1]}} & 8'hf6)^
({8{data_in[178][2]}} & 8'hc7)^
({8{data_in[178][3]}} & 8'ha5)^

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({8{data_in[178][4]}} & 8'h61)^
({8{data_in[178][5]}} & 8'hc2)^
({8{data_in[178][6]}} & 8'haf)^
({8{data_in[178][7]}} & 8'h75)^
({8{data_in[179][0]}} & 8'hbd)^
({8{data_in[179][1]}} & 8'h51)^
({8{data_in[179][2]}} & 8'ha2)^
({8{data_in[179][3]}} & 8'h6f)^
({8{data_in[179][4]}} & 8'hde)^
({8{data_in[179][5]}} & 8'h97)^
({8{data_in[179][6]}} & 8'h5)^
({8{data_in[179][7]}} & 8'ha)^
({8{data_in[180][0]}} & 8'h9f)^
({8{data_in[180][1]}} & 8'h15)^
({8{data_in[180][2]}} & 8'h2a)^
({8{data_in[180][3]}} & 8'h54)^
({8{data_in[180][4]}} & 8'ha8)^
({8{data_in[180][5]}} & 8'h7b)^
({8{data_in[180][6]}} & 8'hf6)^
({8{data_in[180][7]}} & 8'hc7)^
({8{data_in[181][0]}} & 8'hde)^
({8{data_in[181][1]}} & 8'h97)^
({8{data_in[181][2]}} & 8'h5)^
({8{data_in[181][3]}} & 8'ha)^
({8{data_in[181][4]}} & 8'h14)^
({8{data_in[181][5]}} & 8'h28)^
({8{data_in[181][6]}} & 8'h50)^
({8{data_in[181][7]}} & 8'ha0)^
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({8{data_in[182][1]}} & 8'h49)^
({8{data_in[182][2]}} & 8'h92)^
({8{data_in[182][3]}} & 8'hf)^
({8{data_in[182][4]}} & 8'h1e)^
({8{data_in[182][5]}} & 8'h3c)^
({8{data_in[182][6]}} & 8'h78)^
({8{data_in[182][7]}} & 8'hf0)^
({8{data_in[183][0]}} & 8'hee)^
({8{data_in[183][1]}} & 8'hf7)^
({8{data_in[183][2]}} & 8'hc5)^
({8{data_in[183][3]}} & 8'ha1)^
({8{data_in[183][4]}} & 8'h69)^
({8{data_in[183][5]}} & 8'hd2)^
({8{data_in[183][6]}} & 8'h8f)^
({8{data_in[183][7]}} & 8'h35)^
({8{data_in[184][0]}} & 8'h1a)^
({8{data_in[184][1]}} & 8'h34)^
({8{data_in[184][2]}} & 8'h68)^
({8{data_in[184][3]}} & 8'hd0)^
({8{data_in[184][4]}} & 8'h8b)^
({8{data_in[184][5]}} & 8'h3d)^
({8{data_in[184][6]}} & 8'h7a)^
({8{data_in[184][7]}} & 8'hf4)^
({8{data_in[185][0]}} & 8'he0)^
({8{data_in[185][1]}} & 8'heb)^
({8{data_in[185][2]}} & 8'hfd)^
({8{data_in[185][3]}} & 8'hd1)^
({8{data_in[185][4]}} & 8'h89)^
({8{data_in[185][5]}} & 8'h39)^
({8{data_in[185][6]}} & 8'h72)^
({8{data_in[185][7]}} & 8'he4)^
({8{data_in[186][0]}} & 8'hd8)^
({8{data_in[186][1]}} & 8'h9b)^
({8{data_in[186][2]}} & 8'h1d)^
({8{data_in[186][3]}} & 8'h3a)^
({8{data_in[186][4]}} & 8'h74)^

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({8{data_in[186][5]}} & 8'he8)^
({8{data_in[186][6]}} & 8'hfb)^
({8{data_in[186][7]}} & 8'hdd)^
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({8{data_in[187][1]}} & 8'h3e)^
({8{data_in[187][2]}} & 8'h7c)^
({8{data_in[187][3]}} & 8'hf8)^
({8{data_in[187][4]}} & 8'hdb)^
({8{data_in[187][5]}} & 8'h9d)^
({8{data_in[187][6]}} & 8'h11)^
({8{data_in[187][7]}} & 8'h22)^
({8{data_in[188][0]}} & 8'hb4)^
({8{data_in[188][1]}} & 8'h43)^
({8{data_in[188][2]}} & 8'h86)^
({8{data_in[188][3]}} & 8'h27)^
({8{data_in[188][4]}} & 8'h4e)^
({8{data_in[188][5]}} & 8'h9c)^
({8{data_in[188][6]}} & 8'h13)^
({8{data_in[188][7]}} & 8'h26)^
({8{data_in[189][0]}} & 8'hf5)^
({8{data_in[189][1]}} & 8'hc1)^
({8{data_in[189][2]}} & 8'ha9)^
({8{data_in[189][3]}} & 8'h79)^
({8{data_in[189][4]}} & 8'hf2)^
({8{data_in[189][5]}} & 8'hcf)^
({8{data_in[189][6]}} & 8'hb5)^
({8{data_in[189][7]}} & 8'h41)^
({8{data_in[190][0]}} & 8'haf)^
({8{data_in[190][1]}} & 8'h75)^
({8{data_in[190][2]}} & 8'hea)^
({8{data_in[190][3]}} & 8'hff)^
({8{data_in[190][4]}} & 8'hd5)^
({8{data_in[190][5]}} & 8'h81)^
({8{data_in[190][6]}} & 8'h29)^
({8{data_in[190][7]}} & 8'h52)^
({8{data_in[191][0]}} & 8'hfe)^
({8{data_in[191][1]}} & 8'hd7)^
({8{data_in[191][2]}} & 8'h85)^
({8{data_in[191][3]}} & 8'h21)^
({8{data_in[191][4]}} & 8'h42)^
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({8{data_in[192][0]}} & 8'hb2)^
({8{data_in[192][1]}} & 8'h4f)^
({8{data_in[192][2]}} & 8'h9e)^
({8{data_in[192][3]}} & 8'h17)^
({8{data_in[192][4]}} & 8'h2e)^
({8{data_in[192][5]}} & 8'h5c)^
({8{data_in[192][6]}} & 8'hb8)^
({8{data_in[192][7]}} & 8'h5b)^
({8{data_in[193][0]}} & 8'h15)^
({8{data_in[193][1]}} & 8'h2a)^
({8{data_in[193][2]}} & 8'h54)^
({8{data_in[193][3]}} & 8'ha8)^
({8{data_in[193][4]}} & 8'h7b)^
({8{data_in[193][5]}} & 8'hf6)^
({8{data_in[193][6]}} & 8'hc7)^
({8{data_in[193][7]}} & 8'ha5)^
({8{data_in[194][0]}} & 8'h11)^
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({8{data_in[194][2]}} & 8'h44)^
({8{data_in[194][3]}} & 8'h88)^
({8{data_in[194][4]}} & 8'h3b)^
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({8{data_in[195][5]}} & 8'hc1)^
({8{data_in[195][6]}} & 8'ha9)^
({8{data_in[195][7]}} & 8'h79)^
({8{data_in[196][0]}} & 8'hb6)^
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({8{data_in[196][2]}} & 8'h8e)^
({8{data_in[196][3]}} & 8'h37)^
({8{data_in[196][4]}} & 8'h6e)^
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({8{data_in[196][6]}} & 8'h93)^
({8{data_in[196][7]}} & 8'hd)^
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({8{data_in[197][1]}} & 8'h39)^
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({8{data_in[197][3]}} & 8'he4)^
({8{data_in[197][4]}} & 8'he3)^
({8{data_in[197][5]}} & 8'hed)^
({8{data_in[197][6]}} & 8'hf1)^
({8{data_in[197][7]}} & 8'hc9)^
({8{data_in[198][0]}} & 8'h1f)^
({8{data_in[198][1]}} & 8'h3e)^
({8{data_in[198][2]}} & 8'h7c)^
({8{data_in[198][3]}} & 8'hf8)^
({8{data_in[198][4]}} & 8'hdb)^
({8{data_in[198][5]}} & 8'h9d)^
({8{data_in[198][6]}} & 8'h11)^
({8{data_in[198][7]}} & 8'h22)^
({8{data_in[199][0]}} & 8'h28)^
({8{data_in[199][1]}} & 8'h50)^
({8{data_in[199][2]}} & 8'ha0)^
({8{data_in[199][3]}} & 8'h6b)^
({8{data_in[199][4]}} & 8'hd6)^
({8{data_in[199][5]}} & 8'h87)^
({8{data_in[199][6]}} & 8'h25)^
({8{data_in[199][7]}} & 8'h4a)^
({8{data_in[200][0]}} & 8'h46)^
({8{data_in[200][1]}} & 8'h8c)^
({8{data_in[200][2]}} & 8'h33)^
({8{data_in[200][3]}} & 8'h66)^
({8{data_in[200][4]}} & 8'hcc)^
({8{data_in[200][5]}} & 8'hb3)^
({8{data_in[200][6]}} & 8'h4d)^
({8{data_in[200][7]}} & 8'h9a)^
({8{data_in[201][0]}} & 8'hf8)^
({8{data_in[201][1]}} & 8'hdb)^
({8{data_in[201][2]}} & 8'h9d)^
({8{data_in[201][3]}} & 8'h11)^
({8{data_in[201][4]}} & 8'h22)^
({8{data_in[201][5]}} & 8'h44)^
({8{data_in[201][6]}} & 8'h88)^
({8{data_in[201][7]}} & 8'h3b)^
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({8{data_in[202][1]}} & 8'h59)^
({8{data_in[202][2]}} & 8'hb2)^
({8{data_in[202][3]}} & 8'h4f)^
({8{data_in[202][4]}} & 8'h9e)^
({8{data_in[202][5]}} & 8'h17)^
({8{data_in[202][6]}} & 8'h2e)^

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({8{data_in[202][7]}} & 8'h5c)^
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({8{data_in[203][2]}} & 8'hf6)^
({8{data_in[203][3]}} & 8'hc7)^
({8{data_in[203][4]}} & 8'ha5)^
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({8{data_in[203][6]}} & 8'hc2)^
({8{data_in[203][7]}} & 8'haf)^
({8{data_in[204][0]}} & 8'h9a)^
({8{data_in[204][1]}} & 8'h1f)^
({8{data_in[204][2]}} & 8'h3e)^
({8{data_in[204][3]}} & 8'h7c)^
({8{data_in[204][4]}} & 8'hf8)^
({8{data_in[204][5]}} & 8'hdb)^
({8{data_in[204][6]}} & 8'h9d)^
({8{data_in[204][7]}} & 8'h11)^
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({8{data_in[205][2]}} & 8'hcb)^
({8{data_in[205][3]}} & 8'hbd)^
({8{data_in[205][4]}} & 8'h51)^
({8{data_in[205][5]}} & 8'ha2)^
({8{data_in[205][6]}} & 8'h6f)^
({8{data_in[205][7]}} & 8'hde)^
({8{data_in[206][0]}} & 8'h84)^
({8{data_in[206][1]}} & 8'h23)^
({8{data_in[206][2]}} & 8'h46)^
({8{data_in[206][3]}} & 8'h8c)^
({8{data_in[206][4]}} & 8'h33)^
({8{data_in[206][5]}} & 8'h66)^
({8{data_in[206][6]}} & 8'hcc)^
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({8{data_in[207][1]}} & 8'hf2)^
({8{data_in[207][2]}} & 8'hcf)^
({8{data_in[207][3]}} & 8'hb5)^
({8{data_in[207][4]}} & 8'h41)^
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({8{data_in[207][6]}} & 8'h2f)^
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({8{data_in[208][2]}} & 8'h57)^
({8{data_in[208][3]}} & 8'hae)^
({8{data_in[208][4]}} & 8'h77)^
({8{data_in[208][5]}} & 8'hee)^
({8{data_in[208][6]}} & 8'hf7)^
({8{data_in[208][7]}} & 8'hc5)^
({8{data_in[209][0]}} & 8'h9e)^
({8{data_in[209][1]}} & 8'h17)^
({8{data_in[209][2]}} & 8'h2e)^
({8{data_in[209][3]}} & 8'h5c)^
({8{data_in[209][4]}} & 8'hb8)^
({8{data_in[209][5]}} & 8'h5b)^
({8{data_in[209][6]}} & 8'hb6)^
({8{data_in[209][7]}} & 8'h47)^
({8{data_in[210][0]}} & 8'h14)^
({8{data_in[210][1]}} & 8'h28)^
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({8{data_in[210][4]}} & 8'h6b)^
({8{data_in[210][5]}} & 8'hd6)^
({8{data_in[210][6]}} & 8'h87)^
({8{data_in[210][7]}} & 8'h25)^

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({8{data_in[211][0]}} & 8'hdc)^
({8{data_in[211][1]}} & 8'h93)^
({8{data_in[211][2]}} & 8'hd)^
({8{data_in[211][3]}} & 8'h1a)^
({8{data_in[211][4]}} & 8'h34)^
({8{data_in[211][5]}} & 8'h68)^
({8{data_in[211][6]}} & 8'hd0)^
({8{data_in[211][7]}} & 8'h8b)^
({8{data_in[212][0]}} & 8'he0)^
({8{data_in[212][1]}} & 8'heb)^
({8{data_in[212][2]}} & 8'hfd)^
({8{data_in[212][3]}} & 8'hd1)^
({8{data_in[212][4]}} & 8'h89)^
({8{data_in[212][5]}} & 8'h39)^
({8{data_in[212][6]}} & 8'h72)^
({8{data_in[212][7]}} & 8'he4)^
({8{data_in[213][0]}} & 8'h94)^
({8{data_in[213][1]}} & 8'h3)^
({8{data_in[213][2]}} & 8'h6)^
({8{data_in[213][3]}} & 8'hc)^
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({8{data_in[213][5]}} & 8'h30)^
({8{data_in[213][6]}} & 8'h60)^
({8{data_in[213][7]}} & 8'hc0)^
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({8{data_in[214][1]}} & 8'h76)^
({8{data_in[214][2]}} & 8'hec)^
({8{data_in[214][3]}} & 8'hf3)^
({8{data_in[214][4]}} & 8'hcd)^
({8{data_in[214][5]}} & 8'hb1)^
({8{data_in[214][6]}} & 8'h49)^
({8{data_in[214][7]}} & 8'h92)^
({8{data_in[215][0]}} & 8'h46)^
({8{data_in[215][1]}} & 8'h8c)^
({8{data_in[215][2]}} & 8'h33)^
({8{data_in[215][3]}} & 8'h66)^
({8{data_in[215][4]}} & 8'hcc)^
({8{data_in[215][5]}} & 8'hb3)^
({8{data_in[215][6]}} & 8'h4d)^
({8{data_in[215][7]}} & 8'h9a)^
({8{data_in[216][0]}} & 8'h8e)^
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({8{data_in[216][2]}} & 8'h6e)^
({8{data_in[216][3]}} & 8'hdc)^
({8{data_in[216][4]}} & 8'h93)^
({8{data_in[216][5]}} & 8'hd)^
({8{data_in[216][6]}} & 8'h1a)^
({8{data_in[216][7]}} & 8'h34)^
({8{data_in[217][0]}} & 8'h54)^
({8{data_in[217][1]}} & 8'ha8)^
({8{data_in[217][2]}} & 8'h7b)^
({8{data_in[217][3]}} & 8'hf6)^
({8{data_in[217][4]}} & 8'hc7)^
({8{data_in[217][5]}} & 8'ha5)^
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({8{data_in[218][2]}} & 8'h6b)^
({8{data_in[218][3]}} & 8'hd6)^
({8{data_in[218][4]}} & 8'h87)^
({8{data_in[218][5]}} & 8'h25)^
({8{data_in[218][6]}} & 8'h4a)^
({8{data_in[218][7]}} & 8'h94)^
({8{data_in[219][0]}} & 8'h8c)^

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({8{data_in[219][1]}} & 8'h33)^
({8{data_in[219][2]}} & 8'h66)^
({8{data_in[219][3]}} & 8'hcc)^
({8{data_in[219][4]}} & 8'hb3)^
({8{data_in[219][5]}} & 8'h4d)^
({8{data_in[219][6]}} & 8'h9a)^
({8{data_in[219][7]}} & 8'h1f)^
({8{data_in[220][0]}} & 8'h2f)^
({8{data_in[220][1]}} & 8'h5e)^
({8{data_in[220][2]}} & 8'hbc)^
({8{data_in[220][3]}} & 8'h53)^
({8{data_in[220][4]}} & 8'ha6)^
({8{data_in[220][5]}} & 8'h67)^
({8{data_in[220][6]}} & 8'hce)^
({8{data_in[220][7]}} & 8'hb7)^
({8{data_in[221][0]}} & 8'ha0)^
({8{data_in[221][1]}} & 8'h6b)^
({8{data_in[221][2]}} & 8'hd6)^
({8{data_in[221][3]}} & 8'h87)^
({8{data_in[221][4]}} & 8'h25)^
({8{data_in[221][5]}} & 8'h4a)^
({8{data_in[221][6]}} & 8'h94)^
({8{data_in[221][7]}} & 8'h3)^
({8{data_in[222][0]}} & 8'h57)^
({8{data_in[222][1]}} & 8'hae)^
({8{data_in[222][2]}} & 8'h77)^
({8{data_in[222][3]}} & 8'hee)^
({8{data_in[222][4]}} & 8'hf7)^
({8{data_in[222][5]}} & 8'hc5)^
({8{data_in[222][6]}} & 8'ha1)^
({8{data_in[222][7]}} & 8'h69)^
({8{data_in[223][0]}} & 8'hd3)^
({8{data_in[223][1]}} & 8'h8d)^
({8{data_in[223][2]}} & 8'h31)^
({8{data_in[223][3]}} & 8'h62)^
({8{data_in[223][4]}} & 8'hc4)^
({8{data_in[223][5]}} & 8'ha3)^
({8{data_in[223][6]}} & 8'h6d)^
({8{data_in[223][7]}} & 8'hda)^
({8{data_in[224][0]}} & 8'h3d)^
({8{data_in[224][1]}} & 8'h7a)^
({8{data_in[224][2]}} & 8'hf4)^
({8{data_in[224][3]}} & 8'hc3)^
({8{data_in[224][4]}} & 8'had)^
({8{data_in[224][5]}} & 8'h71)^
({8{data_in[224][6]}} & 8'he2)^
({8{data_in[224][7]}} & 8'hef)^
({8{data_in[225][0]}} & 8'hcd)^
({8{data_in[225][1]}} & 8'hb1)^
({8{data_in[225][2]}} & 8'h49)^
({8{data_in[225][3]}} & 8'h92)^
({8{data_in[225][4]}} & 8'hf)^
({8{data_in[225][5]}} & 8'h1e)^
({8{data_in[225][6]}} & 8'h3c)^
({8{data_in[225][7]}} & 8'h78)^
({8{data_in[226][0]}} & 8'h5b)^
({8{data_in[226][1]}} & 8'hb6)^
({8{data_in[226][2]}} & 8'h47)^
({8{data_in[226][3]}} & 8'h8e)^
({8{data_in[226][4]}} & 8'h37)^
({8{data_in[226][5]}} & 8'h6e)^
({8{data_in[226][6]}} & 8'hdc)^
({8{data_in[226][7]}} & 8'h93)^
({8{data_in[227][0]}} & 8'h28)^
({8{data_in[227][1]}} & 8'h50)^

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({8{data_in[227][2]}} & 8'ha0)^
({8{data_in[227][3]}} & 8'h6b)^
({8{data_in[227][4]}} & 8'hd6)^
({8{data_in[227][5]}} & 8'h87)^
({8{data_in[227][6]}} & 8'h25)^
({8{data_in[227][7]}} & 8'h4a)^
({8{data_in[228][0]}} & 8'had)^
({8{data_in[228][1]}} & 8'h71)^
({8{data_in[228][2]}} & 8'he2)^
({8{data_in[228][3]}} & 8'hef)^
({8{data_in[228][4]}} & 8'hf5)^
({8{data_in[228][5]}} & 8'hc1)^
({8{data_in[228][6]}} & 8'ha9)^
({8{data_in[228][7]}} & 8'h79)^
({8{data_in[229][0]}} & 8'hd7)^
({8{data_in[229][1]}} & 8'h85)^
({8{data_in[229][2]}} & 8'h21)^
({8{data_in[229][3]}} & 8'h42)^
({8{data_in[229][4]}} & 8'h84)^
({8{data_in[229][5]}} & 8'h23)^
({8{data_in[229][6]}} & 8'h46)^
({8{data_in[229][7]}} & 8'h8c)^
({8{data_in[230][0]}} & 8'hee)^
({8{data_in[230][1]}} & 8'hf7)^
({8{data_in[230][2]}} & 8'hc5)^
({8{data_in[230][3]}} & 8'ha1)^
({8{data_in[230][4]}} & 8'h69)^
({8{data_in[230][5]}} & 8'hd2)^
({8{data_in[230][6]}} & 8'h8f)^
({8{data_in[230][7]}} & 8'h35)^
({8{data_in[231][0]}} & 8'h91)^
({8{data_in[231][1]}} & 8'h9)^
({8{data_in[231][2]}} & 8'h12)^
({8{data_in[231][3]}} & 8'h24)^
({8{data_in[231][4]}} & 8'h48)^
({8{data_in[231][5]}} & 8'h90)^
({8{data_in[231][6]}} & 8'hb)^
({8{data_in[231][7]}} & 8'h16)^
({8{data_in[232][0]}} & 8'h4c)^
({8{data_in[232][1]}} & 8'h98)^
({8{data_in[232][2]}} & 8'h1b)^
({8{data_in[232][3]}} & 8'h36)^
({8{data_in[232][4]}} & 8'h6c)^
({8{data_in[232][5]}} & 8'hd8)^
({8{data_in[232][6]}} & 8'h9b)^
({8{data_in[232][7]}} & 8'h1d)^
({8{data_in[233][0]}} & 8'h4f)^
({8{data_in[233][1]}} & 8'h9e)^
({8{data_in[233][2]}} & 8'h17)^
({8{data_in[233][3]}} & 8'h2e)^
({8{data_in[233][4]}} & 8'h5c)^
({8{data_in[233][5]}} & 8'hb8)^
({8{data_in[233][6]}} & 8'h5b)^
({8{data_in[233][7]}} & 8'hb6)^
({8{data_in[234][0]}} & 8'hf7)^
({8{data_in[234][1]}} & 8'hc5)^
({8{data_in[234][2]}} & 8'ha1)^
({8{data_in[234][3]}} & 8'h69)^
({8{data_in[234][4]}} & 8'hd2)^
({8{data_in[234][5]}} & 8'h8f)^
({8{data_in[234][6]}} & 8'h35)^
({8{data_in[234][7]}} & 8'h6a)^
({8{data_in[235][0]}} & 8'h18)^
({8{data_in[235][1]}} & 8'h30)^
({8{data_in[235][2]}} & 8'h60)^

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```

({8{data_in[235][3]}} & 8'hc0)^
({8{data_in[235][4]}} & 8'hab)^
({8{data_in[235][5]}} & 8'h7d)^
({8{data_in[235][6]}} & 8'hfa)^
({8{data_in[235][7]}} & 8'hdf)^
({8{data_in[236][0]}} & 8'h99)^
({8{data_in[236][1]}} & 8'h19)^
({8{data_in[236][2]}} & 8'h32)^
({8{data_in[236][3]}} & 8'h64)^
({8{data_in[236][4]}} & 8'hc8)^
({8{data_in[236][5]}} & 8'hbb)^
({8{data_in[236][6]}} & 8'h5d)^
({8{data_in[236][7]}} & 8'hba)^
({8{data_in[237][0]}} & 8'h28)^
({8{data_in[237][1]}} & 8'h50)^
({8{data_in[237][2]}} & 8'ha0)^
({8{data_in[237][3]}} & 8'h6b)^
({8{data_in[237][4]}} & 8'hd6)^
({8{data_in[237][5]}} & 8'h87)^
({8{data_in[237][6]}} & 8'h25)^
({8{data_in[237][7]}} & 8'h4a)^
({8{data_in[238][0]}} & 8'hd0)^
({8{data_in[238][1]}} & 8'h8b)^
({8{data_in[238][2]}} & 8'h3d)^
({8{data_in[238][3]}} & 8'h7a)^
({8{data_in[238][4]}} & 8'hf4)^
({8{data_in[238][5]}} & 8'hc3)^
({8{data_in[238][6]}} & 8'had)^
({8{data_in[238][7]}} & 8'h71)^
({8{data_in[239][0]}} & 8'h58)^
({8{data_in[239][1]}} & 8'hb0)^
({8{data_in[239][2]}} & 8'h4b)^
({8{data_in[239][3]}} & 8'h96)^
({8{data_in[239][4]}} & 8'h7)^
({8{data_in[239][5]}} & 8'he)^
({8{data_in[239][6]}} & 8'h1c)^
({8{data_in[239][7]}} & 8'h38)^
({8{data_in[240][0]}} & 8'h9)^
({8{data_in[240][1]}} & 8'h12)^
({8{data_in[240][2]}} & 8'h24)^
({8{data_in[240][3]}} & 8'h48)^
({8{data_in[240][4]}} & 8'h90)^
({8{data_in[240][5]}} & 8'hb)^
({8{data_in[240][6]}} & 8'h16)^
({8{data_in[240][7]}} & 8'h2c)^
({8{data_in[241][0]}} & 8'h68)^
({8{data_in[241][1]}} & 8'hd0)^
({8{data_in[241][2]}} & 8'h8b)^
({8{data_in[241][3]}} & 8'h3d)^
({8{data_in[241][4]}} & 8'h7a)^
({8{data_in[241][5]}} & 8'hf4)^
({8{data_in[241][6]}} & 8'hc3)^
({8{data_in[241][7]}} & 8'had);
data_out[247] = ({8{data_in[0][0]}} & 8'ha7)^
({8{data_in[0][1]}} & 8'h65)^
({8{data_in[0][2]}} & 8'hca)^
({8{data_in[0][3]}} & 8'hbf)^
({8{data_in[0][4]}} & 8'h55)^
({8{data_in[0][5]}} & 8'haa)^
({8{data_in[0][6]}} & 8'h7f)^
({8{data_in[0][7]}} & 8'hfe)^
({8{data_in[1][0]}} & 8'hdb)^
({8{data_in[1][1]}} & 8'h9d)^
({8{data_in[1][2]}} & 8'h11)^
({8{data_in[1][3]}} & 8'h22)^

```

```

({8{data_in[1][4]}} & 8'h44)^
({8{data_in[1][5]}} & 8'h88)^
({8{data_in[1][6]}} & 8'h3b)^
({8{data_in[1][7]}} & 8'h76)^
({8{data_in[2][0]}} & 8'h37)^
({8{data_in[2][1]}} & 8'h6e)^
({8{data_in[2][2]}} & 8'hdc)^
({8{data_in[2][3]}} & 8'h93)^
({8{data_in[2][4]}} & 8'hd)^
({8{data_in[2][5]}} & 8'h1a)^
({8{data_in[2][6]}} & 8'h34)^
({8{data_in[2][7]}} & 8'h68)^
({8{data_in[3][0]}} & 8'hca)^
({8{data_in[3][1]}} & 8'hbf)^
({8{data_in[3][2]}} & 8'h55)^
({8{data_in[3][3]}} & 8'haa)^
({8{data_in[3][4]}} & 8'h7f)^
({8{data_in[3][5]}} & 8'hfe)^
({8{data_in[3][6]}} & 8'hd7)^
({8{data_in[3][7]}} & 8'h85)^
({8{data_in[4][0]}} & 8'h9e)^
({8{data_in[4][1]}} & 8'h17)^
({8{data_in[4][2]}} & 8'h2e)^
({8{data_in[4][3]}} & 8'h5c)^
({8{data_in[4][4]}} & 8'hb8)^
({8{data_in[4][5]}} & 8'h5b)^
({8{data_in[4][6]}} & 8'hb6)^
({8{data_in[4][7]}} & 8'h47)^
({8{data_in[5][0]}} & 8'h27)^
({8{data_in[5][1]}} & 8'h4e)^
({8{data_in[5][2]}} & 8'h9c)^
({8{data_in[5][3]}} & 8'h13)^
({8{data_in[5][4]}} & 8'h26)^
({8{data_in[5][5]}} & 8'h4c)^
({8{data_in[5][6]}} & 8'h98)^
({8{data_in[5][7]}} & 8'h1b)^
({8{data_in[6][0]}} & 8'h8b)^
({8{data_in[6][1]}} & 8'h3d)^
({8{data_in[6][2]}} & 8'h7a)^
({8{data_in[6][3]}} & 8'hf4)^
({8{data_in[6][4]}} & 8'hc3)^
({8{data_in[6][5]}} & 8'had)^
({8{data_in[6][6]}} & 8'h71)^
({8{data_in[6][7]}} & 8'he2)^
({8{data_in[7][0]}} & 8'h28)^
({8{data_in[7][1]}} & 8'h50)^
({8{data_in[7][2]}} & 8'ha0)^
({8{data_in[7][3]}} & 8'h6b)^
({8{data_in[7][4]}} & 8'hd6)^
({8{data_in[7][5]}} & 8'h87)^
({8{data_in[7][6]}} & 8'h25)^
({8{data_in[7][7]}} & 8'h4a)^
({8{data_in[8][0]}} & 8'h15)^
({8{data_in[8][1]}} & 8'h2a)^
({8{data_in[8][2]}} & 8'h54)^
({8{data_in[8][3]}} & 8'ha8)^
({8{data_in[8][4]}} & 8'h7b)^
({8{data_in[8][5]}} & 8'hf6)^
({8{data_in[8][6]}} & 8'hc7)^
({8{data_in[8][7]}} & 8'ha5)^
({8{data_in[9][0]}} & 8'h86)^
({8{data_in[9][1]}} & 8'h27)^
({8{data_in[9][2]}} & 8'h4e)^
({8{data_in[9][3]}} & 8'h9c)^
({8{data_in[9][4]}} & 8'h13)^

```



```

({8{data_in[9][5]}} & 8'h26)^
({8{data_in[9][6]}} & 8'h4c)^
({8{data_in[9][7]}} & 8'h98)^
({8{data_in[10][0]}} & 8'h17)^
({8{data_in[10][1]}} & 8'h2e)^
({8{data_in[10][2]}} & 8'h5c)^
({8{data_in[10][3]}} & 8'hb8)^
({8{data_in[10][4]}} & 8'h5b)^
({8{data_in[10][5]}} & 8'hb6)^
({8{data_in[10][6]}} & 8'h47)^
({8{data_in[10][7]}} & 8'h8e)^
({8{data_in[11][0]}} & 8'h90)^
({8{data_in[11][1]}} & 8'hb)^
({8{data_in[11][2]}} & 8'h16)^
({8{data_in[11][3]}} & 8'h2c)^
({8{data_in[11][4]}} & 8'h58)^
({8{data_in[11][5]}} & 8'hb0)^
({8{data_in[11][6]}} & 8'h4b)^
({8{data_in[11][7]}} & 8'h96)^
({8{data_in[12][0]}} & 8'h86)^
({8{data_in[12][1]}} & 8'h27)^
({8{data_in[12][2]}} & 8'h4e)^
({8{data_in[12][3]}} & 8'h9c)^
({8{data_in[12][4]}} & 8'h13)^
({8{data_in[12][5]}} & 8'h26)^
({8{data_in[12][6]}} & 8'h4c)^
({8{data_in[12][7]}} & 8'h98)^
({8{data_in[13][0]}} & 8'h2c)^
({8{data_in[13][1]}} & 8'h58)^
({8{data_in[13][2]}} & 8'hb0)^
({8{data_in[13][3]}} & 8'h4b)^
({8{data_in[13][4]}} & 8'h96)^
({8{data_in[13][5]}} & 8'h7)^
({8{data_in[13][6]}} & 8'he)^
({8{data_in[13][7]}} & 8'h1c)^
({8{data_in[14][0]}} & 8'hb)^
({8{data_in[14][1]}} & 8'h16)^
({8{data_in[14][2]}} & 8'h2c)^
({8{data_in[14][3]}} & 8'h58)^
({8{data_in[14][4]}} & 8'hb0)^
({8{data_in[14][5]}} & 8'h4b)^
({8{data_in[14][6]}} & 8'h96)^
({8{data_in[14][7]}} & 8'h7)^
({8{data_in[15][0]}} & 8'h7)^
({8{data_in[15][1]}} & 8'he)^
({8{data_in[15][2]}} & 8'h1c)^
({8{data_in[15][3]}} & 8'h38)^
({8{data_in[15][4]}} & 8'h70)^
({8{data_in[15][5]}} & 8'he0)^
({8{data_in[15][6]}} & 8'heb)^
({8{data_in[15][7]}} & 8'hfd)^
({8{data_in[16][0]}} & 8'h5f)^
({8{data_in[16][1]}} & 8'hbe)^
({8{data_in[16][2]}} & 8'h57)^
({8{data_in[16][3]}} & 8'hae)^
({8{data_in[16][4]}} & 8'h77)^
({8{data_in[16][5]}} & 8'hee)^
({8{data_in[16][6]}} & 8'hf7)^
({8{data_in[16][7]}} & 8'hc5)^
({8{data_in[17][0]}} & 8'h49)^
({8{data_in[17][1]}} & 8'h92)^
({8{data_in[17][2]}} & 8'hf)^
({8{data_in[17][3]}} & 8'h1e)^
({8{data_in[17][4]}} & 8'h3c)^
({8{data_in[17][5]}} & 8'h78)^

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```

({8{data_in[17][6]}} & 8'hf0)^
({8{data_in[17][7]}} & 8'hcb)^
({8{data_in[18][0]}} & 8'h6)^
({8{data_in[18][1]}} & 8'hc)^
({8{data_in[18][2]}} & 8'h18)^
({8{data_in[18][3]}} & 8'h30)^
({8{data_in[18][4]}} & 8'h60)^
({8{data_in[18][5]}} & 8'hc0)^
({8{data_in[18][6]}} & 8'hab)^
({8{data_in[18][7]}} & 8'h7d)^
({8{data_in[19][0]}} & 8'h63)^
({8{data_in[19][1]}} & 8'hc6)^
({8{data_in[19][2]}} & 8'ha7)^
({8{data_in[19][3]}} & 8'h65)^
({8{data_in[19][4]}} & 8'hca)^
({8{data_in[19][5]}} & 8'hbf)^
({8{data_in[19][6]}} & 8'h55)^
({8{data_in[19][7]}} & 8'haa)^
({8{data_in[20][0]}} & 8'h7f)^
({8{data_in[20][1]}} & 8'hfe)^
({8{data_in[20][2]}} & 8'hd7)^
({8{data_in[20][3]}} & 8'h85)^
({8{data_in[20][4]}} & 8'h21)^
({8{data_in[20][5]}} & 8'h42)^
({8{data_in[20][6]}} & 8'h84)^
({8{data_in[20][7]}} & 8'h23)^
({8{data_in[21][0]}} & 8'h5e)^
({8{data_in[21][1]}} & 8'hbc)^
({8{data_in[21][2]}} & 8'h53)^
({8{data_in[21][3]}} & 8'ha6)^
({8{data_in[21][4]}} & 8'h67)^
({8{data_in[21][5]}} & 8'hce)^
({8{data_in[21][6]}} & 8'hb7)^
({8{data_in[21][7]}} & 8'h45)^
({8{data_in[22][0]}} & 8'h84)^
({8{data_in[22][1]}} & 8'h23)^
({8{data_in[22][2]}} & 8'h46)^
({8{data_in[22][3]}} & 8'h8c)^
({8{data_in[22][4]}} & 8'h33)^
({8{data_in[22][5]}} & 8'h66)^
({8{data_in[22][6]}} & 8'hcc)^
({8{data_in[22][7]}} & 8'hb3)^
({8{data_in[23][0]}} & 8'hf5)^
({8{data_in[23][1]}} & 8'hc1)^
({8{data_in[23][2]}} & 8'ha9)^
({8{data_in[23][3]}} & 8'h79)^
({8{data_in[23][4]}} & 8'hf2)^
({8{data_in[23][5]}} & 8'hcf)^
({8{data_in[23][6]}} & 8'hb5)^
({8{data_in[23][7]}} & 8'h41)^
({8{data_in[24][0]}} & 8'ha4)^
({8{data_in[24][1]}} & 8'h63)^
({8{data_in[24][2]}} & 8'hc6)^
({8{data_in[24][3]}} & 8'ha7)^
({8{data_in[24][4]}} & 8'h65)^
({8{data_in[24][5]}} & 8'hca)^
({8{data_in[24][6]}} & 8'hbf)^
({8{data_in[24][7]}} & 8'h55)^
({8{data_in[25][0]}} & 8'h6f)^
({8{data_in[25][1]}} & 8'hde)^
({8{data_in[25][2]}} & 8'h97)^
({8{data_in[25][3]}} & 8'h5)^
({8{data_in[25][4]}} & 8'ha)^
({8{data_in[25][5]}} & 8'h14)^
({8{data_in[25][6]}} & 8'h28)^

```

```

({8{data_in[25][7]}} & 8'h50)^
({8{data_in[26][0]}} & 8'h72)^
({8{data_in[26][1]}} & 8'he4)^
({8{data_in[26][2]}} & 8'he3)^
({8{data_in[26][3]}} & 8'hed)^
({8{data_in[26][4]}} & 8'hf1)^
({8{data_in[26][5]}} & 8'hc9)^
({8{data_in[26][6]}} & 8'hb9)^
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({8{data_in[27][1]}} & 8'h50)^
({8{data_in[27][2]}} & 8'ha0)^
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({8{data_in[27][4]}} & 8'hd6)^
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({8{data_in[27][6]}} & 8'h25)^
({8{data_in[27][7]}} & 8'h4a)^
({8{data_in[28][0]}} & 8'h6f)^
({8{data_in[28][1]}} & 8'hde)^
({8{data_in[28][2]}} & 8'h97)^
({8{data_in[28][3]}} & 8'h5)^
({8{data_in[28][4]}} & 8'ha)^
({8{data_in[28][5]}} & 8'h14)^
({8{data_in[28][6]}} & 8'h28)^
({8{data_in[28][7]}} & 8'h50)^
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({8{data_in[29][1]}} & 8'h1c)^
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({8{data_in[29][3]}} & 8'h70)^
({8{data_in[29][4]}} & 8'he0)^
({8{data_in[29][5]}} & 8'heb)^
({8{data_in[29][6]}} & 8'hfd)^
({8{data_in[29][7]}} & 8'hd1)^
({8{data_in[30][0]}} & 8'h97)^
({8{data_in[30][1]}} & 8'h5)^
({8{data_in[30][2]}} & 8'ha)^
({8{data_in[30][3]}} & 8'h14)^
({8{data_in[30][4]}} & 8'h28)^
({8{data_in[30][5]}} & 8'h50)^
({8{data_in[30][6]}} & 8'ha0)^
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({8{data_in[31][2]}} & 8'hab)^
({8{data_in[31][3]}} & 8'h7d)^
({8{data_in[31][4]}} & 8'hfa)^
({8{data_in[31][5]}} & 8'hdf)^
({8{data_in[31][6]}} & 8'h95)^
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({8{data_in[32][0]}} & 8'hae)^
({8{data_in[32][1]}} & 8'h77)^
({8{data_in[32][2]}} & 8'hee)^
({8{data_in[32][3]}} & 8'hf7)^
({8{data_in[32][4]}} & 8'hc5)^
({8{data_in[32][5]}} & 8'ha1)^
({8{data_in[32][6]}} & 8'h69)^
({8{data_in[32][7]}} & 8'hd2)^
({8{data_in[33][0]}} & 8'h16)^
({8{data_in[33][1]}} & 8'h2c)^
({8{data_in[33][2]}} & 8'h58)^
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({8{data_in[33][4]}} & 8'h4b)^
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({8{data_in[33][6]}} & 8'h7)^
({8{data_in[33][7]}} & 8'he)^

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({8{data_in[34][4]}} & 8'ha5)^
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({8{data_in[34][7]}} & 8'haf)^
({8{data_in[35][0]}} & 8'h61)^
({8{data_in[35][1]}} & 8'hc2)^
({8{data_in[35][2]}} & 8'haf)^
({8{data_in[35][3]}} & 8'h75)^
({8{data_in[35][4]}} & 8'hea)^
({8{data_in[35][5]}} & 8'hff)^
({8{data_in[35][6]}} & 8'hd5)^
({8{data_in[35][7]}} & 8'h81)^
({8{data_in[36][0]}} & 8'h69)^
({8{data_in[36][1]}} & 8'hd2)^
({8{data_in[36][2]}} & 8'h8f)^
({8{data_in[36][3]}} & 8'h35)^
({8{data_in[36][4]}} & 8'h6a)^
({8{data_in[36][5]}} & 8'hd4)^
({8{data_in[36][6]}} & 8'h83)^
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({8{data_in[37][0]}} & 8'h77)^
({8{data_in[37][1]}} & 8'hee)^
({8{data_in[37][2]}} & 8'hf7)^
({8{data_in[37][3]}} & 8'hc5)^
({8{data_in[37][4]}} & 8'ha1)^
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({8{data_in[37][6]}} & 8'hd2)^
({8{data_in[37][7]}} & 8'h8f)^
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({8{data_in[38][1]}} & 8'h51)^
({8{data_in[38][2]}} & 8'ha2)^
({8{data_in[38][3]}} & 8'h6f)^
({8{data_in[38][4]}} & 8'hde)^
({8{data_in[38][5]}} & 8'h97)^
({8{data_in[38][6]}} & 8'h5)^
({8{data_in[38][7]}} & 8'ha)^
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({8{data_in[39][2]}} & 8'h1b)^
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({8{data_in[39][4]}} & 8'h6c)^
({8{data_in[39][5]}} & 8'hd8)^
({8{data_in[39][6]}} & 8'h9b)^
({8{data_in[39][7]}} & 8'h1d)^
({8{data_in[40][0]}} & 8'hc0)^
({8{data_in[40][1]}} & 8'hab)^
({8{data_in[40][2]}} & 8'h7d)^
({8{data_in[40][3]}} & 8'hfa)^
({8{data_in[40][4]}} & 8'hdf)^
({8{data_in[40][5]}} & 8'h95)^
({8{data_in[40][6]}} & 8'h1)^
({8{data_in[40][7]}} & 8'h2)^
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({8{data_in[41][1]}} & 8'he4)^
({8{data_in[41][2]}} & 8'he3)^
({8{data_in[41][3]}} & 8'hed)^
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({8{data_in[41][5]}} & 8'hc9)^
({8{data_in[41][6]}} & 8'hb9)^
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({8{data_in[42][5]}} & 8'h91)^
({8{data_in[42][6]}} & 8'h9)^
({8{data_in[42][7]}} & 8'h12)^
({8{data_in[43][0]}} & 8'ha1)^
({8{data_in[43][1]}} & 8'h69)^
({8{data_in[43][2]}} & 8'hd2)^
({8{data_in[43][3]}} & 8'h8f)^
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({8{data_in[43][5]}} & 8'h6a)^
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({8{data_in[43][7]}} & 8'h83)^
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({8{data_in[46][2]}} & 8'hcf)^
({8{data_in[46][3]}} & 8'hb5)^
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({8{data_in[46][5]}} & 8'h82)^
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({8{data_in[47][3]}} & 8'h6a)^
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({8{data_in[47][5]}} & 8'h83)^
({8{data_in[47][6]}} & 8'h2d)^
({8{data_in[47][7]}} & 8'h5a)^
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({8{data_in[48][4]}} & 8'h8b)^
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({8{data_in[48][6]}} & 8'h7a)^
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({8{data_in[50][1]}} & 8'ha6)^

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({8{data_in[52][2]}} & 8'hc7)^
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({8{data_in[52][4]}} & 8'h61)^
({8{data_in[52][5]}} & 8'hc2)^
({8{data_in[52][6]}} & 8'haf)^
({8{data_in[52][7]}} & 8'h75)^
({8{data_in[53][0]}} & 8'h51)^
({8{data_in[53][1]}} & 8'ha2)^
({8{data_in[53][2]}} & 8'h6f)^
({8{data_in[53][3]}} & 8'hde)^
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({8{data_in[54][6]}} & 8'haf)^
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({8{data_in[55][0]}} & 8'hc7)^
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({8{data_in[56][3]}} & 8'h2)^
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({8{data_in[57][2]}} & 8'h30)^
({8{data_in[57][3]}} & 8'h60)^
({8{data_in[57][4]}} & 8'hc0)^
({8{data_in[57][5]}} & 8'hab)^
({8{data_in[57][6]}} & 8'h7d)^
({8{data_in[57][7]}} & 8'hfa)^
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({8{data_in[59][1]}} & 8'hf1)^
({8{data_in[59][2]}} & 8'hc9)^
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({8{data_in[59][5]}} & 8'hb2)^
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({8{data_in[60][2]}} & 8'h6f)^
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({8{data_in[60][7]}} & 8'h14)^
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({8{data_in[62][7]}} & 8'h60)^
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({8{data_in[63][1]}} & 8'h3e)^
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({8{data_in[63][3]}} & 8'hf8)^
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({8{data_in[64][7]}} & 8'h7e)^
({8{data_in[65][0]}} & 8'hba)^
({8{data_in[65][1]}} & 8'h5f)^
({8{data_in[65][2]}} & 8'hbe)^
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({8{data_in[65][4]}} & 8'hae)^
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({8{data_in[67][3]}} & 8'hc7)^
({8{data_in[67][4]}} & 8'ha5)^
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({8{data_in[67][7]}} & 8'haf)^
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({8{data_in[76][6]}} & 8'h5f)^
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({8{data_in[77][2]}} & 8'h4f)^
({8{data_in[77][3]}} & 8'h9e)^
({8{data_in[77][4]}} & 8'h17)^
({8{data_in[77][5]}} & 8'h2e)^
({8{data_in[77][6]}} & 8'h5c)^
({8{data_in[77][7]}} & 8'hb8)^
({8{data_in[78][0]}} & 8'h47)^
({8{data_in[78][1]}} & 8'h8e)^
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({8{data_in[78][3]}} & 8'h6e)^
({8{data_in[78][4]}} & 8'hdc)^
({8{data_in[78][5]}} & 8'h93)^
({8{data_in[78][6]}} & 8'hd)^
({8{data_in[78][7]}} & 8'h1a)^
({8{data_in[79][0]}} & 8'h5d)^
({8{data_in[79][1]}} & 8'hba)^
({8{data_in[79][2]}} & 8'h5f)^
({8{data_in[79][3]}} & 8'hbe)^
({8{data_in[79][4]}} & 8'h57)^
({8{data_in[79][5]}} & 8'hae)^
({8{data_in[79][6]}} & 8'h77)^
({8{data_in[79][7]}} & 8'hee)^
({8{data_in[80][0]}} & 8'h6c)^
({8{data_in[80][1]}} & 8'hd8)^
({8{data_in[80][2]}} & 8'h9b)^
({8{data_in[80][3]}} & 8'h1d)^
({8{data_in[80][4]}} & 8'h3a)^
({8{data_in[80][5]}} & 8'h74)^
({8{data_in[80][6]}} & 8'he8)^
({8{data_in[80][7]}} & 8'hfb)^
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({8{data_in[81][1]}} & 8'he5)^
({8{data_in[81][2]}} & 8'he1)^
({8{data_in[81][3]}} & 8'h9e)^
({8{data_in[81][4]}} & 8'hf9)^
({8{data_in[81][5]}} & 8'hd9)^
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({8{data_in[81][7]}} & 8'h19)^
({8{data_in[82][0]}} & 8'h96)^
({8{data_in[82][1]}} & 8'h7)^
({8{data_in[82][2]}} & 8'he)^
({8{data_in[82][3]}} & 8'h1c)^
({8{data_in[82][4]}} & 8'h38)^
({8{data_in[82][5]}} & 8'h70)^

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({8{data_in[83][2]}} & 8'h3b)^
({8{data_in[83][3]}} & 8'h76)^
({8{data_in[83][4]}} & 8'hec)^
({8{data_in[83][5]}} & 8'hf3)^
({8{data_in[83][6]}} & 8'hcd)^
({8{data_in[83][7]}} & 8'hb1)^
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({8{data_in[84][1]}} & 8'ha6)^
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({8{data_in[84][3]}} & 8'hce)^
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({8{data_in[86][4]}} & 8'h83)^
({8{data_in[86][5]}} & 8'h2d)^
({8{data_in[86][6]}} & 8'h5a)^
({8{data_in[86][7]}} & 8'hb4)^
({8{data_in[87][0]}} & 8'h8c)^
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({8{data_in[87][2]}} & 8'h66)^
({8{data_in[87][3]}} & 8'hcc)^
({8{data_in[87][4]}} & 8'hb3)^
({8{data_in[87][5]}} & 8'h4d)^
({8{data_in[87][6]}} & 8'h9a)^
({8{data_in[87][7]}} & 8'h1f)^
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({8{data_in[88][1]}} & 8'h66)^
({8{data_in[88][2]}} & 8'hcc)^
({8{data_in[88][3]}} & 8'hb3)^
({8{data_in[88][4]}} & 8'h4d)^
({8{data_in[88][5]}} & 8'h9a)^
({8{data_in[88][6]}} & 8'h1f)^
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({8{data_in[89][1]}} & 8'h60)^
({8{data_in[89][2]}} & 8'hc0)^
({8{data_in[89][3]}} & 8'hab)^
({8{data_in[89][4]}} & 8'h7d)^
({8{data_in[89][5]}} & 8'hfa)^
({8{data_in[89][6]}} & 8'hdf)^
({8{data_in[89][7]}} & 8'h95)^
({8{data_in[90][0]}} & 8'h48)^
({8{data_in[90][1]}} & 8'h90)^
({8{data_in[90][2]}} & 8'hb)^
({8{data_in[90][3]}} & 8'h16)^
({8{data_in[90][4]}} & 8'h2c)^
({8{data_in[90][5]}} & 8'h58)^
({8{data_in[90][6]}} & 8'hb0)^

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({8{data_in[91][3]}} & 8'hc0)^
({8{data_in[91][4]}} & 8'hab)^
({8{data_in[91][5]}} & 8'h7d)^
({8{data_in[91][6]}} & 8'hfa)^
({8{data_in[91][7]}} & 8'hdf)^
({8{data_in[92][0]}} & 8'he8)^
({8{data_in[92][1]}} & 8'hfb)^
({8{data_in[92][2]}} & 8'hdd)^
({8{data_in[92][3]}} & 8'h91)^
({8{data_in[92][4]}} & 8'h9)^
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({8{data_in[93][1]}} & 8'h7f)^
({8{data_in[93][2]}} & 8'hfe)^
({8{data_in[93][3]}} & 8'hd7)^
({8{data_in[93][4]}} & 8'h85)^
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({8{data_in[94][1]}} & 8'h8b)^
({8{data_in[94][2]}} & 8'h3d)^
({8{data_in[94][3]}} & 8'h7a)^
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({8{data_in[94][5]}} & 8'hc3)^
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({8{data_in[95][1]}} & 8'h7e)^
({8{data_in[95][2]}} & 8'hfc)^
({8{data_in[95][3]}} & 8'hd3)^
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({8{data_in[96][3]}} & 8'h2c)^
({8{data_in[96][4]}} & 8'h58)^
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({8{data_in[96][6]}} & 8'h4b)^
({8{data_in[96][7]}} & 8'h96)^
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({8{data_in[97][4]}} & 8'h9b)^
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({8{data_in[97][6]}} & 8'h3a)^
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({8{data_in[98][4]}} & 8'h16)^
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({8{data_in[99][7]}} & 8'hd9)^
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({8{data_in[100][1]}} & 8'hd)^
({8{data_in[100][2]}} & 8'h1a)^
({8{data_in[100][3]}} & 8'h34)^
({8{data_in[100][4]}} & 8'h68)^
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({8{data_in[100][7]}} & 8'h3d)^
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({8{data_in[101][2]}} & 8'h7a)^
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({8{data_in[101][4]}} & 8'hc3)^
({8{data_in[101][5]}} & 8'had)^
({8{data_in[101][6]}} & 8'h71)^
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({8{data_in[103][1]}} & 8'h6e)^
({8{data_in[103][2]}} & 8'hdc)^
({8{data_in[103][3]}} & 8'h93)^
({8{data_in[103][4]}} & 8'hd)^
({8{data_in[103][5]}} & 8'h1a)^
({8{data_in[103][6]}} & 8'h34)^
({8{data_in[103][7]}} & 8'h68)^
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({8{data_in[104][4]}} & 8'h62)^
({8{data_in[104][5]}} & 8'hc4)^
({8{data_in[104][6]}} & 8'ha3)^
({8{data_in[104][7]}} & 8'h6d)^
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({8{data_in[105][2]}} & 8'h74)^
({8{data_in[105][3]}} & 8'he8)^
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({8{data_in[105][6]}} & 8'h91)^
({8{data_in[105][7]}} & 8'h9)^
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({8{data_in[107][7]}} & 8'h83)^
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({8{data_in[108][6]}} & 8'hf1)^
({8{data_in[108][7]}} & 8'hc9)^
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({8{data_in[109][2]}} & 8'hdf)^
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({8{data_in[109][4]}} & 8'h1)^
({8{data_in[109][5]}} & 8'h2)^
({8{data_in[109][6]}} & 8'h4)^
({8{data_in[109][7]}} & 8'h8)^
({8{data_in[110][0]}} & 8'hc0)^
({8{data_in[110][1]}} & 8'hab)^
({8{data_in[110][2]}} & 8'h7d)^
({8{data_in[110][3]}} & 8'hfa)^
({8{data_in[110][4]}} & 8'hdf)^
({8{data_in[110][5]}} & 8'h95)^
({8{data_in[110][6]}} & 8'h1)^
({8{data_in[110][7]}} & 8'h2)^
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({8{data_in[111][2]}} & 8'hab)^
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({8{data_in[111][4]}} & 8'hfa)^
({8{data_in[111][5]}} & 8'hdf)^
({8{data_in[111][6]}} & 8'h95)^
({8{data_in[111][7]}} & 8'h1)^
({8{data_in[112][0]}} & 8'h7e)^
({8{data_in[112][1]}} & 8'hfc)^
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({8{data_in[112][5]}} & 8'h62)^
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({8{data_in[113][7]}} & 8'hcd)^
({8{data_in[114][0]}} & 8'he4)^
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({8{data_in[114][2]}} & 8'hed)^
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({8{data_in[118][3]}} & 8'h8b)^
({8{data_in[118][4]}} & 8'h3d)^
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({8{data_in[119][5]}} & 8'h6e)^
({8{data_in[119][6]}} & 8'hdc)^
({8{data_in[119][7]}} & 8'h93)^
({8{data_in[120][0]}} & 8'hd8)^
({8{data_in[120][1]}} & 8'h9b)^
({8{data_in[120][2]}} & 8'h1d)^
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({8{data_in[120][6]}} & 8'hfb)^
({8{data_in[120][7]}} & 8'hdd)^
({8{data_in[121][0]}} & 8'hb9)^
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({8{data_in[121][2]}} & 8'hb2)^
({8{data_in[121][3]}} & 8'h4f)^
({8{data_in[121][4]}} & 8'h9e)^
({8{data_in[121][5]}} & 8'h17)^
({8{data_in[121][6]}} & 8'h2e)^
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({8{data_in[122][2]}} & 8'h8e)^
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({8{data_in[122][4]}} & 8'h6e)^
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({8{data_in[122][6]}} & 8'h93)^
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({8{data_in[124][5]}} & 8'hb)^
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({8{data_in[124][7]}} & 8'h2c)^
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({8{data_in[125][2]}} & 8'h10)^
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({8{data_in[126][3]}} & 8'hf7)^
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({8{data_in[126][6]}} & 8'h69)^
({8{data_in[126][7]}} & 8'hd2)^
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({8{data_in[127][3]}} & 8'h3d)^
({8{data_in[127][4]}} & 8'h7a)^
({8{data_in[127][5]}} & 8'hf4)^
({8{data_in[127][6]}} & 8'hc3)^
({8{data_in[127][7]}} & 8'had)^
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({8{data_in[128][1]}} & 8'h1b)^
({8{data_in[128][2]}} & 8'h36)^
({8{data_in[128][3]}} & 8'h6c)^
({8{data_in[128][4]}} & 8'hd8)^
({8{data_in[128][5]}} & 8'h9b)^
({8{data_in[128][6]}} & 8'h1d)^
({8{data_in[128][7]}} & 8'h3a)^
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({8{data_in[129][1]}} & 8'h75)^
({8{data_in[129][2]}} & 8'hea)^
({8{data_in[129][3]}} & 8'hff)^
({8{data_in[129][4]}} & 8'hd5)^
({8{data_in[129][5]}} & 8'h81)^
({8{data_in[129][6]}} & 8'h29)^
({8{data_in[129][7]}} & 8'h52)^
({8{data_in[130][0]}} & 8'h11)^
({8{data_in[130][1]}} & 8'h22)^
({8{data_in[130][2]}} & 8'h44)^
({8{data_in[130][3]}} & 8'h88)^
({8{data_in[130][4]}} & 8'h3b)^
({8{data_in[130][5]}} & 8'h76)^
({8{data_in[130][6]}} & 8'hec)^
({8{data_in[130][7]}} & 8'hf3)^
({8{data_in[131][0]}} & 8'h34)^
({8{data_in[131][1]}} & 8'h68)^
({8{data_in[131][2]}} & 8'hd0)^
({8{data_in[131][3]}} & 8'h8b)^

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({8{data_in[131][5]}} & 8'h7a)^
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({8{data_in[131][7]}} & 8'hc3)^
({8{data_in[132][0]}} & 8'hcb)^
({8{data_in[132][1]}} & 8'hbd)^
({8{data_in[132][2]}} & 8'h51)^
({8{data_in[132][3]}} & 8'ha2)^
({8{data_in[132][4]}} & 8'h6f)^
({8{data_in[132][5]}} & 8'hde)^
({8{data_in[132][6]}} & 8'h97)^
({8{data_in[132][7]}} & 8'h5)^
({8{data_in[133][0]}} & 8'h72)^
({8{data_in[133][1]}} & 8'he4)^
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({8{data_in[133][3]}} & 8'hed)^
({8{data_in[133][4]}} & 8'hf1)^
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({8{data_in[133][6]}} & 8'hb9)^
({8{data_in[133][7]}} & 8'h59)^
({8{data_in[134][0]}} & 8'h55)^
({8{data_in[134][1]}} & 8'haa)^
({8{data_in[134][2]}} & 8'h7f)^
({8{data_in[134][3]}} & 8'hfe)^
({8{data_in[134][4]}} & 8'hd7)^
({8{data_in[134][5]}} & 8'h85)^
({8{data_in[134][6]}} & 8'h21)^
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({8{data_in[135][0]}} & 8'hdf)^
({8{data_in[135][1]}} & 8'h95)^
({8{data_in[135][2]}} & 8'h1)^
({8{data_in[135][3]}} & 8'h2)^
({8{data_in[135][4]}} & 8'h4)^
({8{data_in[135][5]}} & 8'h8)^
({8{data_in[135][6]}} & 8'h10)^
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({8{data_in[136][0]}} & 8'h5c)^
({8{data_in[136][1]}} & 8'hb8)^
({8{data_in[136][2]}} & 8'h5b)^
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({8{data_in[136][4]}} & 8'h47)^
({8{data_in[136][5]}} & 8'h8e)^
({8{data_in[136][6]}} & 8'h37)^
({8{data_in[136][7]}} & 8'h6e)^
({8{data_in[137][0]}} & 8'h2)^
({8{data_in[137][1]}} & 8'h4)^
({8{data_in[137][2]}} & 8'h8)^
({8{data_in[137][3]}} & 8'h10)^
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({8{data_in[138][0]}} & 8'h55)^
({8{data_in[138][1]}} & 8'haa)^
({8{data_in[138][2]}} & 8'h7f)^
({8{data_in[138][3]}} & 8'hfe)^
({8{data_in[138][4]}} & 8'hd7)^
({8{data_in[138][5]}} & 8'h85)^
({8{data_in[138][6]}} & 8'h21)^
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({8{data_in[139][1]}} & 8'h94)^
({8{data_in[139][2]}} & 8'h3)^
({8{data_in[139][3]}} & 8'h6)^
({8{data_in[139][4]}} & 8'hc)^

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({8{data_in[140][1]}} & 8'h59)^
({8{data_in[140][2]}} & 8'hb2)^
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({8{data_in[140][4]}} & 8'h9e)^
({8{data_in[140][5]}} & 8'h17)^
({8{data_in[140][6]}} & 8'h2e)^
({8{data_in[140][7]}} & 8'h5c)^
({8{data_in[141][0]}} & 8'h55)^
({8{data_in[141][1]}} & 8'haa)^
({8{data_in[141][2]}} & 8'h7f)^
({8{data_in[141][3]}} & 8'hfe)^
({8{data_in[141][4]}} & 8'hd7)^
({8{data_in[141][5]}} & 8'h85)^
({8{data_in[141][6]}} & 8'h21)^
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({8{data_in[142][2]}} & 8'h23)^
({8{data_in[142][3]}} & 8'h46)^
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({8{data_in[142][6]}} & 8'h66)^
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({8{data_in[143][2]}} & 8'h6e)^
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({8{data_in[143][4]}} & 8'h93)^
({8{data_in[143][5]}} & 8'hd)^
({8{data_in[143][6]}} & 8'h1a)^
({8{data_in[143][7]}} & 8'h34)^
({8{data_in[144][0]}} & 8'h96)^
({8{data_in[144][1]}} & 8'h7)^
({8{data_in[144][2]}} & 8'he)^
({8{data_in[144][3]}} & 8'h1c)^
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({8{data_in[144][5]}} & 8'h70)^
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({8{data_in[144][7]}} & 8'heb)^
({8{data_in[145][0]}} & 8'hb1)^
({8{data_in[145][1]}} & 8'h49)^
({8{data_in[145][2]}} & 8'h92)^
({8{data_in[145][3]}} & 8'hf)^
({8{data_in[145][4]}} & 8'h1e)^
({8{data_in[145][5]}} & 8'h3c)^
({8{data_in[145][6]}} & 8'h78)^
({8{data_in[145][7]}} & 8'hf0)^
({8{data_in[146][0]}} & 8'h2c)^
({8{data_in[146][1]}} & 8'h58)^
({8{data_in[146][2]}} & 8'hb0)^
({8{data_in[146][3]}} & 8'h4b)^
({8{data_in[146][4]}} & 8'h96)^
({8{data_in[146][5]}} & 8'h7)^
({8{data_in[146][6]}} & 8'he)^
({8{data_in[146][7]}} & 8'h1c)^
({8{data_in[147][0]}} & 8'h8c)^
({8{data_in[147][1]}} & 8'h33)^
({8{data_in[147][2]}} & 8'h66)^
({8{data_in[147][3]}} & 8'hcc)^
({8{data_in[147][4]}} & 8'hb3)^
({8{data_in[147][5]}} & 8'h4d)^

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({8{data_in[147][6]}} & 8'h9a)^
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({8{data_in[148][1]}} & 8'hfa)^
({8{data_in[148][2]}} & 8'hdf)^
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({8{data_in[148][4]}} & 8'h1)^
({8{data_in[148][5]}} & 8'h2)^
({8{data_in[148][6]}} & 8'h4)^
({8{data_in[148][7]}} & 8'h8)^
({8{data_in[149][0]}} & 8'hb1)^
({8{data_in[149][1]}} & 8'h49)^
({8{data_in[149][2]}} & 8'h92)^
({8{data_in[149][3]}} & 8'hf)^
({8{data_in[149][4]}} & 8'h1e)^
({8{data_in[149][5]}} & 8'h3c)^
({8{data_in[149][6]}} & 8'h78)^
({8{data_in[149][7]}} & 8'hf0)^
({8{data_in[150][0]}} & 8'hd6)^
({8{data_in[150][1]}} & 8'h87)^
({8{data_in[150][2]}} & 8'h25)^
({8{data_in[150][3]}} & 8'h4a)^
({8{data_in[150][4]}} & 8'h94)^
({8{data_in[150][5]}} & 8'h3)^
({8{data_in[150][6]}} & 8'h6)^
({8{data_in[150][7]}} & 8'hc)^
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({8{data_in[151][1]}} & 8'hec)^
({8{data_in[151][2]}} & 8'hf3)^
({8{data_in[151][3]}} & 8'hcd)^
({8{data_in[151][4]}} & 8'hb1)^
({8{data_in[151][5]}} & 8'h49)^
({8{data_in[151][6]}} & 8'h92)^
({8{data_in[151][7]}} & 8'hf)^
({8{data_in[152][0]}} & 8'h93)^
({8{data_in[152][1]}} & 8'hd)^
({8{data_in[152][2]}} & 8'h1a)^
({8{data_in[152][3]}} & 8'h34)^
({8{data_in[152][4]}} & 8'h68)^
({8{data_in[152][5]}} & 8'hd0)^
({8{data_in[152][6]}} & 8'h8b)^
({8{data_in[152][7]}} & 8'h3d)^
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({8{data_in[153][2]}} & 8'h5)^
({8{data_in[153][3]}} & 8'ha)^
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({8{data_in[153][5]}} & 8'h28)^
({8{data_in[153][6]}} & 8'h50)^
({8{data_in[153][7]}} & 8'ha0)^
({8{data_in[154][0]}} & 8'h38)^
({8{data_in[154][1]}} & 8'h70)^
({8{data_in[154][2]}} & 8'he0)^
({8{data_in[154][3]}} & 8'heb)^
({8{data_in[154][4]}} & 8'hfd)^
({8{data_in[154][5]}} & 8'hd1)^
({8{data_in[154][6]}} & 8'h89)^
({8{data_in[154][7]}} & 8'h39)^
({8{data_in[155][0]}} & 8'hb8)^
({8{data_in[155][1]}} & 8'h5b)^
({8{data_in[155][2]}} & 8'hb6)^
({8{data_in[155][3]}} & 8'h47)^
({8{data_in[155][4]}} & 8'h8e)^
({8{data_in[155][5]}} & 8'h37)^
({8{data_in[155][6]}} & 8'h6e)^

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({8{data_in[156][2]}} & 8'h9f)^
({8{data_in[156][3]}} & 8'h15)^
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({8{data_in[156][6]}} & 8'ha8)^
({8{data_in[156][7]}} & 8'h7b)^
({8{data_in[157][0]}} & 8'hf8)^
({8{data_in[157][1]}} & 8'hdb)^
({8{data_in[157][2]}} & 8'h9d)^
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({8{data_in[158][1]}} & 8'he8)^
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({8{data_in[158][3]}} & 8'hdd)^
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({8{data_in[158][5]}} & 8'h9)^
({8{data_in[158][6]}} & 8'h12)^
({8{data_in[158][7]}} & 8'h24)^
({8{data_in[159][0]}} & 8'hd3)^
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({8{data_in[159][3]}} & 8'h62)^
({8{data_in[159][4]}} & 8'hc4)^
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({8{data_in[159][6]}} & 8'h6d)^
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({8{data_in[160][1]}} & 8'hf5)^
({8{data_in[160][2]}} & 8'hc1)^
({8{data_in[160][3]}} & 8'ha9)^
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({8{data_in[160][6]}} & 8'hcf)^
({8{data_in[160][7]}} & 8'hb5)^
({8{data_in[161][0]}} & 8'hc0)^
({8{data_in[161][1]}} & 8'hab)^
({8{data_in[161][2]}} & 8'h7d)^
({8{data_in[161][3]}} & 8'hfa)^
({8{data_in[161][4]}} & 8'hdf)^
({8{data_in[161][5]}} & 8'h95)^
({8{data_in[161][6]}} & 8'h1)^
({8{data_in[161][7]}} & 8'h2)^
({8{data_in[162][0]}} & 8'hd2)^
({8{data_in[162][1]}} & 8'h8f)^
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({8{data_in[162][4]}} & 8'hd4)^
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({8{data_in[162][6]}} & 8'h2d)^
({8{data_in[162][7]}} & 8'h5a)^
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({8{data_in[163][1]}} & 8'h46)^
({8{data_in[163][2]}} & 8'h8c)^
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({8{data_in[163][4]}} & 8'h66)^
({8{data_in[163][5]}} & 8'hcc)^
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({8{data_in[163][7]}} & 8'h4d)^

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({8{data_in[164][4]}} & 8'h8b)^
({8{data_in[164][5]}} & 8'h3d)^
({8{data_in[164][6]}} & 8'h7a)^
({8{data_in[164][7]}} & 8'hf4)^
({8{data_in[165][0]}} & 8'h2a)^
({8{data_in[165][1]}} & 8'h54)^
({8{data_in[165][2]}} & 8'ha8)^
({8{data_in[165][3]}} & 8'h7b)^
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({8{data_in[165][7]}} & 8'h61)^
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({8{data_in[166][2]}} & 8'h7a)^
({8{data_in[166][3]}} & 8'hf4)^
({8{data_in[166][4]}} & 8'hc3)^
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({8{data_in[167][0]}} & 8'h7)^
({8{data_in[167][1]}} & 8'he)^
({8{data_in[167][2]}} & 8'h1c)^
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({8{data_in[167][4]}} & 8'h70)^
({8{data_in[167][5]}} & 8'he0)^
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({8{data_in[167][7]}} & 8'hfd)^
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({8{data_in[168][2]}} & 8'h9b)^
({8{data_in[168][3]}} & 8'h1d)^
({8{data_in[168][4]}} & 8'h3a)^
({8{data_in[168][5]}} & 8'h74)^
({8{data_in[168][6]}} & 8'he8)^
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({8{data_in[169][2]}} & 8'h92)^
({8{data_in[169][3]}} & 8'hf)^
({8{data_in[169][4]}} & 8'h1e)^
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({8{data_in[170][0]}} & 8'h5f)^
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({8{data_in[170][3]}} & 8'hae)^
({8{data_in[170][4]}} & 8'h77)^
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({8{data_in[173][2]}} & 8'h91)^
({8{data_in[173][3]}} & 8'h9)^
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({8{data_in[175][4]}} & 8'hb7)^
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({8{data_in[175][6]}} & 8'h8a)^
({8{data_in[175][7]}} & 8'h3f)^
({8{data_in[176][0]}} & 8'h9e)^
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({8{data_in[176][2]}} & 8'h2e)^
({8{data_in[176][3]}} & 8'h5c)^
({8{data_in[176][4]}} & 8'hb8)^
({8{data_in[176][5]}} & 8'h5b)^
({8{data_in[176][6]}} & 8'hb6)^
({8{data_in[176][7]}} & 8'h47)^
({8{data_in[177][0]}} & 8'h9f)^
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({8{data_in[177][5]}} & 8'h7b)^
({8{data_in[177][6]}} & 8'hf6)^
({8{data_in[177][7]}} & 8'hc7)^
({8{data_in[178][0]}} & 8'h8e)^
({8{data_in[178][1]}} & 8'h37)^
({8{data_in[178][2]}} & 8'h6e)^
({8{data_in[178][3]}} & 8'hdc)^
({8{data_in[178][4]}} & 8'h93)^
({8{data_in[178][5]}} & 8'hd)^
({8{data_in[178][6]}} & 8'h1a)^
({8{data_in[178][7]}} & 8'h34)^
({8{data_in[179][0]}} & 8'hf9)^
({8{data_in[179][1]}} & 8'hd9)^
({8{data_in[179][2]}} & 8'h99)^
({8{data_in[179][3]}} & 8'h19)^
({8{data_in[179][4]}} & 8'h32)^
({8{data_in[179][5]}} & 8'h64)^
({8{data_in[179][6]}} & 8'hc8)^
({8{data_in[179][7]}} & 8'hbb)^
({8{data_in[180][0]}} & 8'h38)^
({8{data_in[180][1]}} & 8'h70)^

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({8{data_in[180][2]}} & 8'he0)^
({8{data_in[180][3]}} & 8'heb)^
({8{data_in[180][4]}} & 8'hfd)^
({8{data_in[180][5]}} & 8'hd1)^
({8{data_in[180][6]}} & 8'h89)^
({8{data_in[180][7]}} & 8'h39)^
({8{data_in[181][0]}} & 8'hf7)^
({8{data_in[181][1]}} & 8'hc5)^
({8{data_in[181][2]}} & 8'ha1)^
({8{data_in[181][3]}} & 8'h69)^
({8{data_in[181][4]}} & 8'hd2)^
({8{data_in[181][5]}} & 8'h8f)^
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({8{data_in[181][7]}} & 8'h6a)^
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({8{data_in[182][2]}} & 8'h61)^
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({8{data_in[182][6]}} & 8'hea)^
({8{data_in[182][7]}} & 8'hff)^
({8{data_in[183][0]}} & 8'he8)^
({8{data_in[183][1]}} & 8'hfb)^
({8{data_in[183][2]}} & 8'hdd)^
({8{data_in[183][3]}} & 8'h91)^
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({8{data_in[183][6]}} & 8'h24)^
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({8{data_in[184][1]}} & 8'ha6)^
({8{data_in[184][2]}} & 8'h67)^
({8{data_in[184][3]}} & 8'hce)^
({8{data_in[184][4]}} & 8'hb7)^
({8{data_in[184][5]}} & 8'h45)^
({8{data_in[184][6]}} & 8'h8a)^
({8{data_in[184][7]}} & 8'h3f)^
({8{data_in[185][0]}} & 8'he8)^
({8{data_in[185][1]}} & 8'hfb)^
({8{data_in[185][2]}} & 8'hdd)^
({8{data_in[185][3]}} & 8'h91)^
({8{data_in[185][4]}} & 8'h9)^
({8{data_in[185][5]}} & 8'h12)^
({8{data_in[185][6]}} & 8'h24)^
({8{data_in[185][7]}} & 8'h48)^
({8{data_in[186][0]}} & 8'h17)^
({8{data_in[186][1]}} & 8'h2e)^
({8{data_in[186][2]}} & 8'h5c)^
({8{data_in[186][3]}} & 8'hb8)^
({8{data_in[186][4]}} & 8'h5b)^
({8{data_in[186][5]}} & 8'hb6)^
({8{data_in[186][6]}} & 8'h47)^
({8{data_in[186][7]}} & 8'h8e)^
({8{data_in[187][0]}} & 8'h3a)^
({8{data_in[187][1]}} & 8'h74)^
({8{data_in[187][2]}} & 8'he8)^
({8{data_in[187][3]}} & 8'hfb)^
({8{data_in[187][4]}} & 8'hdd)^
({8{data_in[187][5]}} & 8'h91)^
({8{data_in[187][6]}} & 8'h9)^
({8{data_in[187][7]}} & 8'h12)^
({8{data_in[188][0]}} & 8'h3b)^
({8{data_in[188][1]}} & 8'h76)^
({8{data_in[188][2]}} & 8'hec)^

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({8{data_in[188][5]}} & 8'hb1)^
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({8{data_in[188][7]}} & 8'h92)^
({8{data_in[189][0]}} & 8'h46)^
({8{data_in[189][1]}} & 8'h8c)^
({8{data_in[189][2]}} & 8'h33)^
({8{data_in[189][3]}} & 8'h66)^
({8{data_in[189][4]}} & 8'hcc)^
({8{data_in[189][5]}} & 8'hb3)^
({8{data_in[189][6]}} & 8'h4d)^
({8{data_in[189][7]}} & 8'h9a)^
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({8{data_in[190][1]}} & 8'h1)^
({8{data_in[190][2]}} & 8'h2)^
({8{data_in[190][3]}} & 8'h4)^
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({8{data_in[191][1]}} & 8'hcc)^
({8{data_in[191][2]}} & 8'hb3)^
({8{data_in[191][3]}} & 8'h4d)^
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({8{data_in[191][5]}} & 8'h1f)^
({8{data_in[191][6]}} & 8'h3e)^
({8{data_in[191][7]}} & 8'h7c)^
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({8{data_in[192][1]}} & 8'h5d)^
({8{data_in[192][2]}} & 8'hba)^
({8{data_in[192][3]}} & 8'h5f)^
({8{data_in[192][4]}} & 8'hbe)^
({8{data_in[192][5]}} & 8'h57)^
({8{data_in[192][6]}} & 8'hae)^
({8{data_in[192][7]}} & 8'h77)^
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({8{data_in[193][1]}} & 8'h7c)^
({8{data_in[193][2]}} & 8'hf8)^
({8{data_in[193][3]}} & 8'hdb)^
({8{data_in[193][4]}} & 8'h9d)^
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({8{data_in[193][6]}} & 8'h22)^
({8{data_in[193][7]}} & 8'h44)^
({8{data_in[194][0]}} & 8'h40)^
({8{data_in[194][1]}} & 8'h80)^
({8{data_in[194][2]}} & 8'h2b)^
({8{data_in[194][3]}} & 8'h56)^
({8{data_in[194][4]}} & 8'hac)^
({8{data_in[194][5]}} & 8'h73)^
({8{data_in[194][6]}} & 8'he6)^
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({8{data_in[195][2]}} & 8'h87)^
({8{data_in[195][3]}} & 8'h25)^
({8{data_in[195][4]}} & 8'h4a)^
({8{data_in[195][5]}} & 8'h94)^
({8{data_in[195][6]}} & 8'h3)^
({8{data_in[195][7]}} & 8'h6)^
({8{data_in[196][0]}} & 8'h4d)^
({8{data_in[196][1]}} & 8'h9a)^
({8{data_in[196][2]}} & 8'h1f)^
({8{data_in[196][3]}} & 8'h3e)^

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({8{data_in[197][0]}} & 8'h9a)^
({8{data_in[197][1]}} & 8'h1f)^
({8{data_in[197][2]}} & 8'h3e)^
({8{data_in[197][3]}} & 8'h7c)^
({8{data_in[197][4]}} & 8'hf8)^
({8{data_in[197][5]}} & 8'hdb)^
({8{data_in[197][6]}} & 8'h9d)^
({8{data_in[197][7]}} & 8'h11)^
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({8{data_in[198][7]}} & 8'h6c)^
({8{data_in[199][0]}} & 8'h93)^
({8{data_in[199][1]}} & 8'hd)^
({8{data_in[199][2]}} & 8'h1a)^
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({8{data_in[199][4]}} & 8'h68)^
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({8{data_in[199][6]}} & 8'h8b)^
({8{data_in[199][7]}} & 8'h3d)^
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({8{data_in[200][6]}} & 8'h2c)^
({8{data_in[200][7]}} & 8'h58)^
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({8{data_in[202][3]}} & 8'h83)^
({8{data_in[202][4]}} & 8'h2d)^
({8{data_in[202][5]}} & 8'h5a)^
({8{data_in[202][6]}} & 8'hb4)^
({8{data_in[202][7]}} & 8'h43)^
({8{data_in[203][0]}} & 8'hcf)^
({8{data_in[203][1]}} & 8'hb5)^
({8{data_in[203][2]}} & 8'h41)^
({8{data_in[203][3]}} & 8'h82)^
({8{data_in[203][4]}} & 8'h2f)^
({8{data_in[203][5]}} & 8'h5e)^
({8{data_in[203][6]}} & 8'hbc)^
({8{data_in[203][7]}} & 8'h53)^
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({8{data_in[204][1]}} & 8'h25)^
({8{data_in[204][2]}} & 8'h4a)^
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({8{data_in[204][4]}} & 8'h3)^

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({8{data_in[205][5]}} & 8'hef)^
({8{data_in[205][6]}} & 8'hf5)^
({8{data_in[205][7]}} & 8'hc1)^
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({8{data_in[206][1]}} & 8'h53)^
({8{data_in[206][2]}} & 8'ha6)^
({8{data_in[206][3]}} & 8'h67)^
({8{data_in[206][4]}} & 8'hce)^
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({8{data_in[207][4]}} & 8'h80)^
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({8{data_in[208][4]}} & 8'h6c)^
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({8{data_in[208][6]}} & 8'h9b)^
({8{data_in[208][7]}} & 8'h1d)^
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({8{data_in[209][1]}} & 8'ha7)^
({8{data_in[209][2]}} & 8'h65)^
({8{data_in[209][3]}} & 8'hca)^
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({8{data_in[209][5]}} & 8'h55)^
({8{data_in[209][6]}} & 8'haa)^
({8{data_in[209][7]}} & 8'h7f)^
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({8{data_in[210][4]}} & 8'hc4)^
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({8{data_in[213][4]}} & 8'h8e)^
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({8{data_in[213][6]}} & 8'h6e)^
({8{data_in[213][7]}} & 8'hdc)^
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({8{data_in[215][7]}} & 8'hbb)^
({8{data_in[216][0]}} & 8'hcb)^
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({8{data_in[216][7]}} & 8'h5)^
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({8{data_in[219][3]}} & 8'hba)^
({8{data_in[219][4]}} & 8'h5f)^
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({8{data_in[220][1]}} & 8'h93)^
({8{data_in[220][2]}} & 8'hd)^
({8{data_in[220][3]}} & 8'h1a)^
({8{data_in[220][4]}} & 8'h34)^
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({8{data_in[222][4]}} & 8'h44)^
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({8{data_in[223][4]}} & 8'h6e)^
({8{data_in[223][5]}} & 8'hdc)^
({8{data_in[223][6]}} & 8'h93)^
({8{data_in[223][7]}} & 8'hd)^
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({8{data_in[225][1]}} & 8'hef)^
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({8{data_in[225][3]}} & 8'hc1)^
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({8{data_in[226][3]}} & 8'h54)^
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({8{data_in[227][3]}} & 8'h4)^
({8{data_in[227][4]}} & 8'h8)^
({8{data_in[227][5]}} & 8'h10)^
({8{data_in[227][6]}} & 8'h20)^
({8{data_in[227][7]}} & 8'h40)^
({8{data_in[228][0]}} & 8'h9c)^
({8{data_in[228][1]}} & 8'h13)^
({8{data_in[228][2]}} & 8'h26)^
({8{data_in[228][3]}} & 8'h4c)^
({8{data_in[228][4]}} & 8'h98)^
({8{data_in[228][5]}} & 8'h1b)^
({8{data_in[228][6]}} & 8'h36)^
({8{data_in[228][7]}} & 8'h6c)^

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({8{data_in[229][0]}} & 8'h9e)^
({8{data_in[229][1]}} & 8'h17)^
({8{data_in[229][2]}} & 8'h2e)^
({8{data_in[229][3]}} & 8'h5c)^
({8{data_in[229][4]}} & 8'hb8)^
({8{data_in[229][5]}} & 8'h5b)^
({8{data_in[229][6]}} & 8'hb6)^
({8{data_in[229][7]}} & 8'h47)^
({8{data_in[230][0]}} & 8'h1a)^
({8{data_in[230][1]}} & 8'h34)^
({8{data_in[230][2]}} & 8'h68)^
({8{data_in[230][3]}} & 8'hd0)^
({8{data_in[230][4]}} & 8'h8b)^
({8{data_in[230][5]}} & 8'h3d)^
({8{data_in[230][6]}} & 8'h7a)^
({8{data_in[230][7]}} & 8'hf4)^
({8{data_in[231][0]}} & 8'h6f)^
({8{data_in[231][1]}} & 8'hde)^
({8{data_in[231][2]}} & 8'h97)^
({8{data_in[231][3]}} & 8'h5)^
({8{data_in[231][4]}} & 8'ha)^
({8{data_in[231][5]}} & 8'h14)^
({8{data_in[231][6]}} & 8'h28)^
({8{data_in[231][7]}} & 8'h50)^
({8{data_in[232][0]}} & 8'h69)^
({8{data_in[232][1]}} & 8'hd2)^
({8{data_in[232][2]}} & 8'h8f)^
({8{data_in[232][3]}} & 8'h35)^
({8{data_in[232][4]}} & 8'h6a)^
({8{data_in[232][5]}} & 8'hd4)^
({8{data_in[232][6]}} & 8'h83)^
({8{data_in[232][7]}} & 8'h2d)^
({8{data_in[233][0]}} & 8'h8f)^
({8{data_in[233][1]}} & 8'h35)^
({8{data_in[233][2]}} & 8'h6a)^
({8{data_in[233][3]}} & 8'hd4)^
({8{data_in[233][4]}} & 8'h83)^
({8{data_in[233][5]}} & 8'h2d)^
({8{data_in[233][6]}} & 8'h5a)^
({8{data_in[233][7]}} & 8'hb4)^
({8{data_in[234][0]}} & 8'h9e)^
({8{data_in[234][1]}} & 8'h17)^
({8{data_in[234][2]}} & 8'h2e)^
({8{data_in[234][3]}} & 8'h5c)^
({8{data_in[234][4]}} & 8'hb8)^
({8{data_in[234][5]}} & 8'h5b)^
({8{data_in[234][6]}} & 8'hb6)^
({8{data_in[234][7]}} & 8'h47)^
({8{data_in[235][0]}} & 8'h5f)^
({8{data_in[235][1]}} & 8'hbe)^
({8{data_in[235][2]}} & 8'h57)^
({8{data_in[235][3]}} & 8'hae)^
({8{data_in[235][4]}} & 8'h77)^
({8{data_in[235][5]}} & 8'hee)^
({8{data_in[235][6]}} & 8'hf7)^
({8{data_in[235][7]}} & 8'hc5)^
({8{data_in[236][0]}} & 8'h8c)^
({8{data_in[236][1]}} & 8'h33)^
({8{data_in[236][2]}} & 8'h66)^
({8{data_in[236][3]}} & 8'hcc)^
({8{data_in[236][4]}} & 8'hb3)^
({8{data_in[236][5]}} & 8'h4d)^
({8{data_in[236][6]}} & 8'h9a)^
({8{data_in[236][7]}} & 8'h1f)^
({8{data_in[237][0]}} & 8'h17)^

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```

({8{data_in[237][1]}} & 8'h2e)^
({8{data_in[237][2]}} & 8'h5c)^
({8{data_in[237][3]}} & 8'hb8)^
({8{data_in[237][4]}} & 8'h5b)^
({8{data_in[237][5]}} & 8'hb6)^
({8{data_in[237][6]}} & 8'h47)^
({8{data_in[237][7]}} & 8'h8e)^
({8{data_in[238][0]}} & 8'h96)^
({8{data_in[238][1]}} & 8'h7)^
({8{data_in[238][2]}} & 8'he)^
({8{data_in[238][3]}} & 8'h1c)^
({8{data_in[238][4]}} & 8'h38)^
({8{data_in[238][5]}} & 8'h70)^
({8{data_in[238][6]}} & 8'he0)^
({8{data_in[238][7]}} & 8'heb)^
({8{data_in[239][0]}} & 8'h95)^
({8{data_in[239][1]}} & 8'h1)^
({8{data_in[239][2]}} & 8'h2)^
({8{data_in[239][3]}} & 8'h4)^
({8{data_in[239][4]}} & 8'h8)^
({8{data_in[239][5]}} & 8'h10)^
({8{data_in[239][6]}} & 8'h20)^
({8{data_in[239][7]}} & 8'h40)^
({8{data_in[240][0]}} & 8'h89)^
({8{data_in[240][1]}} & 8'h39)^
({8{data_in[240][2]}} & 8'h72)^
({8{data_in[240][3]}} & 8'he4)^
({8{data_in[240][4]}} & 8'he3)^
({8{data_in[240][5]}} & 8'hed)^
({8{data_in[240][6]}} & 8'hf1)^
({8{data_in[240][7]}} & 8'hc9)^
({8{data_in[241][0]}} & 8'hfe)^
({8{data_in[241][1]}} & 8'hd7)^
({8{data_in[241][2]}} & 8'h85)^
({8{data_in[241][3]}} & 8'h21)^
({8{data_in[241][4]}} & 8'h42)^
({8{data_in[241][5]}} & 8'h84)^
({8{data_in[241][6]}} & 8'h23)^
({8{data_in[241][7]}} & 8'h46);
data_out[246] = ({8{data_in[0][0]}} & 8'he9)^
({8{data_in[0][1]}} & 8'hf9)^
({8{data_in[0][2]}} & 8'hd9)^
({8{data_in[0][3]}} & 8'h99)^
({8{data_in[0][4]}} & 8'h19)^
({8{data_in[0][5]}} & 8'h32)^
({8{data_in[0][6]}} & 8'h64)^
({8{data_in[0][7]}} & 8'hc8)^
({8{data_in[1][0]}} & 8'hdf)^
({8{data_in[1][1]}} & 8'h95)^
({8{data_in[1][2]}} & 8'h1)^
({8{data_in[1][3]}} & 8'h2)^
({8{data_in[1][4]}} & 8'h4)^
({8{data_in[1][5]}} & 8'h8)^
({8{data_in[1][6]}} & 8'h10)^
({8{data_in[1][7]}} & 8'h20)^
({8{data_in[2][0]}} & 8'h14)^
({8{data_in[2][1]}} & 8'h28)^
({8{data_in[2][2]}} & 8'h50)^
({8{data_in[2][3]}} & 8'ha0)^
({8{data_in[2][4]}} & 8'h6b)^
({8{data_in[2][5]}} & 8'hd6)^
({8{data_in[2][6]}} & 8'h87)^
({8{data_in[2][7]}} & 8'h25)^
({8{data_in[3][0]}} & 8'h79)^
({8{data_in[3][1]}} & 8'hf2)^

```

```

({8{data_in[3][2]}} & 8'hcf)^
({8{data_in[3][3]}} & 8'hb5)^
({8{data_in[3][4]}} & 8'h41)^
({8{data_in[3][5]}} & 8'h82)^
({8{data_in[3][6]}} & 8'h2f)^
({8{data_in[3][7]}} & 8'h5e)^
({8{data_in[4][0]}} & 8'h92)^
({8{data_in[4][1]}} & 8'hf)^
({8{data_in[4][2]}} & 8'h1e)^
({8{data_in[4][3]}} & 8'h3c)^
({8{data_in[4][4]}} & 8'h78)^
({8{data_in[4][5]}} & 8'hf0)^
({8{data_in[4][6]}} & 8'hcb)^
({8{data_in[4][7]}} & 8'hbd)^
({8{data_in[5][0]}} & 8'h62)^
({8{data_in[5][1]}} & 8'hc4)^
({8{data_in[5][2]}} & 8'ha3)^
({8{data_in[5][3]}} & 8'h6d)^
({8{data_in[5][4]}} & 8'hda)^
({8{data_in[5][5]}} & 8'h9f)^
({8{data_in[5][6]}} & 8'h15)^
({8{data_in[5][7]}} & 8'h2a)^
({8{data_in[6][0]}} & 8'ha9)^
({8{data_in[6][1]}} & 8'h79)^
({8{data_in[6][2]}} & 8'hf2)^
({8{data_in[6][3]}} & 8'hcf)^
({8{data_in[6][4]}} & 8'hb5)^
({8{data_in[6][5]}} & 8'h41)^
({8{data_in[6][6]}} & 8'h82)^
({8{data_in[6][7]}} & 8'h2f)^
({8{data_in[7][0]}} & 8'hec)^
({8{data_in[7][1]}} & 8'hf3)^
({8{data_in[7][2]}} & 8'hcd)^
({8{data_in[7][3]}} & 8'hb1)^
({8{data_in[7][4]}} & 8'h49)^
({8{data_in[7][5]}} & 8'h92)^
({8{data_in[7][6]}} & 8'hf)^
({8{data_in[7][7]}} & 8'h1e)^
({8{data_in[8][0]}} & 8'hbb)^
({8{data_in[8][1]}} & 8'h5d)^
({8{data_in[8][2]}} & 8'hba)^
({8{data_in[8][3]}} & 8'h5f)^
({8{data_in[8][4]}} & 8'hbe)^
({8{data_in[8][5]}} & 8'h57)^
({8{data_in[8][6]}} & 8'hae)^
({8{data_in[8][7]}} & 8'h77)^
({8{data_in[9][0]}} & 8'h37)^
({8{data_in[9][1]}} & 8'h6e)^
({8{data_in[9][2]}} & 8'hdc)^
({8{data_in[9][3]}} & 8'h93)^
({8{data_in[9][4]}} & 8'hd)^
({8{data_in[9][5]}} & 8'h1a)^
({8{data_in[9][6]}} & 8'h34)^
({8{data_in[9][7]}} & 8'h68)^
({8{data_in[10][0]}} & 8'h52)^
({8{data_in[10][1]}} & 8'ha4)^
({8{data_in[10][2]}} & 8'h63)^
({8{data_in[10][3]}} & 8'hc6)^
({8{data_in[10][4]}} & 8'ha7)^
({8{data_in[10][5]}} & 8'h65)^
({8{data_in[10][6]}} & 8'hca)^
({8{data_in[10][7]}} & 8'hbf)^
({8{data_in[11][0]}} & 8'hca)^
({8{data_in[11][1]}} & 8'hbf)^
({8{data_in[11][2]}} & 8'h55)^

```

```

({8{data_in[11][3]}} & 8'haa)^
({8{data_in[11][4]}} & 8'h7f)^
({8{data_in[11][5]}} & 8'hfe)^
({8{data_in[11][6]}} & 8'hd7)^
({8{data_in[11][7]}} & 8'h85)^
({8{data_in[12][0]}} & 8'hd1)^
({8{data_in[12][1]}} & 8'h89)^
({8{data_in[12][2]}} & 8'h39)^
({8{data_in[12][3]}} & 8'h72)^
({8{data_in[12][4]}} & 8'he4)^
({8{data_in[12][5]}} & 8'he3)^
({8{data_in[12][6]}} & 8'hed)^
({8{data_in[12][7]}} & 8'hf1)^
({8{data_in[13][0]}} & 8'h79)^
({8{data_in[13][1]}} & 8'hf2)^
({8{data_in[13][2]}} & 8'hcf)^
({8{data_in[13][3]}} & 8'hb5)^
({8{data_in[13][4]}} & 8'h41)^
({8{data_in[13][5]}} & 8'h82)^
({8{data_in[13][6]}} & 8'h2f)^
({8{data_in[13][7]}} & 8'h5e)^
({8{data_in[14][0]}} & 8'h99)^
({8{data_in[14][1]}} & 8'h19)^
({8{data_in[14][2]}} & 8'h32)^
({8{data_in[14][3]}} & 8'h64)^
({8{data_in[14][4]}} & 8'hc8)^
({8{data_in[14][5]}} & 8'hbb)^
({8{data_in[14][6]}} & 8'h5d)^
({8{data_in[14][7]}} & 8'hba)^
({8{data_in[15][0]}} & 8'he2)^
({8{data_in[15][1]}} & 8'hcf)^
({8{data_in[15][2]}} & 8'hf5)^
({8{data_in[15][3]}} & 8'hc1)^
({8{data_in[15][4]}} & 8'ha9)^
({8{data_in[15][5]}} & 8'h79)^
({8{data_in[15][6]}} & 8'hf2)^
({8{data_in[15][7]}} & 8'hcf)^
({8{data_in[16][0]}} & 8'hac)^
({8{data_in[16][1]}} & 8'h73)^
({8{data_in[16][2]}} & 8'he6)^
({8{data_in[16][3]}} & 8'he7)^
({8{data_in[16][4]}} & 8'he5)^
({8{data_in[16][5]}} & 8'he1)^
({8{data_in[16][6]}} & 8'he9)^
({8{data_in[16][7]}} & 8'hf9)^
({8{data_in[17][0]}} & 8'hcf)^
({8{data_in[17][1]}} & 8'hb5)^
({8{data_in[17][2]}} & 8'h41)^
({8{data_in[17][3]}} & 8'h82)^
({8{data_in[17][4]}} & 8'h2f)^
({8{data_in[17][5]}} & 8'h5e)^
({8{data_in[17][6]}} & 8'hbc)^
({8{data_in[17][7]}} & 8'h53)^
({8{data_in[18][0]}} & 8'h66)^
({8{data_in[18][1]}} & 8'hcc)^
({8{data_in[18][2]}} & 8'hb3)^
({8{data_in[18][3]}} & 8'h4d)^
({8{data_in[18][4]}} & 8'h9a)^
({8{data_in[18][5]}} & 8'h1f)^
({8{data_in[18][6]}} & 8'h3e)^
({8{data_in[18][7]}} & 8'h7c)^
({8{data_in[19][0]}} & 8'h9d)^
({8{data_in[19][1]}} & 8'h11)^
({8{data_in[19][2]}} & 8'h22)^
({8{data_in[19][3]}} & 8'h44)^

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```

({8{data_in[19][4]}} & 8'h88)^
({8{data_in[19][5]}} & 8'h3b)^
({8{data_in[19][6]}} & 8'h76)^
({8{data_in[19][7]}} & 8'hec)^
({8{data_in[20][0]}} & 8'h4f)^
({8{data_in[20][1]}} & 8'h9e)^
({8{data_in[20][2]}} & 8'h17)^
({8{data_in[20][3]}} & 8'h2e)^
({8{data_in[20][4]}} & 8'h5c)^
({8{data_in[20][5]}} & 8'hb8)^
({8{data_in[20][6]}} & 8'h5b)^
({8{data_in[20][7]}} & 8'hb6)^
({8{data_in[21][0]}} & 8'ha4)^
({8{data_in[21][1]}} & 8'h63)^
({8{data_in[21][2]}} & 8'hc6)^
({8{data_in[21][3]}} & 8'ha7)^
({8{data_in[21][4]}} & 8'h65)^
({8{data_in[21][5]}} & 8'hca)^
({8{data_in[21][6]}} & 8'hbf)^
({8{data_in[21][7]}} & 8'h55)^
({8{data_in[22][0]}} & 8'hb2)^
({8{data_in[22][1]}} & 8'h4f)^
({8{data_in[22][2]}} & 8'h9e)^
({8{data_in[22][3]}} & 8'h17)^
({8{data_in[22][4]}} & 8'h2e)^
({8{data_in[22][5]}} & 8'h5c)^
({8{data_in[22][6]}} & 8'hb8)^
({8{data_in[22][7]}} & 8'h5b)^
({8{data_in[23][0]}} & 8'h94)^
({8{data_in[23][1]}} & 8'h3)^
({8{data_in[23][2]}} & 8'h6)^
({8{data_in[23][3]}} & 8'hc)^
({8{data_in[23][4]}} & 8'h18)^
({8{data_in[23][5]}} & 8'h30)^
({8{data_in[23][6]}} & 8'h60)^
({8{data_in[23][7]}} & 8'hc0)^
({8{data_in[24][0]}} & 8'ha9)^
({8{data_in[24][1]}} & 8'h79)^
({8{data_in[24][2]}} & 8'hf2)^
({8{data_in[24][3]}} & 8'hcf)^
({8{data_in[24][4]}} & 8'hb5)^
({8{data_in[24][5]}} & 8'h41)^
({8{data_in[24][6]}} & 8'h82)^
({8{data_in[24][7]}} & 8'h2f)^
({8{data_in[25][0]}} & 8'h14)^
({8{data_in[25][1]}} & 8'h28)^
({8{data_in[25][2]}} & 8'h50)^
({8{data_in[25][3]}} & 8'ha0)^
({8{data_in[25][4]}} & 8'h6b)^
({8{data_in[25][5]}} & 8'hd6)^
({8{data_in[25][6]}} & 8'h87)^
({8{data_in[25][7]}} & 8'h25)^
({8{data_in[26][0]}} & 8'h60)^
({8{data_in[26][1]}} & 8'hc0)^
({8{data_in[26][2]}} & 8'hab)^
({8{data_in[26][3]}} & 8'h7d)^
({8{data_in[26][4]}} & 8'hfa)^
({8{data_in[26][5]}} & 8'hdf)^
({8{data_in[26][6]}} & 8'h95)^
({8{data_in[26][7]}} & 8'h1)^
({8{data_in[27][0]}} & 8'he8)^
({8{data_in[27][1]}} & 8'hfb)^
({8{data_in[27][2]}} & 8'hdd)^
({8{data_in[27][3]}} & 8'h91)^
({8{data_in[27][4]}} & 8'h9)^

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({8{data_in[27][5]}} & 8'h12)^
({8{data_in[27][6]}} & 8'h24)^
({8{data_in[27][7]}} & 8'h48)^
({8{data_in[28][0]}} & 8'h5b)^
({8{data_in[28][1]}} & 8'hb6)^
({8{data_in[28][2]}} & 8'h47)^
({8{data_in[28][3]}} & 8'h8e)^
({8{data_in[28][4]}} & 8'h37)^
({8{data_in[28][5]}} & 8'h6e)^
({8{data_in[28][6]}} & 8'hdc)^
({8{data_in[28][7]}} & 8'h93)^
({8{data_in[29][0]}} & 8'hb)^
({8{data_in[29][1]}} & 8'h16)^
({8{data_in[29][2]}} & 8'h2c)^
({8{data_in[29][3]}} & 8'h58)^
({8{data_in[29][4]}} & 8'hb0)^
({8{data_in[29][5]}} & 8'h4b)^
({8{data_in[29][6]}} & 8'h96)^
({8{data_in[29][7]}} & 8'h7)^
({8{data_in[30][0]}} & 8'ha1)^
({8{data_in[30][1]}} & 8'h69)^
({8{data_in[30][2]}} & 8'hd2)^
({8{data_in[30][3]}} & 8'h8f)^
({8{data_in[30][4]}} & 8'h35)^
({8{data_in[30][5]}} & 8'h6a)^
({8{data_in[30][6]}} & 8'hd4)^
({8{data_in[30][7]}} & 8'h83)^
({8{data_in[31][0]}} & 8'h1c)^
({8{data_in[31][1]}} & 8'h38)^
({8{data_in[31][2]}} & 8'h70)^
({8{data_in[31][3]}} & 8'he0)^
({8{data_in[31][4]}} & 8'heb)^
({8{data_in[31][5]}} & 8'hfd)^
({8{data_in[31][6]}} & 8'hd1)^
({8{data_in[31][7]}} & 8'h89)^
({8{data_in[32][0]}} & 8'hf1)^
({8{data_in[32][1]}} & 8'hc9)^
({8{data_in[32][2]}} & 8'hb9)^
({8{data_in[32][3]}} & 8'h59)^
({8{data_in[32][4]}} & 8'hb2)^
({8{data_in[32][5]}} & 8'h4f)^
({8{data_in[32][6]}} & 8'h9e)^
({8{data_in[32][7]}} & 8'h17)^
({8{data_in[33][0]}} & 8'hde)^
({8{data_in[33][1]}} & 8'h97)^
({8{data_in[33][2]}} & 8'h5)^
({8{data_in[33][3]}} & 8'ha)^
({8{data_in[33][4]}} & 8'h14)^
({8{data_in[33][5]}} & 8'h28)^
({8{data_in[33][6]}} & 8'h50)^
({8{data_in[33][7]}} & 8'ha0)^
({8{data_in[34][0]}} & 8'h9b)^
({8{data_in[34][1]}} & 8'h1d)^
({8{data_in[34][2]}} & 8'h3a)^
({8{data_in[34][3]}} & 8'h74)^
({8{data_in[34][4]}} & 8'he8)^
({8{data_in[34][5]}} & 8'hfb)^
({8{data_in[34][6]}} & 8'hdd)^
({8{data_in[34][7]}} & 8'h91)^
({8{data_in[35][0]}} & 8'hed)^
({8{data_in[35][1]}} & 8'hf1)^
({8{data_in[35][2]}} & 8'hc9)^
({8{data_in[35][3]}} & 8'hb9)^
({8{data_in[35][4]}} & 8'h59)^
({8{data_in[35][5]}} & 8'hb2)^

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({8{data_in[36][1]}} & 8'h3b)^
({8{data_in[36][2]}} & 8'h76)^
({8{data_in[36][3]}} & 8'hec)^
({8{data_in[36][4]}} & 8'hf3)^
({8{data_in[36][5]}} & 8'hcd)^
({8{data_in[36][6]}} & 8'hb1)^
({8{data_in[36][7]}} & 8'h49)^
({8{data_in[37][0]}} & 8'hcb)^
({8{data_in[37][1]}} & 8'hbd)^
({8{data_in[37][2]}} & 8'h51)^
({8{data_in[37][3]}} & 8'ha2)^
({8{data_in[37][4]}} & 8'h6f)^
({8{data_in[37][5]}} & 8'hde)^
({8{data_in[37][6]}} & 8'h97)^
({8{data_in[37][7]}} & 8'h5)^
({8{data_in[38][0]}} & 8'hce)^
({8{data_in[38][1]}} & 8'hb7)^
({8{data_in[38][2]}} & 8'h45)^
({8{data_in[38][3]}} & 8'h8a)^
({8{data_in[38][4]}} & 8'h3f)^
({8{data_in[38][5]}} & 8'h7e)^
({8{data_in[38][6]}} & 8'hfc)^
({8{data_in[38][7]}} & 8'hd3)^
({8{data_in[39][0]}} & 8'hae)^
({8{data_in[39][1]}} & 8'h77)^
({8{data_in[39][2]}} & 8'hee)^
({8{data_in[39][3]}} & 8'hf7)^
({8{data_in[39][4]}} & 8'hc5)^
({8{data_in[39][5]}} & 8'ha1)^
({8{data_in[39][6]}} & 8'h69)^
({8{data_in[39][7]}} & 8'hd2)^
({8{data_in[40][0]}} & 8'ha0)^
({8{data_in[40][1]}} & 8'h6b)^
({8{data_in[40][2]}} & 8'hd6)^
({8{data_in[40][3]}} & 8'h87)^
({8{data_in[40][4]}} & 8'h25)^
({8{data_in[40][5]}} & 8'h4a)^
({8{data_in[40][6]}} & 8'h94)^
({8{data_in[40][7]}} & 8'h3)^
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({8{data_in[41][1]}} & 8'h66)^
({8{data_in[41][2]}} & 8'hcc)^
({8{data_in[41][3]}} & 8'hb3)^
({8{data_in[41][4]}} & 8'h4d)^
({8{data_in[41][5]}} & 8'h9a)^
({8{data_in[41][6]}} & 8'h1f)^
({8{data_in[41][7]}} & 8'h3e)^
({8{data_in[42][0]}} & 8'h38)^
({8{data_in[42][1]}} & 8'h70)^
({8{data_in[42][2]}} & 8'he0)^
({8{data_in[42][3]}} & 8'heb)^
({8{data_in[42][4]}} & 8'hfd)^
({8{data_in[42][5]}} & 8'hd1)^
({8{data_in[42][6]}} & 8'h89)^
({8{data_in[42][7]}} & 8'h39)^
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({8{data_in[43][1]}} & 8'ha0)^
({8{data_in[43][2]}} & 8'h6b)^
({8{data_in[43][3]}} & 8'hd6)^
({8{data_in[43][4]}} & 8'h87)^
({8{data_in[43][5]}} & 8'h25)^
({8{data_in[43][6]}} & 8'h4a)^

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({8{data_in[44][4]}} & 8'h76)^
({8{data_in[44][5]}} & 8'hec)^
({8{data_in[44][6]}} & 8'hf3)^
({8{data_in[44][7]}} & 8'hcd)^
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({8{data_in[45][1]}} & 8'h81)^
({8{data_in[45][2]}} & 8'h29)^
({8{data_in[45][3]}} & 8'h52)^
({8{data_in[45][4]}} & 8'ha4)^
({8{data_in[45][5]}} & 8'h63)^
({8{data_in[45][6]}} & 8'hc6)^
({8{data_in[45][7]}} & 8'ha7)^
({8{data_in[46][0]}} & 8'h43)^
({8{data_in[46][1]}} & 8'h86)^
({8{data_in[46][2]}} & 8'h27)^
({8{data_in[46][3]}} & 8'h4e)^
({8{data_in[46][4]}} & 8'h9c)^
({8{data_in[46][5]}} & 8'h13)^
({8{data_in[46][6]}} & 8'h26)^
({8{data_in[46][7]}} & 8'h4c)^
({8{data_in[47][0]}} & 8'h5d)^
({8{data_in[47][1]}} & 8'hba)^
({8{data_in[47][2]}} & 8'h5f)^
({8{data_in[47][3]}} & 8'hbe)^
({8{data_in[47][4]}} & 8'h57)^
({8{data_in[47][5]}} & 8'hae)^
({8{data_in[47][6]}} & 8'h77)^
({8{data_in[47][7]}} & 8'hee)^
({8{data_in[48][0]}} & 8'h61)^
({8{data_in[48][1]}} & 8'hc2)^
({8{data_in[48][2]}} & 8'haf)^
({8{data_in[48][3]}} & 8'h75)^
({8{data_in[48][4]}} & 8'hea)^
({8{data_in[48][5]}} & 8'hff)^
({8{data_in[48][6]}} & 8'hd5)^
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({8{data_in[49][4]}} & 8'h79)^
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({8{data_in[49][6]}} & 8'hcf)^
({8{data_in[49][7]}} & 8'hb5)^
({8{data_in[50][0]}} & 8'h53)^
({8{data_in[50][1]}} & 8'ha6)^
({8{data_in[50][2]}} & 8'h67)^
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({8{data_in[50][4]}} & 8'hb7)^
({8{data_in[50][5]}} & 8'h45)^
({8{data_in[50][6]}} & 8'h8a)^
({8{data_in[50][7]}} & 8'h3f)^
({8{data_in[51][0]}} & 8'h45)^
({8{data_in[51][1]}} & 8'h8a)^
({8{data_in[51][2]}} & 8'h3f)^
({8{data_in[51][3]}} & 8'h7e)^
({8{data_in[51][4]}} & 8'hfc)^
({8{data_in[51][5]}} & 8'hd3)^
({8{data_in[51][6]}} & 8'h8d)^
({8{data_in[51][7]}} & 8'h31)^

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({8{data_in[52][0]}} & 8'h41)^
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({8{data_in[52][2]}} & 8'h2f)^
({8{data_in[52][3]}} & 8'h5e)^
({8{data_in[52][4]}} & 8'hbc)^
({8{data_in[52][5]}} & 8'h53)^
({8{data_in[52][6]}} & 8'ha6)^
({8{data_in[52][7]}} & 8'h67)^
({8{data_in[53][0]}} & 8'h4b)^
({8{data_in[53][1]}} & 8'h96)^
({8{data_in[53][2]}} & 8'h7)^
({8{data_in[53][3]}} & 8'he)^
({8{data_in[53][4]}} & 8'h1c)^
({8{data_in[53][5]}} & 8'h38)^
({8{data_in[53][6]}} & 8'h70)^
({8{data_in[53][7]}} & 8'he0)^
({8{data_in[54][0]}} & 8'h3f)^
({8{data_in[54][1]}} & 8'h7e)^
({8{data_in[54][2]}} & 8'hfc)^
({8{data_in[54][3]}} & 8'hd3)^
({8{data_in[54][4]}} & 8'h8d)^
({8{data_in[54][5]}} & 8'h31)^
({8{data_in[54][6]}} & 8'h62)^
({8{data_in[54][7]}} & 8'hc4)^
({8{data_in[55][0]}} & 8'hc0)^
({8{data_in[55][1]}} & 8'hab)^
({8{data_in[55][2]}} & 8'h7d)^
({8{data_in[55][3]}} & 8'hfa)^
({8{data_in[55][4]}} & 8'hdf)^
({8{data_in[55][5]}} & 8'h95)^
({8{data_in[55][6]}} & 8'h1)^
({8{data_in[55][7]}} & 8'h2)^
({8{data_in[56][0]}} & 8'h1b)^
({8{data_in[56][1]}} & 8'h36)^
({8{data_in[56][2]}} & 8'h6c)^
({8{data_in[56][3]}} & 8'hd8)^
({8{data_in[56][4]}} & 8'h9b)^
({8{data_in[56][5]}} & 8'h1d)^
({8{data_in[56][6]}} & 8'h3a)^
({8{data_in[56][7]}} & 8'h74)^
({8{data_in[57][0]}} & 8'hf6)^
({8{data_in[57][1]}} & 8'hc7)^
({8{data_in[57][2]}} & 8'ha5)^
({8{data_in[57][3]}} & 8'h61)^
({8{data_in[57][4]}} & 8'hc2)^
({8{data_in[57][5]}} & 8'haf)^
({8{data_in[57][6]}} & 8'h75)^
({8{data_in[57][7]}} & 8'hea)^
({8{data_in[58][0]}} & 8'h9f)^
({8{data_in[58][1]}} & 8'h15)^
({8{data_in[58][2]}} & 8'h2a)^
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({8{data_in[58][4]}} & 8'ha8)^
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({8{data_in[58][7]}} & 8'hc7)^
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({8{data_in[59][2]}} & 8'hcc)^
({8{data_in[59][3]}} & 8'hb3)^
({8{data_in[59][4]}} & 8'h4d)^
({8{data_in[59][5]}} & 8'h9a)^
({8{data_in[59][6]}} & 8'h1f)^
({8{data_in[59][7]}} & 8'h3e)^
({8{data_in[60][0]}} & 8'h9a)^

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({8{data_in[60][4]}} & 8'hf8)^
({8{data_in[60][5]}} & 8'hdb)^
({8{data_in[60][6]}} & 8'h9d)^
({8{data_in[60][7]}} & 8'h11)^
({8{data_in[61][0]}} & 8'h1d)^
({8{data_in[61][1]}} & 8'h3a)^
({8{data_in[61][2]}} & 8'h74)^
({8{data_in[61][3]}} & 8'he8)^
({8{data_in[61][4]}} & 8'hfb)^
({8{data_in[61][5]}} & 8'hdd)^
({8{data_in[61][6]}} & 8'h91)^
({8{data_in[61][7]}} & 8'h9)^
({8{data_in[62][0]}} & 8'h50)^
({8{data_in[62][1]}} & 8'ha0)^
({8{data_in[62][2]}} & 8'h6b)^
({8{data_in[62][3]}} & 8'hd6)^
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({8{data_in[62][5]}} & 8'h25)^
({8{data_in[62][6]}} & 8'h4a)^
({8{data_in[62][7]}} & 8'h94)^
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({8{data_in[64][1]}} & 8'hdd)^
({8{data_in[64][2]}} & 8'h91)^
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({8{data_in[66][1]}} & 8'h9)^
({8{data_in[66][2]}} & 8'h12)^
({8{data_in[66][3]}} & 8'h24)^
({8{data_in[66][4]}} & 8'h48)^
({8{data_in[66][5]}} & 8'h90)^
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({8{data_in[66][7]}} & 8'h16)^
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({8{data_in[67][1]}} & 8'h32)^
({8{data_in[67][2]}} & 8'h64)^
({8{data_in[67][3]}} & 8'hc8)^
({8{data_in[67][4]}} & 8'hbb)^
({8{data_in[67][5]}} & 8'h5d)^
({8{data_in[67][6]}} & 8'hba)^
({8{data_in[67][7]}} & 8'h5f)^
({8{data_in[68][0]}} & 8'h6b)^
({8{data_in[68][1]}} & 8'hd6)^

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({8{data_in[68][7]}} & 8'h6)^
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({8{data_in[72][1]}} & 8'hfc)^
({8{data_in[72][2]}} & 8'hd3)^
({8{data_in[72][3]}} & 8'h8d)^
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({8{data_in[72][5]}} & 8'h62)^
({8{data_in[72][6]}} & 8'hc4)^
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({8{data_in[75][2]}} & 8'h1c)^
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({8{data_in[76][7]}} & 8'hc3)^
({8{data_in[77][0]}} & 8'h1d)^
({8{data_in[77][1]}} & 8'h3a)^
({8{data_in[77][2]}} & 8'h74)^
({8{data_in[77][3]}} & 8'he8)^
({8{data_in[77][4]}} & 8'hfb)^
({8{data_in[77][5]}} & 8'hdd)^
({8{data_in[77][6]}} & 8'h91)^
({8{data_in[77][7]}} & 8'h9)^
({8{data_in[78][0]}} & 8'hb8)^
({8{data_in[78][1]}} & 8'h5b)^
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({8{data_in[79][1]}} & 8'he)^
({8{data_in[79][2]}} & 8'h1c)^
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({8{data_in[79][6]}} & 8'heb)^
({8{data_in[79][7]}} & 8'hfd)^
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({8{data_in[84][3]}} & 8'h9)^

```

```

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({8{data_in[86][3]}} & 8'h5e)^
({8{data_in[86][4]}} & 8'hbc)^
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({8{data_in[86][6]}} & 8'ha6)^
({8{data_in[86][7]}} & 8'h67)^
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({8{data_in[87][2]}} & 8'h21)^
({8{data_in[87][3]}} & 8'h42)^
({8{data_in[87][4]}} & 8'h84)^
({8{data_in[87][5]}} & 8'h23)^
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({8{data_in[87][7]}} & 8'h8c)^
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({8{data_in[88][3]}} & 8'h29)^
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({8{data_in[88][5]}} & 8'ha4)^
({8{data_in[88][6]}} & 8'h63)^
({8{data_in[88][7]}} & 8'hc6)^
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({8{data_in[89][2]}} & 8'ha2)^
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({8{data_in[89][7]}} & 8'ha)^
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({8{data_in[90][1]}} & 8'h93)^
({8{data_in[90][2]}} & 8'hd)^
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({8{data_in[90][5]}} & 8'h68)^
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({8{data_in[91][2]}} & 8'hca)^
({8{data_in[91][3]}} & 8'hbf)^
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({8{data_in[91][5]}} & 8'haa)^
({8{data_in[91][6]}} & 8'h7f)^
({8{data_in[91][7]}} & 8'hfe)^
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({8{data_in[92][1]}} & 8'h96)^
({8{data_in[92][2]}} & 8'h7)^
({8{data_in[92][3]}} & 8'he)^
({8{data_in[92][4]}} & 8'h1c)^

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({8{data_in[93][4]}} & 8'ha9)^
({8{data_in[93][5]}} & 8'h79)^
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({8{data_in[95][2]}} & 8'hd0)^
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({8{data_in[96][7]}} & 8'hdf)^
({8{data_in[97][0]}} & 8'h5d)^
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({8{data_in[97][2]}} & 8'h5f)^
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({8{data_in[97][6]}} & 8'h77)^
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({8{data_in[98][1]}} & 8'hec)^
({8{data_in[98][2]}} & 8'hf3)^
({8{data_in[98][3]}} & 8'hcd)^
({8{data_in[98][4]}} & 8'hb1)^
({8{data_in[98][5]}} & 8'h49)^
({8{data_in[98][6]}} & 8'h92)^
({8{data_in[98][7]}} & 8'hf)^
({8{data_in[99][0]}} & 8'h4c)^
({8{data_in[99][1]}} & 8'h98)^
({8{data_in[99][2]}} & 8'h1b)^
({8{data_in[99][3]}} & 8'h36)^
({8{data_in[99][4]}} & 8'h6c)^
({8{data_in[99][5]}} & 8'hd8)^
({8{data_in[99][6]}} & 8'h9b)^
({8{data_in[99][7]}} & 8'h1d)^
({8{data_in[100][0]}} & 8'h2c)^
({8{data_in[100][1]}} & 8'h58)^
({8{data_in[100][2]}} & 8'hb0)^
({8{data_in[100][3]}} & 8'h4b)^
({8{data_in[100][4]}} & 8'h96)^
({8{data_in[100][5]}} & 8'h7)^

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({8{data_in[101][2]}} & 8'hb7)^
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({8{data_in[101][4]}} & 8'h8a)^
({8{data_in[101][5]}} & 8'h3f)^
({8{data_in[101][6]}} & 8'h7e)^
({8{data_in[101][7]}} & 8'hfc)^
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({8{data_in[103][3]}} & 8'h86)^
({8{data_in[103][4]}} & 8'h27)^
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({8{data_in[105][3]}} & 8'h7f)^
({8{data_in[105][4]}} & 8'hfe)^
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({8{data_in[106][2]}} & 8'he)^
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({8{data_in[107][3]}} & 8'h3c)^
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({8{data_in[107][6]}} & 8'hcb)^
({8{data_in[107][7]}} & 8'hbd)^
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({8{data_in[108][3]}} & 8'hd2)^
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({8{data_in[108][5]}} & 8'h35)^
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({8{data_in[109][6]}} & 8'h12)^
({8{data_in[109][7]}} & 8'h24)^
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({8{data_in[110][1]}} & 8'heb)^
({8{data_in[110][2]}} & 8'hfd)^
({8{data_in[110][3]}} & 8'hd1)^
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({8{data_in[111][3]}} & 8'ha8)^
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({8{data_in[113][3]}} & 8'ha3)^
({8{data_in[113][4]}} & 8'h6d)^
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({8{data_in[113][6]}} & 8'h9f)^
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({8{data_in[114][2]}} & 8'ha2)^
({8{data_in[114][3]}} & 8'h6f)^
({8{data_in[114][4]}} & 8'hde)^
({8{data_in[114][5]}} & 8'h97)^
({8{data_in[114][6]}} & 8'h5)^
({8{data_in[114][7]}} & 8'ha)^
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({8{data_in[115][3]}} & 8'h2d)^
({8{data_in[115][4]}} & 8'h5a)^
({8{data_in[115][5]}} & 8'hb4)^
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({8{data_in[115][7]}} & 8'h86)^
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({8{data_in[116][5]}} & 8'hf2)^
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({8{data_in[117][7]}} & 8'he4)^
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({8{data_in[118][2]}} & 8'ha3)^
({8{data_in[118][3]}} & 8'h6d)^
({8{data_in[118][4]}} & 8'hda)^
({8{data_in[118][5]}} & 8'h9f)^
({8{data_in[118][6]}} & 8'h15)^
({8{data_in[118][7]}} & 8'h2a)^
({8{data_in[119][0]}} & 8'h69)^
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({8{data_in[119][2]}} & 8'h8f)^
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({8{data_in[119][4]}} & 8'h6a)^
({8{data_in[119][5]}} & 8'hd4)^
({8{data_in[119][6]}} & 8'h83)^
({8{data_in[119][7]}} & 8'h2d)^
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({8{data_in[120][2]}} & 8'h63)^
({8{data_in[120][3]}} & 8'hc6)^
({8{data_in[120][4]}} & 8'ha7)^
({8{data_in[120][5]}} & 8'h65)^
({8{data_in[120][6]}} & 8'hca)^
({8{data_in[120][7]}} & 8'hbf)^
({8{data_in[121][0]}} & 8'hd3)^
({8{data_in[121][1]}} & 8'h8d)^
({8{data_in[121][2]}} & 8'h31)^
({8{data_in[121][3]}} & 8'h62)^
({8{data_in[121][4]}} & 8'hc4)^
({8{data_in[121][5]}} & 8'ha3)^
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({8{data_in[122][6]}} & 8'ha6)^
({8{data_in[122][7]}} & 8'h67)^
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({8{data_in[123][3]}} & 8'h7)^
({8{data_in[123][4]}} & 8'he)^
({8{data_in[123][5]}} & 8'h1c)^
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({8{data_in[124][4]}} & 8'ha)^
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({8{data_in[128][3]}} & 8'hcf)^
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({8{data_in[130][6]}} & 8'h9e)^
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({8{data_in[131][1]}} & 8'h63)^
({8{data_in[131][2]}} & 8'hc6)^
({8{data_in[131][3]}} & 8'ha7)^
({8{data_in[131][4]}} & 8'h65)^
({8{data_in[131][5]}} & 8'hca)^
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({8{data_in[134][3]}} & 8'h1e)^
({8{data_in[134][4]}} & 8'h3c)^
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({8{data_in[134][7]}} & 8'hcb)^
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({8{data_in[136][5]}} & 8'hb1)^
({8{data_in[136][6]}} & 8'h49)^
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({8{data_in[137][1]}} & 8'hbe)^
({8{data_in[137][2]}} & 8'h57)^
({8{data_in[137][3]}} & 8'hae)^
({8{data_in[137][4]}} & 8'h77)^
({8{data_in[137][5]}} & 8'hee)^
({8{data_in[137][6]}} & 8'hf7)^
({8{data_in[137][7]}} & 8'hc5)^
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({8{data_in[138][6]}} & 8'hac)^
({8{data_in[138][7]}} & 8'h73)^
({8{data_in[139][0]}} & 8'he7)^
({8{data_in[139][1]}} & 8'he5)^
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({8{data_in[140][1]}} & 8'h19)^
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({8{data_in[140][5]}} & 8'hbb)^
({8{data_in[140][6]}} & 8'h5d)^
({8{data_in[140][7]}} & 8'hba)^
({8{data_in[141][0]}} & 8'hd6)^
({8{data_in[141][1]}} & 8'h87)^
({8{data_in[141][2]}} & 8'h25)^

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({8{data_in[141][7]}} & 8'hc)^
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({8{data_in[142][3]}} & 8'h52)^
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({8{data_in[142][6]}} & 8'hc6)^
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({8{data_in[145][5]}} & 8'h87)^
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({8{data_in[147][1]}} & 8'hb3)^
({8{data_in[147][2]}} & 8'h4d)^
({8{data_in[147][3]}} & 8'h9a)^
({8{data_in[147][4]}} & 8'h1f)^
({8{data_in[147][5]}} & 8'h3e)^
({8{data_in[147][6]}} & 8'h7c)^
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({8{data_in[148][2]}} & 8'hc4)^
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({8{data_in[148][5]}} & 8'hda)^
({8{data_in[148][6]}} & 8'h9f)^
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({8{data_in[150][5]}} & 8'h9b)^
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({8{data_in[156][2]}} & 8'hda)^
({8{data_in[156][3]}} & 8'h9f)^
({8{data_in[156][4]}} & 8'h15)^
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({8{data_in[157][2]}} & 8'h87)^
({8{data_in[157][3]}} & 8'h25)^
({8{data_in[157][4]}} & 8'h4a)^

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({8{data_in[158][5]}} & 8'hcb)^
({8{data_in[158][6]}} & 8'hbd)^
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({8{data_in[160][6]}} & 8'h5c)^
({8{data_in[160][7]}} & 8'hb8)^
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({8{data_in[161][2]}} & 8'h5)^
({8{data_in[161][3]}} & 8'ha)^
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({8{data_in[163][4]}} & 8'h8f)^
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({8{data_in[164][3]}} & 8'h7e)^
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({8{data_in[167][7]}} & 8'hf0)^
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({8{data_in[168][3]}} & 8'ha8)^
({8{data_in[168][4]}} & 8'h7b)^
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({8{data_in[168][6]}} & 8'hc7)^
({8{data_in[168][7]}} & 8'ha5)^
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({8{data_in[170][0]}} & 8'h4f)^
({8{data_in[170][1]}} & 8'h9e)^
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({8{data_in[173][3]}} & 8'h5f)^
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({8{data_in[174][4]}} & 8'hbe)^
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({8{data_in[186][5]}} & 8'h9c)^
({8{data_in[186][6]}} & 8'h13)^
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({8{data_in[187][0]}} & 8'h24)^
({8{data_in[187][1]}} & 8'h48)^
({8{data_in[187][2]}} & 8'h90)^
({8{data_in[187][3]}} & 8'hb)^
({8{data_in[187][4]}} & 8'h16)^
({8{data_in[187][5]}} & 8'h2c)^
({8{data_in[187][6]}} & 8'h58)^
({8{data_in[187][7]}} & 8'hb0)^
({8{data_in[188][0]}} & 8'h69)^
({8{data_in[188][1]}} & 8'hd2)^
({8{data_in[188][2]}} & 8'h8f)^
({8{data_in[188][3]}} & 8'h35)^
({8{data_in[188][4]}} & 8'h6a)^
({8{data_in[188][5]}} & 8'hd4)^
({8{data_in[188][6]}} & 8'h83)^
({8{data_in[188][7]}} & 8'h2d)^
({8{data_in[189][0]}} & 8'h24)^
({8{data_in[189][1]}} & 8'h48)^
({8{data_in[189][2]}} & 8'h90)^
({8{data_in[189][3]}} & 8'hb)^
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({8{data_in[189][5]}} & 8'h2c)^
({8{data_in[189][6]}} & 8'h58)^
({8{data_in[189][7]}} & 8'hb0)^
({8{data_in[190][0]}} & 8'hae)^

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({8{data_in[191][1]}} & 8'h60)^
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({8{data_in[191][5]}} & 8'hfa)^
({8{data_in[191][6]}} & 8'hdf)^
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({8{data_in[192][2]}} & 8'h2)^
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({8{data_in[193][7]}} & 8'h9e)^
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({8{data_in[194][2]}} & 8'hf9)^
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({8{data_in[195][5]}} & 8'h40)^
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({8{data_in[195][7]}} & 8'h2b)^
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({8{data_in[196][1]}} & 8'hbb)^
({8{data_in[196][2]}} & 8'h5d)^
({8{data_in[196][3]}} & 8'hba)^
({8{data_in[196][4]}} & 8'h5f)^
({8{data_in[196][5]}} & 8'hbe)^
({8{data_in[196][6]}} & 8'h57)^
({8{data_in[196][7]}} & 8'hae)^
({8{data_in[197][0]}} & 8'h8)^
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({8{data_in[197][3]}} & 8'h40)^
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({8{data_in[197][6]}} & 8'h56)^
({8{data_in[197][7]}} & 8'hac)^
({8{data_in[198][0]}} & 8'hf5)^
({8{data_in[198][1]}} & 8'hc1)^

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({8{data_in[198][6]}} & 8'hb5)^
({8{data_in[198][7]}} & 8'h41)^
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({8{data_in[199][1]}} & 8'hbf)^
({8{data_in[199][2]}} & 8'h55)^
({8{data_in[199][3]}} & 8'haa)^
({8{data_in[199][4]}} & 8'h7f)^
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({8{data_in[200][4]}} & 8'hea)^
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({8{data_in[200][6]}} & 8'hd5)^
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({8{data_in[202][3]}} & 8'hde)^
({8{data_in[202][4]}} & 8'h97)^
({8{data_in[202][5]}} & 8'h5)^
({8{data_in[202][6]}} & 8'ha)^
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({8{data_in[203][2]}} & 8'h83)^
({8{data_in[203][3]}} & 8'h2d)^
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({8{data_in[204][3]}} & 8'hc6)^
({8{data_in[204][4]}} & 8'ha7)^
({8{data_in[204][5]}} & 8'h65)^
({8{data_in[204][6]}} & 8'hca)^
({8{data_in[204][7]}} & 8'hbf)^
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({8{data_in[205][1]}} & 8'hfe)^
({8{data_in[205][2]}} & 8'hf5)^
({8{data_in[205][3]}} & 8'hc1)^
({8{data_in[205][4]}} & 8'ha9)^
({8{data_in[205][5]}} & 8'h79)^
({8{data_in[205][6]}} & 8'hf2)^
({8{data_in[205][7]}} & 8'hcf)^
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({8{data_in[206][1]}} & 8'h8)^
({8{data_in[206][2]}} & 8'h10)^

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({8{data_in[207][3]}} & 8'h81)^
({8{data_in[207][4]}} & 8'h29)^
({8{data_in[207][5]}} & 8'h52)^
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({8{data_in[208][2]}} & 8'hfc)^
({8{data_in[208][3]}} & 8'hd3)^
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({8{data_in[208][7]}} & 8'hc4)^
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({8{data_in[209][2]}} & 8'hff)^
({8{data_in[209][3]}} & 8'hd5)^
({8{data_in[209][4]}} & 8'h81)^
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({8{data_in[210][7]}} & 8'h4d)^
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({8{data_in[211][1]}} & 8'hd0)^
({8{data_in[211][2]}} & 8'h8b)^
({8{data_in[211][3]}} & 8'h3d)^
({8{data_in[211][4]}} & 8'h7a)^
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({8{data_in[212][2]}} & 8'hc2)^
({8{data_in[212][3]}} & 8'haf)^
({8{data_in[212][4]}} & 8'h75)^
({8{data_in[212][5]}} & 8'hea)^
({8{data_in[212][6]}} & 8'hff)^
({8{data_in[212][7]}} & 8'hd5)^
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({8{data_in[213][1]}} & 8'h74)^
({8{data_in[213][2]}} & 8'he8)^
({8{data_in[213][3]}} & 8'hfb)^
({8{data_in[213][4]}} & 8'hdd)^
({8{data_in[213][5]}} & 8'h91)^
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({8{data_in[215][5]}} & 8'h6)^
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({8{data_in[216][3]}} & 8'he9)^
({8{data_in[216][4]}} & 8'hf9)^
({8{data_in[216][5]}} & 8'hd9)^
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({8{data_in[216][7]}} & 8'h19)^
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({8{data_in[217][2]}} & 8'h69)^
({8{data_in[217][3]}} & 8'hd2)^
({8{data_in[217][4]}} & 8'h8f)^
({8{data_in[217][5]}} & 8'h35)^
({8{data_in[217][6]}} & 8'h6a)^
({8{data_in[217][7]}} & 8'hd4)^
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({8{data_in[218][3]}} & 8'ha0)^
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({8{data_in[218][7]}} & 8'h25)^
({8{data_in[219][0]}} & 8'h6f)^
({8{data_in[219][1]}} & 8'hde)^
({8{data_in[219][2]}} & 8'h97)^
({8{data_in[219][3]}} & 8'h5)^
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({8{data_in[220][7]}} & 8'hc2)^
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({8{data_in[221][2]}} & 8'h75)^
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({8{data_in[221][5]}} & 8'hd5)^
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({8{data_in[223][5]}} & 8'hfa)^
({8{data_in[223][6]}} & 8'hdf)^
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({8{data_in[225][3]}} & 8'h74)^
({8{data_in[225][4]}} & 8'he8)^
({8{data_in[225][5]}} & 8'hfb)^
({8{data_in[225][6]}} & 8'hdd)^
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({8{data_in[226][4]}} & 8'h69)^
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({8{data_in[228][3]}} & 8'h8f)^
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({8{data_in[229][1]}} & 8'h60)^
({8{data_in[229][2]}} & 8'hc0)^
({8{data_in[229][3]}} & 8'hab)^
({8{data_in[229][4]}} & 8'h7d)^
({8{data_in[229][5]}} & 8'hfa)^
({8{data_in[229][6]}} & 8'hdf)^
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({8{data_in[230][3]}} & 8'hab)^
({8{data_in[230][4]}} & 8'h7d)^
({8{data_in[230][5]}} & 8'hfa)^

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({8{data_in[231][5]}} & 8'h5d)^
({8{data_in[231][6]}} & 8'hba)^
({8{data_in[231][7]}} & 8'h5f)^
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({8{data_in[234][7]}} & 8'h73)^
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({8{data_in[235][4]}} & 8'he8)^
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({8{data_in[236][2]}} & 8'hcc)^
({8{data_in[236][3]}} & 8'hb3)^
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({8{data_in[236][5]}} & 8'h9a)^
({8{data_in[236][6]}} & 8'h1f)^
({8{data_in[236][7]}} & 8'h3e)^
({8{data_in[237][0]}} & 8'h4a)^
({8{data_in[237][1]}} & 8'h94)^
({8{data_in[237][2]}} & 8'h3)^
({8{data_in[237][3]}} & 8'h6)^
({8{data_in[237][4]}} & 8'hc)^
({8{data_in[237][5]}} & 8'h18)^
({8{data_in[237][6]}} & 8'h30)^
({8{data_in[237][7]}} & 8'h60)^
({8{data_in[238][0]}} & 8'ha5)^
({8{data_in[238][1]}} & 8'h61)^
({8{data_in[238][2]}} & 8'hc2)^
({8{data_in[238][3]}} & 8'haf)^
({8{data_in[238][4]}} & 8'h75)^
({8{data_in[238][5]}} & 8'hea)^
({8{data_in[238][6]}} & 8'hff)^

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({8{data_in[238][7]}} & 8'hd5)^
({8{data_in[239][0]}} & 8'h65)^
({8{data_in[239][1]}} & 8'hca)^
({8{data_in[239][2]}} & 8'hbf)^
({8{data_in[239][3]}} & 8'h55)^
({8{data_in[239][4]}} & 8'haa)^
({8{data_in[239][5]}} & 8'h7f)^
({8{data_in[239][6]}} & 8'hfe)^
({8{data_in[239][7]}} & 8'hd7)^
({8{data_in[240][0]}} & 8'ha)^
({8{data_in[240][1]}} & 8'h14)^
({8{data_in[240][2]}} & 8'h28)^
({8{data_in[240][3]}} & 8'h50)^
({8{data_in[240][4]}} & 8'ha0)^
({8{data_in[240][5]}} & 8'h6b)^
({8{data_in[240][6]}} & 8'hd6)^
({8{data_in[240][7]}} & 8'h87)^
({8{data_in[241][0]}} & 8'hd5)^
({8{data_in[241][1]}} & 8'h81)^
({8{data_in[241][2]}} & 8'h29)^
({8{data_in[241][3]}} & 8'h52)^
({8{data_in[241][4]}} & 8'ha4)^
({8{data_in[241][5]}} & 8'h63)^
({8{data_in[241][6]}} & 8'hc6)^
({8{data_in[241][7]}} & 8'ha7);
data_out[245] = ({8{data_in[0][0]}} & 8'h1a)^
({8{data_in[0][1]}} & 8'h34)^
({8{data_in[0][2]}} & 8'h68)^
({8{data_in[0][3]}} & 8'hd0)^
({8{data_in[0][4]}} & 8'h8b)^
({8{data_in[0][5]}} & 8'h3d)^
({8{data_in[0][6]}} & 8'h7a)^
({8{data_in[0][7]}} & 8'hf4)^
({8{data_in[1][0]}} & 8'hcb)^
({8{data_in[1][1]}} & 8'hbd)^
({8{data_in[1][2]}} & 8'h51)^
({8{data_in[1][3]}} & 8'ha2)^
({8{data_in[1][4]}} & 8'h6f)^
({8{data_in[1][5]}} & 8'hde)^
({8{data_in[1][6]}} & 8'h97)^
({8{data_in[1][7]}} & 8'h5)^
({8{data_in[2][0]}} & 8'h54)^
({8{data_in[2][1]}} & 8'ha8)^
({8{data_in[2][2]}} & 8'h7b)^
({8{data_in[2][3]}} & 8'hf6)^
({8{data_in[2][4]}} & 8'hc7)^
({8{data_in[2][5]}} & 8'ha5)^
({8{data_in[2][6]}} & 8'h61)^
({8{data_in[2][7]}} & 8'hc2)^
({8{data_in[3][0]}} & 8'hb5)^
({8{data_in[3][1]}} & 8'h41)^
({8{data_in[3][2]}} & 8'h82)^
({8{data_in[3][3]}} & 8'h2f)^
({8{data_in[3][4]}} & 8'h5e)^
({8{data_in[3][5]}} & 8'hbc)^
({8{data_in[3][6]}} & 8'h53)^
({8{data_in[3][7]}} & 8'ha6)^
({8{data_in[4][0]}} & 8'h43)^
({8{data_in[4][1]}} & 8'h86)^
({8{data_in[4][2]}} & 8'h27)^
({8{data_in[4][3]}} & 8'h4e)^
({8{data_in[4][4]}} & 8'h9c)^
({8{data_in[4][5]}} & 8'h13)^
({8{data_in[4][6]}} & 8'h26)^
({8{data_in[4][7]}} & 8'h4c)^

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```

({8{data_in[5][0]}} & 8'hd3)^
({8{data_in[5][1]}} & 8'h8d)^
({8{data_in[5][2]}} & 8'h31)^
({8{data_in[5][3]}} & 8'h62)^
({8{data_in[5][4]}} & 8'hc4)^
({8{data_in[5][5]}} & 8'ha3)^
({8{data_in[5][6]}} & 8'h6d)^
({8{data_in[5][7]}} & 8'hda)^
({8{data_in[6][0]}} & 8'h93)^
({8{data_in[6][1]}} & 8'hd)^
({8{data_in[6][2]}} & 8'h1a)^
({8{data_in[6][3]}} & 8'h34)^
({8{data_in[6][4]}} & 8'h68)^
({8{data_in[6][5]}} & 8'hd0)^
({8{data_in[6][6]}} & 8'h8b)^
({8{data_in[6][7]}} & 8'h3d)^
({8{data_in[7][0]}} & 8'h5c)^
({8{data_in[7][1]}} & 8'hb8)^
({8{data_in[7][2]}} & 8'h5b)^
({8{data_in[7][3]}} & 8'hb6)^
({8{data_in[7][4]}} & 8'h47)^
({8{data_in[7][5]}} & 8'h8e)^
({8{data_in[7][6]}} & 8'h37)^
({8{data_in[7][7]}} & 8'h6e)^
({8{data_in[8][0]}} & 8'h5e)^
({8{data_in[8][1]}} & 8'hbc)^
({8{data_in[8][2]}} & 8'h53)^
({8{data_in[8][3]}} & 8'ha6)^
({8{data_in[8][4]}} & 8'h67)^
({8{data_in[8][5]}} & 8'hce)^
({8{data_in[8][6]}} & 8'hb7)^
({8{data_in[8][7]}} & 8'h45)^
({8{data_in[9][0]}} & 8'h37)^
({8{data_in[9][1]}} & 8'h6e)^
({8{data_in[9][2]}} & 8'hdc)^
({8{data_in[9][3]}} & 8'h93)^
({8{data_in[9][4]}} & 8'hd)^
({8{data_in[9][5]}} & 8'h1a)^
({8{data_in[9][6]}} & 8'h34)^
({8{data_in[9][7]}} & 8'h68)^
({8{data_in[10][0]}} & 8'h68)^
({8{data_in[10][1]}} & 8'hd0)^
({8{data_in[10][2]}} & 8'h8b)^
({8{data_in[10][3]}} & 8'h3d)^
({8{data_in[10][4]}} & 8'h7a)^
({8{data_in[10][5]}} & 8'hf4)^
({8{data_in[10][6]}} & 8'hc3)^
({8{data_in[10][7]}} & 8'had)^
({8{data_in[11][0]}} & 8'h41)^
({8{data_in[11][1]}} & 8'h82)^
({8{data_in[11][2]}} & 8'h2f)^
({8{data_in[11][3]}} & 8'h5e)^
({8{data_in[11][4]}} & 8'hbc)^
({8{data_in[11][5]}} & 8'h53)^
({8{data_in[11][6]}} & 8'ha6)^
({8{data_in[11][7]}} & 8'h67)^
({8{data_in[12][0]}} & 8'hb0)^
({8{data_in[12][1]}} & 8'h4b)^
({8{data_in[12][2]}} & 8'h96)^
({8{data_in[12][3]}} & 8'h7)^
({8{data_in[12][4]}} & 8'he)^
({8{data_in[12][5]}} & 8'h1c)^
({8{data_in[12][6]}} & 8'h38)^
({8{data_in[12][7]}} & 8'h70)^
({8{data_in[13][0]}} & 8'h4e)^

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({8{data_in[13][1]}} & 8'h9c)^
({8{data_in[13][2]}} & 8'h13)^
({8{data_in[13][3]}} & 8'h26)^
({8{data_in[13][4]}} & 8'h4c)^
({8{data_in[13][5]}} & 8'h98)^
({8{data_in[13][6]}} & 8'h1b)^
({8{data_in[13][7]}} & 8'h36)^
({8{data_in[14][0]}} & 8'h44)^
({8{data_in[14][1]}} & 8'h88)^
({8{data_in[14][2]}} & 8'h3b)^
({8{data_in[14][3]}} & 8'h76)^
({8{data_in[14][4]}} & 8'hec)^
({8{data_in[14][5]}} & 8'hf3)^
({8{data_in[14][6]}} & 8'hcd)^
({8{data_in[14][7]}} & 8'hb1)^
({8{data_in[15][0]}} & 8'h9d)^
({8{data_in[15][1]}} & 8'h11)^
({8{data_in[15][2]}} & 8'h22)^
({8{data_in[15][3]}} & 8'h44)^
({8{data_in[15][4]}} & 8'h88)^
({8{data_in[15][5]}} & 8'h3b)^
({8{data_in[15][6]}} & 8'h76)^
({8{data_in[15][7]}} & 8'hec)^
({8{data_in[16][0]}} & 8'h42)^
({8{data_in[16][1]}} & 8'h84)^
({8{data_in[16][2]}} & 8'h23)^
({8{data_in[16][3]}} & 8'h46)^
({8{data_in[16][4]}} & 8'h8c)^
({8{data_in[16][5]}} & 8'h33)^
({8{data_in[16][6]}} & 8'h66)^
({8{data_in[16][7]}} & 8'hcc)^
({8{data_in[17][0]}} & 8'hc0)^
({8{data_in[17][1]}} & 8'hab)^
({8{data_in[17][2]}} & 8'h7d)^
({8{data_in[17][3]}} & 8'hfa)^
({8{data_in[17][4]}} & 8'hdf)^
({8{data_in[17][5]}} & 8'h95)^
({8{data_in[17][6]}} & 8'h1)^
({8{data_in[17][7]}} & 8'h2)^
({8{data_in[18][0]}} & 8'hc)^
({8{data_in[18][1]}} & 8'h18)^
({8{data_in[18][2]}} & 8'h30)^
({8{data_in[18][3]}} & 8'h60)^
({8{data_in[18][4]}} & 8'hc0)^
({8{data_in[18][5]}} & 8'hab)^
({8{data_in[18][6]}} & 8'h7d)^
({8{data_in[18][7]}} & 8'hfa)^
({8{data_in[19][0]}} & 8'hd2)^
({8{data_in[19][1]}} & 8'h8f)^
({8{data_in[19][2]}} & 8'h35)^
({8{data_in[19][3]}} & 8'h6a)^
({8{data_in[19][4]}} & 8'hd4)^
({8{data_in[19][5]}} & 8'h83)^
({8{data_in[19][6]}} & 8'h2d)^
({8{data_in[19][7]}} & 8'h5a)^
({8{data_in[20][0]}} & 8'h80)^
({8{data_in[20][1]}} & 8'h2b)^
({8{data_in[20][2]}} & 8'h56)^
({8{data_in[20][3]}} & 8'hac)^
({8{data_in[20][4]}} & 8'h73)^
({8{data_in[20][5]}} & 8'he6)^
({8{data_in[20][6]}} & 8'he7)^
({8{data_in[20][7]}} & 8'he5)^
({8{data_in[21][0]}} & 8'hcb)^
({8{data_in[21][1]}} & 8'hbd)^

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({8{data_in[21][2]}} & 8'h51)^
({8{data_in[21][3]}} & 8'ha2)^
({8{data_in[21][4]}} & 8'h6f)^
({8{data_in[21][5]}} & 8'hde)^
({8{data_in[21][6]}} & 8'h97)^
({8{data_in[21][7]}} & 8'h5)^
({8{data_in[22][0]}} & 8'he9)^
({8{data_in[22][1]}} & 8'hf9)^
({8{data_in[22][2]}} & 8'hd9)^
({8{data_in[22][3]}} & 8'h99)^
({8{data_in[22][4]}} & 8'h19)^
({8{data_in[22][5]}} & 8'h32)^
({8{data_in[22][6]}} & 8'h64)^
({8{data_in[22][7]}} & 8'hc8)^
({8{data_in[23][0]}} & 8'hbe)^
({8{data_in[23][1]}} & 8'h57)^
({8{data_in[23][2]}} & 8'hae)^
({8{data_in[23][3]}} & 8'h77)^
({8{data_in[23][4]}} & 8'hee)^
({8{data_in[23][5]}} & 8'hf7)^
({8{data_in[23][6]}} & 8'hc5)^
({8{data_in[23][7]}} & 8'ha1)^
({8{data_in[24][0]}} & 8'had)^
({8{data_in[24][1]}} & 8'h71)^
({8{data_in[24][2]}} & 8'he2)^
({8{data_in[24][3]}} & 8'hef)^
({8{data_in[24][4]}} & 8'hf5)^
({8{data_in[24][5]}} & 8'hc1)^
({8{data_in[24][6]}} & 8'ha9)^
({8{data_in[24][7]}} & 8'h79)^
({8{data_in[25][0]}} & 8'hdd)^
({8{data_in[25][1]}} & 8'h91)^
({8{data_in[25][2]}} & 8'h9)^
({8{data_in[25][3]}} & 8'h12)^
({8{data_in[25][4]}} & 8'h24)^
({8{data_in[25][5]}} & 8'h48)^
({8{data_in[25][6]}} & 8'h90)^
({8{data_in[25][7]}} & 8'hb)^
({8{data_in[26][0]}} & 8'hcf)^
({8{data_in[26][1]}} & 8'hb5)^
({8{data_in[26][2]}} & 8'h41)^
({8{data_in[26][3]}} & 8'h82)^
({8{data_in[26][4]}} & 8'h2f)^
({8{data_in[26][5]}} & 8'h5e)^
({8{data_in[26][6]}} & 8'hbc)^
({8{data_in[26][7]}} & 8'h53)^
({8{data_in[27][0]}} & 8'h9e)^
({8{data_in[27][1]}} & 8'h17)^
({8{data_in[27][2]}} & 8'h2e)^
({8{data_in[27][3]}} & 8'h5c)^
({8{data_in[27][4]}} & 8'hb8)^
({8{data_in[27][5]}} & 8'h5b)^
({8{data_in[27][6]}} & 8'hb6)^
({8{data_in[27][7]}} & 8'h47)^
({8{data_in[28][0]}} & 8'h12)^
({8{data_in[28][1]}} & 8'h24)^
({8{data_in[28][2]}} & 8'h48)^
({8{data_in[28][3]}} & 8'h90)^
({8{data_in[28][4]}} & 8'hb)^
({8{data_in[28][5]}} & 8'h16)^
({8{data_in[28][6]}} & 8'h2c)^
({8{data_in[28][7]}} & 8'h58)^
({8{data_in[29][0]}} & 8'h70)^
({8{data_in[29][1]}} & 8'he0)^
({8{data_in[29][2]}} & 8'heb)^

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({8{data_in[29][3]}} & 8'hfd)^
({8{data_in[29][4]}} & 8'hd1)^
({8{data_in[29][5]}} & 8'h89)^
({8{data_in[29][6]}} & 8'h39)^
({8{data_in[29][7]}} & 8'h72)^
({8{data_in[30][0]}} & 8'ha8)^
({8{data_in[30][1]}} & 8'h7b)^
({8{data_in[30][2]}} & 8'hf6)^
({8{data_in[30][3]}} & 8'hc7)^
({8{data_in[30][4]}} & 8'ha5)^
({8{data_in[30][5]}} & 8'h61)^
({8{data_in[30][6]}} & 8'hc2)^
({8{data_in[30][7]}} & 8'haf)^
({8{data_in[31][0]}} & 8'h19)^
({8{data_in[31][1]}} & 8'h32)^
({8{data_in[31][2]}} & 8'h64)^
({8{data_in[31][3]}} & 8'hc8)^
({8{data_in[31][4]}} & 8'hbb)^
({8{data_in[31][5]}} & 8'h5d)^
({8{data_in[31][6]}} & 8'hba)^
({8{data_in[31][7]}} & 8'h5f)^
({8{data_in[32][0]}} & 8'h3a)^
({8{data_in[32][1]}} & 8'h74)^
({8{data_in[32][2]}} & 8'he8)^
({8{data_in[32][3]}} & 8'hfb)^
({8{data_in[32][4]}} & 8'hdd)^
({8{data_in[32][5]}} & 8'h91)^
({8{data_in[32][6]}} & 8'h9)^
({8{data_in[32][7]}} & 8'h12)^
({8{data_in[33][0]}} & 8'hd5)^
({8{data_in[33][1]}} & 8'h81)^
({8{data_in[33][2]}} & 8'h29)^
({8{data_in[33][3]}} & 8'h52)^
({8{data_in[33][4]}} & 8'ha4)^
({8{data_in[33][5]}} & 8'h63)^
({8{data_in[33][6]}} & 8'hc6)^
({8{data_in[33][7]}} & 8'ha7)^
({8{data_in[34][0]}} & 8'hf1)^
({8{data_in[34][1]}} & 8'hc9)^
({8{data_in[34][2]}} & 8'hb9)^
({8{data_in[34][3]}} & 8'h59)^
({8{data_in[34][4]}} & 8'hb2)^
({8{data_in[34][5]}} & 8'h4f)^
({8{data_in[34][6]}} & 8'h9e)^
({8{data_in[34][7]}} & 8'h17)^
({8{data_in[35][0]}} & 8'he2)^
({8{data_in[35][1]}} & 8'hcf)^
({8{data_in[35][2]}} & 8'hf5)^
({8{data_in[35][3]}} & 8'hc1)^
({8{data_in[35][4]}} & 8'ha9)^
({8{data_in[35][5]}} & 8'h79)^
({8{data_in[35][6]}} & 8'hf2)^
({8{data_in[35][7]}} & 8'hcf)^
({8{data_in[36][0]}} & 8'he9)^
({8{data_in[36][1]}} & 8'hf9)^
({8{data_in[36][2]}} & 8'hd9)^
({8{data_in[36][3]}} & 8'h99)^
({8{data_in[36][4]}} & 8'h19)^
({8{data_in[36][5]}} & 8'h32)^
({8{data_in[36][6]}} & 8'h64)^
({8{data_in[36][7]}} & 8'hc8)^
({8{data_in[37][0]}} & 8'h64)^
({8{data_in[37][1]}} & 8'hc8)^
({8{data_in[37][2]}} & 8'hbb)^
({8{data_in[37][3]}} & 8'h5d)^

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({8{data_in[37][4]}} & 8'hba)^
({8{data_in[37][5]}} & 8'h5f)^
({8{data_in[37][6]}} & 8'hbe)^
({8{data_in[37][7]}} & 8'h57)^
({8{data_in[38][0]}} & 8'hf3)^
({8{data_in[38][1]}} & 8'hcd)^
({8{data_in[38][2]}} & 8'hb1)^
({8{data_in[38][3]}} & 8'h49)^
({8{data_in[38][4]}} & 8'h92)^
({8{data_in[38][5]}} & 8'hf)^
({8{data_in[38][6]}} & 8'h1e)^
({8{data_in[38][7]}} & 8'h3c)^
({8{data_in[39][0]}} & 8'h1c)^
({8{data_in[39][1]}} & 8'h38)^
({8{data_in[39][2]}} & 8'h70)^
({8{data_in[39][3]}} & 8'he0)^
({8{data_in[39][4]}} & 8'heb)^
({8{data_in[39][5]}} & 8'hfd)^
({8{data_in[39][6]}} & 8'hd1)^
({8{data_in[39][7]}} & 8'h89)^
({8{data_in[40][0]}} & 8'he3)^
({8{data_in[40][1]}} & 8'hed)^
({8{data_in[40][2]}} & 8'hf1)^
({8{data_in[40][3]}} & 8'hc9)^
({8{data_in[40][4]}} & 8'hb9)^
({8{data_in[40][5]}} & 8'h59)^
({8{data_in[40][6]}} & 8'hb2)^
({8{data_in[40][7]}} & 8'h4f)^
({8{data_in[41][0]}} & 8'h3a)^
({8{data_in[41][1]}} & 8'h74)^
({8{data_in[41][2]}} & 8'he8)^
({8{data_in[41][3]}} & 8'hfb)^
({8{data_in[41][4]}} & 8'hdd)^
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({8{data_in[41][6]}} & 8'h9)^
({8{data_in[41][7]}} & 8'h12)^
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({8{data_in[42][1]}} & 8'h9)^
({8{data_in[42][2]}} & 8'h12)^
({8{data_in[42][3]}} & 8'h24)^
({8{data_in[42][4]}} & 8'h48)^
({8{data_in[42][5]}} & 8'h90)^
({8{data_in[42][6]}} & 8'hb)^
({8{data_in[42][7]}} & 8'h16)^
({8{data_in[43][0]}} & 8'h82)^
({8{data_in[43][1]}} & 8'h2f)^
({8{data_in[43][2]}} & 8'h5e)^
({8{data_in[43][3]}} & 8'hbc)^
({8{data_in[43][4]}} & 8'h53)^
({8{data_in[43][5]}} & 8'ha6)^
({8{data_in[43][6]}} & 8'h67)^
({8{data_in[43][7]}} & 8'hce)^
({8{data_in[44][0]}} & 8'hee)^
({8{data_in[44][1]}} & 8'hf7)^
({8{data_in[44][2]}} & 8'hc5)^
({8{data_in[44][3]}} & 8'ha1)^
({8{data_in[44][4]}} & 8'h69)^
({8{data_in[44][5]}} & 8'hd2)^
({8{data_in[44][6]}} & 8'h8f)^
({8{data_in[44][7]}} & 8'h35)^
({8{data_in[45][0]}} & 8'he9)^
({8{data_in[45][1]}} & 8'hf9)^
({8{data_in[45][2]}} & 8'hd9)^
({8{data_in[45][3]}} & 8'h99)^
({8{data_in[45][4]}} & 8'h19)^

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({8{data_in[45][5]}} & 8'h32)^
({8{data_in[45][6]}} & 8'h64)^
({8{data_in[45][7]}} & 8'hc8)^
({8{data_in[46][0]}} & 8'hb4)^
({8{data_in[46][1]}} & 8'h43)^
({8{data_in[46][2]}} & 8'h86)^
({8{data_in[46][3]}} & 8'h27)^
({8{data_in[46][4]}} & 8'h4e)^
({8{data_in[46][5]}} & 8'h9c)^
({8{data_in[46][6]}} & 8'h13)^
({8{data_in[46][7]}} & 8'h26)^
({8{data_in[47][0]}} & 8'h58)^
({8{data_in[47][1]}} & 8'hb0)^
({8{data_in[47][2]}} & 8'h4b)^
({8{data_in[47][3]}} & 8'h96)^
({8{data_in[47][4]}} & 8'h7)^
({8{data_in[47][5]}} & 8'he)^
({8{data_in[47][6]}} & 8'h1c)^
({8{data_in[47][7]}} & 8'h38)^
({8{data_in[48][0]}} & 8'hf7)^
({8{data_in[48][1]}} & 8'hc5)^
({8{data_in[48][2]}} & 8'ha1)^
({8{data_in[48][3]}} & 8'h69)^
({8{data_in[48][4]}} & 8'hd2)^
({8{data_in[48][5]}} & 8'h8f)^
({8{data_in[48][6]}} & 8'h35)^
({8{data_in[48][7]}} & 8'h6a)^
({8{data_in[49][0]}} & 8'h6c)^
({8{data_in[49][1]}} & 8'hd8)^
({8{data_in[49][2]}} & 8'h9b)^
({8{data_in[49][3]}} & 8'h1d)^
({8{data_in[49][4]}} & 8'h3a)^
({8{data_in[49][5]}} & 8'h74)^
({8{data_in[49][6]}} & 8'he8)^
({8{data_in[49][7]}} & 8'hfb)^
({8{data_in[50][0]}} & 8'hbb)^
({8{data_in[50][1]}} & 8'h5d)^
({8{data_in[50][2]}} & 8'hba)^
({8{data_in[50][3]}} & 8'h5f)^
({8{data_in[50][4]}} & 8'hbe)^
({8{data_in[50][5]}} & 8'h57)^
({8{data_in[50][6]}} & 8'hae)^
({8{data_in[50][7]}} & 8'h77)^
({8{data_in[51][0]}} & 8'hc8)^
({8{data_in[51][1]}} & 8'hbb)^
({8{data_in[51][2]}} & 8'h5d)^
({8{data_in[51][3]}} & 8'hba)^
({8{data_in[51][4]}} & 8'h5f)^
({8{data_in[51][5]}} & 8'hbe)^
({8{data_in[51][6]}} & 8'h57)^
({8{data_in[51][7]}} & 8'hae)^
({8{data_in[52][0]}} & 8'hb9)^
({8{data_in[52][1]}} & 8'h59)^
({8{data_in[52][2]}} & 8'hb2)^
({8{data_in[52][3]}} & 8'h4f)^
({8{data_in[52][4]}} & 8'h9e)^
({8{data_in[52][5]}} & 8'h17)^
({8{data_in[52][6]}} & 8'h2e)^
({8{data_in[52][7]}} & 8'h5c)^
({8{data_in[53][0]}} & 8'h55)^
({8{data_in[53][1]}} & 8'haa)^
({8{data_in[53][2]}} & 8'h7f)^
({8{data_in[53][3]}} & 8'hfe)^
({8{data_in[53][4]}} & 8'hd7)^
({8{data_in[53][5]}} & 8'h85)^

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({8{data_in[54][1]}} & 8'hcf)^
({8{data_in[54][2]}} & 8'hb5)^
({8{data_in[54][3]}} & 8'h41)^
({8{data_in[54][4]}} & 8'h82)^
({8{data_in[54][5]}} & 8'h2f)^
({8{data_in[54][6]}} & 8'h5e)^
({8{data_in[54][7]}} & 8'hbc)^
({8{data_in[55][0]}} & 8'h93)^
({8{data_in[55][1]}} & 8'hd)^
({8{data_in[55][2]}} & 8'h1a)^
({8{data_in[55][3]}} & 8'h34)^
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({8{data_in[55][5]}} & 8'hd0)^
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({8{data_in[55][7]}} & 8'h3d)^
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({8{data_in[56][6]}} & 8'h5d)^
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({8{data_in[57][4]}} & 8'h5c)^
({8{data_in[57][5]}} & 8'hb8)^
({8{data_in[57][6]}} & 8'h5b)^
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({8{data_in[58][1]}} & 8'h88)^
({8{data_in[58][2]}} & 8'h3b)^
({8{data_in[58][3]}} & 8'h76)^
({8{data_in[58][4]}} & 8'hec)^
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({8{data_in[58][6]}} & 8'hcd)^
({8{data_in[58][7]}} & 8'hb1)^
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({8{data_in[59][2]}} & 8'h82)^
({8{data_in[59][3]}} & 8'h2f)^
({8{data_in[59][4]}} & 8'h5e)^
({8{data_in[59][5]}} & 8'hbc)^
({8{data_in[59][6]}} & 8'h53)^
({8{data_in[59][7]}} & 8'ha6)^
({8{data_in[60][0]}} & 8'hca)^
({8{data_in[60][1]}} & 8'hbf)^
({8{data_in[60][2]}} & 8'h55)^
({8{data_in[60][3]}} & 8'haa)^
({8{data_in[60][4]}} & 8'h7f)^
({8{data_in[60][5]}} & 8'hfe)^
({8{data_in[60][6]}} & 8'hd7)^
({8{data_in[60][7]}} & 8'h85)^
({8{data_in[61][0]}} & 8'haf)^
({8{data_in[61][1]}} & 8'h75)^
({8{data_in[61][2]}} & 8'hea)^
({8{data_in[61][3]}} & 8'hff)^
({8{data_in[61][4]}} & 8'hd5)^
({8{data_in[61][5]}} & 8'h81)^
({8{data_in[61][6]}} & 8'h29)^

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({8{data_in[62][2]}} & 8'h62)^
({8{data_in[62][3]}} & 8'hc4)^
({8{data_in[62][4]}} & 8'ha3)^
({8{data_in[62][5]}} & 8'h6d)^
({8{data_in[62][6]}} & 8'hda)^
({8{data_in[62][7]}} & 8'h9f)^
({8{data_in[63][0]}} & 8'h67)^
({8{data_in[63][1]}} & 8'hce)^
({8{data_in[63][2]}} & 8'hb7)^
({8{data_in[63][3]}} & 8'h45)^
({8{data_in[63][4]}} & 8'h8a)^
({8{data_in[63][5]}} & 8'h3f)^
({8{data_in[63][6]}} & 8'h7e)^
({8{data_in[63][7]}} & 8'hfc)^
({8{data_in[64][0]}} & 8'hac)^
({8{data_in[64][1]}} & 8'h73)^
({8{data_in[64][2]}} & 8'he6)^
({8{data_in[64][3]}} & 8'he7)^
({8{data_in[64][4]}} & 8'he5)^
({8{data_in[64][5]}} & 8'he1)^
({8{data_in[64][6]}} & 8'he9)^
({8{data_in[64][7]}} & 8'hf9)^
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({8{data_in[65][1]}} & 8'h2c)^
({8{data_in[65][2]}} & 8'h58)^
({8{data_in[65][3]}} & 8'hb0)^
({8{data_in[65][4]}} & 8'h4b)^
({8{data_in[65][5]}} & 8'h96)^
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({8{data_in[65][7]}} & 8'he)^
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({8{data_in[67][1]}} & 8'h9c)^
({8{data_in[67][2]}} & 8'h13)^
({8{data_in[67][3]}} & 8'h26)^
({8{data_in[67][4]}} & 8'h4c)^
({8{data_in[67][5]}} & 8'h98)^
({8{data_in[67][6]}} & 8'h1b)^
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({8{data_in[68][3]}} & 8'h14)^
({8{data_in[68][4]}} & 8'h28)^
({8{data_in[68][5]}} & 8'h50)^
({8{data_in[68][6]}} & 8'ha0)^
({8{data_in[68][7]}} & 8'h6b)^
({8{data_in[69][0]}} & 8'hbb)^
({8{data_in[69][1]}} & 8'h5d)^
({8{data_in[69][2]}} & 8'hba)^
({8{data_in[69][3]}} & 8'h5f)^
({8{data_in[69][4]}} & 8'hbe)^
({8{data_in[69][5]}} & 8'h57)^
({8{data_in[69][6]}} & 8'hae)^
({8{data_in[69][7]}} & 8'h77)^

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({8{data_in[70][5]}} & 8'hc5)^
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({8{data_in[70][7]}} & 8'h69)^
({8{data_in[71][0]}} & 8'h9e)^
({8{data_in[71][1]}} & 8'h17)^
({8{data_in[71][2]}} & 8'h2e)^
({8{data_in[71][3]}} & 8'h5c)^
({8{data_in[71][4]}} & 8'hb8)^
({8{data_in[71][5]}} & 8'h5b)^
({8{data_in[71][6]}} & 8'hb6)^
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({8{data_in[72][2]}} & 8'hbf)^
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({8{data_in[72][4]}} & 8'haa)^
({8{data_in[72][5]}} & 8'h7f)^
({8{data_in[72][6]}} & 8'hfe)^
({8{data_in[72][7]}} & 8'hd7)^
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({8{data_in[73][2]}} & 8'he8)^
({8{data_in[73][3]}} & 8'hfb)^
({8{data_in[73][4]}} & 8'hdd)^
({8{data_in[73][5]}} & 8'h91)^
({8{data_in[73][6]}} & 8'h9)^
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({8{data_in[74][1]}} & 8'h7e)^
({8{data_in[74][2]}} & 8'hfc)^
({8{data_in[74][3]}} & 8'hd3)^
({8{data_in[74][4]}} & 8'h8d)^
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({8{data_in[75][5]}} & 8'h3)^
({8{data_in[75][6]}} & 8'h6)^
({8{data_in[75][7]}} & 8'hc)^
({8{data_in[76][0]}} & 8'h59)^
({8{data_in[76][1]}} & 8'hb2)^
({8{data_in[76][2]}} & 8'h4f)^
({8{data_in[76][3]}} & 8'h9e)^
({8{data_in[76][4]}} & 8'h17)^
({8{data_in[76][5]}} & 8'h2e)^
({8{data_in[76][6]}} & 8'h5c)^
({8{data_in[76][7]}} & 8'hb8)^
({8{data_in[77][0]}} & 8'hf7)^
({8{data_in[77][1]}} & 8'hc5)^
({8{data_in[77][2]}} & 8'ha1)^
({8{data_in[77][3]}} & 8'h69)^
({8{data_in[77][4]}} & 8'hd2)^
({8{data_in[77][5]}} & 8'h8f)^
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({8{data_in[77][7]}} & 8'h6a)^
({8{data_in[78][0]}} & 8'h1f)^

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({8{data_in[78][7]}} & 8'h22)^
({8{data_in[79][0]}} & 8'h88)^
({8{data_in[79][1]}} & 8'h3b)^
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({8{data_in[79][3]}} & 8'hec)^
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({8{data_in[79][5]}} & 8'hcd)^
({8{data_in[79][6]}} & 8'hb1)^
({8{data_in[79][7]}} & 8'h49)^
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({8{data_in[80][1]}} & 8'h3d)^
({8{data_in[80][2]}} & 8'h7a)^
({8{data_in[80][3]}} & 8'hf4)^
({8{data_in[80][4]}} & 8'hc3)^
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({8{data_in[80][6]}} & 8'h71)^
({8{data_in[80][7]}} & 8'he2)^
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({8{data_in[81][1]}} & 8'h4)^
({8{data_in[81][2]}} & 8'h8)^
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({8{data_in[82][2]}} & 8'hc4)^
({8{data_in[82][3]}} & 8'ha3)^
({8{data_in[82][4]}} & 8'h6d)^
({8{data_in[82][5]}} & 8'hda)^
({8{data_in[82][6]}} & 8'h9f)^
({8{data_in[82][7]}} & 8'h15)^
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({8{data_in[83][1]}} & 8'hbf)^
({8{data_in[83][2]}} & 8'h55)^
({8{data_in[83][3]}} & 8'haa)^
({8{data_in[83][4]}} & 8'h7f)^
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({8{data_in[83][6]}} & 8'hd7)^
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({8{data_in[84][1]}} & 8'h7)^
({8{data_in[84][2]}} & 8'he)^
({8{data_in[84][3]}} & 8'h1c)^
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({8{data_in[85][4]}} & 8'h7d)^
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({8{data_in[85][6]}} & 8'hdf)^
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({8{data_in[87][2]}} & 8'hc7)^
({8{data_in[87][3]}} & 8'ha5)^
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({8{data_in[87][7]}} & 8'h75)^
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({8{data_in[88][6]}} & 8'h4e)^
({8{data_in[88][7]}} & 8'h9c)^
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({8{data_in[89][1]}} & 8'h1c)^
({8{data_in[89][2]}} & 8'h38)^
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({8{data_in[89][5]}} & 8'heb)^
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({8{data_in[90][1]}} & 8'hcb)^
({8{data_in[90][2]}} & 8'hbd)^
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({8{data_in[90][6]}} & 8'hde)^
({8{data_in[90][7]}} & 8'h97)^
({8{data_in[91][0]}} & 8'h4f)^
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({8{data_in[91][4]}} & 8'h5c)^
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({8{data_in[91][6]}} & 8'h5b)^
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({8{data_in[92][2]}} & 8'h3f)^
({8{data_in[92][3]}} & 8'h7e)^
({8{data_in[92][4]}} & 8'hfc)^
({8{data_in[92][5]}} & 8'hd3)^
({8{data_in[92][6]}} & 8'h8d)^
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({8{data_in[93][2]}} & 8'h19)^
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({8{data_in[93][6]}} & 8'hbb)^
({8{data_in[93][7]}} & 8'h5d)^
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({8{data_in[95][4]}} & 8'h90)^
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({8{data_in[95][7]}} & 8'h2c)^
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({8{data_in[96][1]}} & 8'hc9)^
({8{data_in[96][2]}} & 8'hb9)^
({8{data_in[96][3]}} & 8'h59)^
({8{data_in[96][4]}} & 8'hb2)^
({8{data_in[96][5]}} & 8'h4f)^
({8{data_in[96][6]}} & 8'h9e)^
({8{data_in[96][7]}} & 8'h17)^
({8{data_in[97][0]}} & 8'h7)^
({8{data_in[97][1]}} & 8'he)^
({8{data_in[97][2]}} & 8'h1c)^
({8{data_in[97][3]}} & 8'h38)^
({8{data_in[97][4]}} & 8'h70)^
({8{data_in[97][5]}} & 8'he0)^
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({8{data_in[98][2]}} & 8'he8)^
({8{data_in[98][3]}} & 8'hfb)^
({8{data_in[98][4]}} & 8'hdd)^
({8{data_in[98][5]}} & 8'h91)^
({8{data_in[98][6]}} & 8'h9)^
({8{data_in[98][7]}} & 8'h12)^
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({8{data_in[99][4]}} & 8'h7)^
({8{data_in[99][5]}} & 8'he)^
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({8{data_in[99][7]}} & 8'h38)^
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({8{data_in[100][1]}} & 8'h7d)^
({8{data_in[100][2]}} & 8'hfa)^
({8{data_in[100][3]}} & 8'hdf)^
({8{data_in[100][4]}} & 8'h95)^
({8{data_in[100][5]}} & 8'h1)^
({8{data_in[100][6]}} & 8'h2)^
({8{data_in[100][7]}} & 8'h4)^
({8{data_in[101][0]}} & 8'h6b)^
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({8{data_in[101][2]}} & 8'h87)^
({8{data_in[101][3]}} & 8'h25)^
({8{data_in[101][4]}} & 8'h4a)^
({8{data_in[101][5]}} & 8'h94)^
({8{data_in[101][6]}} & 8'h3)^
({8{data_in[101][7]}} & 8'h6)^
({8{data_in[102][0]}} & 8'hd6)^
({8{data_in[102][1]}} & 8'h87)^
({8{data_in[102][2]}} & 8'h25)^
({8{data_in[102][3]}} & 8'h4a)^

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({8{data_in[102][6]}} & 8'h6)^
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({8{data_in[103][1]}} & 8'hb0)^
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({8{data_in[103][3]}} & 8'h96)^
({8{data_in[103][4]}} & 8'h7)^
({8{data_in[103][5]}} & 8'he)^
({8{data_in[103][6]}} & 8'h1c)^
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({8{data_in[104][3]}} & 8'h2c)^
({8{data_in[104][4]}} & 8'h58)^
({8{data_in[104][5]}} & 8'hb0)^
({8{data_in[104][6]}} & 8'h4b)^
({8{data_in[104][7]}} & 8'h96)^
({8{data_in[105][0]}} & 8'hea)^
({8{data_in[105][1]}} & 8'hff)^
({8{data_in[105][2]}} & 8'hd5)^
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({8{data_in[106][3]}} & 8'h38)^
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({8{data_in[106][5]}} & 8'he0)^
({8{data_in[106][6]}} & 8'heb)^
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({8{data_in[107][5]}} & 8'hc8)^
({8{data_in[107][6]}} & 8'hbb)^
({8{data_in[107][7]}} & 8'h5d)^
({8{data_in[108][0]}} & 8'hb9)^
({8{data_in[108][1]}} & 8'h59)^
({8{data_in[108][2]}} & 8'hb2)^
({8{data_in[108][3]}} & 8'h4f)^
({8{data_in[108][4]}} & 8'h9e)^
({8{data_in[108][5]}} & 8'h17)^
({8{data_in[108][6]}} & 8'h2e)^
({8{data_in[108][7]}} & 8'h5c)^
({8{data_in[109][0]}} & 8'hce)^
({8{data_in[109][1]}} & 8'hb7)^
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({8{data_in[109][3]}} & 8'h8a)^
({8{data_in[109][4]}} & 8'h3f)^
({8{data_in[109][5]}} & 8'h7e)^
({8{data_in[109][6]}} & 8'hfc)^
({8{data_in[109][7]}} & 8'hd3)^
({8{data_in[110][0]}} & 8'h57)^
({8{data_in[110][1]}} & 8'hae)^
({8{data_in[110][2]}} & 8'h77)^
({8{data_in[110][3]}} & 8'hee)^
({8{data_in[110][4]}} & 8'hf7)^

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({8{data_in[111][4]}} & 8'hf2)^
({8{data_in[111][5]}} & 8'hcf)^
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({8{data_in[111][7]}} & 8'h41)^
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({8{data_in[113][2]}} & 8'h1)^
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({8{data_in[116][1]}} & 8'h90)^
({8{data_in[116][2]}} & 8'hb)^
({8{data_in[116][3]}} & 8'h16)^
({8{data_in[116][4]}} & 8'h2c)^
({8{data_in[116][5]}} & 8'h58)^
({8{data_in[116][6]}} & 8'hb0)^
({8{data_in[116][7]}} & 8'h4b)^
({8{data_in[117][0]}} & 8'h63)^
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({8{data_in[117][2]}} & 8'ha7)^
({8{data_in[117][3]}} & 8'h65)^
({8{data_in[117][4]}} & 8'hca)^
({8{data_in[117][5]}} & 8'hbf)^
({8{data_in[117][6]}} & 8'h55)^
({8{data_in[117][7]}} & 8'haa)^
({8{data_in[118][0]}} & 8'h57)^
({8{data_in[118][1]}} & 8'hae)^
({8{data_in[118][2]}} & 8'h77)^
({8{data_in[118][3]}} & 8'hee)^
({8{data_in[118][4]}} & 8'hf7)^
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({8{data_in[119][6]}} & 8'hec)^
({8{data_in[119][7]}} & 8'hf3)^
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({8{data_in[121][2]}} & 8'h7e)^
({8{data_in[121][3]}} & 8'hfc)^
({8{data_in[121][4]}} & 8'hd3)^
({8{data_in[121][5]}} & 8'h8d)^
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({8{data_in[121][7]}} & 8'h62)^
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({8{data_in[123][3]}} & 8'h4a)^
({8{data_in[123][4]}} & 8'h94)^
({8{data_in[123][5]}} & 8'h3)^
({8{data_in[123][6]}} & 8'h6)^
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({8{data_in[124][3]}} & 8'h66)^
({8{data_in[124][4]}} & 8'hcc)^
({8{data_in[124][5]}} & 8'hb3)^
({8{data_in[124][6]}} & 8'h4d)^
({8{data_in[124][7]}} & 8'h9a)^
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({8{data_in[125][1]}} & 8'h6f)^
({8{data_in[125][2]}} & 8'hde)^
({8{data_in[125][3]}} & 8'h97)^
({8{data_in[125][4]}} & 8'h5)^
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({8{data_in[125][7]}} & 8'h28)^
({8{data_in[126][0]}} & 8'h45)^
({8{data_in[126][1]}} & 8'h8a)^
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({8{data_in[126][3]}} & 8'h7e)^
({8{data_in[126][4]}} & 8'hfc)^
({8{data_in[126][5]}} & 8'hd3)^
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({8{data_in[127][6]}} & 8'h3d)^
({8{data_in[127][7]}} & 8'h7a)^
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({8{data_in[128][1]}} & 8'h4b)^
({8{data_in[128][2]}} & 8'h96)^
({8{data_in[128][3]}} & 8'h7)^
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({8{data_in[129][7]}} & 8'h8)^
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({8{data_in[130][1]}} & 8'h5c)^
({8{data_in[130][2]}} & 8'hb8)^
({8{data_in[130][3]}} & 8'h5b)^
({8{data_in[130][4]}} & 8'hb6)^
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({8{data_in[131][1]}} & 8'hb3)^
({8{data_in[131][2]}} & 8'h4d)^
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({8{data_in[131][6]}} & 8'h7c)^
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({8{data_in[132][7]}} & 8'he1)^
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({8{data_in[133][3]}} & 8'h5a)^
({8{data_in[133][4]}} & 8'hb4)^
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({8{data_in[133][6]}} & 8'h86)^
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({8{data_in[134][4]}} & 8'h16)^
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({8{data_in[136][1]}} & 8'hb4)^
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({8{data_in[136][5]}} & 8'h4e)^
({8{data_in[136][6]}} & 8'h9c)^
({8{data_in[136][7]}} & 8'h13)^
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({8{data_in[138][3]}} & 8'hc2)^
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({8{data_in[139][6]}} & 8'hde)^
({8{data_in[139][7]}} & 8'h97)^
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({8{data_in[140][6]}} & 8'hfe)^
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({8{data_in[141][4]}} & 8'h5a)^
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({8{data_in[145][4]}} & 8'hb)^
({8{data_in[145][5]}} & 8'h16)^
({8{data_in[145][6]}} & 8'h2c)^
({8{data_in[145][7]}} & 8'h58)^
({8{data_in[146][0]}} & 8'h18)^
({8{data_in[146][1]}} & 8'h30)^
({8{data_in[146][2]}} & 8'h60)^
({8{data_in[146][3]}} & 8'hc0)^
({8{data_in[146][4]}} & 8'hab)^
({8{data_in[146][5]}} & 8'h7d)^
({8{data_in[146][6]}} & 8'hfa)^
({8{data_in[146][7]}} & 8'hdf)^
({8{data_in[147][0]}} & 8'hb9)^
({8{data_in[147][1]}} & 8'h59)^
({8{data_in[147][2]}} & 8'hb2)^
({8{data_in[147][3]}} & 8'h4f)^
({8{data_in[147][4]}} & 8'h9e)^
({8{data_in[147][5]}} & 8'h17)^
({8{data_in[147][6]}} & 8'h2e)^
({8{data_in[147][7]}} & 8'h5c)^
({8{data_in[148][0]}} & 8'hf7)^
({8{data_in[148][1]}} & 8'hc5)^
({8{data_in[148][2]}} & 8'ha1)^
({8{data_in[148][3]}} & 8'h69)^
({8{data_in[148][4]}} & 8'hd2)^
({8{data_in[148][5]}} & 8'h8f)^
({8{data_in[148][6]}} & 8'h35)^
({8{data_in[148][7]}} & 8'h6a)^
({8{data_in[149][0]}} & 8'h6a)^
({8{data_in[149][1]}} & 8'hd4)^
({8{data_in[149][2]}} & 8'h83)^
({8{data_in[149][3]}} & 8'h2d)^
({8{data_in[149][4]}} & 8'h5a)^
({8{data_in[149][5]}} & 8'hb4)^
({8{data_in[149][6]}} & 8'h43)^
({8{data_in[149][7]}} & 8'h86)^
({8{data_in[150][0]}} & 8'h14)^
({8{data_in[150][1]}} & 8'h28)^
({8{data_in[150][2]}} & 8'h50)^
({8{data_in[150][3]}} & 8'ha0)^
({8{data_in[150][4]}} & 8'h6b)^
({8{data_in[150][5]}} & 8'hd6)^
({8{data_in[150][6]}} & 8'h87)^
({8{data_in[150][7]}} & 8'h25)^
({8{data_in[151][0]}} & 8'hc8)^
({8{data_in[151][1]}} & 8'hbb)^

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({8{data_in[151][2]}} & 8'h5d)^
({8{data_in[151][3]}} & 8'hba)^
({8{data_in[151][4]}} & 8'h5f)^
({8{data_in[151][5]}} & 8'hbe)^
({8{data_in[151][6]}} & 8'h57)^
({8{data_in[151][7]}} & 8'hae)^
({8{data_in[152][0]}} & 8'h8a)^
({8{data_in[152][1]}} & 8'h3f)^
({8{data_in[152][2]}} & 8'h7e)^
({8{data_in[152][3]}} & 8'hfc)^
({8{data_in[152][4]}} & 8'hd3)^
({8{data_in[152][5]}} & 8'h8d)^
({8{data_in[152][6]}} & 8'h31)^
({8{data_in[152][7]}} & 8'h62)^
({8{data_in[153][0]}} & 8'ha8)^
({8{data_in[153][1]}} & 8'h7b)^
({8{data_in[153][2]}} & 8'hf6)^
({8{data_in[153][3]}} & 8'hc7)^
({8{data_in[153][4]}} & 8'ha5)^
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({8{data_in[153][6]}} & 8'hc2)^
({8{data_in[153][7]}} & 8'haf)^
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({8{data_in[154][3]}} & 8'h2e)^
({8{data_in[154][4]}} & 8'h5c)^
({8{data_in[154][5]}} & 8'hb8)^
({8{data_in[154][6]}} & 8'h5b)^
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({8{data_in[155][5]}} & 8'h4c)^
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({8{data_in[156][1]}} & 8'h5)^
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({8{data_in[156][5]}} & 8'h50)^
({8{data_in[156][6]}} & 8'ha0)^
({8{data_in[156][7]}} & 8'h6b)^
({8{data_in[157][0]}} & 8'h34)^
({8{data_in[157][1]}} & 8'h68)^
({8{data_in[157][2]}} & 8'hd0)^
({8{data_in[157][3]}} & 8'h8b)^
({8{data_in[157][4]}} & 8'h3d)^
({8{data_in[157][5]}} & 8'h7a)^
({8{data_in[157][6]}} & 8'hf4)^
({8{data_in[157][7]}} & 8'hc3)^
({8{data_in[158][0]}} & 8'hf2)^
({8{data_in[158][1]}} & 8'hcf)^
({8{data_in[158][2]}} & 8'hb5)^
({8{data_in[158][3]}} & 8'h41)^
({8{data_in[158][4]}} & 8'h82)^
({8{data_in[158][5]}} & 8'h2f)^
({8{data_in[158][6]}} & 8'h5e)^
({8{data_in[158][7]}} & 8'hbc)^
({8{data_in[159][0]}} & 8'h96)^
({8{data_in[159][1]}} & 8'h7)^
({8{data_in[159][2]}} & 8'he)^

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({8{data_in[159][3]}} & 8'h1c)^
({8{data_in[159][4]}} & 8'h38)^
({8{data_in[159][5]}} & 8'h70)^
({8{data_in[159][6]}} & 8'he0)^
({8{data_in[159][7]}} & 8'heb)^
({8{data_in[160][0]}} & 8'h71)^
({8{data_in[160][1]}} & 8'he2)^
({8{data_in[160][2]}} & 8'hef)^
({8{data_in[160][3]}} & 8'hf5)^
({8{data_in[160][4]}} & 8'hc1)^
({8{data_in[160][5]}} & 8'ha9)^
({8{data_in[160][6]}} & 8'h79)^
({8{data_in[160][7]}} & 8'hf2)^
({8{data_in[161][0]}} & 8'h7)^
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({8{data_in[161][7]}} & 8'hfd)^
({8{data_in[162][0]}} & 8'h94)^
({8{data_in[162][1]}} & 8'h3)^
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({8{data_in[162][3]}} & 8'hc)^
({8{data_in[162][4]}} & 8'h18)^
({8{data_in[162][5]}} & 8'h30)^
({8{data_in[162][6]}} & 8'h60)^
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({8{data_in[163][1]}} & 8'h20)^
({8{data_in[163][2]}} & 8'h40)^
({8{data_in[163][3]}} & 8'h80)^
({8{data_in[163][4]}} & 8'h2b)^
({8{data_in[163][5]}} & 8'h56)^
({8{data_in[163][6]}} & 8'hac)^
({8{data_in[163][7]}} & 8'h73)^
({8{data_in[164][0]}} & 8'h7a)^
({8{data_in[164][1]}} & 8'hf4)^
({8{data_in[164][2]}} & 8'hc3)^
({8{data_in[164][3]}} & 8'had)^
({8{data_in[164][4]}} & 8'h71)^
({8{data_in[164][5]}} & 8'he2)^
({8{data_in[164][6]}} & 8'hef)^
({8{data_in[164][7]}} & 8'hf5)^
({8{data_in[165][0]}} & 8'hf)^
({8{data_in[165][1]}} & 8'h1e)^
({8{data_in[165][2]}} & 8'h3c)^
({8{data_in[165][3]}} & 8'h78)^
({8{data_in[165][4]}} & 8'hf0)^
({8{data_in[165][5]}} & 8'hcb)^
({8{data_in[165][6]}} & 8'hbd)^
({8{data_in[165][7]}} & 8'h51)^
({8{data_in[166][0]}} & 8'h20)^
({8{data_in[166][1]}} & 8'h40)^
({8{data_in[166][2]}} & 8'h80)^
({8{data_in[166][3]}} & 8'h2b)^
({8{data_in[166][4]}} & 8'h56)^
({8{data_in[166][5]}} & 8'hac)^
({8{data_in[166][6]}} & 8'h73)^
({8{data_in[166][7]}} & 8'he6)^
({8{data_in[167][0]}} & 8'h11)^
({8{data_in[167][1]}} & 8'h22)^
({8{data_in[167][2]}} & 8'h44)^
({8{data_in[167][3]}} & 8'h88)^

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({8{data_in[168][4]}} & 8'hc6)^
({8{data_in[168][5]}} & 8'ha7)^
({8{data_in[168][6]}} & 8'h65)^
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({8{data_in[169][0]}} & 8'h7c)^
({8{data_in[169][1]}} & 8'hf8)^
({8{data_in[169][2]}} & 8'hdb)^
({8{data_in[169][3]}} & 8'h9d)^
({8{data_in[169][4]}} & 8'h11)^
({8{data_in[169][5]}} & 8'h22)^
({8{data_in[169][6]}} & 8'h44)^
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({8{data_in[171][3]}} & 8'h55)^
({8{data_in[171][4]}} & 8'haa)^
({8{data_in[171][5]}} & 8'h7f)^
({8{data_in[171][6]}} & 8'hfe)^
({8{data_in[171][7]}} & 8'hd7)^
({8{data_in[172][0]}} & 8'h1e)^
({8{data_in[172][1]}} & 8'h3c)^
({8{data_in[172][2]}} & 8'h78)^
({8{data_in[172][3]}} & 8'hf0)^
({8{data_in[172][4]}} & 8'hcb)^
({8{data_in[172][5]}} & 8'hbd)^
({8{data_in[172][6]}} & 8'h51)^
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({8{data_in[173][1]}} & 8'ha8)^
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({8{data_in[173][3]}} & 8'hf6)^
({8{data_in[173][4]}} & 8'hc7)^
({8{data_in[173][5]}} & 8'ha5)^
({8{data_in[173][6]}} & 8'h61)^
({8{data_in[173][7]}} & 8'hc2)^
({8{data_in[174][0]}} & 8'h8b)^
({8{data_in[174][1]}} & 8'h3d)^
({8{data_in[174][2]}} & 8'h7a)^
({8{data_in[174][3]}} & 8'hf4)^
({8{data_in[174][4]}} & 8'hc3)^
({8{data_in[174][5]}} & 8'had)^
({8{data_in[174][6]}} & 8'h71)^
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({8{data_in[175][4]}} & 8'h46)^

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({8{data_in[176][6]}} & 8'h90)^
({8{data_in[176][7]}} & 8'hb)^
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({8{data_in[177][4]}} & 8'h1b)^
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({8{data_in[178][0]}} & 8'h10)^
({8{data_in[178][1]}} & 8'h20)^
({8{data_in[178][2]}} & 8'h40)^
({8{data_in[178][3]}} & 8'h80)^
({8{data_in[178][4]}} & 8'h2b)^
({8{data_in[178][5]}} & 8'h56)^
({8{data_in[178][6]}} & 8'hac)^
({8{data_in[178][7]}} & 8'h73)^
({8{data_in[179][0]}} & 8'h2)^
({8{data_in[179][1]}} & 8'h4)^
({8{data_in[179][2]}} & 8'h8)^
({8{data_in[179][3]}} & 8'h10)^
({8{data_in[179][4]}} & 8'h20)^
({8{data_in[179][5]}} & 8'h40)^
({8{data_in[179][6]}} & 8'h80)^
({8{data_in[179][7]}} & 8'h2b)^
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({8{data_in[180][1]}} & 8'ha6)^
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({8{data_in[180][3]}} & 8'hce)^
({8{data_in[180][4]}} & 8'hb7)^
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({8{data_in[180][6]}} & 8'h8a)^
({8{data_in[180][7]}} & 8'h3f)^
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({8{data_in[181][1]}} & 8'h48)^
({8{data_in[181][2]}} & 8'h90)^
({8{data_in[181][3]}} & 8'hb)^
({8{data_in[181][4]}} & 8'h16)^
({8{data_in[181][5]}} & 8'h2c)^
({8{data_in[181][6]}} & 8'h58)^
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({8{data_in[182][2]}} & 8'hf3)^
({8{data_in[182][3]}} & 8'hcd)^
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({8{data_in[183][2]}} & 8'h2f)^
({8{data_in[183][3]}} & 8'h5e)^
({8{data_in[183][4]}} & 8'hbc)^
({8{data_in[183][5]}} & 8'h53)^

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({8{data_in[184][2]}} & 8'haa)^
({8{data_in[184][3]}} & 8'h7f)^
({8{data_in[184][4]}} & 8'hfe)^
({8{data_in[184][5]}} & 8'hd7)^
({8{data_in[184][6]}} & 8'h85)^
({8{data_in[184][7]}} & 8'h21)^
({8{data_in[185][0]}} & 8'h55)^
({8{data_in[185][1]}} & 8'haa)^
({8{data_in[185][2]}} & 8'h7f)^
({8{data_in[185][3]}} & 8'hfe)^
({8{data_in[185][4]}} & 8'hd7)^
({8{data_in[185][5]}} & 8'h85)^
({8{data_in[185][6]}} & 8'h21)^
({8{data_in[185][7]}} & 8'h42)^
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({8{data_in[186][2]}} & 8'hff)^
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({8{data_in[186][5]}} & 8'h29)^
({8{data_in[186][6]}} & 8'h52)^
({8{data_in[186][7]}} & 8'ha4)^
({8{data_in[187][0]}} & 8'h7e)^
({8{data_in[187][1]}} & 8'hfc)^
({8{data_in[187][2]}} & 8'hd3)^
({8{data_in[187][3]}} & 8'h8d)^
({8{data_in[187][4]}} & 8'h31)^
({8{data_in[187][5]}} & 8'h62)^
({8{data_in[187][6]}} & 8'hc4)^
({8{data_in[187][7]}} & 8'ha3)^
({8{data_in[188][0]}} & 8'hc6)^
({8{data_in[188][1]}} & 8'ha7)^
({8{data_in[188][2]}} & 8'h65)^
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({8{data_in[188][4]}} & 8'hbf)^
({8{data_in[188][5]}} & 8'h55)^
({8{data_in[188][6]}} & 8'haa)^
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({8{data_in[189][1]}} & 8'h57)^
({8{data_in[189][2]}} & 8'hae)^
({8{data_in[189][3]}} & 8'h77)^
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({8{data_in[189][6]}} & 8'hc5)^
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({8{data_in[190][0]}} & 8'h6a)^
({8{data_in[190][1]}} & 8'hd4)^
({8{data_in[190][2]}} & 8'h83)^
({8{data_in[190][3]}} & 8'h2d)^
({8{data_in[190][4]}} & 8'h5a)^
({8{data_in[190][5]}} & 8'hb4)^
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({8{data_in[192][6]}} & 8'h42)^
({8{data_in[192][7]}} & 8'h84)^
({8{data_in[193][0]}} & 8'h3e)^
({8{data_in[193][1]}} & 8'h7c)^
({8{data_in[193][2]}} & 8'hf8)^
({8{data_in[193][3]}} & 8'hdb)^
({8{data_in[193][4]}} & 8'h9d)^
({8{data_in[193][5]}} & 8'h11)^
({8{data_in[193][6]}} & 8'h22)^
({8{data_in[193][7]}} & 8'h44)^
({8{data_in[194][0]}} & 8'h6a)^
({8{data_in[194][1]}} & 8'hd4)^
({8{data_in[194][2]}} & 8'h83)^
({8{data_in[194][3]}} & 8'h2d)^
({8{data_in[194][4]}} & 8'h5a)^
({8{data_in[194][5]}} & 8'hb4)^
({8{data_in[194][6]}} & 8'h43)^
({8{data_in[194][7]}} & 8'h86)^
({8{data_in[195][0]}} & 8'h45)^
({8{data_in[195][1]}} & 8'h8a)^
({8{data_in[195][2]}} & 8'h3f)^
({8{data_in[195][3]}} & 8'h7e)^
({8{data_in[195][4]}} & 8'hfc)^
({8{data_in[195][5]}} & 8'hd3)^
({8{data_in[195][6]}} & 8'h8d)^
({8{data_in[195][7]}} & 8'h31)^
({8{data_in[196][0]}} & 8'ha4)^
({8{data_in[196][1]}} & 8'h63)^
({8{data_in[196][2]}} & 8'hc6)^
({8{data_in[196][3]}} & 8'ha7)^
({8{data_in[196][4]}} & 8'h65)^
({8{data_in[196][5]}} & 8'hca)^
({8{data_in[196][6]}} & 8'hbf)^
({8{data_in[196][7]}} & 8'h55)^
({8{data_in[197][0]}} & 8'hb1)^
({8{data_in[197][1]}} & 8'h49)^
({8{data_in[197][2]}} & 8'h92)^
({8{data_in[197][3]}} & 8'hf)^
({8{data_in[197][4]}} & 8'h1e)^
({8{data_in[197][5]}} & 8'h3c)^
({8{data_in[197][6]}} & 8'h78)^
({8{data_in[197][7]}} & 8'hf0)^
({8{data_in[198][0]}} & 8'hfb)^
({8{data_in[198][1]}} & 8'hdd)^
({8{data_in[198][2]}} & 8'h91)^
({8{data_in[198][3]}} & 8'h9)^
({8{data_in[198][4]}} & 8'h12)^
({8{data_in[198][5]}} & 8'h24)^
({8{data_in[198][6]}} & 8'h48)^
({8{data_in[198][7]}} & 8'h90)^
({8{data_in[199][0]}} & 8'h5e)^
({8{data_in[199][1]}} & 8'hbc)^
({8{data_in[199][2]}} & 8'h53)^
({8{data_in[199][3]}} & 8'ha6)^
({8{data_in[199][4]}} & 8'h67)^
({8{data_in[199][5]}} & 8'hce)^
({8{data_in[199][6]}} & 8'hb7)^
({8{data_in[199][7]}} & 8'h45)^

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({8{data_in[200][0]}} & 8'h1a)^
({8{data_in[200][1]}} & 8'h34)^
({8{data_in[200][2]}} & 8'h68)^
({8{data_in[200][3]}} & 8'hd0)^
({8{data_in[200][4]}} & 8'h8b)^
({8{data_in[200][5]}} & 8'h3d)^
({8{data_in[200][6]}} & 8'h7a)^
({8{data_in[200][7]}} & 8'hf4)^
({8{data_in[201][0]}} & 8'h3)^
({8{data_in[201][1]}} & 8'h6)^
({8{data_in[201][2]}} & 8'hc)^
({8{data_in[201][3]}} & 8'h18)^
({8{data_in[201][4]}} & 8'h30)^
({8{data_in[201][5]}} & 8'h60)^
({8{data_in[201][6]}} & 8'hc0)^
({8{data_in[201][7]}} & 8'hab)^
({8{data_in[202][0]}} & 8'h9a)^
({8{data_in[202][1]}} & 8'h1f)^
({8{data_in[202][2]}} & 8'h3e)^
({8{data_in[202][3]}} & 8'h7c)^
({8{data_in[202][4]}} & 8'hf8)^
({8{data_in[202][5]}} & 8'hdb)^
({8{data_in[202][6]}} & 8'h9d)^
({8{data_in[202][7]}} & 8'h11)^
({8{data_in[203][0]}} & 8'hb6)^
({8{data_in[203][1]}} & 8'h47)^
({8{data_in[203][2]}} & 8'h8e)^
({8{data_in[203][3]}} & 8'h37)^
({8{data_in[203][4]}} & 8'h6e)^
({8{data_in[203][5]}} & 8'hdc)^
({8{data_in[203][6]}} & 8'h93)^
({8{data_in[203][7]}} & 8'hd)^
({8{data_in[204][0]}} & 8'h49)^
({8{data_in[204][1]}} & 8'h92)^
({8{data_in[204][2]}} & 8'hf)^
({8{data_in[204][3]}} & 8'h1e)^
({8{data_in[204][4]}} & 8'h3c)^
({8{data_in[204][5]}} & 8'h78)^
({8{data_in[204][6]}} & 8'hf0)^
({8{data_in[204][7]}} & 8'hcb)^
({8{data_in[205][0]}} & 8'h33)^
({8{data_in[205][1]}} & 8'h66)^
({8{data_in[205][2]}} & 8'hcc)^
({8{data_in[205][3]}} & 8'hb3)^
({8{data_in[205][4]}} & 8'h4d)^
({8{data_in[205][5]}} & 8'h9a)^
({8{data_in[205][6]}} & 8'h1f)^
({8{data_in[205][7]}} & 8'h3e)^
({8{data_in[206][0]}} & 8'ha6)^
({8{data_in[206][1]}} & 8'h67)^
({8{data_in[206][2]}} & 8'hce)^
({8{data_in[206][3]}} & 8'hb7)^
({8{data_in[206][4]}} & 8'h45)^
({8{data_in[206][5]}} & 8'h8a)^
({8{data_in[206][6]}} & 8'h3f)^
({8{data_in[206][7]}} & 8'h7e)^
({8{data_in[207][0]}} & 8'haf)^
({8{data_in[207][1]}} & 8'h75)^
({8{data_in[207][2]}} & 8'hea)^
({8{data_in[207][3]}} & 8'hff)^
({8{data_in[207][4]}} & 8'hd5)^
({8{data_in[207][5]}} & 8'h81)^
({8{data_in[207][6]}} & 8'h29)^
({8{data_in[207][7]}} & 8'h52)^
({8{data_in[208][0]}} & 8'h23)^

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({8{data_in[208][1]}} & 8'h46)^
({8{data_in[208][2]}} & 8'h8c)^
({8{data_in[208][3]}} & 8'h33)^
({8{data_in[208][4]}} & 8'h66)^
({8{data_in[208][5]}} & 8'hcc)^
({8{data_in[208][6]}} & 8'hb3)^
({8{data_in[208][7]}} & 8'h4d)^
({8{data_in[209][0]}} & 8'h67)^
({8{data_in[209][1]}} & 8'hce)^
({8{data_in[209][2]}} & 8'hb7)^
({8{data_in[209][3]}} & 8'h45)^
({8{data_in[209][4]}} & 8'h8a)^
({8{data_in[209][5]}} & 8'h3f)^
({8{data_in[209][6]}} & 8'h7e)^
({8{data_in[209][7]}} & 8'hfc)^
({8{data_in[210][0]}} & 8'h74)^
({8{data_in[210][1]}} & 8'he8)^
({8{data_in[210][2]}} & 8'hfb)^
({8{data_in[210][3]}} & 8'hdd)^
({8{data_in[210][4]}} & 8'h91)^
({8{data_in[210][5]}} & 8'h9)^
({8{data_in[210][6]}} & 8'h12)^
({8{data_in[210][7]}} & 8'h24)^
({8{data_in[211][0]}} & 8'h8f)^
({8{data_in[211][1]}} & 8'h35)^
({8{data_in[211][2]}} & 8'h6a)^
({8{data_in[211][3]}} & 8'hd4)^
({8{data_in[211][4]}} & 8'h83)^
({8{data_in[211][5]}} & 8'h2d)^
({8{data_in[211][6]}} & 8'h5a)^
({8{data_in[211][7]}} & 8'hb4)^
({8{data_in[212][0]}} & 8'h65)^
({8{data_in[212][1]}} & 8'hca)^
({8{data_in[212][2]}} & 8'hbf)^
({8{data_in[212][3]}} & 8'h55)^
({8{data_in[212][4]}} & 8'haa)^
({8{data_in[212][5]}} & 8'h7f)^
({8{data_in[212][6]}} & 8'hfe)^
({8{data_in[212][7]}} & 8'hd7)^
({8{data_in[213][0]}} & 8'h57)^
({8{data_in[213][1]}} & 8'hae)^
({8{data_in[213][2]}} & 8'h77)^
({8{data_in[213][3]}} & 8'hee)^
({8{data_in[213][4]}} & 8'hf7)^
({8{data_in[213][5]}} & 8'hc5)^
({8{data_in[213][6]}} & 8'ha1)^
({8{data_in[213][7]}} & 8'h69)^
({8{data_in[214][0]}} & 8'hf2)^
({8{data_in[214][1]}} & 8'hcf)^
({8{data_in[214][2]}} & 8'hb5)^
({8{data_in[214][3]}} & 8'h41)^
({8{data_in[214][4]}} & 8'h82)^
({8{data_in[214][5]}} & 8'h2f)^
({8{data_in[214][6]}} & 8'h5e)^
({8{data_in[214][7]}} & 8'hbc)^
({8{data_in[215][0]}} & 8'h9a)^
({8{data_in[215][1]}} & 8'h1f)^
({8{data_in[215][2]}} & 8'h3e)^
({8{data_in[215][3]}} & 8'h7c)^
({8{data_in[215][4]}} & 8'hf8)^
({8{data_in[215][5]}} & 8'hdb)^
({8{data_in[215][6]}} & 8'h9d)^
({8{data_in[215][7]}} & 8'h11)^
({8{data_in[216][0]}} & 8'h1a)^
({8{data_in[216][1]}} & 8'h34)^

```

```

({8{data_in[216][2]}} & 8'h68)^
({8{data_in[216][3]}} & 8'hd0)^
({8{data_in[216][4]}} & 8'h8b)^
({8{data_in[216][5]}} & 8'h3d)^
({8{data_in[216][6]}} & 8'h7a)^
({8{data_in[216][7]}} & 8'hf4)^
({8{data_in[217][0]}} & 8'h76)^
({8{data_in[217][1]}} & 8'hec)^
({8{data_in[217][2]}} & 8'hf3)^
({8{data_in[217][3]}} & 8'hcd)^
({8{data_in[217][4]}} & 8'hb1)^
({8{data_in[217][5]}} & 8'h49)^
({8{data_in[217][6]}} & 8'h92)^
({8{data_in[217][7]}} & 8'hf)^
({8{data_in[218][0]}} & 8'h46)^
({8{data_in[218][1]}} & 8'h8c)^
({8{data_in[218][2]}} & 8'h33)^
({8{data_in[218][3]}} & 8'h66)^
({8{data_in[218][4]}} & 8'hcc)^
({8{data_in[218][5]}} & 8'hb3)^
({8{data_in[218][6]}} & 8'h4d)^
({8{data_in[218][7]}} & 8'h9a)^
({8{data_in[219][0]}} & 8'h7a)^
({8{data_in[219][1]}} & 8'hf4)^
({8{data_in[219][2]}} & 8'hc3)^
({8{data_in[219][3]}} & 8'had)^
({8{data_in[219][4]}} & 8'h71)^
({8{data_in[219][5]}} & 8'he2)^
({8{data_in[219][6]}} & 8'hef)^
({8{data_in[219][7]}} & 8'hf5)^
({8{data_in[220][0]}} & 8'h6)^
({8{data_in[220][1]}} & 8'hc)^
({8{data_in[220][2]}} & 8'h18)^
({8{data_in[220][3]}} & 8'h30)^
({8{data_in[220][4]}} & 8'h60)^
({8{data_in[220][5]}} & 8'hc0)^
({8{data_in[220][6]}} & 8'hab)^
({8{data_in[220][7]}} & 8'h7d)^
({8{data_in[221][0]}} & 8'hc9)^
({8{data_in[221][1]}} & 8'hb9)^
({8{data_in[221][2]}} & 8'h59)^
({8{data_in[221][3]}} & 8'hb2)^
({8{data_in[221][4]}} & 8'h4f)^
({8{data_in[221][5]}} & 8'h9e)^
({8{data_in[221][6]}} & 8'h17)^
({8{data_in[221][7]}} & 8'h2e)^
({8{data_in[222][0]}} & 8'h50)^
({8{data_in[222][1]}} & 8'ha0)^
({8{data_in[222][2]}} & 8'h6b)^
({8{data_in[222][3]}} & 8'hd6)^
({8{data_in[222][4]}} & 8'h87)^
({8{data_in[222][5]}} & 8'h25)^
({8{data_in[222][6]}} & 8'h4a)^
({8{data_in[222][7]}} & 8'h94)^
({8{data_in[223][0]}} & 8'h38)^
({8{data_in[223][1]}} & 8'h70)^
({8{data_in[223][2]}} & 8'he0)^
({8{data_in[223][3]}} & 8'heb)^
({8{data_in[223][4]}} & 8'hfd)^
({8{data_in[223][5]}} & 8'hd1)^
({8{data_in[223][6]}} & 8'h89)^
({8{data_in[223][7]}} & 8'h39)^
({8{data_in[224][0]}} & 8'hae)^
({8{data_in[224][1]}} & 8'h77)^
({8{data_in[224][2]}} & 8'hee)^

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```

({8{data_in[224][3]}} & 8'hf7)^
({8{data_in[224][4]}} & 8'hc5)^
({8{data_in[224][5]}} & 8'ha1)^
({8{data_in[224][6]}} & 8'h69)^
({8{data_in[224][7]}} & 8'hd2)^
({8{data_in[225][0]}} & 8'h34)^
({8{data_in[225][1]}} & 8'h68)^
({8{data_in[225][2]}} & 8'hd0)^
({8{data_in[225][3]}} & 8'h8b)^
({8{data_in[225][4]}} & 8'h3d)^
({8{data_in[225][5]}} & 8'h7a)^
({8{data_in[225][6]}} & 8'hf4)^
({8{data_in[225][7]}} & 8'hc3)^
({8{data_in[226][0]}} & 8'h9e)^
({8{data_in[226][1]}} & 8'h17)^
({8{data_in[226][2]}} & 8'h2e)^
({8{data_in[226][3]}} & 8'h5c)^
({8{data_in[226][4]}} & 8'hb8)^
({8{data_in[226][5]}} & 8'h5b)^
({8{data_in[226][6]}} & 8'hb6)^
({8{data_in[226][7]}} & 8'h47)^
({8{data_in[227][0]}} & 8'h5d)^
({8{data_in[227][1]}} & 8'hba)^
({8{data_in[227][2]}} & 8'h5f)^
({8{data_in[227][3]}} & 8'hbe)^
({8{data_in[227][4]}} & 8'h57)^
({8{data_in[227][5]}} & 8'hae)^
({8{data_in[227][6]}} & 8'h77)^
({8{data_in[227][7]}} & 8'hee)^
({8{data_in[228][0]}} & 8'hfe)^
({8{data_in[228][1]}} & 8'hd7)^
({8{data_in[228][2]}} & 8'h85)^
({8{data_in[228][3]}} & 8'h21)^
({8{data_in[228][4]}} & 8'h42)^
({8{data_in[228][5]}} & 8'h84)^
({8{data_in[228][6]}} & 8'h23)^
({8{data_in[228][7]}} & 8'h46)^
({8{data_in[229][0]}} & 8'hdc)^
({8{data_in[229][1]}} & 8'h93)^
({8{data_in[229][2]}} & 8'hd)^
({8{data_in[229][3]}} & 8'h1a)^
({8{data_in[229][4]}} & 8'h34)^
({8{data_in[229][5]}} & 8'h68)^
({8{data_in[229][6]}} & 8'hd0)^
({8{data_in[229][7]}} & 8'h8b)^
({8{data_in[230][0]}} & 8'hd9)^
({8{data_in[230][1]}} & 8'h99)^
({8{data_in[230][2]}} & 8'h19)^
({8{data_in[230][3]}} & 8'h32)^
({8{data_in[230][4]}} & 8'h64)^
({8{data_in[230][5]}} & 8'hc8)^
({8{data_in[230][6]}} & 8'hbb)^
({8{data_in[230][7]}} & 8'h5d)^
({8{data_in[231][0]}} & 8'hee)^
({8{data_in[231][1]}} & 8'hf7)^
({8{data_in[231][2]}} & 8'hc5)^
({8{data_in[231][3]}} & 8'ha1)^
({8{data_in[231][4]}} & 8'h69)^
({8{data_in[231][5]}} & 8'hd2)^
({8{data_in[231][6]}} & 8'h8f)^
({8{data_in[231][7]}} & 8'h35)^
({8{data_in[232][0]}} & 8'h39)^
({8{data_in[232][1]}} & 8'h72)^
({8{data_in[232][2]}} & 8'he4)^
({8{data_in[232][3]}} & 8'he3)^

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({8{data_in[232][4]}} & 8'hed)^
({8{data_in[232][5]}} & 8'hf1)^
({8{data_in[232][6]}} & 8'hc9)^
({8{data_in[232][7]}} & 8'hb9)^
({8{data_in[233][0]}} & 8'h30)^
({8{data_in[233][1]}} & 8'h60)^
({8{data_in[233][2]}} & 8'hc0)^
({8{data_in[233][3]}} & 8'hab)^
({8{data_in[233][4]}} & 8'h7d)^
({8{data_in[233][5]}} & 8'hfa)^
({8{data_in[233][6]}} & 8'hdf)^
({8{data_in[233][7]}} & 8'h95)^
({8{data_in[234][0]}} & 8'hfd)^
({8{data_in[234][1]}} & 8'hd1)^
({8{data_in[234][2]}} & 8'h89)^
({8{data_in[234][3]}} & 8'h39)^
({8{data_in[234][4]}} & 8'h72)^
({8{data_in[234][5]}} & 8'he4)^
({8{data_in[234][6]}} & 8'he3)^
({8{data_in[234][7]}} & 8'hed)^
({8{data_in[235][0]}} & 8'h59)^
({8{data_in[235][1]}} & 8'hb2)^
({8{data_in[235][2]}} & 8'h4f)^
({8{data_in[235][3]}} & 8'h9e)^
({8{data_in[235][4]}} & 8'h17)^
({8{data_in[235][5]}} & 8'h2e)^
({8{data_in[235][6]}} & 8'h5c)^
({8{data_in[235][7]}} & 8'hb8)^
({8{data_in[236][0]}} & 8'hb6)^
({8{data_in[236][1]}} & 8'h47)^
({8{data_in[236][2]}} & 8'h8e)^
({8{data_in[236][3]}} & 8'h37)^
({8{data_in[236][4]}} & 8'h6e)^
({8{data_in[236][5]}} & 8'hdc)^
({8{data_in[236][6]}} & 8'h93)^
({8{data_in[236][7]}} & 8'hd)^
({8{data_in[237][0]}} & 8'hf4)^
({8{data_in[237][1]}} & 8'hc3)^
({8{data_in[237][2]}} & 8'had)^
({8{data_in[237][3]}} & 8'h71)^
({8{data_in[237][4]}} & 8'he2)^
({8{data_in[237][5]}} & 8'hef)^
({8{data_in[237][6]}} & 8'hf5)^
({8{data_in[237][7]}} & 8'hc1)^
({8{data_in[238][0]}} & 8'haa)^
({8{data_in[238][1]}} & 8'h7f)^
({8{data_in[238][2]}} & 8'hfe)^
({8{data_in[238][3]}} & 8'hd7)^
({8{data_in[238][4]}} & 8'h85)^
({8{data_in[238][5]}} & 8'h21)^
({8{data_in[238][6]}} & 8'h42)^
({8{data_in[238][7]}} & 8'h84)^
({8{data_in[239][0]}} & 8'h3f)^
({8{data_in[239][1]}} & 8'h7e)^
({8{data_in[239][2]}} & 8'hfc)^
({8{data_in[239][3]}} & 8'hd3)^
({8{data_in[239][4]}} & 8'h8d)^
({8{data_in[239][5]}} & 8'h31)^
({8{data_in[239][6]}} & 8'h62)^
({8{data_in[239][7]}} & 8'hc4)^
({8{data_in[240][0]}} & 8'hd2)^
({8{data_in[240][1]}} & 8'h8f)^
({8{data_in[240][2]}} & 8'h35)^
({8{data_in[240][3]}} & 8'h6a)^
({8{data_in[240][4]}} & 8'hd4)^

```



```

({8{data_in[240][5]}} & 8'h83)^
({8{data_in[240][6]}} & 8'h2d)^
({8{data_in[240][7]}} & 8'h5a)^
({8{data_in[241][0]}} & 8'h33)^
({8{data_in[241][1]}} & 8'h66)^
({8{data_in[241][2]}} & 8'hcc)^
({8{data_in[241][3]}} & 8'hb3)^
({8{data_in[241][4]}} & 8'h4d)^
({8{data_in[241][5]}} & 8'h9a)^
({8{data_in[241][6]}} & 8'h1f)^
({8{data_in[241][7]}} & 8'h3e);
data_out[244] = ({8{data_in[0][0]}} & 8'hc3)^
({8{data_in[0][1]}} & 8'had)^
({8{data_in[0][2]}} & 8'h71)^
({8{data_in[0][3]}} & 8'he2)^
({8{data_in[0][4]}} & 8'hcf)^
({8{data_in[0][5]}} & 8'hf5)^
({8{data_in[0][6]}} & 8'hc1)^
({8{data_in[0][7]}} & 8'ha9)^
({8{data_in[1][0]}} & 8'h1f)^
({8{data_in[1][1]}} & 8'h3e)^
({8{data_in[1][2]}} & 8'h7c)^
({8{data_in[1][3]}} & 8'hf8)^
({8{data_in[1][4]}} & 8'hdb)^
({8{data_in[1][5]}} & 8'h9d)^
({8{data_in[1][6]}} & 8'h11)^
({8{data_in[1][7]}} & 8'h22)^
({8{data_in[2][0]}} & 8'hac)^
({8{data_in[2][1]}} & 8'h73)^
({8{data_in[2][2]}} & 8'he6)^
({8{data_in[2][3]}} & 8'he7)^
({8{data_in[2][4]}} & 8'he5)^
({8{data_in[2][5]}} & 8'he1)^
({8{data_in[2][6]}} & 8'he9)^
({8{data_in[2][7]}} & 8'hf9)^
({8{data_in[3][0]}} & 8'h8c)^
({8{data_in[3][1]}} & 8'h33)^
({8{data_in[3][2]}} & 8'h66)^
({8{data_in[3][3]}} & 8'hcc)^
({8{data_in[3][4]}} & 8'hb3)^
({8{data_in[3][5]}} & 8'h4d)^
({8{data_in[3][6]}} & 8'h9a)^
({8{data_in[3][7]}} & 8'h1f)^
({8{data_in[4][0]}} & 8'h55)^
({8{data_in[4][1]}} & 8'haa)^
({8{data_in[4][2]}} & 8'h7f)^
({8{data_in[4][3]}} & 8'hfe)^
({8{data_in[4][4]}} & 8'hd7)^
({8{data_in[4][5]}} & 8'h85)^
({8{data_in[4][6]}} & 8'h21)^
({8{data_in[4][7]}} & 8'h42)^
({8{data_in[5][0]}} & 8'hdc)^
({8{data_in[5][1]}} & 8'h93)^
({8{data_in[5][2]}} & 8'hd)^
({8{data_in[5][3]}} & 8'h1a)^
({8{data_in[5][4]}} & 8'h34)^
({8{data_in[5][5]}} & 8'h68)^
({8{data_in[5][6]}} & 8'hd0)^
({8{data_in[5][7]}} & 8'h8b)^
({8{data_in[6][0]}} & 8'h3)^
({8{data_in[6][1]}} & 8'h6)^
({8{data_in[6][2]}} & 8'hc)^
({8{data_in[6][3]}} & 8'h18)^
({8{data_in[6][4]}} & 8'h30)^
({8{data_in[6][5]}} & 8'h60)^

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({8{data_in[6][7]}} & 8'hab)^
({8{data_in[7][0]}} & 8'h1e)^
({8{data_in[7][1]}} & 8'h3c)^
({8{data_in[7][2]}} & 8'h78)^
({8{data_in[7][3]}} & 8'hf0)^
({8{data_in[7][4]}} & 8'hcb)^
({8{data_in[7][5]}} & 8'hbd)^
({8{data_in[7][6]}} & 8'h51)^
({8{data_in[7][7]}} & 8'ha2)^
({8{data_in[8][0]}} & 8'ha8)^
({8{data_in[8][1]}} & 8'h7b)^
({8{data_in[8][2]}} & 8'hf6)^
({8{data_in[8][3]}} & 8'hc7)^
({8{data_in[8][4]}} & 8'ha5)^
({8{data_in[8][5]}} & 8'h61)^
({8{data_in[8][6]}} & 8'hc2)^
({8{data_in[8][7]}} & 8'haf)^
({8{data_in[9][0]}} & 8'h17)^
({8{data_in[9][1]}} & 8'h2e)^
({8{data_in[9][2]}} & 8'h5c)^
({8{data_in[9][3]}} & 8'hb8)^
({8{data_in[9][4]}} & 8'h5b)^
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({8{data_in[9][7]}} & 8'h8e)^
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({8{data_in[10][1]}} & 8'h7d)^
({8{data_in[10][2]}} & 8'hfa)^
({8{data_in[10][3]}} & 8'hdf)^
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({8{data_in[10][5]}} & 8'h1)^
({8{data_in[10][6]}} & 8'h2)^
({8{data_in[10][7]}} & 8'h4)^
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({8{data_in[11][2]}} & 8'h3b)^
({8{data_in[11][3]}} & 8'h76)^
({8{data_in[11][4]}} & 8'hec)^
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({8{data_in[11][6]}} & 8'hcd)^
({8{data_in[11][7]}} & 8'hb1)^
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({8{data_in[12][1]}} & 8'hc7)^
({8{data_in[12][2]}} & 8'ha5)^
({8{data_in[12][3]}} & 8'h61)^
({8{data_in[12][4]}} & 8'hc2)^
({8{data_in[12][5]}} & 8'haf)^
({8{data_in[12][6]}} & 8'h75)^
({8{data_in[12][7]}} & 8'hea)^
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({8{data_in[13][1]}} & 8'h81)^
({8{data_in[13][2]}} & 8'h29)^
({8{data_in[13][3]}} & 8'h52)^
({8{data_in[13][4]}} & 8'ha4)^
({8{data_in[13][5]}} & 8'h63)^
({8{data_in[13][6]}} & 8'hc6)^
({8{data_in[13][7]}} & 8'ha7)^
({8{data_in[14][0]}} & 8'h80)^
({8{data_in[14][1]}} & 8'h2b)^
({8{data_in[14][2]}} & 8'h56)^
({8{data_in[14][3]}} & 8'hac)^
({8{data_in[14][4]}} & 8'h73)^
({8{data_in[14][5]}} & 8'he6)^
({8{data_in[14][6]}} & 8'he7)^

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({8{data_in[15][3]}} & 8'hc8)^
({8{data_in[15][4]}} & 8'hbb)^
({8{data_in[15][5]}} & 8'h5d)^
({8{data_in[15][6]}} & 8'hba)^
({8{data_in[15][7]}} & 8'h5f)^
({8{data_in[16][0]}} & 8'h8d)^
({8{data_in[16][1]}} & 8'h31)^
({8{data_in[16][2]}} & 8'h62)^
({8{data_in[16][3]}} & 8'hc4)^
({8{data_in[16][4]}} & 8'ha3)^
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({8{data_in[17][2]}} & 8'h1b)^
({8{data_in[17][3]}} & 8'h36)^
({8{data_in[17][4]}} & 8'h6c)^
({8{data_in[17][5]}} & 8'hd8)^
({8{data_in[17][6]}} & 8'h9b)^
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({8{data_in[18][0]}} & 8'h4a)^
({8{data_in[18][1]}} & 8'h94)^
({8{data_in[18][2]}} & 8'h3)^
({8{data_in[18][3]}} & 8'h6)^
({8{data_in[18][4]}} & 8'hc)^
({8{data_in[18][5]}} & 8'h18)^
({8{data_in[18][6]}} & 8'h30)^
({8{data_in[18][7]}} & 8'h60)^
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({8{data_in[19][1]}} & 8'h3c)^
({8{data_in[19][2]}} & 8'h78)^
({8{data_in[19][3]}} & 8'hf0)^
({8{data_in[19][4]}} & 8'hcb)^
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({8{data_in[20][0]}} & 8'ha2)^
({8{data_in[20][1]}} & 8'h6f)^
({8{data_in[20][2]}} & 8'hde)^
({8{data_in[20][3]}} & 8'h97)^
({8{data_in[20][4]}} & 8'h5)^
({8{data_in[20][5]}} & 8'ha)^
({8{data_in[20][6]}} & 8'h14)^
({8{data_in[20][7]}} & 8'h28)^
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({8{data_in[21][2]}} & 8'he7)^
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({8{data_in[22][2]}} & 8'h9a)^
({8{data_in[22][3]}} & 8'h1f)^
({8{data_in[22][4]}} & 8'h3e)^
({8{data_in[22][5]}} & 8'h7c)^
({8{data_in[22][6]}} & 8'hf8)^
({8{data_in[22][7]}} & 8'hdb)^

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({8{data_in[23][5]}} & 8'heb)^
({8{data_in[23][6]}} & 8'hfd)^
({8{data_in[23][7]}} & 8'hd1)^
({8{data_in[24][0]}} & 8'h2d)^
({8{data_in[24][1]}} & 8'h5a)^
({8{data_in[24][2]}} & 8'hb4)^
({8{data_in[24][3]}} & 8'h43)^
({8{data_in[24][4]}} & 8'h86)^
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({8{data_in[24][7]}} & 8'h9c)^
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({8{data_in[25][1]}} & 8'h8c)^
({8{data_in[25][2]}} & 8'h33)^
({8{data_in[25][3]}} & 8'h66)^
({8{data_in[25][4]}} & 8'hcc)^
({8{data_in[25][5]}} & 8'hb3)^
({8{data_in[25][6]}} & 8'h4d)^
({8{data_in[25][7]}} & 8'h9a)^
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({8{data_in[26][1]}} & 8'h4f)^
({8{data_in[26][2]}} & 8'h9e)^
({8{data_in[26][3]}} & 8'h17)^
({8{data_in[26][4]}} & 8'h2e)^
({8{data_in[26][5]}} & 8'h5c)^
({8{data_in[26][6]}} & 8'hb8)^
({8{data_in[26][7]}} & 8'h5b)^
({8{data_in[27][0]}} & 8'h8b)^
({8{data_in[27][1]}} & 8'h3d)^
({8{data_in[27][2]}} & 8'h7a)^
({8{data_in[27][3]}} & 8'hf4)^
({8{data_in[27][4]}} & 8'hc3)^
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({8{data_in[27][6]}} & 8'h71)^
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({8{data_in[28][4]}} & 8'h3)^
({8{data_in[28][5]}} & 8'h6)^
({8{data_in[28][6]}} & 8'hc)^
({8{data_in[28][7]}} & 8'h18)^
({8{data_in[29][0]}} & 8'h65)^
({8{data_in[29][1]}} & 8'hca)^
({8{data_in[29][2]}} & 8'hbf)^
({8{data_in[29][3]}} & 8'h55)^
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({8{data_in[29][5]}} & 8'h7f)^
({8{data_in[29][6]}} & 8'hfe)^
({8{data_in[29][7]}} & 8'hd7)^
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({8{data_in[30][1]}} & 8'h26)^
({8{data_in[30][2]}} & 8'h4c)^
({8{data_in[30][3]}} & 8'h98)^
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({8{data_in[30][6]}} & 8'h6c)^
({8{data_in[30][7]}} & 8'hd8)^
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({8{data_in[31][5]}} & 8'hae)^
({8{data_in[31][6]}} & 8'h77)^
({8{data_in[31][7]}} & 8'hee)^
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({8{data_in[32][1]}} & 8'h82)^
({8{data_in[32][2]}} & 8'h2f)^
({8{data_in[32][3]}} & 8'h5e)^
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({8{data_in[33][4]}} & 8'h64)^
({8{data_in[33][5]}} & 8'hc8)^
({8{data_in[33][6]}} & 8'hbb)^
({8{data_in[33][7]}} & 8'h5d)^
({8{data_in[34][0]}} & 8'hff)^
({8{data_in[34][1]}} & 8'hd5)^
({8{data_in[34][2]}} & 8'h81)^
({8{data_in[34][3]}} & 8'h29)^
({8{data_in[34][4]}} & 8'h52)^
({8{data_in[34][5]}} & 8'ha4)^
({8{data_in[34][6]}} & 8'h63)^
({8{data_in[34][7]}} & 8'hc6)^
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({8{data_in[35][1]}} & 8'h6a)^
({8{data_in[35][2]}} & 8'hd4)^
({8{data_in[35][3]}} & 8'h83)^
({8{data_in[35][4]}} & 8'h2d)^
({8{data_in[35][5]}} & 8'h5a)^
({8{data_in[35][6]}} & 8'hb4)^
({8{data_in[35][7]}} & 8'h43)^
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({8{data_in[36][1]}} & 8'h55)^
({8{data_in[36][2]}} & 8'haa)^
({8{data_in[36][3]}} & 8'h7f)^
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({8{data_in[36][5]}} & 8'hd7)^
({8{data_in[36][6]}} & 8'h85)^
({8{data_in[36][7]}} & 8'h21)^
({8{data_in[37][0]}} & 8'h49)^
({8{data_in[37][1]}} & 8'h92)^
({8{data_in[37][2]}} & 8'hf)^
({8{data_in[37][3]}} & 8'h1e)^
({8{data_in[37][4]}} & 8'h3c)^
({8{data_in[37][5]}} & 8'h78)^
({8{data_in[37][6]}} & 8'hf0)^
({8{data_in[37][7]}} & 8'hcb)^
({8{data_in[38][0]}} & 8'h3f)^
({8{data_in[38][1]}} & 8'h7e)^
({8{data_in[38][2]}} & 8'hfc)^
({8{data_in[38][3]}} & 8'hd3)^
({8{data_in[38][4]}} & 8'h8d)^
({8{data_in[38][5]}} & 8'h31)^
({8{data_in[38][6]}} & 8'h62)^
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({8{data_in[39][0]}} & 8'hee)^
({8{data_in[39][1]}} & 8'hf7)^

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({8{data_in[40][1]}} & 8'hc8)^
({8{data_in[40][2]}} & 8'hbb)^
({8{data_in[40][3]}} & 8'h5d)^
({8{data_in[40][4]}} & 8'hba)^
({8{data_in[40][5]}} & 8'h5f)^
({8{data_in[40][6]}} & 8'hbe)^
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({8{data_in[41][2]}} & 8'h4c)^
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({8{data_in[41][4]}} & 8'h1b)^
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({8{data_in[41][6]}} & 8'h6c)^
({8{data_in[41][7]}} & 8'hd8)^
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({8{data_in[42][2]}} & 8'h12)^
({8{data_in[42][3]}} & 8'h24)^
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({8{data_in[43][2]}} & 8'h1)^
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({8{data_in[43][4]}} & 8'h4)^
({8{data_in[43][5]}} & 8'h8)^
({8{data_in[43][6]}} & 8'h10)^
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({8{data_in[44][4]}} & 8'h48)^
({8{data_in[44][5]}} & 8'h90)^
({8{data_in[44][6]}} & 8'hb)^
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({8{data_in[45][1]}} & 8'h97)^
({8{data_in[45][2]}} & 8'h5)^
({8{data_in[45][3]}} & 8'ha)^
({8{data_in[45][4]}} & 8'h14)^
({8{data_in[45][5]}} & 8'h28)^
({8{data_in[45][6]}} & 8'h50)^
({8{data_in[45][7]}} & 8'ha0)^
({8{data_in[46][0]}} & 8'hc8)^
({8{data_in[46][1]}} & 8'hbb)^
({8{data_in[46][2]}} & 8'h5d)^
({8{data_in[46][3]}} & 8'hba)^
({8{data_in[46][4]}} & 8'h5f)^
({8{data_in[46][5]}} & 8'hbe)^
({8{data_in[46][6]}} & 8'h57)^
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({8{data_in[48][1]}} & 8'h92)^
({8{data_in[48][2]}} & 8'hf)^
({8{data_in[48][3]}} & 8'h1e)^
({8{data_in[48][4]}} & 8'h3c)^
({8{data_in[48][5]}} & 8'h78)^
({8{data_in[48][6]}} & 8'hf0)^
({8{data_in[48][7]}} & 8'hcb)^
({8{data_in[49][0]}} & 8'hd8)^
({8{data_in[49][1]}} & 8'h9b)^
({8{data_in[49][2]}} & 8'h1d)^
({8{data_in[49][3]}} & 8'h3a)^
({8{data_in[49][4]}} & 8'h74)^
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({8{data_in[49][6]}} & 8'hfb)^
({8{data_in[49][7]}} & 8'hdd)^
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({8{data_in[50][3]}} & 8'he3)^
({8{data_in[50][4]}} & 8'hed)^
({8{data_in[50][5]}} & 8'hf1)^
({8{data_in[50][6]}} & 8'hc9)^
({8{data_in[50][7]}} & 8'hb9)^
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({8{data_in[51][2]}} & 8'h5a)^
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({8{data_in[51][7]}} & 8'h4e)^
({8{data_in[52][0]}} & 8'h37)^
({8{data_in[52][1]}} & 8'h6e)^
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({8{data_in[52][3]}} & 8'h93)^
({8{data_in[52][4]}} & 8'hd)^
({8{data_in[52][5]}} & 8'h1a)^
({8{data_in[52][6]}} & 8'h34)^
({8{data_in[52][7]}} & 8'h68)^
({8{data_in[53][0]}} & 8'hbb)^
({8{data_in[53][1]}} & 8'h5d)^
({8{data_in[53][2]}} & 8'hba)^
({8{data_in[53][3]}} & 8'h5f)^
({8{data_in[53][4]}} & 8'hbe)^
({8{data_in[53][5]}} & 8'h57)^
({8{data_in[53][6]}} & 8'hae)^
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({8{data_in[54][2]}} & 8'h8b)^
({8{data_in[54][3]}} & 8'h3d)^
({8{data_in[54][4]}} & 8'h7a)^
({8{data_in[54][5]}} & 8'hf4)^
({8{data_in[54][6]}} & 8'hc3)^
({8{data_in[54][7]}} & 8'had)^
({8{data_in[55][0]}} & 8'h5)^
({8{data_in[55][1]}} & 8'ha)^
({8{data_in[55][2]}} & 8'h14)^
({8{data_in[55][3]}} & 8'h28)^

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({8{data_in[55][4]}} & 8'h50)^
({8{data_in[55][5]}} & 8'ha0)^
({8{data_in[55][6]}} & 8'h6b)^
({8{data_in[55][7]}} & 8'hd6)^
({8{data_in[56][0]}} & 8'he9)^
({8{data_in[56][1]}} & 8'hf9)^
({8{data_in[56][2]}} & 8'hd9)^
({8{data_in[56][3]}} & 8'h99)^
({8{data_in[56][4]}} & 8'h19)^
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({8{data_in[56][6]}} & 8'h64)^
({8{data_in[56][7]}} & 8'hc8)^
({8{data_in[57][0]}} & 8'hcc)^
({8{data_in[57][1]}} & 8'hb3)^
({8{data_in[57][2]}} & 8'h4d)^
({8{data_in[57][3]}} & 8'h9a)^
({8{data_in[57][4]}} & 8'h1f)^
({8{data_in[57][5]}} & 8'h3e)^
({8{data_in[57][6]}} & 8'h7c)^
({8{data_in[57][7]}} & 8'hf8)^
({8{data_in[58][0]}} & 8'hbb)^
({8{data_in[58][1]}} & 8'h5d)^
({8{data_in[58][2]}} & 8'hba)^
({8{data_in[58][3]}} & 8'h5f)^
({8{data_in[58][4]}} & 8'hbe)^
({8{data_in[58][5]}} & 8'h57)^
({8{data_in[58][6]}} & 8'hae)^
({8{data_in[58][7]}} & 8'h77)^
({8{data_in[59][0]}} & 8'hfb)^
({8{data_in[59][1]}} & 8'hdd)^
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({8{data_in[59][3]}} & 8'h9)^
({8{data_in[59][4]}} & 8'h12)^
({8{data_in[59][5]}} & 8'h24)^
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({8{data_in[60][0]}} & 8'hdf)^
({8{data_in[60][1]}} & 8'h95)^
({8{data_in[60][2]}} & 8'h1)^
({8{data_in[60][3]}} & 8'h2)^
({8{data_in[60][4]}} & 8'h4)^
({8{data_in[60][5]}} & 8'h8)^
({8{data_in[60][6]}} & 8'h10)^
({8{data_in[60][7]}} & 8'h20)^
({8{data_in[61][0]}} & 8'hbe)^
({8{data_in[61][1]}} & 8'h57)^
({8{data_in[61][2]}} & 8'hae)^
({8{data_in[61][3]}} & 8'h77)^
({8{data_in[61][4]}} & 8'hee)^
({8{data_in[61][5]}} & 8'hf7)^
({8{data_in[61][6]}} & 8'hc5)^
({8{data_in[61][7]}} & 8'ha1)^
({8{data_in[62][0]}} & 8'h5e)^
({8{data_in[62][1]}} & 8'hbc)^
({8{data_in[62][2]}} & 8'h53)^
({8{data_in[62][3]}} & 8'ha6)^
({8{data_in[62][4]}} & 8'h67)^
({8{data_in[62][5]}} & 8'hce)^
({8{data_in[62][6]}} & 8'hb7)^
({8{data_in[62][7]}} & 8'h45)^
({8{data_in[63][0]}} & 8'h42)^
({8{data_in[63][1]}} & 8'h84)^
({8{data_in[63][2]}} & 8'h23)^
({8{data_in[63][3]}} & 8'h46)^
({8{data_in[63][4]}} & 8'h8c)^

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({8{data_in[64][3]}} & 8'h19)^
({8{data_in[64][4]}} & 8'h32)^
({8{data_in[64][5]}} & 8'h64)^
({8{data_in[64][6]}} & 8'hc8)^
({8{data_in[64][7]}} & 8'hbb)^
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({8{data_in[65][2]}} & 8'h6d)^
({8{data_in[65][3]}} & 8'hda)^
({8{data_in[65][4]}} & 8'h9f)^
({8{data_in[65][5]}} & 8'h15)^
({8{data_in[65][6]}} & 8'h2a)^
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({8{data_in[66][3]}} & 8'ha9)^
({8{data_in[66][4]}} & 8'h79)^
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({8{data_in[66][6]}} & 8'hcf)^
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({8{data_in[67][4]}} & 8'h1b)^
({8{data_in[67][5]}} & 8'h36)^
({8{data_in[67][6]}} & 8'h6c)^
({8{data_in[67][7]}} & 8'hd8)^
({8{data_in[68][0]}} & 8'hbc)^
({8{data_in[68][1]}} & 8'h53)^
({8{data_in[68][2]}} & 8'ha6)^
({8{data_in[68][3]}} & 8'h67)^
({8{data_in[68][4]}} & 8'hce)^
({8{data_in[68][5]}} & 8'hb7)^
({8{data_in[68][6]}} & 8'h45)^
({8{data_in[68][7]}} & 8'h8a)^
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({8{data_in[69][1]}} & 8'h62)^
({8{data_in[69][2]}} & 8'hc4)^
({8{data_in[69][3]}} & 8'ha3)^
({8{data_in[69][4]}} & 8'h6d)^
({8{data_in[69][5]}} & 8'hda)^
({8{data_in[69][6]}} & 8'h9f)^
({8{data_in[69][7]}} & 8'h15)^
({8{data_in[70][0]}} & 8'hca)^
({8{data_in[70][1]}} & 8'hbf)^
({8{data_in[70][2]}} & 8'h55)^
({8{data_in[70][3]}} & 8'haa)^
({8{data_in[70][4]}} & 8'h7f)^
({8{data_in[70][5]}} & 8'hfe)^
({8{data_in[70][6]}} & 8'hd7)^
({8{data_in[70][7]}} & 8'h85)^
({8{data_in[71][0]}} & 8'hec)^
({8{data_in[71][1]}} & 8'hf3)^
({8{data_in[71][2]}} & 8'hcd)^
({8{data_in[71][3]}} & 8'hb1)^
({8{data_in[71][4]}} & 8'h49)^
({8{data_in[71][5]}} & 8'h92)^

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({8{data_in[72][2]}} & 8'ha8)^
({8{data_in[72][3]}} & 8'h7b)^
({8{data_in[72][4]}} & 8'hf6)^
({8{data_in[72][5]}} & 8'hc7)^
({8{data_in[72][6]}} & 8'ha5)^
({8{data_in[72][7]}} & 8'h61)^
({8{data_in[73][0]}} & 8'h6f)^
({8{data_in[73][1]}} & 8'hde)^
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({8{data_in[73][3]}} & 8'h5)^
({8{data_in[73][4]}} & 8'ha)^
({8{data_in[73][5]}} & 8'h14)^
({8{data_in[73][6]}} & 8'h28)^
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({8{data_in[76][0]}} & 8'h82)^
({8{data_in[76][1]}} & 8'h2f)^
({8{data_in[76][2]}} & 8'h5e)^
({8{data_in[76][3]}} & 8'hbc)^
({8{data_in[76][4]}} & 8'h53)^
({8{data_in[76][5]}} & 8'ha6)^
({8{data_in[76][6]}} & 8'h67)^
({8{data_in[76][7]}} & 8'hce)^
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({8{data_in[77][3]}} & 8'h62)^
({8{data_in[77][4]}} & 8'hc4)^
({8{data_in[77][5]}} & 8'ha3)^
({8{data_in[77][6]}} & 8'h6d)^
({8{data_in[77][7]}} & 8'hda)^
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({8{data_in[78][2]}} & 8'h1b)^
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({8{data_in[78][4]}} & 8'h6c)^
({8{data_in[78][5]}} & 8'hd8)^
({8{data_in[78][6]}} & 8'h9b)^
({8{data_in[78][7]}} & 8'h1d)^
({8{data_in[79][0]}} & 8'hfe)^
({8{data_in[79][1]}} & 8'hd7)^
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({8{data_in[79][3]}} & 8'h21)^
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({8{data_in[79][6]}} & 8'h23)^

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({8{data_in[80][3]}} & 8'hf2)^
({8{data_in[80][4]}} & 8'hcf)^
({8{data_in[80][5]}} & 8'hb5)^
({8{data_in[80][6]}} & 8'h41)^
({8{data_in[80][7]}} & 8'h82)^
({8{data_in[81][0]}} & 8'h12)^
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({8{data_in[81][2]}} & 8'h48)^
({8{data_in[81][3]}} & 8'h90)^
({8{data_in[81][4]}} & 8'hb)^
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({8{data_in[81][7]}} & 8'h58)^
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({8{data_in[82][1]}} & 8'h8f)^
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({8{data_in[82][3]}} & 8'h6a)^
({8{data_in[82][4]}} & 8'hd4)^
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({8{data_in[82][6]}} & 8'h2d)^
({8{data_in[82][7]}} & 8'h5a)^
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({8{data_in[83][3]}} & 8'hc6)^
({8{data_in[83][4]}} & 8'ha7)^
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({8{data_in[83][6]}} & 8'hca)^
({8{data_in[83][7]}} & 8'hbf)^
({8{data_in[84][0]}} & 8'h4e)^
({8{data_in[84][1]}} & 8'h9c)^
({8{data_in[84][2]}} & 8'h13)^
({8{data_in[84][3]}} & 8'h26)^
({8{data_in[84][4]}} & 8'h4c)^
({8{data_in[84][5]}} & 8'h98)^
({8{data_in[84][6]}} & 8'h1b)^
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({8{data_in[85][3]}} & 8'hb7)^
({8{data_in[85][4]}} & 8'h45)^
({8{data_in[85][5]}} & 8'h8a)^
({8{data_in[85][6]}} & 8'h3f)^
({8{data_in[85][7]}} & 8'h7e)^
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({8{data_in[86][1]}} & 8'h15)^
({8{data_in[86][2]}} & 8'h2a)^
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({8{data_in[87][3]}} & 8'ha9)^
({8{data_in[87][4]}} & 8'h79)^
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({8{data_in[87][6]}} & 8'hcf)^
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({8{data_in[89][1]}} & 8'hd1)^
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({8{data_in[90][3]}} & 8'hcd)^
({8{data_in[90][4]}} & 8'hb1)^
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({8{data_in[91][2]}} & 8'he3)^
({8{data_in[91][3]}} & 8'hed)^
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({8{data_in[91][5]}} & 8'hc9)^
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({8{data_in[92][2]}} & 8'h9a)^
({8{data_in[92][3]}} & 8'h1f)^
({8{data_in[92][4]}} & 8'h3e)^
({8{data_in[92][5]}} & 8'h7c)^
({8{data_in[92][6]}} & 8'hf8)^
({8{data_in[92][7]}} & 8'hdb)^
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({8{data_in[93][1]}} & 8'h1e)^
({8{data_in[93][2]}} & 8'h3c)^
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({8{data_in[94][7]}} & 8'h5a)^
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({8{data_in[95][3]}} & 8'hbb)^
({8{data_in[95][4]}} & 8'h5d)^
({8{data_in[95][5]}} & 8'hba)^
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({8{data_in[99][0]}} & 8'hd8)^
({8{data_in[99][1]}} & 8'h9b)^
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({8{data_in[100][4]}} & 8'h6d)^
({8{data_in[100][5]}} & 8'hda)^
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({8{data_in[100][7]}} & 8'h15)^
({8{data_in[101][0]}} & 8'hd2)^
({8{data_in[101][1]}} & 8'h8f)^
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({8{data_in[105][0]}} & 8'h8b)^
({8{data_in[105][1]}} & 8'h3d)^
({8{data_in[105][2]}} & 8'h7a)^
({8{data_in[105][3]}} & 8'hf4)^
({8{data_in[105][4]}} & 8'hc3)^
({8{data_in[105][5]}} & 8'had)^
({8{data_in[105][6]}} & 8'h71)^
({8{data_in[105][7]}} & 8'he2)^
({8{data_in[106][0]}} & 8'h1f)^
({8{data_in[106][1]}} & 8'h3e)^
({8{data_in[106][2]}} & 8'h7c)^
({8{data_in[106][3]}} & 8'hf8)^
({8{data_in[106][4]}} & 8'hdb)^
({8{data_in[106][5]}} & 8'h9d)^
({8{data_in[106][6]}} & 8'h11)^
({8{data_in[106][7]}} & 8'h22)^
({8{data_in[107][0]}} & 8'hc4)^
({8{data_in[107][1]}} & 8'ha3)^
({8{data_in[107][2]}} & 8'h6d)^
({8{data_in[107][3]}} & 8'hda)^
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({8{data_in[108][1]}} & 8'h77)^
({8{data_in[108][2]}} & 8'hee)^
({8{data_in[108][3]}} & 8'hf7)^
({8{data_in[108][4]}} & 8'hc5)^
({8{data_in[108][5]}} & 8'ha1)^
({8{data_in[108][6]}} & 8'h69)^
({8{data_in[108][7]}} & 8'hd2)^
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({8{data_in[109][1]}} & 8'he0)^
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({8{data_in[109][4]}} & 8'hd1)^
({8{data_in[109][5]}} & 8'h89)^
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({8{data_in[109][7]}} & 8'h72)^
({8{data_in[110][0]}} & 8'h3)^
({8{data_in[110][1]}} & 8'h6)^
({8{data_in[110][2]}} & 8'hc)^
({8{data_in[110][3]}} & 8'h18)^
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({8{data_in[110][5]}} & 8'h60)^
({8{data_in[110][6]}} & 8'hc0)^
({8{data_in[110][7]}} & 8'hab)^
({8{data_in[111][0]}} & 8'h9d)^
({8{data_in[111][1]}} & 8'h11)^
({8{data_in[111][2]}} & 8'h22)^
({8{data_in[111][3]}} & 8'h44)^
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({8{data_in[111][5]}} & 8'h3b)^
({8{data_in[111][6]}} & 8'h76)^
({8{data_in[111][7]}} & 8'hec)^
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({8{data_in[112][1]}} & 8'h2b)^
({8{data_in[112][2]}} & 8'h56)^

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({8{data_in[112][7]}} & 8'he5)^
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({8{data_in[113][2]}} & 8'hc6)^
({8{data_in[113][3]}} & 8'ha7)^
({8{data_in[113][4]}} & 8'h65)^
({8{data_in[113][5]}} & 8'hca)^
({8{data_in[113][6]}} & 8'hbf)^
({8{data_in[113][7]}} & 8'h55)^
({8{data_in[114][0]}} & 8'hbe)^
({8{data_in[114][1]}} & 8'h57)^
({8{data_in[114][2]}} & 8'hae)^
({8{data_in[114][3]}} & 8'h77)^
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({8{data_in[114][7]}} & 8'ha1)^
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({8{data_in[116][3]}} & 8'h1e)^
({8{data_in[116][4]}} & 8'h3c)^
({8{data_in[116][5]}} & 8'h78)^
({8{data_in[116][6]}} & 8'hf0)^
({8{data_in[116][7]}} & 8'hcb)^
({8{data_in[117][0]}} & 8'h1)^
({8{data_in[117][1]}} & 8'h2)^
({8{data_in[117][2]}} & 8'h4)^
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({8{data_in[118][5]}} & 8'h40)^
({8{data_in[118][6]}} & 8'h80)^
({8{data_in[118][7]}} & 8'h2b)^
({8{data_in[119][0]}} & 8'h92)^
({8{data_in[119][1]}} & 8'hf)^
({8{data_in[119][2]}} & 8'h1e)^
({8{data_in[119][3]}} & 8'h3c)^
({8{data_in[119][4]}} & 8'h78)^
({8{data_in[119][5]}} & 8'hf0)^
({8{data_in[119][6]}} & 8'hcb)^
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({8{data_in[120][1]}} & 8'h69)^
({8{data_in[120][2]}} & 8'hd2)^
({8{data_in[120][3]}} & 8'h8f)^

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({8{data_in[121][1]}} & 8'h72)^
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({8{data_in[121][5]}} & 8'hf1)^
({8{data_in[121][6]}} & 8'hc9)^
({8{data_in[121][7]}} & 8'hb9)^
({8{data_in[122][0]}} & 8'h66)^
({8{data_in[122][1]}} & 8'hcc)^
({8{data_in[122][2]}} & 8'hb3)^
({8{data_in[122][3]}} & 8'h4d)^
({8{data_in[122][4]}} & 8'h9a)^
({8{data_in[122][5]}} & 8'h1f)^
({8{data_in[122][6]}} & 8'h3e)^
({8{data_in[122][7]}} & 8'h7c)^
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({8{data_in[123][4]}} & 8'h4c)^
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({8{data_in[124][7]}} & 8'h4a)^
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({8{data_in[126][3]}} & 8'h91)^
({8{data_in[126][4]}} & 8'h9)^
({8{data_in[126][5]}} & 8'h12)^
({8{data_in[126][6]}} & 8'h24)^
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({8{data_in[127][3]}} & 8'h9b)^
({8{data_in[127][4]}} & 8'h1d)^
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({8{data_in[128][1]}} & 8'ha6)^
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({8{data_in[128][3]}} & 8'hce)^
({8{data_in[128][4]}} & 8'hb7)^

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({8{data_in[129][1]}} & 8'h96)^
({8{data_in[129][2]}} & 8'h7)^
({8{data_in[129][3]}} & 8'he)^
({8{data_in[129][4]}} & 8'h1c)^
({8{data_in[129][5]}} & 8'h38)^
({8{data_in[129][6]}} & 8'h70)^
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({8{data_in[130][4]}} & 8'h8c)^
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({8{data_in[130][6]}} & 8'h66)^
({8{data_in[130][7]}} & 8'hcc)^
({8{data_in[131][0]}} & 8'he0)^
({8{data_in[131][1]}} & 8'heb)^
({8{data_in[131][2]}} & 8'hfd)^
({8{data_in[131][3]}} & 8'hd1)^
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({8{data_in[138][1]}} & 8'h77)^
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({8{data_in[143][3]}} & 8'h5c)^
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({8{data_in[145][3]}} & 8'h7e)^
({8{data_in[145][4]}} & 8'hfc)^
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({8{data_in[146][6]}} & 8'h1e)^
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({8{data_in[149][3]}} & 8'h9b)^
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({8{data_in[149][7]}} & 8'he8)^
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({8{data_in[150][2]}} & 8'hfc)^
({8{data_in[150][3]}} & 8'hd3)^
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({8{data_in[151][2]}} & 8'h70)^
({8{data_in[151][3]}} & 8'he0)^
({8{data_in[151][4]}} & 8'heb)^
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({8{data_in[154][5]}} & 8'h6d)^
({8{data_in[154][6]}} & 8'hda)^
({8{data_in[154][7]}} & 8'h9f)^
({8{data_in[155][0]}} & 8'hb0)^
({8{data_in[155][1]}} & 8'h4b)^
({8{data_in[155][2]}} & 8'h96)^
({8{data_in[155][3]}} & 8'h7)^
({8{data_in[155][4]}} & 8'he)^
({8{data_in[155][5]}} & 8'h1c)^
({8{data_in[155][6]}} & 8'h38)^
({8{data_in[155][7]}} & 8'h70)^
({8{data_in[156][0]}} & 8'hdc)^
({8{data_in[156][1]}} & 8'h93)^
({8{data_in[156][2]}} & 8'hd)^
({8{data_in[156][3]}} & 8'h1a)^
({8{data_in[156][4]}} & 8'h34)^
({8{data_in[156][5]}} & 8'h68)^
({8{data_in[156][6]}} & 8'hd0)^
({8{data_in[156][7]}} & 8'h8b)^
({8{data_in[157][0]}} & 8'h48)^
({8{data_in[157][1]}} & 8'h90)^
({8{data_in[157][2]}} & 8'hb)^
({8{data_in[157][3]}} & 8'h16)^
({8{data_in[157][4]}} & 8'h2c)^
({8{data_in[157][5]}} & 8'h58)^
({8{data_in[157][6]}} & 8'hb0)^
({8{data_in[157][7]}} & 8'h4b)^
({8{data_in[158][0]}} & 8'hb7)^
({8{data_in[158][1]}} & 8'h45)^
({8{data_in[158][2]}} & 8'h8a)^
({8{data_in[158][3]}} & 8'h3f)^
({8{data_in[158][4]}} & 8'h7e)^
({8{data_in[158][5]}} & 8'hfc)^
({8{data_in[158][6]}} & 8'hd3)^
({8{data_in[158][7]}} & 8'h8d)^
({8{data_in[159][0]}} & 8'h71)^
({8{data_in[159][1]}} & 8'he2)^
({8{data_in[159][2]}} & 8'hef)^
({8{data_in[159][3]}} & 8'hf5)^
({8{data_in[159][4]}} & 8'hc1)^
({8{data_in[159][5]}} & 8'ha9)^
({8{data_in[159][6]}} & 8'h79)^
({8{data_in[159][7]}} & 8'hf2)^
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({8{data_in[160][1]}} & 8'h6a)^
({8{data_in[160][2]}} & 8'hd4)^
({8{data_in[160][3]}} & 8'h83)^
({8{data_in[160][4]}} & 8'h2d)^
({8{data_in[160][5]}} & 8'h5a)^
({8{data_in[160][6]}} & 8'hb4)^
({8{data_in[160][7]}} & 8'h43)^
({8{data_in[161][0]}} & 8'h25)^

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({8{data_in[161][4]}} & 8'h6)^
({8{data_in[161][5]}} & 8'hc)^
({8{data_in[161][6]}} & 8'h18)^
({8{data_in[161][7]}} & 8'h30)^
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({8{data_in[162][1]}} & 8'ha2)^
({8{data_in[162][2]}} & 8'h6f)^
({8{data_in[162][3]}} & 8'hde)^
({8{data_in[162][4]}} & 8'h97)^
({8{data_in[162][5]}} & 8'h5)^
({8{data_in[162][6]}} & 8'ha)^
({8{data_in[162][7]}} & 8'h14)^
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({8{data_in[163][1]}} & 8'hdf)^
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({8{data_in[163][3]}} & 8'h1)^
({8{data_in[163][4]}} & 8'h2)^
({8{data_in[163][5]}} & 8'h4)^
({8{data_in[163][6]}} & 8'h8)^
({8{data_in[163][7]}} & 8'h10)^
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({8{data_in[164][2]}} & 8'h51)^
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({8{data_in[164][4]}} & 8'h6f)^
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({8{data_in[164][6]}} & 8'h97)^
({8{data_in[164][7]}} & 8'h5)^
({8{data_in[165][0]}} & 8'h2c)^
({8{data_in[165][1]}} & 8'h58)^
({8{data_in[165][2]}} & 8'hb0)^
({8{data_in[165][3]}} & 8'h4b)^
({8{data_in[165][4]}} & 8'h96)^
({8{data_in[165][5]}} & 8'h7)^
({8{data_in[165][6]}} & 8'he)^
({8{data_in[165][7]}} & 8'h1c)^
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({8{data_in[166][1]}} & 8'h4e)^
({8{data_in[166][2]}} & 8'h9c)^
({8{data_in[166][3]}} & 8'h13)^
({8{data_in[166][4]}} & 8'h26)^
({8{data_in[166][5]}} & 8'h4c)^
({8{data_in[166][6]}} & 8'h98)^
({8{data_in[166][7]}} & 8'h1b)^
({8{data_in[167][0]}} & 8'h68)^
({8{data_in[167][1]}} & 8'hd0)^
({8{data_in[167][2]}} & 8'h8b)^
({8{data_in[167][3]}} & 8'h3d)^
({8{data_in[167][4]}} & 8'h7a)^
({8{data_in[167][5]}} & 8'hf4)^
({8{data_in[167][6]}} & 8'hc3)^
({8{data_in[167][7]}} & 8'had)^
({8{data_in[168][0]}} & 8'h5a)^
({8{data_in[168][1]}} & 8'hb4)^
({8{data_in[168][2]}} & 8'h43)^
({8{data_in[168][3]}} & 8'h86)^
({8{data_in[168][4]}} & 8'h27)^
({8{data_in[168][5]}} & 8'h4e)^
({8{data_in[168][6]}} & 8'h9c)^
({8{data_in[168][7]}} & 8'h13)^
({8{data_in[169][0]}} & 8'hb2)^
({8{data_in[169][1]}} & 8'h4f)^

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({8{data_in[169][4]}} & 8'h2e)^
({8{data_in[169][5]}} & 8'h5c)^
({8{data_in[169][6]}} & 8'hb8)^
({8{data_in[169][7]}} & 8'h5b)^
({8{data_in[170][0]}} & 8'h4f)^
({8{data_in[170][1]}} & 8'h9e)^
({8{data_in[170][2]}} & 8'h17)^
({8{data_in[170][3]}} & 8'h2e)^
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({8{data_in[170][7]}} & 8'hb6)^
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({8{data_in[171][1]}} & 8'h1f)^
({8{data_in[171][2]}} & 8'h3e)^
({8{data_in[171][3]}} & 8'h7c)^
({8{data_in[171][4]}} & 8'hf8)^
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({8{data_in[171][6]}} & 8'h9d)^
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({8{data_in[172][0]}} & 8'h9c)^
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({8{data_in[172][2]}} & 8'h26)^
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({8{data_in[173][1]}} & 8'h5c)^
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({8{data_in[173][3]}} & 8'h5b)^
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({8{data_in[173][5]}} & 8'h47)^
({8{data_in[173][6]}} & 8'h8e)^
({8{data_in[173][7]}} & 8'h37)^
({8{data_in[174][0]}} & 8'hb5)^
({8{data_in[174][1]}} & 8'h41)^
({8{data_in[174][2]}} & 8'h82)^
({8{data_in[174][3]}} & 8'h2f)^
({8{data_in[174][4]}} & 8'h5e)^
({8{data_in[174][5]}} & 8'hbc)^
({8{data_in[174][6]}} & 8'h53)^
({8{data_in[174][7]}} & 8'ha6)^
({8{data_in[175][0]}} & 8'h7b)^
({8{data_in[175][1]}} & 8'hf6)^
({8{data_in[175][2]}} & 8'hc7)^
({8{data_in[175][3]}} & 8'ha5)^
({8{data_in[175][4]}} & 8'h61)^
({8{data_in[175][5]}} & 8'hc2)^
({8{data_in[175][6]}} & 8'haf)^
({8{data_in[175][7]}} & 8'h75)^
({8{data_in[176][0]}} & 8'h9e)^
({8{data_in[176][1]}} & 8'h17)^
({8{data_in[176][2]}} & 8'h2e)^
({8{data_in[176][3]}} & 8'h5c)^
({8{data_in[176][4]}} & 8'hb8)^
({8{data_in[176][5]}} & 8'h5b)^
({8{data_in[176][6]}} & 8'hb6)^
({8{data_in[176][7]}} & 8'h47)^
({8{data_in[177][0]}} & 8'hec)^
({8{data_in[177][1]}} & 8'hf3)^
({8{data_in[177][2]}} & 8'hcd)^

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({8{data_in[178][1]}} & 8'h88)^
({8{data_in[178][2]}} & 8'h3b)^
({8{data_in[178][3]}} & 8'h76)^
({8{data_in[178][4]}} & 8'hec)^
({8{data_in[178][5]}} & 8'hf3)^
({8{data_in[178][6]}} & 8'hcd)^
({8{data_in[178][7]}} & 8'hb1)^
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({8{data_in[180][1]}} & 8'he3)^
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({8{data_in[180][3]}} & 8'hf1)^
({8{data_in[180][4]}} & 8'hc9)^
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({8{data_in[180][7]}} & 8'hb2)^
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({8{data_in[181][1]}} & 8'hc0)^
({8{data_in[181][2]}} & 8'hab)^
({8{data_in[181][3]}} & 8'h7d)^
({8{data_in[181][4]}} & 8'hfa)^
({8{data_in[181][5]}} & 8'hdf)^
({8{data_in[181][6]}} & 8'h95)^
({8{data_in[181][7]}} & 8'h1)^
({8{data_in[182][0]}} & 8'h6b)^
({8{data_in[182][1]}} & 8'hd6)^
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({8{data_in[182][6]}} & 8'h3)^
({8{data_in[182][7]}} & 8'h6)^
({8{data_in[183][0]}} & 8'hd2)^
({8{data_in[183][1]}} & 8'h8f)^
({8{data_in[183][2]}} & 8'h35)^
({8{data_in[183][3]}} & 8'h6a)^
({8{data_in[183][4]}} & 8'hd4)^
({8{data_in[183][5]}} & 8'h83)^
({8{data_in[183][6]}} & 8'h2d)^
({8{data_in[183][7]}} & 8'h5a)^
({8{data_in[184][0]}} & 8'hf)^
({8{data_in[184][1]}} & 8'h1e)^
({8{data_in[184][2]}} & 8'h3c)^
({8{data_in[184][3]}} & 8'h78)^
({8{data_in[184][4]}} & 8'hf0)^
({8{data_in[184][5]}} & 8'hcb)^
({8{data_in[184][6]}} & 8'hbd)^
({8{data_in[184][7]}} & 8'h51)^
({8{data_in[185][0]}} & 8'h37)^
({8{data_in[185][1]}} & 8'h6e)^
({8{data_in[185][2]}} & 8'hdc)^
({8{data_in[185][3]}} & 8'h93)^

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({8{data_in[186][1]}} & 8'ha7)^
({8{data_in[186][2]}} & 8'h65)^
({8{data_in[186][3]}} & 8'hca)^
({8{data_in[186][4]}} & 8'hbf)^
({8{data_in[186][5]}} & 8'h55)^
({8{data_in[186][6]}} & 8'haa)^
({8{data_in[186][7]}} & 8'h7f)^
({8{data_in[187][0]}} & 8'h8d)^
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({8{data_in[187][2]}} & 8'h62)^
({8{data_in[187][3]}} & 8'hc4)^
({8{data_in[187][4]}} & 8'ha3)^
({8{data_in[187][5]}} & 8'h6d)^
({8{data_in[187][6]}} & 8'hda)^
({8{data_in[187][7]}} & 8'h9f)^
({8{data_in[188][0]}} & 8'h82)^
({8{data_in[188][1]}} & 8'h2f)^
({8{data_in[188][2]}} & 8'h5e)^
({8{data_in[188][3]}} & 8'hbc)^
({8{data_in[188][4]}} & 8'h53)^
({8{data_in[188][5]}} & 8'ha6)^
({8{data_in[188][6]}} & 8'h67)^
({8{data_in[188][7]}} & 8'hce)^
({8{data_in[189][0]}} & 8'h4e)^
({8{data_in[189][1]}} & 8'h9c)^
({8{data_in[189][2]}} & 8'h13)^
({8{data_in[189][3]}} & 8'h26)^
({8{data_in[189][4]}} & 8'h4c)^
({8{data_in[189][5]}} & 8'h98)^
({8{data_in[189][6]}} & 8'h1b)^
({8{data_in[189][7]}} & 8'h36)^
({8{data_in[190][0]}} & 8'hb5)^
({8{data_in[190][1]}} & 8'h41)^
({8{data_in[190][2]}} & 8'h82)^
({8{data_in[190][3]}} & 8'h2f)^
({8{data_in[190][4]}} & 8'h5e)^
({8{data_in[190][5]}} & 8'hbc)^
({8{data_in[190][6]}} & 8'h53)^
({8{data_in[190][7]}} & 8'ha6)^
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({8{data_in[191][2]}} & 8'h27)^
({8{data_in[191][3]}} & 8'h4e)^
({8{data_in[191][4]}} & 8'h9c)^
({8{data_in[191][5]}} & 8'h13)^
({8{data_in[191][6]}} & 8'h26)^
({8{data_in[191][7]}} & 8'h4c)^
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({8{data_in[192][1]}} & 8'hde)^
({8{data_in[192][2]}} & 8'h97)^
({8{data_in[192][3]}} & 8'h5)^
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({8{data_in[194][4]}} & 8'hee)^
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({8{data_in[194][6]}} & 8'hc5)^
({8{data_in[194][7]}} & 8'ha1)^
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({8{data_in[195][2]}} & 8'h9c)^
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({8{data_in[195][5]}} & 8'h4c)^
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({8{data_in[195][7]}} & 8'h1b)^
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({8{data_in[196][2]}} & 8'h9a)^
({8{data_in[196][3]}} & 8'h1f)^
({8{data_in[196][4]}} & 8'h3e)^
({8{data_in[196][5]}} & 8'h7c)^
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({8{data_in[196][7]}} & 8'hdb)^
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({8{data_in[197][1]}} & 8'hc0)^
({8{data_in[197][2]}} & 8'hab)^
({8{data_in[197][3]}} & 8'h7d)^
({8{data_in[197][4]}} & 8'hfa)^
({8{data_in[197][5]}} & 8'hdf)^
({8{data_in[197][6]}} & 8'h95)^
({8{data_in[197][7]}} & 8'h1)^
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({8{data_in[198][1]}} & 8'h9f)^
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({8{data_in[198][4]}} & 8'h54)^
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({8{data_in[199][3]}} & 8'h3b)^
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({8{data_in[199][5]}} & 8'hec)^
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({8{data_in[199][7]}} & 8'hcd)^
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({8{data_in[200][1]}} & 8'hdb)^
({8{data_in[200][2]}} & 8'h9d)^
({8{data_in[200][3]}} & 8'h11)^
({8{data_in[200][4]}} & 8'h22)^
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({8{data_in[201][1]}} & 8'h90)^
({8{data_in[201][2]}} & 8'hb)^
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({8{data_in[201][4]}} & 8'h2c)^
({8{data_in[201][5]}} & 8'h58)^

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({8{data_in[202][4]}} & 8'h1d)^
({8{data_in[202][5]}} & 8'h3a)^
({8{data_in[202][6]}} & 8'h74)^
({8{data_in[202][7]}} & 8'he8)^
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({8{data_in[203][1]}} & 8'hcd)^
({8{data_in[203][2]}} & 8'hb1)^
({8{data_in[203][3]}} & 8'h49)^
({8{data_in[203][4]}} & 8'h92)^
({8{data_in[203][5]}} & 8'hf)^
({8{data_in[203][6]}} & 8'h1e)^
({8{data_in[203][7]}} & 8'h3c)^
({8{data_in[204][0]}} & 8'h7b)^
({8{data_in[204][1]}} & 8'hf6)^
({8{data_in[204][2]}} & 8'hc7)^
({8{data_in[204][3]}} & 8'ha5)^
({8{data_in[204][4]}} & 8'h61)^
({8{data_in[204][5]}} & 8'hc2)^
({8{data_in[204][6]}} & 8'haf)^
({8{data_in[204][7]}} & 8'h75)^
({8{data_in[205][0]}} & 8'h68)^
({8{data_in[205][1]}} & 8'hd0)^
({8{data_in[205][2]}} & 8'h8b)^
({8{data_in[205][3]}} & 8'h3d)^
({8{data_in[205][4]}} & 8'h7a)^
({8{data_in[205][5]}} & 8'hf4)^
({8{data_in[205][6]}} & 8'hc3)^
({8{data_in[205][7]}} & 8'had)^
({8{data_in[206][0]}} & 8'h39)^
({8{data_in[206][1]}} & 8'h72)^
({8{data_in[206][2]}} & 8'he4)^
({8{data_in[206][3]}} & 8'he3)^
({8{data_in[206][4]}} & 8'hed)^
({8{data_in[206][5]}} & 8'hf1)^
({8{data_in[206][6]}} & 8'hc9)^
({8{data_in[206][7]}} & 8'hb9)^
({8{data_in[207][0]}} & 8'h7f)^
({8{data_in[207][1]}} & 8'hfe)^
({8{data_in[207][2]}} & 8'hd7)^
({8{data_in[207][3]}} & 8'h85)^
({8{data_in[207][4]}} & 8'h21)^
({8{data_in[207][5]}} & 8'h42)^
({8{data_in[207][6]}} & 8'h84)^
({8{data_in[207][7]}} & 8'h23)^
({8{data_in[208][0]}} & 8'h24)^
({8{data_in[208][1]}} & 8'h48)^
({8{data_in[208][2]}} & 8'h90)^
({8{data_in[208][3]}} & 8'hb)^
({8{data_in[208][4]}} & 8'h16)^
({8{data_in[208][5]}} & 8'h2c)^
({8{data_in[208][6]}} & 8'h58)^
({8{data_in[208][7]}} & 8'hb0)^
({8{data_in[209][0]}} & 8'h91)^
({8{data_in[209][1]}} & 8'h9)^
({8{data_in[209][2]}} & 8'h12)^
({8{data_in[209][3]}} & 8'h24)^
({8{data_in[209][4]}} & 8'h48)^
({8{data_in[209][5]}} & 8'h90)^
({8{data_in[209][6]}} & 8'hb)^

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({8{data_in[209][7]}} & 8'h16)^
({8{data_in[210][0]}} & 8'haf)^
({8{data_in[210][1]}} & 8'h75)^
({8{data_in[210][2]}} & 8'hea)^
({8{data_in[210][3]}} & 8'hff)^
({8{data_in[210][4]}} & 8'hd5)^
({8{data_in[210][5]}} & 8'h81)^
({8{data_in[210][6]}} & 8'h29)^
({8{data_in[210][7]}} & 8'h52)^
({8{data_in[211][0]}} & 8'h83)^
({8{data_in[211][1]}} & 8'h2d)^
({8{data_in[211][2]}} & 8'h5a)^
({8{data_in[211][3]}} & 8'hb4)^
({8{data_in[211][4]}} & 8'h43)^
({8{data_in[211][5]}} & 8'h86)^
({8{data_in[211][6]}} & 8'h27)^
({8{data_in[211][7]}} & 8'h4e)^
({8{data_in[212][0]}} & 8'ha0)^
({8{data_in[212][1]}} & 8'h6b)^
({8{data_in[212][2]}} & 8'hd6)^
({8{data_in[212][3]}} & 8'h87)^
({8{data_in[212][4]}} & 8'h25)^
({8{data_in[212][5]}} & 8'h4a)^
({8{data_in[212][6]}} & 8'h94)^
({8{data_in[212][7]}} & 8'h3)^
({8{data_in[213][0]}} & 8'hc6)^
({8{data_in[213][1]}} & 8'ha7)^
({8{data_in[213][2]}} & 8'h65)^
({8{data_in[213][3]}} & 8'hca)^
({8{data_in[213][4]}} & 8'hbf)^
({8{data_in[213][5]}} & 8'h55)^
({8{data_in[213][6]}} & 8'haa)^
({8{data_in[213][7]}} & 8'h7f)^
({8{data_in[214][0]}} & 8'h14)^
({8{data_in[214][1]}} & 8'h28)^
({8{data_in[214][2]}} & 8'h50)^
({8{data_in[214][3]}} & 8'ha0)^
({8{data_in[214][4]}} & 8'h6b)^
({8{data_in[214][5]}} & 8'hd6)^
({8{data_in[214][6]}} & 8'h87)^
({8{data_in[214][7]}} & 8'h25)^
({8{data_in[215][0]}} & 8'hea)^
({8{data_in[215][1]}} & 8'hff)^
({8{data_in[215][2]}} & 8'hd5)^
({8{data_in[215][3]}} & 8'h81)^
({8{data_in[215][4]}} & 8'h29)^
({8{data_in[215][5]}} & 8'h52)^
({8{data_in[215][6]}} & 8'ha4)^
({8{data_in[215][7]}} & 8'h63)^
({8{data_in[216][0]}} & 8'h44)^
({8{data_in[216][1]}} & 8'h88)^
({8{data_in[216][2]}} & 8'h3b)^
({8{data_in[216][3]}} & 8'h76)^
({8{data_in[216][4]}} & 8'hec)^
({8{data_in[216][5]}} & 8'hf3)^
({8{data_in[216][6]}} & 8'hcd)^
({8{data_in[216][7]}} & 8'hb1)^
({8{data_in[217][0]}} & 8'h23)^
({8{data_in[217][1]}} & 8'h46)^
({8{data_in[217][2]}} & 8'h8c)^
({8{data_in[217][3]}} & 8'h33)^
({8{data_in[217][4]}} & 8'h66)^
({8{data_in[217][5]}} & 8'hcc)^
({8{data_in[217][6]}} & 8'hb3)^
({8{data_in[217][7]}} & 8'h4d)^

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({8{data_in[218][0]}} & 8'hab)^
({8{data_in[218][1]}} & 8'h7d)^
({8{data_in[218][2]}} & 8'hfa)^
({8{data_in[218][3]}} & 8'hdf)^
({8{data_in[218][4]}} & 8'h95)^
({8{data_in[218][5]}} & 8'h1)^
({8{data_in[218][6]}} & 8'h2)^
({8{data_in[218][7]}} & 8'h4)^
({8{data_in[219][0]}} & 8'hf3)^
({8{data_in[219][1]}} & 8'hcd)^
({8{data_in[219][2]}} & 8'hb1)^
({8{data_in[219][3]}} & 8'h49)^
({8{data_in[219][4]}} & 8'h92)^
({8{data_in[219][5]}} & 8'hf)^
({8{data_in[219][6]}} & 8'h1e)^
({8{data_in[219][7]}} & 8'h3c)^
({8{data_in[220][0]}} & 8'he1)^
({8{data_in[220][1]}} & 8'he9)^
({8{data_in[220][2]}} & 8'hf9)^
({8{data_in[220][3]}} & 8'hd9)^
({8{data_in[220][4]}} & 8'h99)^
({8{data_in[220][5]}} & 8'h19)^
({8{data_in[220][6]}} & 8'h32)^
({8{data_in[220][7]}} & 8'h64)^
({8{data_in[221][0]}} & 8'hd8)^
({8{data_in[221][1]}} & 8'h9b)^
({8{data_in[221][2]}} & 8'h1d)^
({8{data_in[221][3]}} & 8'h3a)^
({8{data_in[221][4]}} & 8'h74)^
({8{data_in[221][5]}} & 8'he8)^
({8{data_in[221][6]}} & 8'hfb)^
({8{data_in[221][7]}} & 8'hdd)^
({8{data_in[222][0]}} & 8'h83)^
({8{data_in[222][1]}} & 8'h2d)^
({8{data_in[222][2]}} & 8'h5a)^
({8{data_in[222][3]}} & 8'hb4)^
({8{data_in[222][4]}} & 8'h43)^
({8{data_in[222][5]}} & 8'h86)^
({8{data_in[222][6]}} & 8'h27)^
({8{data_in[222][7]}} & 8'h4e)^
({8{data_in[223][0]}} & 8'ha1)^
({8{data_in[223][1]}} & 8'h69)^
({8{data_in[223][2]}} & 8'hd2)^
({8{data_in[223][3]}} & 8'h8f)^
({8{data_in[223][4]}} & 8'h35)^
({8{data_in[223][5]}} & 8'h6a)^
({8{data_in[223][6]}} & 8'hd4)^
({8{data_in[223][7]}} & 8'h83)^
({8{data_in[224][0]}} & 8'h95)^
({8{data_in[224][1]}} & 8'h1)^
({8{data_in[224][2]}} & 8'h2)^
({8{data_in[224][3]}} & 8'h4)^
({8{data_in[224][4]}} & 8'h8)^
({8{data_in[224][5]}} & 8'h10)^
({8{data_in[224][6]}} & 8'h20)^
({8{data_in[224][7]}} & 8'h40)^
({8{data_in[225][0]}} & 8'he5)^
({8{data_in[225][1]}} & 8'he1)^
({8{data_in[225][2]}} & 8'he9)^
({8{data_in[225][3]}} & 8'hf9)^
({8{data_in[225][4]}} & 8'hd9)^
({8{data_in[225][5]}} & 8'h99)^
({8{data_in[225][6]}} & 8'h19)^
({8{data_in[225][7]}} & 8'h32)^
({8{data_in[226][0]}} & 8'ha1)^

```

```

({8{data_in[226][1]}} & 8'h69)^
({8{data_in[226][2]}} & 8'hd2)^
({8{data_in[226][3]}} & 8'h8f)^
({8{data_in[226][4]}} & 8'h35)^
({8{data_in[226][5]}} & 8'h6a)^
({8{data_in[226][6]}} & 8'hd4)^
({8{data_in[226][7]}} & 8'h83)^
({8{data_in[227][0]}} & 8'ha2)^
({8{data_in[227][1]}} & 8'h6f)^
({8{data_in[227][2]}} & 8'hde)^
({8{data_in[227][3]}} & 8'h97)^
({8{data_in[227][4]}} & 8'h5)^
({8{data_in[227][5]}} & 8'ha)^
({8{data_in[227][6]}} & 8'h14)^
({8{data_in[227][7]}} & 8'h28)^
({8{data_in[228][0]}} & 8'h2c)^
({8{data_in[228][1]}} & 8'h58)^
({8{data_in[228][2]}} & 8'hb0)^
({8{data_in[228][3]}} & 8'h4b)^
({8{data_in[228][4]}} & 8'h96)^
({8{data_in[228][5]}} & 8'h7)^
({8{data_in[228][6]}} & 8'he)^
({8{data_in[228][7]}} & 8'h1c)^
({8{data_in[229][0]}} & 8'h67)^
({8{data_in[229][1]}} & 8'hce)^
({8{data_in[229][2]}} & 8'hb7)^
({8{data_in[229][3]}} & 8'h45)^
({8{data_in[229][4]}} & 8'h8a)^
({8{data_in[229][5]}} & 8'h3f)^
({8{data_in[229][6]}} & 8'h7e)^
({8{data_in[229][7]}} & 8'hfc)^
({8{data_in[230][0]}} & 8'he9)^
({8{data_in[230][1]}} & 8'hf9)^
({8{data_in[230][2]}} & 8'hd9)^
({8{data_in[230][3]}} & 8'h99)^
({8{data_in[230][4]}} & 8'h19)^
({8{data_in[230][5]}} & 8'h32)^
({8{data_in[230][6]}} & 8'h64)^
({8{data_in[230][7]}} & 8'hc8)^
({8{data_in[231][0]}} & 8'h23)^
({8{data_in[231][1]}} & 8'h46)^
({8{data_in[231][2]}} & 8'h8c)^
({8{data_in[231][3]}} & 8'h33)^
({8{data_in[231][4]}} & 8'h66)^
({8{data_in[231][5]}} & 8'hcc)^
({8{data_in[231][6]}} & 8'hb3)^
({8{data_in[231][7]}} & 8'h4d)^
({8{data_in[232][0]}} & 8'h50)^
({8{data_in[232][1]}} & 8'ha0)^
({8{data_in[232][2]}} & 8'h6b)^
({8{data_in[232][3]}} & 8'hd6)^
({8{data_in[232][4]}} & 8'h87)^
({8{data_in[232][5]}} & 8'h25)^
({8{data_in[232][6]}} & 8'h4a)^
({8{data_in[232][7]}} & 8'h94)^
({8{data_in[233][0]}} & 8'h26)^
({8{data_in[233][1]}} & 8'h4c)^
({8{data_in[233][2]}} & 8'h98)^
({8{data_in[233][3]}} & 8'h1b)^
({8{data_in[233][4]}} & 8'h36)^
({8{data_in[233][5]}} & 8'h6c)^
({8{data_in[233][6]}} & 8'hd8)^
({8{data_in[233][7]}} & 8'h9b)^
({8{data_in[234][0]}} & 8'h51)^
({8{data_in[234][1]}} & 8'ha2)^

```

```

({8{data_in[234][2]}} & 8'h6f)^
({8{data_in[234][3]}} & 8'hde)^
({8{data_in[234][4]}} & 8'h97)^
({8{data_in[234][5]}} & 8'h5)^
({8{data_in[234][6]}} & 8'ha)^
({8{data_in[234][7]}} & 8'h14)^
({8{data_in[235][0]}} & 8'hd8)^
({8{data_in[235][1]}} & 8'h9b)^
({8{data_in[235][2]}} & 8'h1d)^
({8{data_in[235][3]}} & 8'h3a)^
({8{data_in[235][4]}} & 8'h74)^
({8{data_in[235][5]}} & 8'he8)^
({8{data_in[235][6]}} & 8'hfb)^
({8{data_in[235][7]}} & 8'hdd)^
({8{data_in[236][0]}} & 8'hc8)^
({8{data_in[236][1]}} & 8'hbb)^
({8{data_in[236][2]}} & 8'h5d)^
({8{data_in[236][3]}} & 8'hba)^
({8{data_in[236][4]}} & 8'h5f)^
({8{data_in[236][5]}} & 8'hbe)^
({8{data_in[236][6]}} & 8'h57)^
({8{data_in[236][7]}} & 8'hae)^
({8{data_in[237][0]}} & 8'h61)^
({8{data_in[237][1]}} & 8'hc2)^
({8{data_in[237][2]}} & 8'haf)^
({8{data_in[237][3]}} & 8'h75)^
({8{data_in[237][4]}} & 8'hea)^
({8{data_in[237][5]}} & 8'hff)^
({8{data_in[237][6]}} & 8'hd5)^
({8{data_in[237][7]}} & 8'h81)^
({8{data_in[238][0]}} & 8'hb3)^
({8{data_in[238][1]}} & 8'h4d)^
({8{data_in[238][2]}} & 8'h9a)^
({8{data_in[238][3]}} & 8'h1f)^
({8{data_in[238][4]}} & 8'h3e)^
({8{data_in[238][5]}} & 8'h7c)^
({8{data_in[238][6]}} & 8'hf8)^
({8{data_in[238][7]}} & 8'hdb)^
({8{data_in[239][0]}} & 8'h5a)^
({8{data_in[239][1]}} & 8'hb4)^
({8{data_in[239][2]}} & 8'h43)^
({8{data_in[239][3]}} & 8'h86)^
({8{data_in[239][4]}} & 8'h27)^
({8{data_in[239][5]}} & 8'h4e)^
({8{data_in[239][6]}} & 8'h9c)^
({8{data_in[239][7]}} & 8'h13)^
({8{data_in[240][0]}} & 8'h5e)^
({8{data_in[240][1]}} & 8'hbc)^
({8{data_in[240][2]}} & 8'h53)^
({8{data_in[240][3]}} & 8'ha6)^
({8{data_in[240][4]}} & 8'h67)^
({8{data_in[240][5]}} & 8'hce)^
({8{data_in[240][6]}} & 8'hb7)^
({8{data_in[240][7]}} & 8'h45)^
({8{data_in[241][0]}} & 8'h41)^
({8{data_in[241][1]}} & 8'h82)^
({8{data_in[241][2]}} & 8'h2f)^
({8{data_in[241][3]}} & 8'h5e)^
({8{data_in[241][4]}} & 8'hbc)^
({8{data_in[241][5]}} & 8'h53)^
({8{data_in[241][6]}} & 8'ha6)^
({8{data_in[241][7]}} & 8'h67);
data_out[243] = ({8{data_in[0][0]}} & 8'h3b)^
({8{data_in[0][1]}} & 8'h76)^
({8{data_in[0][2]}} & 8'hec)^

```

```

({8{data_in[0][3]}} & 8'hf3)^
({8{data_in[0][4]}} & 8'hcd)^
({8{data_in[0][5]}} & 8'hb1)^
({8{data_in[0][6]}} & 8'h49)^
({8{data_in[0][7]}} & 8'h92)^
({8{data_in[1][0]}} & 8'h29)^
({8{data_in[1][1]}} & 8'h52)^
({8{data_in[1][2]}} & 8'ha4)^
({8{data_in[1][3]}} & 8'h63)^
({8{data_in[1][4]}} & 8'hc6)^
({8{data_in[1][5]}} & 8'ha7)^
({8{data_in[1][6]}} & 8'h65)^
({8{data_in[1][7]}} & 8'hca)^
({8{data_in[2][0]}} & 8'h11)^
({8{data_in[2][1]}} & 8'h22)^
({8{data_in[2][2]}} & 8'h44)^
({8{data_in[2][3]}} & 8'h88)^
({8{data_in[2][4]}} & 8'h3b)^
({8{data_in[2][5]}} & 8'h76)^
({8{data_in[2][6]}} & 8'hec)^
({8{data_in[2][7]}} & 8'hf3)^
({8{data_in[3][0]}} & 8'h56)^
({8{data_in[3][1]}} & 8'hac)^
({8{data_in[3][2]}} & 8'h73)^
({8{data_in[3][3]}} & 8'he6)^
({8{data_in[3][4]}} & 8'he7)^
({8{data_in[3][5]}} & 8'he5)^
({8{data_in[3][6]}} & 8'he1)^
({8{data_in[3][7]}} & 8'he9)^
({8{data_in[4][0]}} & 8'h44)^
({8{data_in[4][1]}} & 8'h88)^
({8{data_in[4][2]}} & 8'h3b)^
({8{data_in[4][3]}} & 8'h76)^
({8{data_in[4][4]}} & 8'hec)^
({8{data_in[4][5]}} & 8'hf3)^
({8{data_in[4][6]}} & 8'hcd)^
({8{data_in[4][7]}} & 8'hb1)^
({8{data_in[5][0]}} & 8'hec)^
({8{data_in[5][1]}} & 8'hf3)^
({8{data_in[5][2]}} & 8'hcd)^
({8{data_in[5][3]}} & 8'hb1)^
({8{data_in[5][4]}} & 8'h49)^
({8{data_in[5][5]}} & 8'h92)^
({8{data_in[5][6]}} & 8'hf)^
({8{data_in[5][7]}} & 8'h1e)^
({8{data_in[6][0]}} & 8'hea)^
({8{data_in[6][1]}} & 8'hff)^
({8{data_in[6][2]}} & 8'hd5)^
({8{data_in[6][3]}} & 8'h81)^
({8{data_in[6][4]}} & 8'h29)^
({8{data_in[6][5]}} & 8'h52)^
({8{data_in[6][6]}} & 8'ha4)^
({8{data_in[6][7]}} & 8'h63)^
({8{data_in[7][0]}} & 8'h3a)^
({8{data_in[7][1]}} & 8'h74)^
({8{data_in[7][2]}} & 8'he8)^
({8{data_in[7][3]}} & 8'hfb)^
({8{data_in[7][4]}} & 8'hdd)^
({8{data_in[7][5]}} & 8'h91)^
({8{data_in[7][6]}} & 8'h9)^
({8{data_in[7][7]}} & 8'h12)^
({8{data_in[8][0]}} & 8'h3)^
({8{data_in[8][1]}} & 8'h6)^
({8{data_in[8][2]}} & 8'hc)^
({8{data_in[8][3]}} & 8'h18)^

```

```

({8{data_in[8][4]}} & 8'h30)^
({8{data_in[8][5]}} & 8'h60)^
({8{data_in[8][6]}} & 8'hc0)^
({8{data_in[8][7]}} & 8'hab)^
({8{data_in[9][0]}} & 8'h72)^
({8{data_in[9][1]}} & 8'he4)^
({8{data_in[9][2]}} & 8'he3)^
({8{data_in[9][3]}} & 8'hed)^
({8{data_in[9][4]}} & 8'hf1)^
({8{data_in[9][5]}} & 8'hc9)^
({8{data_in[9][6]}} & 8'hb9)^
({8{data_in[9][7]}} & 8'h59)^
({8{data_in[10][0]}} & 8'h11)^
({8{data_in[10][1]}} & 8'h22)^
({8{data_in[10][2]}} & 8'h44)^
({8{data_in[10][3]}} & 8'h88)^
({8{data_in[10][4]}} & 8'h3b)^
({8{data_in[10][5]}} & 8'h76)^
({8{data_in[10][6]}} & 8'hec)^
({8{data_in[10][7]}} & 8'hf3)^
({8{data_in[11][0]}} & 8'h4c)^
({8{data_in[11][1]}} & 8'h98)^
({8{data_in[11][2]}} & 8'h1b)^
({8{data_in[11][3]}} & 8'h36)^
({8{data_in[11][4]}} & 8'h6c)^
({8{data_in[11][5]}} & 8'hd8)^
({8{data_in[11][6]}} & 8'h9b)^
({8{data_in[11][7]}} & 8'h1d)^
({8{data_in[12][0]}} & 8'h7c)^
({8{data_in[12][1]}} & 8'hf8)^
({8{data_in[12][2]}} & 8'hdb)^
({8{data_in[12][3]}} & 8'h9d)^
({8{data_in[12][4]}} & 8'h11)^
({8{data_in[12][5]}} & 8'h22)^
({8{data_in[12][6]}} & 8'h44)^
({8{data_in[12][7]}} & 8'h88)^
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({8{data_in[14][3]}} & 8'hb2)^
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({8{data_in[14][5]}} & 8'h9e)^
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({8{data_in[63][1]}} & 8'hb7)^
({8{data_in[63][2]}} & 8'h45)^
({8{data_in[63][3]}} & 8'h8a)^
({8{data_in[63][4]}} & 8'h3f)^
({8{data_in[63][5]}} & 8'h7e)^
({8{data_in[63][6]}} & 8'hfc)^
({8{data_in[63][7]}} & 8'hd3)^
({8{data_in[64][0]}} & 8'h77)^
({8{data_in[64][1]}} & 8'hee)^
({8{data_in[64][2]}} & 8'hf7)^
({8{data_in[64][3]}} & 8'hc5)^
({8{data_in[64][4]}} & 8'ha1)^
({8{data_in[64][5]}} & 8'h69)^
({8{data_in[64][6]}} & 8'hd2)^
({8{data_in[64][7]}} & 8'h8f)^
({8{data_in[65][0]}} & 8'he2)^
({8{data_in[65][1]}} & 8'hef)^
({8{data_in[65][2]}} & 8'hf5)^

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({8{data_in[65][4]}} & 8'ha9)^
({8{data_in[65][5]}} & 8'h79)^
({8{data_in[65][6]}} & 8'hf2)^
({8{data_in[65][7]}} & 8'hcf)^
({8{data_in[66][0]}} & 8'hff)^
({8{data_in[66][1]}} & 8'hd5)^
({8{data_in[66][2]}} & 8'h81)^
({8{data_in[66][3]}} & 8'h29)^
({8{data_in[66][4]}} & 8'h52)^
({8{data_in[66][5]}} & 8'ha4)^
({8{data_in[66][6]}} & 8'h63)^
({8{data_in[66][7]}} & 8'hc6)^
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({8{data_in[67][1]}} & 8'h44)^
({8{data_in[67][2]}} & 8'h88)^
({8{data_in[67][3]}} & 8'h3b)^
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({8{data_in[67][5]}} & 8'hec)^
({8{data_in[67][6]}} & 8'hf3)^
({8{data_in[67][7]}} & 8'hcd)^
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({8{data_in[69][1]}} & 8'h5d)^
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({8{data_in[69][3]}} & 8'h5f)^
({8{data_in[69][4]}} & 8'hbe)^
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({8{data_in[70][6]}} & 8'hbb)^
({8{data_in[70][7]}} & 8'h5d)^
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({8{data_in[71][3]}} & 8'h96)^
({8{data_in[71][4]}} & 8'h7)^
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({8{data_in[71][7]}} & 8'h38)^
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({8{data_in[72][1]}} & 8'hd6)^
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({8{data_in[72][5]}} & 8'h94)^
({8{data_in[72][6]}} & 8'h3)^
({8{data_in[72][7]}} & 8'h6)^
({8{data_in[73][0]}} & 8'hd5)^
({8{data_in[73][1]}} & 8'h81)^
({8{data_in[73][2]}} & 8'h29)^
({8{data_in[73][3]}} & 8'h52)^

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({8{data_in[74][4]}} & 8'h9a)^
({8{data_in[74][5]}} & 8'h1f)^
({8{data_in[74][6]}} & 8'h3e)^
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({8{data_in[77][4]}} & 8'hda)^
({8{data_in[77][5]}} & 8'h9f)^
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({8{data_in[78][2]}} & 8'h2f)^
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({8{data_in[79][7]}} & 8'hac)^
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({8{data_in[80][3]}} & 8'hb)^
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({8{data_in[87][2]}} & 8'h77)^
({8{data_in[87][3]}} & 8'hee)^
({8{data_in[87][4]}} & 8'hf7)^
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({8{data_in[87][6]}} & 8'ha1)^
({8{data_in[87][7]}} & 8'h69)^
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({8{data_in[88][1]}} & 8'h38)^
({8{data_in[88][2]}} & 8'h70)^
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({8{data_in[88][4]}} & 8'heb)^
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({8{data_in[93][5]}} & 8'h3)^
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({8{data_in[96][4]}} & 8'h5c)^
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({8{data_in[98][7]}} & 8'h7c)^
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({8{data_in[108][7]}} & 8'h5e)^
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({8{data_in[109][4]}} & 8'h4a)^
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({8{data_in[113][2]}} & 8'h18)^
({8{data_in[113][3]}} & 8'h30)^
({8{data_in[113][4]}} & 8'h60)^
({8{data_in[113][5]}} & 8'hc0)^
({8{data_in[113][6]}} & 8'hab)^
({8{data_in[113][7]}} & 8'h7d)^
({8{data_in[114][0]}} & 8'hff)^

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({8{data_in[114][1]}} & 8'hd5)^
({8{data_in[114][2]}} & 8'h81)^
({8{data_in[114][3]}} & 8'h29)^
({8{data_in[114][4]}} & 8'h52)^
({8{data_in[114][5]}} & 8'ha4)^
({8{data_in[114][6]}} & 8'h63)^
({8{data_in[114][7]}} & 8'hc6)^
({8{data_in[115][0]}} & 8'h88)^
({8{data_in[115][1]}} & 8'h3b)^
({8{data_in[115][2]}} & 8'h76)^
({8{data_in[115][3]}} & 8'hec)^
({8{data_in[115][4]}} & 8'hf3)^
({8{data_in[115][5]}} & 8'hcd)^
({8{data_in[115][6]}} & 8'hb1)^
({8{data_in[115][7]}} & 8'h49)^
({8{data_in[116][0]}} & 8'hbc)^
({8{data_in[116][1]}} & 8'h53)^
({8{data_in[116][2]}} & 8'ha6)^
({8{data_in[116][3]}} & 8'h67)^
({8{data_in[116][4]}} & 8'hce)^
({8{data_in[116][5]}} & 8'hb7)^
({8{data_in[116][6]}} & 8'h45)^
({8{data_in[116][7]}} & 8'h8a)^
({8{data_in[117][0]}} & 8'h93)^
({8{data_in[117][1]}} & 8'hd)^
({8{data_in[117][2]}} & 8'h1a)^
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({8{data_in[117][4]}} & 8'h68)^
({8{data_in[117][5]}} & 8'hd0)^
({8{data_in[117][6]}} & 8'h8b)^
({8{data_in[117][7]}} & 8'h3d)^
({8{data_in[118][0]}} & 8'h5a)^
({8{data_in[118][1]}} & 8'hb4)^
({8{data_in[118][2]}} & 8'h43)^
({8{data_in[118][3]}} & 8'h86)^
({8{data_in[118][4]}} & 8'h27)^
({8{data_in[118][5]}} & 8'h4e)^
({8{data_in[118][6]}} & 8'h9c)^
({8{data_in[118][7]}} & 8'h13)^
({8{data_in[119][0]}} & 8'h6d)^
({8{data_in[119][1]}} & 8'hda)^
({8{data_in[119][2]}} & 8'h9f)^
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({8{data_in[119][4]}} & 8'h2a)^
({8{data_in[119][5]}} & 8'h54)^
({8{data_in[119][6]}} & 8'ha8)^
({8{data_in[119][7]}} & 8'h7b)^
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({8{data_in[120][1]}} & 8'he6)^
({8{data_in[120][2]}} & 8'he7)^
({8{data_in[120][3]}} & 8'he5)^
({8{data_in[120][4]}} & 8'he1)^
({8{data_in[120][5]}} & 8'he9)^
({8{data_in[120][6]}} & 8'hf9)^
({8{data_in[120][7]}} & 8'hd9)^
({8{data_in[121][0]}} & 8'hb8)^
({8{data_in[121][1]}} & 8'h5b)^
({8{data_in[121][2]}} & 8'hb6)^
({8{data_in[121][3]}} & 8'h47)^
({8{data_in[121][4]}} & 8'h8e)^
({8{data_in[121][5]}} & 8'h37)^
({8{data_in[121][6]}} & 8'h6e)^
({8{data_in[121][7]}} & 8'hdc)^
({8{data_in[122][0]}} & 8'h5b)^
({8{data_in[122][1]}} & 8'hb6)^

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({8{data_in[122][2]}} & 8'h47)^
({8{data_in[122][3]}} & 8'h8e)^
({8{data_in[122][4]}} & 8'h37)^
({8{data_in[122][5]}} & 8'h6e)^
({8{data_in[122][6]}} & 8'hdc)^
({8{data_in[122][7]}} & 8'h93)^
({8{data_in[123][0]}} & 8'h45)^
({8{data_in[123][1]}} & 8'h8a)^
({8{data_in[123][2]}} & 8'h3f)^
({8{data_in[123][3]}} & 8'h7e)^
({8{data_in[123][4]}} & 8'hfc)^
({8{data_in[123][5]}} & 8'hd3)^
({8{data_in[123][6]}} & 8'h8d)^
({8{data_in[123][7]}} & 8'h31)^
({8{data_in[124][0]}} & 8'hac)^
({8{data_in[124][1]}} & 8'h73)^
({8{data_in[124][2]}} & 8'he6)^
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({8{data_in[124][4]}} & 8'he5)^
({8{data_in[124][5]}} & 8'he1)^
({8{data_in[124][6]}} & 8'he9)^
({8{data_in[124][7]}} & 8'hf9)^
({8{data_in[125][0]}} & 8'h4b)^
({8{data_in[125][1]}} & 8'h96)^
({8{data_in[125][2]}} & 8'h7)^
({8{data_in[125][3]}} & 8'he)^
({8{data_in[125][4]}} & 8'h1c)^
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({8{data_in[125][6]}} & 8'h70)^
({8{data_in[125][7]}} & 8'he0)^
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({8{data_in[126][1]}} & 8'hc1)^
({8{data_in[126][2]}} & 8'ha9)^
({8{data_in[126][3]}} & 8'h79)^
({8{data_in[126][4]}} & 8'hf2)^
({8{data_in[126][5]}} & 8'hcf)^
({8{data_in[126][6]}} & 8'hb5)^
({8{data_in[126][7]}} & 8'h41)^
({8{data_in[127][0]}} & 8'h33)^
({8{data_in[127][1]}} & 8'h66)^
({8{data_in[127][2]}} & 8'hcc)^
({8{data_in[127][3]}} & 8'hb3)^
({8{data_in[127][4]}} & 8'h4d)^
({8{data_in[127][5]}} & 8'h9a)^
({8{data_in[127][6]}} & 8'h1f)^
({8{data_in[127][7]}} & 8'h3e)^
({8{data_in[128][0]}} & 8'h86)^
({8{data_in[128][1]}} & 8'h27)^
({8{data_in[128][2]}} & 8'h4e)^
({8{data_in[128][3]}} & 8'h9c)^
({8{data_in[128][4]}} & 8'h13)^
({8{data_in[128][5]}} & 8'h26)^
({8{data_in[128][6]}} & 8'h4c)^
({8{data_in[128][7]}} & 8'h98)^
({8{data_in[129][0]}} & 8'h5b)^
({8{data_in[129][1]}} & 8'hb6)^
({8{data_in[129][2]}} & 8'h47)^
({8{data_in[129][3]}} & 8'h8e)^
({8{data_in[129][4]}} & 8'h37)^
({8{data_in[129][5]}} & 8'h6e)^
({8{data_in[129][6]}} & 8'hdc)^
({8{data_in[129][7]}} & 8'h93)^
({8{data_in[130][0]}} & 8'ha0)^
({8{data_in[130][1]}} & 8'h6b)^
({8{data_in[130][2]}} & 8'hd6)^

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({8{data_in[130][3]}} & 8'h87)^
({8{data_in[130][4]}} & 8'h25)^
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({8{data_in[130][7]}} & 8'h3)^
({8{data_in[131][0]}} & 8'h5e)^
({8{data_in[131][1]}} & 8'hbc)^
({8{data_in[131][2]}} & 8'h53)^
({8{data_in[131][3]}} & 8'ha6)^
({8{data_in[131][4]}} & 8'h67)^
({8{data_in[131][5]}} & 8'hce)^
({8{data_in[131][6]}} & 8'hb7)^
({8{data_in[131][7]}} & 8'h45)^
({8{data_in[132][0]}} & 8'he1)^
({8{data_in[132][1]}} & 8'he9)^
({8{data_in[132][2]}} & 8'hf9)^
({8{data_in[132][3]}} & 8'hd9)^
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({8{data_in[132][7]}} & 8'h64)^
({8{data_in[133][0]}} & 8'h50)^
({8{data_in[133][1]}} & 8'ha0)^
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({8{data_in[133][3]}} & 8'hd6)^
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({8{data_in[133][6]}} & 8'h4a)^
({8{data_in[133][7]}} & 8'h94)^
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({8{data_in[134][1]}} & 8'h3c)^
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({8{data_in[134][3]}} & 8'hf0)^
({8{data_in[134][4]}} & 8'hcb)^
({8{data_in[134][5]}} & 8'hbd)^
({8{data_in[134][6]}} & 8'h51)^
({8{data_in[134][7]}} & 8'ha2)^
({8{data_in[135][0]}} & 8'h55)^
({8{data_in[135][1]}} & 8'haa)^
({8{data_in[135][2]}} & 8'h7f)^
({8{data_in[135][3]}} & 8'hfe)^
({8{data_in[135][4]}} & 8'hd7)^
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({8{data_in[135][7]}} & 8'h42)^
({8{data_in[136][0]}} & 8'h83)^
({8{data_in[136][1]}} & 8'h2d)^
({8{data_in[136][2]}} & 8'h5a)^
({8{data_in[136][3]}} & 8'hb4)^
({8{data_in[136][4]}} & 8'h43)^
({8{data_in[136][5]}} & 8'h86)^
({8{data_in[136][6]}} & 8'h27)^
({8{data_in[136][7]}} & 8'h4e)^
({8{data_in[137][0]}} & 8'h60)^
({8{data_in[137][1]}} & 8'hc0)^
({8{data_in[137][2]}} & 8'hab)^
({8{data_in[137][3]}} & 8'h7d)^
({8{data_in[137][4]}} & 8'hfa)^
({8{data_in[137][5]}} & 8'hdf)^
({8{data_in[137][6]}} & 8'h95)^
({8{data_in[137][7]}} & 8'h1)^
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({8{data_in[138][1]}} & 8'h92)^
({8{data_in[138][2]}} & 8'hf)^
({8{data_in[138][3]}} & 8'h1e)^

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({8{data_in[139][1]}} & 8'hf1)^
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({8{data_in[139][3]}} & 8'hb9)^
({8{data_in[139][4]}} & 8'h59)^
({8{data_in[139][5]}} & 8'hb2)^
({8{data_in[139][6]}} & 8'h4f)^
({8{data_in[139][7]}} & 8'h9e)^
({8{data_in[140][0]}} & 8'hd9)^
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({8{data_in[140][2]}} & 8'h19)^
({8{data_in[140][3]}} & 8'h32)^
({8{data_in[140][4]}} & 8'h64)^
({8{data_in[140][5]}} & 8'hc8)^
({8{data_in[140][6]}} & 8'hbb)^
({8{data_in[140][7]}} & 8'h5d)^
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({8{data_in[141][1]}} & 8'h82)^
({8{data_in[141][2]}} & 8'h2f)^
({8{data_in[141][3]}} & 8'h5e)^
({8{data_in[141][4]}} & 8'hbc)^
({8{data_in[141][5]}} & 8'h53)^
({8{data_in[141][6]}} & 8'ha6)^
({8{data_in[141][7]}} & 8'h67)^
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({8{data_in[142][1]}} & 8'h27)^
({8{data_in[142][2]}} & 8'h4e)^
({8{data_in[142][3]}} & 8'h9c)^
({8{data_in[142][4]}} & 8'h13)^
({8{data_in[142][5]}} & 8'h26)^
({8{data_in[142][6]}} & 8'h4c)^
({8{data_in[142][7]}} & 8'h98)^
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({8{data_in[143][1]}} & 8'h4)^
({8{data_in[143][2]}} & 8'h8)^
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({8{data_in[143][7]}} & 8'h2b)^
({8{data_in[144][0]}} & 8'h8)^
({8{data_in[144][1]}} & 8'h10)^
({8{data_in[144][2]}} & 8'h20)^
({8{data_in[144][3]}} & 8'h40)^
({8{data_in[144][4]}} & 8'h80)^
({8{data_in[144][5]}} & 8'h2b)^
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({8{data_in[145][2]}} & 8'hb6)^
({8{data_in[145][3]}} & 8'h47)^
({8{data_in[145][4]}} & 8'h8e)^
({8{data_in[145][5]}} & 8'h37)^
({8{data_in[145][6]}} & 8'h6e)^
({8{data_in[145][7]}} & 8'hdc)^
({8{data_in[146][0]}} & 8'h1)^
({8{data_in[146][1]}} & 8'h2)^
({8{data_in[146][2]}} & 8'h4)^
({8{data_in[146][3]}} & 8'h8)^
({8{data_in[146][4]}} & 8'h10)^

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({8{data_in[147][3]}} & 8'he8)^
({8{data_in[147][4]}} & 8'hfb)^
({8{data_in[147][5]}} & 8'hdd)^
({8{data_in[147][6]}} & 8'h91)^
({8{data_in[147][7]}} & 8'h9)^
({8{data_in[148][0]}} & 8'h74)^
({8{data_in[148][1]}} & 8'he8)^
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({8{data_in[148][3]}} & 8'hdd)^
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({8{data_in[148][7]}} & 8'h24)^
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({8{data_in[149][1]}} & 8'h1b)^
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({8{data_in[149][3]}} & 8'h6c)^
({8{data_in[149][4]}} & 8'hd8)^
({8{data_in[149][5]}} & 8'h9b)^
({8{data_in[149][6]}} & 8'h1d)^
({8{data_in[149][7]}} & 8'h3a)^
({8{data_in[150][0]}} & 8'hf5)^
({8{data_in[150][1]}} & 8'hc1)^
({8{data_in[150][2]}} & 8'ha9)^
({8{data_in[150][3]}} & 8'h79)^
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({8{data_in[151][4]}} & 8'h92)^
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({8{data_in[151][6]}} & 8'h1e)^
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({8{data_in[152][6]}} & 8'h8)^
({8{data_in[152][7]}} & 8'h10)^
({8{data_in[153][0]}} & 8'he0)^
({8{data_in[153][1]}} & 8'heb)^
({8{data_in[153][2]}} & 8'hfd)^
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({8{data_in[154][2]}} & 8'h18)^
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({8{data_in[155][3]}} & 8'hc9)^
({8{data_in[155][4]}} & 8'hb9)^
({8{data_in[155][5]}} & 8'h59)^
({8{data_in[155][6]}} & 8'hb2)^
({8{data_in[155][7]}} & 8'h4f)^
({8{data_in[156][0]}} & 8'hb8)^
({8{data_in[156][1]}} & 8'h5b)^
({8{data_in[156][2]}} & 8'hb6)^
({8{data_in[156][3]}} & 8'h47)^
({8{data_in[156][4]}} & 8'h8e)^
({8{data_in[156][5]}} & 8'h37)^
({8{data_in[156][6]}} & 8'h6e)^
({8{data_in[156][7]}} & 8'hdc)^
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({8{data_in[158][1]}} & 8'hfb)^
({8{data_in[158][2]}} & 8'hdd)^
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({8{data_in[159][1]}} & 8'h2e)^
({8{data_in[159][2]}} & 8'h5c)^
({8{data_in[159][3]}} & 8'hb8)^
({8{data_in[159][4]}} & 8'h5b)^
({8{data_in[159][5]}} & 8'hb6)^
({8{data_in[159][6]}} & 8'h47)^
({8{data_in[159][7]}} & 8'h8e)^
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({8{data_in[160][2]}} & 8'h26)^
({8{data_in[160][3]}} & 8'h4c)^
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({8{data_in[160][5]}} & 8'h1b)^
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({8{data_in[160][7]}} & 8'h6c)^
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({8{data_in[162][3]}} & 8'h9e)^
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({8{data_in[162][5]}} & 8'h2e)^
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({8{data_in[163][1]}} & 8'h3e)^
({8{data_in[163][2]}} & 8'h7c)^
({8{data_in[163][3]}} & 8'hf8)^
({8{data_in[163][4]}} & 8'hdb)^
({8{data_in[163][5]}} & 8'h9d)^
({8{data_in[163][6]}} & 8'h11)^
({8{data_in[163][7]}} & 8'h22)^
({8{data_in[164][0]}} & 8'hbf)^
({8{data_in[164][1]}} & 8'h55)^
({8{data_in[164][2]}} & 8'haa)^
({8{data_in[164][3]}} & 8'h7f)^
({8{data_in[164][4]}} & 8'hfe)^
({8{data_in[164][5]}} & 8'hd7)^
({8{data_in[164][6]}} & 8'h85)^
({8{data_in[164][7]}} & 8'h21)^
({8{data_in[165][0]}} & 8'hb7)^
({8{data_in[165][1]}} & 8'h45)^
({8{data_in[165][2]}} & 8'h8a)^
({8{data_in[165][3]}} & 8'h3f)^
({8{data_in[165][4]}} & 8'h7e)^
({8{data_in[165][5]}} & 8'hfc)^
({8{data_in[165][6]}} & 8'hd3)^
({8{data_in[165][7]}} & 8'h8d)^
({8{data_in[166][0]}} & 8'had)^
({8{data_in[166][1]}} & 8'h71)^
({8{data_in[166][2]}} & 8'he2)^
({8{data_in[166][3]}} & 8'hef)^
({8{data_in[166][4]}} & 8'hf5)^
({8{data_in[166][5]}} & 8'hc1)^
({8{data_in[166][6]}} & 8'ha9)^
({8{data_in[166][7]}} & 8'h79)^
({8{data_in[167][0]}} & 8'h71)^
({8{data_in[167][1]}} & 8'he2)^
({8{data_in[167][2]}} & 8'hef)^
({8{data_in[167][3]}} & 8'hf5)^
({8{data_in[167][4]}} & 8'hc1)^
({8{data_in[167][5]}} & 8'ha9)^
({8{data_in[167][6]}} & 8'h79)^
({8{data_in[167][7]}} & 8'hf2)^
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({8{data_in[168][1]}} & 8'h29)^
({8{data_in[168][2]}} & 8'h52)^
({8{data_in[168][3]}} & 8'ha4)^
({8{data_in[168][4]}} & 8'h63)^
({8{data_in[168][5]}} & 8'hc6)^
({8{data_in[168][6]}} & 8'ha7)^
({8{data_in[168][7]}} & 8'h65)^
({8{data_in[169][0]}} & 8'h85)^
({8{data_in[169][1]}} & 8'h21)^
({8{data_in[169][2]}} & 8'h42)^
({8{data_in[169][3]}} & 8'h84)^
({8{data_in[169][4]}} & 8'h23)^
({8{data_in[169][5]}} & 8'h46)^
({8{data_in[169][6]}} & 8'h8c)^
({8{data_in[169][7]}} & 8'h33)^
({8{data_in[170][0]}} & 8'hf3)^
({8{data_in[170][1]}} & 8'hcd)^
({8{data_in[170][2]}} & 8'hb1)^
({8{data_in[170][3]}} & 8'h49)^
({8{data_in[170][4]}} & 8'h92)^
({8{data_in[170][5]}} & 8'hf)^
({8{data_in[170][6]}} & 8'h1e)^
({8{data_in[170][7]}} & 8'h3c)^

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({8{data_in[171][0]}} & 8'hbb)^
({8{data_in[171][1]}} & 8'h5d)^
({8{data_in[171][2]}} & 8'hba)^
({8{data_in[171][3]}} & 8'h5f)^
({8{data_in[171][4]}} & 8'hbe)^
({8{data_in[171][5]}} & 8'h57)^
({8{data_in[171][6]}} & 8'hae)^
({8{data_in[171][7]}} & 8'h77)^
({8{data_in[172][0]}} & 8'ha1)^
({8{data_in[172][1]}} & 8'h69)^
({8{data_in[172][2]}} & 8'hd2)^
({8{data_in[172][3]}} & 8'h8f)^
({8{data_in[172][4]}} & 8'h35)^
({8{data_in[172][5]}} & 8'h6a)^
({8{data_in[172][6]}} & 8'hd4)^
({8{data_in[172][7]}} & 8'h83)^
({8{data_in[173][0]}} & 8'h62)^
({8{data_in[173][1]}} & 8'hc4)^
({8{data_in[173][2]}} & 8'ha3)^
({8{data_in[173][3]}} & 8'h6d)^
({8{data_in[173][4]}} & 8'hda)^
({8{data_in[173][5]}} & 8'h9f)^
({8{data_in[173][6]}} & 8'h15)^
({8{data_in[173][7]}} & 8'h2a)^
({8{data_in[174][0]}} & 8'h6a)^
({8{data_in[174][1]}} & 8'hd4)^
({8{data_in[174][2]}} & 8'h83)^
({8{data_in[174][3]}} & 8'h2d)^
({8{data_in[174][4]}} & 8'h5a)^
({8{data_in[174][5]}} & 8'hb4)^
({8{data_in[174][6]}} & 8'h43)^
({8{data_in[174][7]}} & 8'h86)^
({8{data_in[175][0]}} & 8'hce)^
({8{data_in[175][1]}} & 8'hb7)^
({8{data_in[175][2]}} & 8'h45)^
({8{data_in[175][3]}} & 8'h8a)^
({8{data_in[175][4]}} & 8'h3f)^
({8{data_in[175][5]}} & 8'h7e)^
({8{data_in[175][6]}} & 8'hfc)^
({8{data_in[175][7]}} & 8'hd3)^
({8{data_in[176][0]}} & 8'h8f)^
({8{data_in[176][1]}} & 8'h35)^
({8{data_in[176][2]}} & 8'h6a)^
({8{data_in[176][3]}} & 8'hd4)^
({8{data_in[176][4]}} & 8'h83)^
({8{data_in[176][5]}} & 8'h2d)^
({8{data_in[176][6]}} & 8'h5a)^
({8{data_in[176][7]}} & 8'hb4)^
({8{data_in[177][0]}} & 8'hec)^
({8{data_in[177][1]}} & 8'hf3)^
({8{data_in[177][2]}} & 8'hcd)^
({8{data_in[177][3]}} & 8'hb1)^
({8{data_in[177][4]}} & 8'h49)^
({8{data_in[177][5]}} & 8'h92)^
({8{data_in[177][6]}} & 8'hf)^
({8{data_in[177][7]}} & 8'h1e)^
({8{data_in[178][0]}} & 8'h1c)^
({8{data_in[178][1]}} & 8'h38)^
({8{data_in[178][2]}} & 8'h70)^
({8{data_in[178][3]}} & 8'he0)^
({8{data_in[178][4]}} & 8'heb)^
({8{data_in[178][5]}} & 8'hfd)^
({8{data_in[178][6]}} & 8'hd1)^
({8{data_in[178][7]}} & 8'h89)^
({8{data_in[179][0]}} & 8'h80)^

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({8{data_in[179][1]}} & 8'h2b)^
({8{data_in[179][2]}} & 8'h56)^
({8{data_in[179][3]}} & 8'hac)^
({8{data_in[179][4]}} & 8'h73)^
({8{data_in[179][5]}} & 8'he6)^
({8{data_in[179][6]}} & 8'he7)^
({8{data_in[179][7]}} & 8'he5)^
({8{data_in[180][0]}} & 8'h7e)^
({8{data_in[180][1]}} & 8'hfc)^
({8{data_in[180][2]}} & 8'hd3)^
({8{data_in[180][3]}} & 8'h8d)^
({8{data_in[180][4]}} & 8'h31)^
({8{data_in[180][5]}} & 8'h62)^
({8{data_in[180][6]}} & 8'hc4)^
({8{data_in[180][7]}} & 8'ha3)^
({8{data_in[181][0]}} & 8'ha5)^
({8{data_in[181][1]}} & 8'h61)^
({8{data_in[181][2]}} & 8'hc2)^
({8{data_in[181][3]}} & 8'haf)^
({8{data_in[181][4]}} & 8'h75)^
({8{data_in[181][5]}} & 8'hea)^
({8{data_in[181][6]}} & 8'hff)^
({8{data_in[181][7]}} & 8'hd5)^
({8{data_in[182][0]}} & 8'hef)^
({8{data_in[182][1]}} & 8'hf5)^
({8{data_in[182][2]}} & 8'hc1)^
({8{data_in[182][3]}} & 8'ha9)^
({8{data_in[182][4]}} & 8'h79)^
({8{data_in[182][5]}} & 8'hf2)^
({8{data_in[182][6]}} & 8'hcf)^
({8{data_in[182][7]}} & 8'hb5)^
({8{data_in[183][0]}} & 8'h5f)^
({8{data_in[183][1]}} & 8'hbe)^
({8{data_in[183][2]}} & 8'h57)^
({8{data_in[183][3]}} & 8'hae)^
({8{data_in[183][4]}} & 8'h77)^
({8{data_in[183][5]}} & 8'hee)^
({8{data_in[183][6]}} & 8'hf7)^
({8{data_in[183][7]}} & 8'hc5)^
({8{data_in[184][0]}} & 8'hd1)^
({8{data_in[184][1]}} & 8'h89)^
({8{data_in[184][2]}} & 8'h39)^
({8{data_in[184][3]}} & 8'h72)^
({8{data_in[184][4]}} & 8'he4)^
({8{data_in[184][5]}} & 8'he3)^
({8{data_in[184][6]}} & 8'hed)^
({8{data_in[184][7]}} & 8'hf1)^
({8{data_in[185][0]}} & 8'hdc)^
({8{data_in[185][1]}} & 8'h93)^
({8{data_in[185][2]}} & 8'hd)^
({8{data_in[185][3]}} & 8'h1a)^
({8{data_in[185][4]}} & 8'h34)^
({8{data_in[185][5]}} & 8'h68)^
({8{data_in[185][6]}} & 8'hd0)^
({8{data_in[185][7]}} & 8'h8b)^
({8{data_in[186][0]}} & 8'h24)^
({8{data_in[186][1]}} & 8'h48)^
({8{data_in[186][2]}} & 8'h90)^
({8{data_in[186][3]}} & 8'hb)^
({8{data_in[186][4]}} & 8'h16)^
({8{data_in[186][5]}} & 8'h2c)^
({8{data_in[186][6]}} & 8'h58)^
({8{data_in[186][7]}} & 8'hb0)^
({8{data_in[187][0]}} & 8'h71)^
({8{data_in[187][1]}} & 8'he2)^

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({8{data_in[187][2]}} & 8'hef)^
({8{data_in[187][3]}} & 8'hf5)^
({8{data_in[187][4]}} & 8'hc1)^
({8{data_in[187][5]}} & 8'ha9)^
({8{data_in[187][6]}} & 8'h79)^
({8{data_in[187][7]}} & 8'hf2)^
({8{data_in[188][0]}} & 8'h5c)^
({8{data_in[188][1]}} & 8'hb8)^
({8{data_in[188][2]}} & 8'h5b)^
({8{data_in[188][3]}} & 8'hb6)^
({8{data_in[188][4]}} & 8'h47)^
({8{data_in[188][5]}} & 8'h8e)^
({8{data_in[188][6]}} & 8'h37)^
({8{data_in[188][7]}} & 8'h6e)^
({8{data_in[189][0]}} & 8'h51)^
({8{data_in[189][1]}} & 8'ha2)^
({8{data_in[189][2]}} & 8'h6f)^
({8{data_in[189][3]}} & 8'hde)^
({8{data_in[189][4]}} & 8'h97)^
({8{data_in[189][5]}} & 8'h5)^
({8{data_in[189][6]}} & 8'ha)^
({8{data_in[189][7]}} & 8'h14)^
({8{data_in[190][0]}} & 8'h3d)^
({8{data_in[190][1]}} & 8'h7a)^
({8{data_in[190][2]}} & 8'hf4)^
({8{data_in[190][3]}} & 8'hc3)^
({8{data_in[190][4]}} & 8'had)^
({8{data_in[190][5]}} & 8'h71)^
({8{data_in[190][6]}} & 8'he2)^
({8{data_in[190][7]}} & 8'hef)^
({8{data_in[191][0]}} & 8'hb8)^
({8{data_in[191][1]}} & 8'h5b)^
({8{data_in[191][2]}} & 8'hb6)^
({8{data_in[191][3]}} & 8'h47)^
({8{data_in[191][4]}} & 8'h8e)^
({8{data_in[191][5]}} & 8'h37)^
({8{data_in[191][6]}} & 8'h6e)^
({8{data_in[191][7]}} & 8'hdc)^
({8{data_in[192][0]}} & 8'h38)^
({8{data_in[192][1]}} & 8'h70)^
({8{data_in[192][2]}} & 8'he0)^
({8{data_in[192][3]}} & 8'heb)^
({8{data_in[192][4]}} & 8'hfd)^
({8{data_in[192][5]}} & 8'hd1)^
({8{data_in[192][6]}} & 8'h89)^
({8{data_in[192][7]}} & 8'h39)^
({8{data_in[193][0]}} & 8'h19)^
({8{data_in[193][1]}} & 8'h32)^
({8{data_in[193][2]}} & 8'h64)^
({8{data_in[193][3]}} & 8'hc8)^
({8{data_in[193][4]}} & 8'hbb)^
({8{data_in[193][5]}} & 8'h5d)^
({8{data_in[193][6]}} & 8'hba)^
({8{data_in[193][7]}} & 8'h5f)^
({8{data_in[194][0]}} & 8'h6c)^
({8{data_in[194][1]}} & 8'hd8)^
({8{data_in[194][2]}} & 8'h9b)^
({8{data_in[194][3]}} & 8'h1d)^
({8{data_in[194][4]}} & 8'h3a)^
({8{data_in[194][5]}} & 8'h74)^
({8{data_in[194][6]}} & 8'he8)^
({8{data_in[194][7]}} & 8'hfb)^
({8{data_in[195][0]}} & 8'h2)^
({8{data_in[195][1]}} & 8'h4)^
({8{data_in[195][2]}} & 8'h8)^

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({8{data_in[195][3]}} & 8'h10)^
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({8{data_in[195][6]}} & 8'h80)^
({8{data_in[195][7]}} & 8'h2b)^
({8{data_in[196][0]}} & 8'h9)^
({8{data_in[196][1]}} & 8'h12)^
({8{data_in[196][2]}} & 8'h24)^
({8{data_in[196][3]}} & 8'h48)^
({8{data_in[196][4]}} & 8'h90)^
({8{data_in[196][5]}} & 8'hb)^
({8{data_in[196][6]}} & 8'h16)^
({8{data_in[196][7]}} & 8'h2c)^
({8{data_in[197][0]}} & 8'h50)^
({8{data_in[197][1]}} & 8'ha0)^
({8{data_in[197][2]}} & 8'h6b)^
({8{data_in[197][3]}} & 8'hd6)^
({8{data_in[197][4]}} & 8'h87)^
({8{data_in[197][5]}} & 8'h25)^
({8{data_in[197][6]}} & 8'h4a)^
({8{data_in[197][7]}} & 8'h94)^
({8{data_in[198][0]}} & 8'hc4)^
({8{data_in[198][1]}} & 8'ha3)^
({8{data_in[198][2]}} & 8'h6d)^
({8{data_in[198][3]}} & 8'hda)^
({8{data_in[198][4]}} & 8'h9f)^
({8{data_in[198][5]}} & 8'h15)^
({8{data_in[198][6]}} & 8'h2a)^
({8{data_in[198][7]}} & 8'h54)^
({8{data_in[199][0]}} & 8'hac)^
({8{data_in[199][1]}} & 8'h73)^
({8{data_in[199][2]}} & 8'he6)^
({8{data_in[199][3]}} & 8'he7)^
({8{data_in[199][4]}} & 8'he5)^
({8{data_in[199][5]}} & 8'he1)^
({8{data_in[199][6]}} & 8'he9)^
({8{data_in[199][7]}} & 8'hf9)^
({8{data_in[200][0]}} & 8'h25)^
({8{data_in[200][1]}} & 8'h4a)^
({8{data_in[200][2]}} & 8'h94)^
({8{data_in[200][3]}} & 8'h3)^
({8{data_in[200][4]}} & 8'h6)^
({8{data_in[200][5]}} & 8'hc)^
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({8{data_in[200][7]}} & 8'h30)^
({8{data_in[201][0]}} & 8'he2)^
({8{data_in[201][1]}} & 8'hcf)^
({8{data_in[201][2]}} & 8'hf5)^
({8{data_in[201][3]}} & 8'hc1)^
({8{data_in[201][4]}} & 8'ha9)^
({8{data_in[201][5]}} & 8'h79)^
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({8{data_in[201][7]}} & 8'hcf)^
({8{data_in[202][0]}} & 8'h5c)^
({8{data_in[202][1]}} & 8'hb8)^
({8{data_in[202][2]}} & 8'h5b)^
({8{data_in[202][3]}} & 8'hb6)^
({8{data_in[202][4]}} & 8'h47)^
({8{data_in[202][5]}} & 8'h8e)^
({8{data_in[202][6]}} & 8'h37)^
({8{data_in[202][7]}} & 8'h6e)^
({8{data_in[203][0]}} & 8'ha1)^
({8{data_in[203][1]}} & 8'h69)^
({8{data_in[203][2]}} & 8'hd2)^
({8{data_in[203][3]}} & 8'h8f)^

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({8{data_in[203][4]}} & 8'h35)^
({8{data_in[203][5]}} & 8'h6a)^
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({8{data_in[203][7]}} & 8'h83)^
({8{data_in[204][0]}} & 8'h50)^
({8{data_in[204][1]}} & 8'ha0)^
({8{data_in[204][2]}} & 8'h6b)^
({8{data_in[204][3]}} & 8'hd6)^
({8{data_in[204][4]}} & 8'h87)^
({8{data_in[204][5]}} & 8'h25)^
({8{data_in[204][6]}} & 8'h4a)^
({8{data_in[204][7]}} & 8'h94)^
({8{data_in[205][0]}} & 8'hba)^
({8{data_in[205][1]}} & 8'h5f)^
({8{data_in[205][2]}} & 8'hbe)^
({8{data_in[205][3]}} & 8'h57)^
({8{data_in[205][4]}} & 8'hae)^
({8{data_in[205][5]}} & 8'h77)^
({8{data_in[205][6]}} & 8'hee)^
({8{data_in[205][7]}} & 8'hf7)^
({8{data_in[206][0]}} & 8'h97)^
({8{data_in[206][1]}} & 8'h5)^
({8{data_in[206][2]}} & 8'ha)^
({8{data_in[206][3]}} & 8'h14)^
({8{data_in[206][4]}} & 8'h28)^
({8{data_in[206][5]}} & 8'h50)^
({8{data_in[206][6]}} & 8'ha0)^
({8{data_in[206][7]}} & 8'h6b)^
({8{data_in[207][0]}} & 8'h4f)^
({8{data_in[207][1]}} & 8'h9e)^
({8{data_in[207][2]}} & 8'h17)^
({8{data_in[207][3]}} & 8'h2e)^
({8{data_in[207][4]}} & 8'h5c)^
({8{data_in[207][5]}} & 8'hb8)^
({8{data_in[207][6]}} & 8'h5b)^
({8{data_in[207][7]}} & 8'hb6)^
({8{data_in[208][0]}} & 8'h13)^
({8{data_in[208][1]}} & 8'h26)^
({8{data_in[208][2]}} & 8'h4c)^
({8{data_in[208][3]}} & 8'h98)^
({8{data_in[208][4]}} & 8'h1b)^
({8{data_in[208][5]}} & 8'h36)^
({8{data_in[208][6]}} & 8'h6c)^
({8{data_in[208][7]}} & 8'hd8)^
({8{data_in[209][0]}} & 8'hf6)^
({8{data_in[209][1]}} & 8'hc7)^
({8{data_in[209][2]}} & 8'ha5)^
({8{data_in[209][3]}} & 8'h61)^
({8{data_in[209][4]}} & 8'hc2)^
({8{data_in[209][5]}} & 8'haf)^
({8{data_in[209][6]}} & 8'h75)^
({8{data_in[209][7]}} & 8'hea)^
({8{data_in[210][0]}} & 8'hd8)^
({8{data_in[210][1]}} & 8'h9b)^
({8{data_in[210][2]}} & 8'h1d)^
({8{data_in[210][3]}} & 8'h3a)^
({8{data_in[210][4]}} & 8'h74)^
({8{data_in[210][5]}} & 8'he8)^
({8{data_in[210][6]}} & 8'hfb)^
({8{data_in[210][7]}} & 8'hdd)^
({8{data_in[211][0]}} & 8'hd)^
({8{data_in[211][1]}} & 8'h1a)^
({8{data_in[211][2]}} & 8'h34)^
({8{data_in[211][3]}} & 8'h68)^
({8{data_in[211][4]}} & 8'hd0)^

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({8{data_in[211][5]}} & 8'h8b)^
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({8{data_in[211][7]}} & 8'h7a)^
({8{data_in[212][0]}} & 8'hdb)^
({8{data_in[212][1]}} & 8'h9d)^
({8{data_in[212][2]}} & 8'h11)^
({8{data_in[212][3]}} & 8'h22)^
({8{data_in[212][4]}} & 8'h44)^
({8{data_in[212][5]}} & 8'h88)^
({8{data_in[212][6]}} & 8'h3b)^
({8{data_in[212][7]}} & 8'h76)^
({8{data_in[213][0]}} & 8'h4d)^
({8{data_in[213][1]}} & 8'h9a)^
({8{data_in[213][2]}} & 8'h1f)^
({8{data_in[213][3]}} & 8'h3e)^
({8{data_in[213][4]}} & 8'h7c)^
({8{data_in[213][5]}} & 8'hf8)^
({8{data_in[213][6]}} & 8'hdb)^
({8{data_in[213][7]}} & 8'h9d)^
({8{data_in[214][0]}} & 8'he3)^
({8{data_in[214][1]}} & 8'hed)^
({8{data_in[214][2]}} & 8'hf1)^
({8{data_in[214][3]}} & 8'hc9)^
({8{data_in[214][4]}} & 8'hb9)^
({8{data_in[214][5]}} & 8'h59)^
({8{data_in[214][6]}} & 8'hb2)^
({8{data_in[214][7]}} & 8'h4f)^
({8{data_in[215][0]}} & 8'h6b)^
({8{data_in[215][1]}} & 8'hd6)^
({8{data_in[215][2]}} & 8'h87)^
({8{data_in[215][3]}} & 8'h25)^
({8{data_in[215][4]}} & 8'h4a)^
({8{data_in[215][5]}} & 8'h94)^
({8{data_in[215][6]}} & 8'h3)^
({8{data_in[215][7]}} & 8'h6)^
({8{data_in[216][0]}} & 8'h45)^
({8{data_in[216][1]}} & 8'h8a)^
({8{data_in[216][2]}} & 8'h3f)^
({8{data_in[216][3]}} & 8'h7e)^
({8{data_in[216][4]}} & 8'hfc)^
({8{data_in[216][5]}} & 8'hd3)^
({8{data_in[216][6]}} & 8'h8d)^
({8{data_in[216][7]}} & 8'h31)^
({8{data_in[217][0]}} & 8'hfa)^
({8{data_in[217][1]}} & 8'hdf)^
({8{data_in[217][2]}} & 8'h95)^
({8{data_in[217][3]}} & 8'h1)^
({8{data_in[217][4]}} & 8'h2)^
({8{data_in[217][5]}} & 8'h4)^
({8{data_in[217][6]}} & 8'h8)^
({8{data_in[217][7]}} & 8'h10)^
({8{data_in[218][0]}} & 8'h33)^
({8{data_in[218][1]}} & 8'h66)^
({8{data_in[218][2]}} & 8'hcc)^
({8{data_in[218][3]}} & 8'hb3)^
({8{data_in[218][4]}} & 8'h4d)^
({8{data_in[218][5]}} & 8'h9a)^
({8{data_in[218][6]}} & 8'h1f)^
({8{data_in[218][7]}} & 8'h3e)^
({8{data_in[219][0]}} & 8'ha0)^
({8{data_in[219][1]}} & 8'h6b)^
({8{data_in[219][2]}} & 8'hd6)^
({8{data_in[219][3]}} & 8'h87)^
({8{data_in[219][4]}} & 8'h25)^
({8{data_in[219][5]}} & 8'h4a)^

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({8{data_in[219][6]}} & 8'h94)^
({8{data_in[219][7]}} & 8'h3)^
({8{data_in[220][0]}} & 8'h2c)^
({8{data_in[220][1]}} & 8'h58)^
({8{data_in[220][2]}} & 8'hb0)^
({8{data_in[220][3]}} & 8'h4b)^
({8{data_in[220][4]}} & 8'h96)^
({8{data_in[220][5]}} & 8'h7)^
({8{data_in[220][6]}} & 8'he)^
({8{data_in[220][7]}} & 8'h1c)^
({8{data_in[221][0]}} & 8'h4e)^
({8{data_in[221][1]}} & 8'h9c)^
({8{data_in[221][2]}} & 8'h13)^
({8{data_in[221][3]}} & 8'h26)^
({8{data_in[221][4]}} & 8'h4c)^
({8{data_in[221][5]}} & 8'h98)^
({8{data_in[221][6]}} & 8'h1b)^
({8{data_in[221][7]}} & 8'h36)^
({8{data_in[222][0]}} & 8'hbd)^
({8{data_in[222][1]}} & 8'h51)^
({8{data_in[222][2]}} & 8'ha2)^
({8{data_in[222][3]}} & 8'h6f)^
({8{data_in[222][4]}} & 8'hde)^
({8{data_in[222][5]}} & 8'h97)^
({8{data_in[222][6]}} & 8'h5)^
({8{data_in[222][7]}} & 8'ha)^
({8{data_in[223][0]}} & 8'h74)^
({8{data_in[223][1]}} & 8'he8)^
({8{data_in[223][2]}} & 8'hfb)^
({8{data_in[223][3]}} & 8'hdd)^
({8{data_in[223][4]}} & 8'h91)^
({8{data_in[223][5]}} & 8'h9)^
({8{data_in[223][6]}} & 8'h12)^
({8{data_in[223][7]}} & 8'h24)^
({8{data_in[224][0]}} & 8'hd5)^
({8{data_in[224][1]}} & 8'h81)^
({8{data_in[224][2]}} & 8'h29)^
({8{data_in[224][3]}} & 8'h52)^
({8{data_in[224][4]}} & 8'ha4)^
({8{data_in[224][5]}} & 8'h63)^
({8{data_in[224][6]}} & 8'hc6)^
({8{data_in[224][7]}} & 8'ha7)^
({8{data_in[225][0]}} & 8'h7c)^
({8{data_in[225][1]}} & 8'hf8)^
({8{data_in[225][2]}} & 8'hdb)^
({8{data_in[225][3]}} & 8'h9d)^
({8{data_in[225][4]}} & 8'h11)^
({8{data_in[225][5]}} & 8'h22)^
({8{data_in[225][6]}} & 8'h44)^
({8{data_in[225][7]}} & 8'h88)^
({8{data_in[226][0]}} & 8'ha3)^
({8{data_in[226][1]}} & 8'h6d)^
({8{data_in[226][2]}} & 8'hda)^
({8{data_in[226][3]}} & 8'h9f)^
({8{data_in[226][4]}} & 8'h15)^
({8{data_in[226][5]}} & 8'h2a)^
({8{data_in[226][6]}} & 8'h54)^
({8{data_in[226][7]}} & 8'ha8)^
({8{data_in[227][0]}} & 8'hf5)^
({8{data_in[227][1]}} & 8'hc1)^
({8{data_in[227][2]}} & 8'ha9)^
({8{data_in[227][3]}} & 8'h79)^
({8{data_in[227][4]}} & 8'hf2)^
({8{data_in[227][5]}} & 8'hcf)^
({8{data_in[227][6]}} & 8'hb5)^

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({8{data_in[227][7]}} & 8'h41)^
({8{data_in[228][0]}} & 8'h4a)^
({8{data_in[228][1]}} & 8'h94)^
({8{data_in[228][2]}} & 8'h3)^
({8{data_in[228][3]}} & 8'h6)^
({8{data_in[228][4]}} & 8'hc)^
({8{data_in[228][5]}} & 8'h18)^
({8{data_in[228][6]}} & 8'h30)^
({8{data_in[228][7]}} & 8'h60)^
({8{data_in[229][0]}} & 8'hc0)^
({8{data_in[229][1]}} & 8'hab)^
({8{data_in[229][2]}} & 8'h7d)^
({8{data_in[229][3]}} & 8'hfa)^
({8{data_in[229][4]}} & 8'hdf)^
({8{data_in[229][5]}} & 8'h95)^
({8{data_in[229][6]}} & 8'h1)^
({8{data_in[229][7]}} & 8'h2)^
({8{data_in[230][0]}} & 8'h73)^
({8{data_in[230][1]}} & 8'he6)^
({8{data_in[230][2]}} & 8'he7)^
({8{data_in[230][3]}} & 8'he5)^
({8{data_in[230][4]}} & 8'he1)^
({8{data_in[230][5]}} & 8'he9)^
({8{data_in[230][6]}} & 8'hf9)^
({8{data_in[230][7]}} & 8'hd9)^
({8{data_in[231][0]}} & 8'h6d)^
({8{data_in[231][1]}} & 8'hda)^
({8{data_in[231][2]}} & 8'h9f)^
({8{data_in[231][3]}} & 8'h15)^
({8{data_in[231][4]}} & 8'h2a)^
({8{data_in[231][5]}} & 8'h54)^
({8{data_in[231][6]}} & 8'ha8)^
({8{data_in[231][7]}} & 8'h7b)^
({8{data_in[232][0]}} & 8'h5b)^
({8{data_in[232][1]}} & 8'hb6)^
({8{data_in[232][2]}} & 8'h47)^
({8{data_in[232][3]}} & 8'h8e)^
({8{data_in[232][4]}} & 8'h37)^
({8{data_in[232][5]}} & 8'h6e)^
({8{data_in[232][6]}} & 8'hdc)^
({8{data_in[232][7]}} & 8'h93)^
({8{data_in[233][0]}} & 8'hf6)^
({8{data_in[233][1]}} & 8'hc7)^
({8{data_in[233][2]}} & 8'ha5)^
({8{data_in[233][3]}} & 8'h61)^
({8{data_in[233][4]}} & 8'hc2)^
({8{data_in[233][5]}} & 8'haf)^
({8{data_in[233][6]}} & 8'h75)^
({8{data_in[233][7]}} & 8'hea)^
({8{data_in[234][0]}} & 8'h7d)^
({8{data_in[234][1]}} & 8'hfa)^
({8{data_in[234][2]}} & 8'hdf)^
({8{data_in[234][3]}} & 8'h95)^
({8{data_in[234][4]}} & 8'h1)^
({8{data_in[234][5]}} & 8'h2)^
({8{data_in[234][6]}} & 8'h4)^
({8{data_in[234][7]}} & 8'h8)^
({8{data_in[235][0]}} & 8'hf6)^
({8{data_in[235][1]}} & 8'hc7)^
({8{data_in[235][2]}} & 8'ha5)^
({8{data_in[235][3]}} & 8'h61)^
({8{data_in[235][4]}} & 8'hc2)^
({8{data_in[235][5]}} & 8'haf)^
({8{data_in[235][6]}} & 8'h75)^
({8{data_in[235][7]}} & 8'hea)^

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({8{data_in[236][0]}} & 8'hf8)^
({8{data_in[236][1]}} & 8'hdb)^
({8{data_in[236][2]}} & 8'h9d)^
({8{data_in[236][3]}} & 8'h11)^
({8{data_in[236][4]}} & 8'h22)^
({8{data_in[236][5]}} & 8'h44)^
({8{data_in[236][6]}} & 8'h88)^
({8{data_in[236][7]}} & 8'h3b)^
({8{data_in[237][0]}} & 8'h27)^
({8{data_in[237][1]}} & 8'h4e)^
({8{data_in[237][2]}} & 8'h9c)^
({8{data_in[237][3]}} & 8'h13)^
({8{data_in[237][4]}} & 8'h26)^
({8{data_in[237][5]}} & 8'h4c)^
({8{data_in[237][6]}} & 8'h98)^
({8{data_in[237][7]}} & 8'h1b)^
({8{data_in[238][0]}} & 8'h22)^
({8{data_in[238][1]}} & 8'h44)^
({8{data_in[238][2]}} & 8'h88)^
({8{data_in[238][3]}} & 8'h3b)^
({8{data_in[238][4]}} & 8'h76)^
({8{data_in[238][5]}} & 8'hec)^
({8{data_in[238][6]}} & 8'hf3)^
({8{data_in[238][7]}} & 8'hcd)^
({8{data_in[239][0]}} & 8'hc8)^
({8{data_in[239][1]}} & 8'hbb)^
({8{data_in[239][2]}} & 8'h5d)^
({8{data_in[239][3]}} & 8'hba)^
({8{data_in[239][4]}} & 8'h5f)^
({8{data_in[239][5]}} & 8'hbe)^
({8{data_in[239][6]}} & 8'h57)^
({8{data_in[239][7]}} & 8'hae)^
({8{data_in[240][0]}} & 8'h1)^
({8{data_in[240][1]}} & 8'h2)^
({8{data_in[240][2]}} & 8'h4)^
({8{data_in[240][3]}} & 8'h8)^
({8{data_in[240][4]}} & 8'h10)^
({8{data_in[240][5]}} & 8'h20)^
({8{data_in[240][6]}} & 8'h40)^
({8{data_in[240][7]}} & 8'h80)^
({8{data_in[241][0]}} & 8'h4d)^
({8{data_in[241][1]}} & 8'h9a)^
({8{data_in[241][2]}} & 8'h1f)^
({8{data_in[241][3]}} & 8'h3e)^
({8{data_in[241][4]}} & 8'h7c)^
({8{data_in[241][5]}} & 8'hf8)^
({8{data_in[241][6]}} & 8'hdb)^
({8{data_in[241][7]}} & 8'h9d);
data_out[242] = ({8{data_in[0][0]}} & 8'hb)^
({8{data_in[0][1]}} & 8'h16)^
({8{data_in[0][2]}} & 8'h2c)^
({8{data_in[0][3]}} & 8'h58)^
({8{data_in[0][4]}} & 8'hb0)^
({8{data_in[0][5]}} & 8'h4b)^
({8{data_in[0][6]}} & 8'h96)^
({8{data_in[0][7]}} & 8'h7)^
({8{data_in[1][0]}} & 8'h6b)^
({8{data_in[1][1]}} & 8'hd6)^
({8{data_in[1][2]}} & 8'h87)^
({8{data_in[1][3]}} & 8'h25)^
({8{data_in[1][4]}} & 8'h4a)^
({8{data_in[1][5]}} & 8'h94)^
({8{data_in[1][6]}} & 8'h3)^
({8{data_in[1][7]}} & 8'h6)^
({8{data_in[2][0]}} & 8'ha3)^

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({8{data_in[2][1]}} & 8'h6d)^
({8{data_in[2][2]}} & 8'hda)^
({8{data_in[2][3]}} & 8'h9f)^
({8{data_in[2][4]}} & 8'h15)^
({8{data_in[2][5]}} & 8'h2a)^
({8{data_in[2][6]}} & 8'h54)^
({8{data_in[2][7]}} & 8'ha8)^
({8{data_in[3][0]}} & 8'h65)^
({8{data_in[3][1]}} & 8'hca)^
({8{data_in[3][2]}} & 8'hbf)^
({8{data_in[3][3]}} & 8'h55)^
({8{data_in[3][4]}} & 8'haa)^
({8{data_in[3][5]}} & 8'h7f)^
({8{data_in[3][6]}} & 8'hfe)^
({8{data_in[3][7]}} & 8'hd7)^
({8{data_in[4][0]}} & 8'hdf)^
({8{data_in[4][1]}} & 8'h95)^
({8{data_in[4][2]}} & 8'h1)^
({8{data_in[4][3]}} & 8'h2)^
({8{data_in[4][4]}} & 8'h4)^
({8{data_in[4][5]}} & 8'h8)^
({8{data_in[4][6]}} & 8'h10)^
({8{data_in[4][7]}} & 8'h20)^
({8{data_in[5][0]}} & 8'hec)^
({8{data_in[5][1]}} & 8'hf3)^
({8{data_in[5][2]}} & 8'hcd)^
({8{data_in[5][3]}} & 8'hb1)^
({8{data_in[5][4]}} & 8'h49)^
({8{data_in[5][5]}} & 8'h92)^
({8{data_in[5][6]}} & 8'hf)^
({8{data_in[5][7]}} & 8'h1e)^
({8{data_in[6][0]}} & 8'h18)^
({8{data_in[6][1]}} & 8'h30)^
({8{data_in[6][2]}} & 8'h60)^
({8{data_in[6][3]}} & 8'hc0)^
({8{data_in[6][4]}} & 8'hab)^
({8{data_in[6][5]}} & 8'h7d)^
({8{data_in[6][6]}} & 8'hfa)^
({8{data_in[6][7]}} & 8'hdf)^
({8{data_in[7][0]}} & 8'h49)^
({8{data_in[7][1]}} & 8'h92)^
({8{data_in[7][2]}} & 8'hf)^
({8{data_in[7][3]}} & 8'h1e)^
({8{data_in[7][4]}} & 8'h3c)^
({8{data_in[7][5]}} & 8'h78)^
({8{data_in[7][6]}} & 8'hf0)^
({8{data_in[7][7]}} & 8'hcb)^
({8{data_in[8][0]}} & 8'hd8)^
({8{data_in[8][1]}} & 8'h9b)^
({8{data_in[8][2]}} & 8'h1d)^
({8{data_in[8][3]}} & 8'h3a)^
({8{data_in[8][4]}} & 8'h74)^
({8{data_in[8][5]}} & 8'he8)^
({8{data_in[8][6]}} & 8'hfb)^
({8{data_in[8][7]}} & 8'hdd)^
({8{data_in[9][0]}} & 8'h3f)^
({8{data_in[9][1]}} & 8'h7e)^
({8{data_in[9][2]}} & 8'hfc)^
({8{data_in[9][3]}} & 8'hd3)^
({8{data_in[9][4]}} & 8'h8d)^
({8{data_in[9][5]}} & 8'h31)^
({8{data_in[9][6]}} & 8'h62)^
({8{data_in[9][7]}} & 8'hc4)^
({8{data_in[10][0]}} & 8'hea)^
({8{data_in[10][1]}} & 8'hff)^

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({8{data_in[10][2]}} & 8'hd5)^
({8{data_in[10][3]}} & 8'h81)^
({8{data_in[10][4]}} & 8'h29)^
({8{data_in[10][5]}} & 8'h52)^
({8{data_in[10][6]}} & 8'ha4)^
({8{data_in[10][7]}} & 8'h63)^
({8{data_in[11][0]}} & 8'h87)^
({8{data_in[11][1]}} & 8'h25)^
({8{data_in[11][2]}} & 8'h4a)^
({8{data_in[11][3]}} & 8'h94)^
({8{data_in[11][4]}} & 8'h3)^
({8{data_in[11][5]}} & 8'h6)^
({8{data_in[11][6]}} & 8'hc)^
({8{data_in[11][7]}} & 8'h18)^
({8{data_in[12][0]}} & 8'h32)^
({8{data_in[12][1]}} & 8'h64)^
({8{data_in[12][2]}} & 8'hc8)^
({8{data_in[12][3]}} & 8'hbb)^
({8{data_in[12][4]}} & 8'h5d)^
({8{data_in[12][5]}} & 8'hba)^
({8{data_in[12][6]}} & 8'h5f)^
({8{data_in[12][7]}} & 8'hbe)^
({8{data_in[13][0]}} & 8'hd6)^
({8{data_in[13][1]}} & 8'h87)^
({8{data_in[13][2]}} & 8'h25)^
({8{data_in[13][3]}} & 8'h4a)^
({8{data_in[13][4]}} & 8'h94)^
({8{data_in[13][5]}} & 8'h3)^
({8{data_in[13][6]}} & 8'h6)^
({8{data_in[13][7]}} & 8'hc)^
({8{data_in[14][0]}} & 8'hf4)^
({8{data_in[14][1]}} & 8'hc3)^
({8{data_in[14][2]}} & 8'had)^
({8{data_in[14][3]}} & 8'h71)^
({8{data_in[14][4]}} & 8'he2)^
({8{data_in[14][5]}} & 8'hef)^
({8{data_in[14][6]}} & 8'hf5)^
({8{data_in[14][7]}} & 8'hc1)^
({8{data_in[15][0]}} & 8'h9e)^
({8{data_in[15][1]}} & 8'h17)^
({8{data_in[15][2]}} & 8'h2e)^
({8{data_in[15][3]}} & 8'h5c)^
({8{data_in[15][4]}} & 8'hb8)^
({8{data_in[15][5]}} & 8'h5b)^
({8{data_in[15][6]}} & 8'hb6)^
({8{data_in[15][7]}} & 8'h47)^
({8{data_in[16][0]}} & 8'ha4)^
({8{data_in[16][1]}} & 8'h63)^
({8{data_in[16][2]}} & 8'hc6)^
({8{data_in[16][3]}} & 8'ha7)^
({8{data_in[16][4]}} & 8'h65)^
({8{data_in[16][5]}} & 8'hca)^
({8{data_in[16][6]}} & 8'hbf)^
({8{data_in[16][7]}} & 8'h55)^
({8{data_in[17][0]}} & 8'h4a)^
({8{data_in[17][1]}} & 8'h94)^
({8{data_in[17][2]}} & 8'h3)^
({8{data_in[17][3]}} & 8'h6)^
({8{data_in[17][4]}} & 8'hc)^
({8{data_in[17][5]}} & 8'h18)^
({8{data_in[17][6]}} & 8'h30)^
({8{data_in[17][7]}} & 8'h60)^
({8{data_in[18][0]}} & 8'hc7)^
({8{data_in[18][1]}} & 8'ha5)^
({8{data_in[18][2]}} & 8'h61)^

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```

({8{data_in[18][3]}} & 8'hc2)^
({8{data_in[18][4]}} & 8'haf)^
({8{data_in[18][5]}} & 8'h75)^
({8{data_in[18][6]}} & 8'hea)^
({8{data_in[18][7]}} & 8'hff)^
({8{data_in[19][0]}} & 8'ha7)^
({8{data_in[19][1]}} & 8'h65)^
({8{data_in[19][2]}} & 8'hca)^
({8{data_in[19][3]}} & 8'hbf)^
({8{data_in[19][4]}} & 8'h55)^
({8{data_in[19][5]}} & 8'haa)^
({8{data_in[19][6]}} & 8'h7f)^
({8{data_in[19][7]}} & 8'hfe)^
({8{data_in[20][0]}} & 8'h1b)^
({8{data_in[20][1]}} & 8'h36)^
({8{data_in[20][2]}} & 8'h6c)^
({8{data_in[20][3]}} & 8'hd8)^
({8{data_in[20][4]}} & 8'h9b)^
({8{data_in[20][5]}} & 8'h1d)^
({8{data_in[20][6]}} & 8'h3a)^
({8{data_in[20][7]}} & 8'h74)^
({8{data_in[21][0]}} & 8'he8)^
({8{data_in[21][1]}} & 8'hfb)^
({8{data_in[21][2]}} & 8'hdd)^
({8{data_in[21][3]}} & 8'h91)^
({8{data_in[21][4]}} & 8'h9)^
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({8{data_in[22][3]}} & 8'h93)^
({8{data_in[22][4]}} & 8'hd)^
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({8{data_in[23][7]}} & 8'hcd)^
({8{data_in[24][0]}} & 8'hca)^
({8{data_in[24][1]}} & 8'hbf)^
({8{data_in[24][2]}} & 8'h55)^
({8{data_in[24][3]}} & 8'haa)^
({8{data_in[24][4]}} & 8'h7f)^
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({8{data_in[24][6]}} & 8'hd7)^
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({8{data_in[25][2]}} & 8'h90)^
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({8{data_in[27][3]}} & 8'h3d)^
({8{data_in[27][4]}} & 8'h7a)^
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({8{data_in[33][1]}} & 8'h2)^
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({8{data_in[49][3]}} & 8'h5f)^
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({8{data_in[53][5]}} & 8'h1e)^
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({8{data_in[54][1]}} & 8'h79)^
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({8{data_in[55][5]}} & 8'h3c)^
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({8{data_in[57][4]}} & 8'h3c)^
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({8{data_in[59][7]}} & 8'hb7)^
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({8{data_in[63][6]}} & 8'h3e)^
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({8{data_in[72][3]}} & 8'hb0)^
({8{data_in[72][4]}} & 8'h4b)^
({8{data_in[72][5]}} & 8'h96)^
({8{data_in[72][6]}} & 8'h7)^
({8{data_in[72][7]}} & 8'he)^
({8{data_in[73][0]}} & 8'hcb)^
({8{data_in[73][1]}} & 8'hbd)^
({8{data_in[73][2]}} & 8'h51)^
({8{data_in[73][3]}} & 8'ha2)^
({8{data_in[73][4]}} & 8'h6f)^
({8{data_in[73][5]}} & 8'hde)^
({8{data_in[73][6]}} & 8'h97)^
({8{data_in[73][7]}} & 8'h5)^
({8{data_in[74][0]}} & 8'h1a)^
({8{data_in[74][1]}} & 8'h34)^
({8{data_in[74][2]}} & 8'h68)^
({8{data_in[74][3]}} & 8'hd0)^
({8{data_in[74][4]}} & 8'h8b)^
({8{data_in[74][5]}} & 8'h3d)^
({8{data_in[74][6]}} & 8'h7a)^
({8{data_in[74][7]}} & 8'hf4)^
({8{data_in[75][0]}} & 8'h2)^
({8{data_in[75][1]}} & 8'h4)^

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({8{data_in[75][7]}} & 8'h2b)^
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({8{data_in[76][2]}} & 8'h3)^
({8{data_in[76][3]}} & 8'h6)^
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({8{data_in[78][3]}} & 8'h1a)^
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({8{data_in[79][1]}} & 8'h9b)^
({8{data_in[79][2]}} & 8'h1d)^
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({8{data_in[79][4]}} & 8'h74)^
({8{data_in[79][5]}} & 8'he8)^
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({8{data_in[80][4]}} & 8'h3d)^
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({8{data_in[80][6]}} & 8'hf4)^
({8{data_in[80][7]}} & 8'hc3)^
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({8{data_in[81][1]}} & 8'hf3)^
({8{data_in[81][2]}} & 8'hcd)^
({8{data_in[81][3]}} & 8'hb1)^
({8{data_in[81][4]}} & 8'h49)^
({8{data_in[81][5]}} & 8'h92)^
({8{data_in[81][6]}} & 8'hf)^
({8{data_in[81][7]}} & 8'h1e)^
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({8{data_in[82][4]}} & 8'h64)^
({8{data_in[82][5]}} & 8'hc8)^
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({8{data_in[82][7]}} & 8'h5d)^
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({8{data_in[83][1]}} & 8'hf7)^
({8{data_in[83][2]}} & 8'hc5)^

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({8{data_in[84][4]}} & 8'he3)^
({8{data_in[84][5]}} & 8'hed)^
({8{data_in[84][6]}} & 8'hf1)^
({8{data_in[84][7]}} & 8'hc9)^
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({8{data_in[85][7]}} & 8'he9)^
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({8{data_in[87][3]}} & 8'h3d)^
({8{data_in[87][4]}} & 8'h7a)^
({8{data_in[87][5]}} & 8'hf4)^
({8{data_in[87][6]}} & 8'hc3)^
({8{data_in[87][7]}} & 8'had)^
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({8{data_in[88][5]}} & 8'h92)^
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({8{data_in[88][7]}} & 8'h1e)^
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({8{data_in[89][1]}} & 8'hfb)^
({8{data_in[89][2]}} & 8'hdd)^
({8{data_in[89][3]}} & 8'h91)^
({8{data_in[89][4]}} & 8'h9)^
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({8{data_in[89][7]}} & 8'h48)^
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({8{data_in[90][7]}} & 8'hd6)^
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({8{data_in[92][4]}} & 8'hfd)^
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({8{data_in[92][6]}} & 8'h89)^
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({8{data_in[94][4]}} & 8'h9a)^
({8{data_in[94][5]}} & 8'h1f)^
({8{data_in[94][6]}} & 8'h3e)^
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({8{data_in[98][2]}} & 8'hf0)^
({8{data_in[98][3]}} & 8'hcb)^
({8{data_in[98][4]}} & 8'hbd)^
({8{data_in[98][5]}} & 8'h51)^
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({8{data_in[100][5]}} & 8'h5)^
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({8{data_in[105][1]}} & 8'hcb)^
({8{data_in[105][2]}} & 8'hbd)^
({8{data_in[105][3]}} & 8'h51)^
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({8{data_in[105][7]}} & 8'h97)^
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({8{data_in[108][7]}} & 8'hb3)^
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({8{data_in[109][3]}} & 8'h9b)^
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({8{data_in[114][2]}} & 8'hcd)^
({8{data_in[114][3]}} & 8'hb1)^
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({8{data_in[117][3]}} & 8'hac)^
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({8{data_in[118][6]}} & 8'h2f)^
({8{data_in[118][7]}} & 8'h5e)^
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({8{data_in[121][6]}} & 8'h1d)^
({8{data_in[121][7]}} & 8'h3a)^
({8{data_in[122][0]}} & 8'hf1)^
({8{data_in[122][1]}} & 8'hc9)^
({8{data_in[122][2]}} & 8'hb9)^
({8{data_in[122][3]}} & 8'h59)^
({8{data_in[122][4]}} & 8'hb2)^
({8{data_in[122][5]}} & 8'h4f)^
({8{data_in[122][6]}} & 8'h9e)^
({8{data_in[122][7]}} & 8'h17)^
({8{data_in[123][0]}} & 8'h5f)^
({8{data_in[123][1]}} & 8'hbe)^
({8{data_in[123][2]}} & 8'h57)^
({8{data_in[123][3]}} & 8'hae)^
({8{data_in[123][4]}} & 8'h77)^
({8{data_in[123][5]}} & 8'hee)^
({8{data_in[123][6]}} & 8'hf7)^
({8{data_in[123][7]}} & 8'hc5)^

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({8{data_in[124][0]}} & 8'h7)^
({8{data_in[124][1]}} & 8'he)^
({8{data_in[124][2]}} & 8'h1c)^
({8{data_in[124][3]}} & 8'h38)^
({8{data_in[124][4]}} & 8'h70)^
({8{data_in[124][5]}} & 8'he0)^
({8{data_in[124][6]}} & 8'heb)^
({8{data_in[124][7]}} & 8'hfd)^
({8{data_in[125][0]}} & 8'h38)^
({8{data_in[125][1]}} & 8'h70)^
({8{data_in[125][2]}} & 8'he0)^
({8{data_in[125][3]}} & 8'heb)^
({8{data_in[125][4]}} & 8'hfd)^
({8{data_in[125][5]}} & 8'hd1)^
({8{data_in[125][6]}} & 8'h89)^
({8{data_in[125][7]}} & 8'h39)^
({8{data_in[126][0]}} & 8'h47)^
({8{data_in[126][1]}} & 8'h8e)^
({8{data_in[126][2]}} & 8'h37)^
({8{data_in[126][3]}} & 8'h6e)^
({8{data_in[126][4]}} & 8'hdc)^
({8{data_in[126][5]}} & 8'h93)^
({8{data_in[126][6]}} & 8'hd)^
({8{data_in[126][7]}} & 8'h1a)^
({8{data_in[127][0]}} & 8'h14)^
({8{data_in[127][1]}} & 8'h28)^
({8{data_in[127][2]}} & 8'h50)^
({8{data_in[127][3]}} & 8'ha0)^
({8{data_in[127][4]}} & 8'h6b)^
({8{data_in[127][5]}} & 8'hd6)^
({8{data_in[127][6]}} & 8'h87)^
({8{data_in[127][7]}} & 8'h25)^
({8{data_in[128][0]}} & 8'h54)^
({8{data_in[128][1]}} & 8'ha8)^
({8{data_in[128][2]}} & 8'h7b)^
({8{data_in[128][3]}} & 8'hf6)^
({8{data_in[128][4]}} & 8'hc7)^
({8{data_in[128][5]}} & 8'ha5)^
({8{data_in[128][6]}} & 8'h61)^
({8{data_in[128][7]}} & 8'hc2)^
({8{data_in[129][0]}} & 8'h94)^
({8{data_in[129][1]}} & 8'h3)^
({8{data_in[129][2]}} & 8'h6)^
({8{data_in[129][3]}} & 8'hc)^
({8{data_in[129][4]}} & 8'h18)^
({8{data_in[129][5]}} & 8'h30)^
({8{data_in[129][6]}} & 8'h60)^
({8{data_in[129][7]}} & 8'hc0)^
({8{data_in[130][0]}} & 8'hd6)^
({8{data_in[130][1]}} & 8'h87)^
({8{data_in[130][2]}} & 8'h25)^
({8{data_in[130][3]}} & 8'h4a)^
({8{data_in[130][4]}} & 8'h94)^
({8{data_in[130][5]}} & 8'h3)^
({8{data_in[130][6]}} & 8'h6)^
({8{data_in[130][7]}} & 8'hc)^
({8{data_in[131][0]}} & 8'h9f)^
({8{data_in[131][1]}} & 8'h15)^
({8{data_in[131][2]}} & 8'h2a)^
({8{data_in[131][3]}} & 8'h54)^
({8{data_in[131][4]}} & 8'ha8)^
({8{data_in[131][5]}} & 8'h7b)^
({8{data_in[131][6]}} & 8'hf6)^
({8{data_in[131][7]}} & 8'hc7)^
({8{data_in[132][0]}} & 8'h83)^

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({8{data_in[132][1]}} & 8'h2d)^
({8{data_in[132][2]}} & 8'h5a)^
({8{data_in[132][3]}} & 8'hb4)^
({8{data_in[132][4]}} & 8'h43)^
({8{data_in[132][5]}} & 8'h86)^
({8{data_in[132][6]}} & 8'h27)^
({8{data_in[132][7]}} & 8'h4e)^
({8{data_in[133][0]}} & 8'h3a)^
({8{data_in[133][1]}} & 8'h74)^
({8{data_in[133][2]}} & 8'he8)^
({8{data_in[133][3]}} & 8'hfb)^
({8{data_in[133][4]}} & 8'hdd)^
({8{data_in[133][5]}} & 8'h91)^
({8{data_in[133][6]}} & 8'h9)^
({8{data_in[133][7]}} & 8'h12)^
({8{data_in[134][0]}} & 8'h9b)^
({8{data_in[134][1]}} & 8'h1d)^
({8{data_in[134][2]}} & 8'h3a)^
({8{data_in[134][3]}} & 8'h74)^
({8{data_in[134][4]}} & 8'he8)^
({8{data_in[134][5]}} & 8'hfb)^
({8{data_in[134][6]}} & 8'hdd)^
({8{data_in[134][7]}} & 8'h91)^
({8{data_in[135][0]}} & 8'h66)^
({8{data_in[135][1]}} & 8'hcc)^
({8{data_in[135][2]}} & 8'hb3)^
({8{data_in[135][3]}} & 8'h4d)^
({8{data_in[135][4]}} & 8'h9a)^
({8{data_in[135][5]}} & 8'h1f)^
({8{data_in[135][6]}} & 8'h3e)^
({8{data_in[135][7]}} & 8'h7c)^
({8{data_in[136][0]}} & 8'hed)^
({8{data_in[136][1]}} & 8'hf1)^
({8{data_in[136][2]}} & 8'hc9)^
({8{data_in[136][3]}} & 8'hb9)^
({8{data_in[136][4]}} & 8'h59)^
({8{data_in[136][5]}} & 8'hb2)^
({8{data_in[136][6]}} & 8'h4f)^
({8{data_in[136][7]}} & 8'h9e)^
({8{data_in[137][0]}} & 8'h81)^
({8{data_in[137][1]}} & 8'h29)^
({8{data_in[137][2]}} & 8'h52)^
({8{data_in[137][3]}} & 8'ha4)^
({8{data_in[137][4]}} & 8'h63)^
({8{data_in[137][5]}} & 8'hc6)^
({8{data_in[137][6]}} & 8'ha7)^
({8{data_in[137][7]}} & 8'h65)^
({8{data_in[138][0]}} & 8'h7c)^
({8{data_in[138][1]}} & 8'hf8)^
({8{data_in[138][2]}} & 8'hdb)^
({8{data_in[138][3]}} & 8'h9d)^
({8{data_in[138][4]}} & 8'h11)^
({8{data_in[138][5]}} & 8'h22)^
({8{data_in[138][6]}} & 8'h44)^
({8{data_in[138][7]}} & 8'h88)^
({8{data_in[139][0]}} & 8'h95)^
({8{data_in[139][1]}} & 8'h1)^
({8{data_in[139][2]}} & 8'h2)^
({8{data_in[139][3]}} & 8'h4)^
({8{data_in[139][4]}} & 8'h8)^
({8{data_in[139][5]}} & 8'h10)^
({8{data_in[139][6]}} & 8'h20)^
({8{data_in[139][7]}} & 8'h40)^
({8{data_in[140][0]}} & 8'h96)^
({8{data_in[140][1]}} & 8'h7)^

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({8{data_in[140][2]}} & 8'he)^
({8{data_in[140][3]}} & 8'h1c)^
({8{data_in[140][4]}} & 8'h38)^
({8{data_in[140][5]}} & 8'h70)^
({8{data_in[140][6]}} & 8'he0)^
({8{data_in[140][7]}} & 8'heb)^
({8{data_in[141][0]}} & 8'h93)^
({8{data_in[141][1]}} & 8'hd)^
({8{data_in[141][2]}} & 8'h1a)^
({8{data_in[141][3]}} & 8'h34)^
({8{data_in[141][4]}} & 8'h68)^
({8{data_in[141][5]}} & 8'hd0)^
({8{data_in[141][6]}} & 8'h8b)^
({8{data_in[141][7]}} & 8'h3d)^
({8{data_in[142][0]}} & 8'h58)^
({8{data_in[142][1]}} & 8'hb0)^
({8{data_in[142][2]}} & 8'h4b)^
({8{data_in[142][3]}} & 8'h96)^
({8{data_in[142][4]}} & 8'h7)^
({8{data_in[142][5]}} & 8'he)^
({8{data_in[142][6]}} & 8'h1c)^
({8{data_in[142][7]}} & 8'h38)^
({8{data_in[143][0]}} & 8'h50)^
({8{data_in[143][1]}} & 8'ha0)^
({8{data_in[143][2]}} & 8'h6b)^
({8{data_in[143][3]}} & 8'hd6)^
({8{data_in[143][4]}} & 8'h87)^
({8{data_in[143][5]}} & 8'h25)^
({8{data_in[143][6]}} & 8'h4a)^
({8{data_in[143][7]}} & 8'h94)^
({8{data_in[144][0]}} & 8'hb5)^
({8{data_in[144][1]}} & 8'h41)^
({8{data_in[144][2]}} & 8'h82)^
({8{data_in[144][3]}} & 8'h2f)^
({8{data_in[144][4]}} & 8'h5e)^
({8{data_in[144][5]}} & 8'hbc)^
({8{data_in[144][6]}} & 8'h53)^
({8{data_in[144][7]}} & 8'ha6)^
({8{data_in[145][0]}} & 8'hdc)^
({8{data_in[145][1]}} & 8'h93)^
({8{data_in[145][2]}} & 8'hd)^
({8{data_in[145][3]}} & 8'h1a)^
({8{data_in[145][4]}} & 8'h34)^
({8{data_in[145][5]}} & 8'h68)^
({8{data_in[145][6]}} & 8'hd0)^
({8{data_in[145][7]}} & 8'h8b)^
({8{data_in[146][0]}} & 8'h21)^
({8{data_in[146][1]}} & 8'h42)^
({8{data_in[146][2]}} & 8'h84)^
({8{data_in[146][3]}} & 8'h23)^
({8{data_in[146][4]}} & 8'h46)^
({8{data_in[146][5]}} & 8'h8c)^
({8{data_in[146][6]}} & 8'h33)^
({8{data_in[146][7]}} & 8'h66)^
({8{data_in[147][0]}} & 8'h58)^
({8{data_in[147][1]}} & 8'hb0)^
({8{data_in[147][2]}} & 8'h4b)^
({8{data_in[147][3]}} & 8'h96)^
({8{data_in[147][4]}} & 8'h7)^
({8{data_in[147][5]}} & 8'he)^
({8{data_in[147][6]}} & 8'h1c)^
({8{data_in[147][7]}} & 8'h38)^
({8{data_in[148][0]}} & 8'hcb)^
({8{data_in[148][1]}} & 8'hbd)^
({8{data_in[148][2]}} & 8'h51)^

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({8{data_in[148][3]}} & 8'ha2)^
({8{data_in[148][4]}} & 8'h6f)^
({8{data_in[148][5]}} & 8'hde)^
({8{data_in[148][6]}} & 8'h97)^
({8{data_in[148][7]}} & 8'h5)^
({8{data_in[149][0]}} & 8'hbb)^
({8{data_in[149][1]}} & 8'h5d)^
({8{data_in[149][2]}} & 8'hba)^
({8{data_in[149][3]}} & 8'h5f)^
({8{data_in[149][4]}} & 8'hbe)^
({8{data_in[149][5]}} & 8'h57)^
({8{data_in[149][6]}} & 8'hae)^
({8{data_in[149][7]}} & 8'h77)^
({8{data_in[150][0]}} & 8'h4f)^
({8{data_in[150][1]}} & 8'h9e)^
({8{data_in[150][2]}} & 8'h17)^
({8{data_in[150][3]}} & 8'h2e)^
({8{data_in[150][4]}} & 8'h5c)^
({8{data_in[150][5]}} & 8'hb8)^
({8{data_in[150][6]}} & 8'h5b)^
({8{data_in[150][7]}} & 8'hb6)^
({8{data_in[151][0]}} & 8'h75)^
({8{data_in[151][1]}} & 8'hea)^
({8{data_in[151][2]}} & 8'hff)^
({8{data_in[151][3]}} & 8'hd5)^
({8{data_in[151][4]}} & 8'h81)^
({8{data_in[151][5]}} & 8'h29)^
({8{data_in[151][6]}} & 8'h52)^
({8{data_in[151][7]}} & 8'ha4)^
({8{data_in[152][0]}} & 8'hb8)^
({8{data_in[152][1]}} & 8'h5b)^
({8{data_in[152][2]}} & 8'hb6)^
({8{data_in[152][3]}} & 8'h47)^
({8{data_in[152][4]}} & 8'h8e)^
({8{data_in[152][5]}} & 8'h37)^
({8{data_in[152][6]}} & 8'h6e)^
({8{data_in[152][7]}} & 8'hdc)^
({8{data_in[153][0]}} & 8'h1c)^
({8{data_in[153][1]}} & 8'h38)^
({8{data_in[153][2]}} & 8'h70)^
({8{data_in[153][3]}} & 8'he0)^
({8{data_in[153][4]}} & 8'heb)^
({8{data_in[153][5]}} & 8'hfd)^
({8{data_in[153][6]}} & 8'hd1)^
({8{data_in[153][7]}} & 8'h89)^
({8{data_in[154][0]}} & 8'he6)^
({8{data_in[154][1]}} & 8'he7)^
({8{data_in[154][2]}} & 8'he5)^
({8{data_in[154][3]}} & 8'he1)^
({8{data_in[154][4]}} & 8'he9)^
({8{data_in[154][5]}} & 8'hf9)^
({8{data_in[154][6]}} & 8'hd9)^
({8{data_in[154][7]}} & 8'h99)^
({8{data_in[155][0]}} & 8'h54)^
({8{data_in[155][1]}} & 8'ha8)^
({8{data_in[155][2]}} & 8'h7b)^
({8{data_in[155][3]}} & 8'hf6)^
({8{data_in[155][4]}} & 8'hc7)^
({8{data_in[155][5]}} & 8'ha5)^
({8{data_in[155][6]}} & 8'h61)^
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({8{data_in[156][0]}} & 8'hf1)^
({8{data_in[156][1]}} & 8'hc9)^
({8{data_in[156][2]}} & 8'hb9)^
({8{data_in[156][3]}} & 8'h59)^

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({8{data_in[156][4]}} & 8'hb2)^
({8{data_in[156][5]}} & 8'h4f)^
({8{data_in[156][6]}} & 8'h9e)^
({8{data_in[156][7]}} & 8'h17)^
({8{data_in[157][0]}} & 8'hbc)^
({8{data_in[157][1]}} & 8'h53)^
({8{data_in[157][2]}} & 8'ha6)^
({8{data_in[157][3]}} & 8'h67)^
({8{data_in[157][4]}} & 8'hce)^
({8{data_in[157][5]}} & 8'hb7)^
({8{data_in[157][6]}} & 8'h45)^
({8{data_in[157][7]}} & 8'h8a)^
({8{data_in[158][0]}} & 8'hbc)^
({8{data_in[158][1]}} & 8'h53)^
({8{data_in[158][2]}} & 8'ha6)^
({8{data_in[158][3]}} & 8'h67)^
({8{data_in[158][4]}} & 8'hce)^
({8{data_in[158][5]}} & 8'hb7)^
({8{data_in[158][6]}} & 8'h45)^
({8{data_in[158][7]}} & 8'h8a)^
({8{data_in[159][0]}} & 8'hab)^
({8{data_in[159][1]}} & 8'h7d)^
({8{data_in[159][2]}} & 8'hfa)^
({8{data_in[159][3]}} & 8'hdf)^
({8{data_in[159][4]}} & 8'h95)^
({8{data_in[159][5]}} & 8'h1)^
({8{data_in[159][6]}} & 8'h2)^
({8{data_in[159][7]}} & 8'h4)^
({8{data_in[160][0]}} & 8'h2)^
({8{data_in[160][1]}} & 8'h4)^
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({8{data_in[160][3]}} & 8'h10)^
({8{data_in[160][4]}} & 8'h20)^
({8{data_in[160][5]}} & 8'h40)^
({8{data_in[160][6]}} & 8'h80)^
({8{data_in[160][7]}} & 8'h2b)^
({8{data_in[161][0]}} & 8'h5b)^
({8{data_in[161][1]}} & 8'hb6)^
({8{data_in[161][2]}} & 8'h47)^
({8{data_in[161][3]}} & 8'h8e)^
({8{data_in[161][4]}} & 8'h37)^
({8{data_in[161][5]}} & 8'h6e)^
({8{data_in[161][6]}} & 8'hdc)^
({8{data_in[161][7]}} & 8'h93)^
({8{data_in[162][0]}} & 8'h9d)^
({8{data_in[162][1]}} & 8'h11)^
({8{data_in[162][2]}} & 8'h22)^
({8{data_in[162][3]}} & 8'h44)^
({8{data_in[162][4]}} & 8'h88)^
({8{data_in[162][5]}} & 8'h3b)^
({8{data_in[162][6]}} & 8'h76)^
({8{data_in[162][7]}} & 8'hec)^
({8{data_in[163][0]}} & 8'h43)^
({8{data_in[163][1]}} & 8'h86)^
({8{data_in[163][2]}} & 8'h27)^
({8{data_in[163][3]}} & 8'h4e)^
({8{data_in[163][4]}} & 8'h9c)^
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({8{data_in[163][6]}} & 8'h26)^
({8{data_in[163][7]}} & 8'h4c)^
({8{data_in[164][0]}} & 8'h5b)^
({8{data_in[164][1]}} & 8'hb6)^
({8{data_in[164][2]}} & 8'h47)^
({8{data_in[164][3]}} & 8'h8e)^
({8{data_in[164][4]}} & 8'h37)^

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({8{data_in[164][5]}} & 8'h6e)^
({8{data_in[164][6]}} & 8'hdc)^
({8{data_in[164][7]}} & 8'h93)^
({8{data_in[165][0]}} & 8'h58)^
({8{data_in[165][1]}} & 8'hb0)^
({8{data_in[165][2]}} & 8'h4b)^
({8{data_in[165][3]}} & 8'h96)^
({8{data_in[165][4]}} & 8'h7)^
({8{data_in[165][5]}} & 8'he)^
({8{data_in[165][6]}} & 8'h1c)^
({8{data_in[165][7]}} & 8'h38)^
({8{data_in[166][0]}} & 8'h61)^
({8{data_in[166][1]}} & 8'hc2)^
({8{data_in[166][2]}} & 8'haf)^
({8{data_in[166][3]}} & 8'h75)^
({8{data_in[166][4]}} & 8'hea)^
({8{data_in[166][5]}} & 8'hff)^
({8{data_in[166][6]}} & 8'hd5)^
({8{data_in[166][7]}} & 8'h81)^
({8{data_in[167][0]}} & 8'h81)^
({8{data_in[167][1]}} & 8'h29)^
({8{data_in[167][2]}} & 8'h52)^
({8{data_in[167][3]}} & 8'ha4)^
({8{data_in[167][4]}} & 8'h63)^
({8{data_in[167][5]}} & 8'hc6)^
({8{data_in[167][6]}} & 8'ha7)^
({8{data_in[167][7]}} & 8'h65)^
({8{data_in[168][0]}} & 8'h6d)^
({8{data_in[168][1]}} & 8'hda)^
({8{data_in[168][2]}} & 8'h9f)^
({8{data_in[168][3]}} & 8'h15)^
({8{data_in[168][4]}} & 8'h2a)^
({8{data_in[168][5]}} & 8'h54)^
({8{data_in[168][6]}} & 8'ha8)^
({8{data_in[168][7]}} & 8'h7b)^
({8{data_in[169][0]}} & 8'h69)^
({8{data_in[169][1]}} & 8'hd2)^
({8{data_in[169][2]}} & 8'h8f)^
({8{data_in[169][3]}} & 8'h35)^
({8{data_in[169][4]}} & 8'h6a)^
({8{data_in[169][5]}} & 8'hd4)^
({8{data_in[169][6]}} & 8'h83)^
({8{data_in[169][7]}} & 8'h2d)^
({8{data_in[170][0]}} & 8'hc8)^
({8{data_in[170][1]}} & 8'hbb)^
({8{data_in[170][2]}} & 8'h5d)^
({8{data_in[170][3]}} & 8'hba)^
({8{data_in[170][4]}} & 8'h5f)^
({8{data_in[170][5]}} & 8'hbe)^
({8{data_in[170][6]}} & 8'h57)^
({8{data_in[170][7]}} & 8'hae)^
({8{data_in[171][0]}} & 8'hd)^
({8{data_in[171][1]}} & 8'h1a)^
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({8{data_in[171][3]}} & 8'h68)^
({8{data_in[171][4]}} & 8'hd0)^
({8{data_in[171][5]}} & 8'h8b)^
({8{data_in[171][6]}} & 8'h3d)^
({8{data_in[171][7]}} & 8'h7a)^
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({8{data_in[172][1]}} & 8'h39)^
({8{data_in[172][2]}} & 8'h72)^
({8{data_in[172][3]}} & 8'he4)^
({8{data_in[172][4]}} & 8'he3)^
({8{data_in[172][5]}} & 8'hed)^

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({8{data_in[172][6]}} & 8'hf1)^
({8{data_in[172][7]}} & 8'hc9)^
({8{data_in[173][0]}} & 8'hdc)^
({8{data_in[173][1]}} & 8'h93)^
({8{data_in[173][2]}} & 8'hd)^
({8{data_in[173][3]}} & 8'h1a)^
({8{data_in[173][4]}} & 8'h34)^
({8{data_in[173][5]}} & 8'h68)^
({8{data_in[173][6]}} & 8'hd0)^
({8{data_in[173][7]}} & 8'h8b)^
({8{data_in[174][0]}} & 8'hfb)^
({8{data_in[174][1]}} & 8'hdd)^
({8{data_in[174][2]}} & 8'h91)^
({8{data_in[174][3]}} & 8'h9)^
({8{data_in[174][4]}} & 8'h12)^
({8{data_in[174][5]}} & 8'h24)^
({8{data_in[174][6]}} & 8'h48)^
({8{data_in[174][7]}} & 8'h90)^
({8{data_in[175][0]}} & 8'hc8)^
({8{data_in[175][1]}} & 8'hbb)^
({8{data_in[175][2]}} & 8'h5d)^
({8{data_in[175][3]}} & 8'hba)^
({8{data_in[175][4]}} & 8'h5f)^
({8{data_in[175][5]}} & 8'hbe)^
({8{data_in[175][6]}} & 8'h57)^
({8{data_in[175][7]}} & 8'hae)^
({8{data_in[176][0]}} & 8'h30)^
({8{data_in[176][1]}} & 8'h60)^
({8{data_in[176][2]}} & 8'hc0)^
({8{data_in[176][3]}} & 8'hab)^
({8{data_in[176][4]}} & 8'h7d)^
({8{data_in[176][5]}} & 8'hfa)^
({8{data_in[176][6]}} & 8'hdf)^
({8{data_in[176][7]}} & 8'h95)^
({8{data_in[177][0]}} & 8'he2)^
({8{data_in[177][1]}} & 8'hcf)^
({8{data_in[177][2]}} & 8'hf5)^
({8{data_in[177][3]}} & 8'hc1)^
({8{data_in[177][4]}} & 8'ha9)^
({8{data_in[177][5]}} & 8'h79)^
({8{data_in[177][6]}} & 8'hf2)^
({8{data_in[177][7]}} & 8'hcf)^
({8{data_in[178][0]}} & 8'h1b)^
({8{data_in[178][1]}} & 8'h36)^
({8{data_in[178][2]}} & 8'h6c)^
({8{data_in[178][3]}} & 8'hd8)^
({8{data_in[178][4]}} & 8'h9b)^
({8{data_in[178][5]}} & 8'h1d)^
({8{data_in[178][6]}} & 8'h3a)^
({8{data_in[178][7]}} & 8'h74)^
({8{data_in[179][0]}} & 8'h8e)^
({8{data_in[179][1]}} & 8'h37)^
({8{data_in[179][2]}} & 8'h6e)^
({8{data_in[179][3]}} & 8'hdc)^
({8{data_in[179][4]}} & 8'h93)^
({8{data_in[179][5]}} & 8'hd)^
({8{data_in[179][6]}} & 8'h1a)^
({8{data_in[179][7]}} & 8'h34)^
({8{data_in[180][0]}} & 8'h69)^
({8{data_in[180][1]}} & 8'hd2)^
({8{data_in[180][2]}} & 8'h8f)^
({8{data_in[180][3]}} & 8'h35)^
({8{data_in[180][4]}} & 8'h6a)^
({8{data_in[180][5]}} & 8'hd4)^
({8{data_in[180][6]}} & 8'h83)^

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({8{data_in[180][7]}} & 8'h2d)^
({8{data_in[181][0]}} & 8'h33)^
({8{data_in[181][1]}} & 8'h66)^
({8{data_in[181][2]}} & 8'hcc)^
({8{data_in[181][3]}} & 8'hb3)^
({8{data_in[181][4]}} & 8'h4d)^
({8{data_in[181][5]}} & 8'h9a)^
({8{data_in[181][6]}} & 8'h1f)^
({8{data_in[181][7]}} & 8'h3e)^
({8{data_in[182][0]}} & 8'hf9)^
({8{data_in[182][1]}} & 8'hd9)^
({8{data_in[182][2]}} & 8'h99)^
({8{data_in[182][3]}} & 8'h19)^
({8{data_in[182][4]}} & 8'h32)^
({8{data_in[182][5]}} & 8'h64)^
({8{data_in[182][6]}} & 8'hc8)^
({8{data_in[182][7]}} & 8'hbb)^
({8{data_in[183][0]}} & 8'h8a)^
({8{data_in[183][1]}} & 8'h3f)^
({8{data_in[183][2]}} & 8'h7e)^
({8{data_in[183][3]}} & 8'hfc)^
({8{data_in[183][4]}} & 8'hd3)^
({8{data_in[183][5]}} & 8'h8d)^
({8{data_in[183][6]}} & 8'h31)^
({8{data_in[183][7]}} & 8'h62)^
({8{data_in[184][0]}} & 8'h13)^
({8{data_in[184][1]}} & 8'h26)^
({8{data_in[184][2]}} & 8'h4c)^
({8{data_in[184][3]}} & 8'h98)^
({8{data_in[184][4]}} & 8'h1b)^
({8{data_in[184][5]}} & 8'h36)^
({8{data_in[184][6]}} & 8'h6c)^
({8{data_in[184][7]}} & 8'hd8)^
({8{data_in[185][0]}} & 8'h22)^
({8{data_in[185][1]}} & 8'h44)^
({8{data_in[185][2]}} & 8'h88)^
({8{data_in[185][3]}} & 8'h3b)^
({8{data_in[185][4]}} & 8'h76)^
({8{data_in[185][5]}} & 8'hec)^
({8{data_in[185][6]}} & 8'hf3)^
({8{data_in[185][7]}} & 8'hcd)^
({8{data_in[186][0]}} & 8'hb4)^
({8{data_in[186][1]}} & 8'h43)^
({8{data_in[186][2]}} & 8'h86)^
({8{data_in[186][3]}} & 8'h27)^
({8{data_in[186][4]}} & 8'h4e)^
({8{data_in[186][5]}} & 8'h9c)^
({8{data_in[186][6]}} & 8'h13)^
({8{data_in[186][7]}} & 8'h26)^
({8{data_in[187][0]}} & 8'h6)^
({8{data_in[187][1]}} & 8'hc)^
({8{data_in[187][2]}} & 8'h18)^
({8{data_in[187][3]}} & 8'h30)^
({8{data_in[187][4]}} & 8'h60)^
({8{data_in[187][5]}} & 8'hc0)^
({8{data_in[187][6]}} & 8'hab)^
({8{data_in[187][7]}} & 8'h7d)^
({8{data_in[188][0]}} & 8'h13)^
({8{data_in[188][1]}} & 8'h26)^
({8{data_in[188][2]}} & 8'h4c)^
({8{data_in[188][3]}} & 8'h98)^
({8{data_in[188][4]}} & 8'h1b)^
({8{data_in[188][5]}} & 8'h36)^
({8{data_in[188][6]}} & 8'h6c)^
({8{data_in[188][7]}} & 8'hd8)^

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({8{data_in[189][0]}} & 8'haf)^
({8{data_in[189][1]}} & 8'h75)^
({8{data_in[189][2]}} & 8'hea)^
({8{data_in[189][3]}} & 8'hff)^
({8{data_in[189][4]}} & 8'hd5)^
({8{data_in[189][5]}} & 8'h81)^
({8{data_in[189][6]}} & 8'h29)^
({8{data_in[189][7]}} & 8'h52)^
({8{data_in[190][0]}} & 8'he1)^
({8{data_in[190][1]}} & 8'he9)^
({8{data_in[190][2]}} & 8'hf9)^
({8{data_in[190][3]}} & 8'hd9)^
({8{data_in[190][4]}} & 8'h99)^
({8{data_in[190][5]}} & 8'h19)^
({8{data_in[190][6]}} & 8'h32)^
({8{data_in[190][7]}} & 8'h64)^
({8{data_in[191][0]}} & 8'hfb)^
({8{data_in[191][1]}} & 8'hdd)^
({8{data_in[191][2]}} & 8'h91)^
({8{data_in[191][3]}} & 8'h9)^
({8{data_in[191][4]}} & 8'h12)^
({8{data_in[191][5]}} & 8'h24)^
({8{data_in[191][6]}} & 8'h48)^
({8{data_in[191][7]}} & 8'h90)^
({8{data_in[192][0]}} & 8'h1a)^
({8{data_in[192][1]}} & 8'h34)^
({8{data_in[192][2]}} & 8'h68)^
({8{data_in[192][3]}} & 8'hd0)^
({8{data_in[192][4]}} & 8'h8b)^
({8{data_in[192][5]}} & 8'h3d)^
({8{data_in[192][6]}} & 8'h7a)^
({8{data_in[192][7]}} & 8'hf4)^
({8{data_in[193][0]}} & 8'h5c)^
({8{data_in[193][1]}} & 8'hb8)^
({8{data_in[193][2]}} & 8'h5b)^
({8{data_in[193][3]}} & 8'hb6)^
({8{data_in[193][4]}} & 8'h47)^
({8{data_in[193][5]}} & 8'h8e)^
({8{data_in[193][6]}} & 8'h37)^
({8{data_in[193][7]}} & 8'h6e)^
({8{data_in[194][0]}} & 8'h6a)^
({8{data_in[194][1]}} & 8'hd4)^
({8{data_in[194][2]}} & 8'h83)^
({8{data_in[194][3]}} & 8'h2d)^
({8{data_in[194][4]}} & 8'h5a)^
({8{data_in[194][5]}} & 8'hb4)^
({8{data_in[194][6]}} & 8'h43)^
({8{data_in[194][7]}} & 8'h86)^
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({8{data_in[195][1]}} & 8'h20)^
({8{data_in[195][2]}} & 8'h40)^
({8{data_in[195][3]}} & 8'h80)^
({8{data_in[195][4]}} & 8'h2b)^
({8{data_in[195][5]}} & 8'h56)^
({8{data_in[195][6]}} & 8'hac)^
({8{data_in[195][7]}} & 8'h73)^
({8{data_in[196][0]}} & 8'hc0)^
({8{data_in[196][1]}} & 8'hab)^
({8{data_in[196][2]}} & 8'h7d)^
({8{data_in[196][3]}} & 8'hfa)^
({8{data_in[196][4]}} & 8'hdf)^
({8{data_in[196][5]}} & 8'h95)^
({8{data_in[196][6]}} & 8'h1)^
({8{data_in[196][7]}} & 8'h2)^
({8{data_in[197][0]}} & 8'h96)^

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({8{data_in[197][1]}} & 8'h7)^
({8{data_in[197][2]}} & 8'he)^
({8{data_in[197][3]}} & 8'h1c)^
({8{data_in[197][4]}} & 8'h38)^
({8{data_in[197][5]}} & 8'h70)^
({8{data_in[197][6]}} & 8'he0)^
({8{data_in[197][7]}} & 8'heb)^
({8{data_in[198][0]}} & 8'h1a)^
({8{data_in[198][1]}} & 8'h34)^
({8{data_in[198][2]}} & 8'h68)^
({8{data_in[198][3]}} & 8'hd0)^
({8{data_in[198][4]}} & 8'h8b)^
({8{data_in[198][5]}} & 8'h3d)^
({8{data_in[198][6]}} & 8'h7a)^
({8{data_in[198][7]}} & 8'hf4)^
({8{data_in[199][0]}} & 8'ha0)^
({8{data_in[199][1]}} & 8'h6b)^
({8{data_in[199][2]}} & 8'hd6)^
({8{data_in[199][3]}} & 8'h87)^
({8{data_in[199][4]}} & 8'h25)^
({8{data_in[199][5]}} & 8'h4a)^
({8{data_in[199][6]}} & 8'h94)^
({8{data_in[199][7]}} & 8'h3)^
({8{data_in[200][0]}} & 8'he9)^
({8{data_in[200][1]}} & 8'hf9)^
({8{data_in[200][2]}} & 8'hd9)^
({8{data_in[200][3]}} & 8'h99)^
({8{data_in[200][4]}} & 8'h19)^
({8{data_in[200][5]}} & 8'h32)^
({8{data_in[200][6]}} & 8'h64)^
({8{data_in[200][7]}} & 8'hc8)^
({8{data_in[201][0]}} & 8'h82)^
({8{data_in[201][1]}} & 8'h2f)^
({8{data_in[201][2]}} & 8'h5e)^
({8{data_in[201][3]}} & 8'hbc)^
({8{data_in[201][4]}} & 8'h53)^
({8{data_in[201][5]}} & 8'ha6)^
({8{data_in[201][6]}} & 8'h67)^
({8{data_in[201][7]}} & 8'hce)^
({8{data_in[202][0]}} & 8'hcf)^
({8{data_in[202][1]}} & 8'hb5)^
({8{data_in[202][2]}} & 8'h41)^
({8{data_in[202][3]}} & 8'h82)^
({8{data_in[202][4]}} & 8'h2f)^
({8{data_in[202][5]}} & 8'h5e)^
({8{data_in[202][6]}} & 8'hbc)^
({8{data_in[202][7]}} & 8'h53)^
({8{data_in[203][0]}} & 8'h36)^
({8{data_in[203][1]}} & 8'h6c)^
({8{data_in[203][2]}} & 8'hd8)^
({8{data_in[203][3]}} & 8'h9b)^
({8{data_in[203][4]}} & 8'h1d)^
({8{data_in[203][5]}} & 8'h3a)^
({8{data_in[203][6]}} & 8'h74)^
({8{data_in[203][7]}} & 8'he8)^
({8{data_in[204][0]}} & 8'hae)^
({8{data_in[204][1]}} & 8'h77)^
({8{data_in[204][2]}} & 8'hee)^
({8{data_in[204][3]}} & 8'hf7)^
({8{data_in[204][4]}} & 8'hc5)^
({8{data_in[204][5]}} & 8'ha1)^
({8{data_in[204][6]}} & 8'h69)^
({8{data_in[204][7]}} & 8'hd2)^
({8{data_in[205][0]}} & 8'h16)^
({8{data_in[205][1]}} & 8'h2c)^

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({8{data_in[205][2]}} & 8'h58)^
({8{data_in[205][3]}} & 8'hb0)^
({8{data_in[205][4]}} & 8'h4b)^
({8{data_in[205][5]}} & 8'h96)^
({8{data_in[205][6]}} & 8'h7)^
({8{data_in[205][7]}} & 8'he)^
({8{data_in[206][0]}} & 8'h1a)^
({8{data_in[206][1]}} & 8'h34)^
({8{data_in[206][2]}} & 8'h68)^
({8{data_in[206][3]}} & 8'hd0)^
({8{data_in[206][4]}} & 8'h8b)^
({8{data_in[206][5]}} & 8'h3d)^
({8{data_in[206][6]}} & 8'h7a)^
({8{data_in[206][7]}} & 8'hf4)^
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({8{data_in[207][2]}} & 8'hb1)^
({8{data_in[207][3]}} & 8'h49)^
({8{data_in[207][4]}} & 8'h92)^
({8{data_in[207][5]}} & 8'hf)^
({8{data_in[207][6]}} & 8'h1e)^
({8{data_in[207][7]}} & 8'h3c)^
({8{data_in[208][0]}} & 8'h8c)^
({8{data_in[208][1]}} & 8'h33)^
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({8{data_in[208][3]}} & 8'hcc)^
({8{data_in[208][4]}} & 8'hb3)^
({8{data_in[208][5]}} & 8'h4d)^
({8{data_in[208][6]}} & 8'h9a)^
({8{data_in[208][7]}} & 8'h1f)^
({8{data_in[209][0]}} & 8'h3d)^
({8{data_in[209][1]}} & 8'h7a)^
({8{data_in[209][2]}} & 8'hf4)^
({8{data_in[209][3]}} & 8'hc3)^
({8{data_in[209][4]}} & 8'had)^
({8{data_in[209][5]}} & 8'h71)^
({8{data_in[209][6]}} & 8'he2)^
({8{data_in[209][7]}} & 8'hef)^
({8{data_in[210][0]}} & 8'ha9)^
({8{data_in[210][1]}} & 8'h79)^
({8{data_in[210][2]}} & 8'hf2)^
({8{data_in[210][3]}} & 8'hcf)^
({8{data_in[210][4]}} & 8'hb5)^
({8{data_in[210][5]}} & 8'h41)^
({8{data_in[210][6]}} & 8'h82)^
({8{data_in[210][7]}} & 8'h2f)^
({8{data_in[211][0]}} & 8'ha)^
({8{data_in[211][1]}} & 8'h14)^
({8{data_in[211][2]}} & 8'h28)^
({8{data_in[211][3]}} & 8'h50)^
({8{data_in[211][4]}} & 8'ha0)^
({8{data_in[211][5]}} & 8'h6b)^
({8{data_in[211][6]}} & 8'hd6)^
({8{data_in[211][7]}} & 8'h87)^
({8{data_in[212][0]}} & 8'hab)^
({8{data_in[212][1]}} & 8'h7d)^
({8{data_in[212][2]}} & 8'hfa)^
({8{data_in[212][3]}} & 8'hdf)^
({8{data_in[212][4]}} & 8'h95)^
({8{data_in[212][5]}} & 8'h1)^
({8{data_in[212][6]}} & 8'h2)^
({8{data_in[212][7]}} & 8'h4)^
({8{data_in[213][0]}} & 8'hce)^
({8{data_in[213][1]}} & 8'hb7)^
({8{data_in[213][2]}} & 8'h45)^

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({8{data_in[213][3]}} & 8'h8a)^
({8{data_in[213][4]}} & 8'h3f)^
({8{data_in[213][5]}} & 8'h7e)^
({8{data_in[213][6]}} & 8'hfc)^
({8{data_in[213][7]}} & 8'hd3)^
({8{data_in[214][0]}} & 8'hd1)^
({8{data_in[214][1]}} & 8'h89)^
({8{data_in[214][2]}} & 8'h39)^
({8{data_in[214][3]}} & 8'h72)^
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({8{data_in[214][5]}} & 8'he3)^
({8{data_in[214][6]}} & 8'hed)^
({8{data_in[214][7]}} & 8'hf1)^
({8{data_in[215][0]}} & 8'h48)^
({8{data_in[215][1]}} & 8'h90)^
({8{data_in[215][2]}} & 8'hb)^
({8{data_in[215][3]}} & 8'h16)^
({8{data_in[215][4]}} & 8'h2c)^
({8{data_in[215][5]}} & 8'h58)^
({8{data_in[215][6]}} & 8'hb0)^
({8{data_in[215][7]}} & 8'h4b)^
({8{data_in[216][0]}} & 8'h7f)^
({8{data_in[216][1]}} & 8'hfe)^
({8{data_in[216][2]}} & 8'hd7)^
({8{data_in[216][3]}} & 8'h85)^
({8{data_in[216][4]}} & 8'h21)^
({8{data_in[216][5]}} & 8'h42)^
({8{data_in[216][6]}} & 8'h84)^
({8{data_in[216][7]}} & 8'h23)^
({8{data_in[217][0]}} & 8'ha8)^
({8{data_in[217][1]}} & 8'h7b)^
({8{data_in[217][2]}} & 8'hf6)^
({8{data_in[217][3]}} & 8'hc7)^
({8{data_in[217][4]}} & 8'ha5)^
({8{data_in[217][5]}} & 8'h61)^
({8{data_in[217][6]}} & 8'hc2)^
({8{data_in[217][7]}} & 8'haf)^
({8{data_in[218][0]}} & 8'hde)^
({8{data_in[218][1]}} & 8'h97)^
({8{data_in[218][2]}} & 8'h5)^
({8{data_in[218][3]}} & 8'ha)^
({8{data_in[218][4]}} & 8'h14)^
({8{data_in[218][5]}} & 8'h28)^
({8{data_in[218][6]}} & 8'h50)^
({8{data_in[218][7]}} & 8'ha0)^
({8{data_in[219][0]}} & 8'h6d)^
({8{data_in[219][1]}} & 8'hda)^
({8{data_in[219][2]}} & 8'h9f)^
({8{data_in[219][3]}} & 8'h15)^
({8{data_in[219][4]}} & 8'h2a)^
({8{data_in[219][5]}} & 8'h54)^
({8{data_in[219][6]}} & 8'ha8)^
({8{data_in[219][7]}} & 8'h7b)^
({8{data_in[220][0]}} & 8'h48)^
({8{data_in[220][1]}} & 8'h90)^
({8{data_in[220][2]}} & 8'hb)^
({8{data_in[220][3]}} & 8'h16)^
({8{data_in[220][4]}} & 8'h2c)^
({8{data_in[220][5]}} & 8'h58)^
({8{data_in[220][6]}} & 8'hb0)^
({8{data_in[220][7]}} & 8'h4b)^
({8{data_in[221][0]}} & 8'h38)^
({8{data_in[221][1]}} & 8'h70)^
({8{data_in[221][2]}} & 8'he0)^
({8{data_in[221][3]}} & 8'heb)^

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({8{data_in[221][4]}} & 8'hfd)^
({8{data_in[221][5]}} & 8'hd1)^
({8{data_in[221][6]}} & 8'h89)^
({8{data_in[221][7]}} & 8'h39)^
({8{data_in[222][0]}} & 8'h42)^
({8{data_in[222][1]}} & 8'h84)^
({8{data_in[222][2]}} & 8'h23)^
({8{data_in[222][3]}} & 8'h46)^
({8{data_in[222][4]}} & 8'h8c)^
({8{data_in[222][5]}} & 8'h33)^
({8{data_in[222][6]}} & 8'h66)^
({8{data_in[222][7]}} & 8'hcc)^
({8{data_in[223][0]}} & 8'hf)^
({8{data_in[223][1]}} & 8'h1e)^
({8{data_in[223][2]}} & 8'h3c)^
({8{data_in[223][3]}} & 8'h78)^
({8{data_in[223][4]}} & 8'hf0)^
({8{data_in[223][5]}} & 8'hcb)^
({8{data_in[223][6]}} & 8'hbd)^
({8{data_in[223][7]}} & 8'h51)^
({8{data_in[224][0]}} & 8'h81)^
({8{data_in[224][1]}} & 8'h29)^
({8{data_in[224][2]}} & 8'h52)^
({8{data_in[224][3]}} & 8'ha4)^
({8{data_in[224][4]}} & 8'h63)^
({8{data_in[224][5]}} & 8'hc6)^
({8{data_in[224][6]}} & 8'ha7)^
({8{data_in[224][7]}} & 8'h65)^
({8{data_in[225][0]}} & 8'hc9)^
({8{data_in[225][1]}} & 8'hb9)^
({8{data_in[225][2]}} & 8'h59)^
({8{data_in[225][3]}} & 8'hb2)^
({8{data_in[225][4]}} & 8'h4f)^
({8{data_in[225][5]}} & 8'h9e)^
({8{data_in[225][6]}} & 8'h17)^
({8{data_in[225][7]}} & 8'h2e)^
({8{data_in[226][0]}} & 8'h74)^
({8{data_in[226][1]}} & 8'he8)^
({8{data_in[226][2]}} & 8'hfb)^
({8{data_in[226][3]}} & 8'hdd)^
({8{data_in[226][4]}} & 8'h91)^
({8{data_in[226][5]}} & 8'h9)^
({8{data_in[226][6]}} & 8'h12)^
({8{data_in[226][7]}} & 8'h24)^
({8{data_in[227][0]}} & 8'h1e)^
({8{data_in[227][1]}} & 8'h3c)^
({8{data_in[227][2]}} & 8'h78)^
({8{data_in[227][3]}} & 8'hf0)^
({8{data_in[227][4]}} & 8'hcb)^
({8{data_in[227][5]}} & 8'hbd)^
({8{data_in[227][6]}} & 8'h51)^
({8{data_in[227][7]}} & 8'ha2)^
({8{data_in[228][0]}} & 8'h34)^
({8{data_in[228][1]}} & 8'h68)^
({8{data_in[228][2]}} & 8'hd0)^
({8{data_in[228][3]}} & 8'h8b)^
({8{data_in[228][4]}} & 8'h3d)^
({8{data_in[228][5]}} & 8'h7a)^
({8{data_in[228][6]}} & 8'hf4)^
({8{data_in[228][7]}} & 8'hc3)^
({8{data_in[229][0]}} & 8'h82)^
({8{data_in[229][1]}} & 8'h2f)^
({8{data_in[229][2]}} & 8'h5e)^
({8{data_in[229][3]}} & 8'hbc)^
({8{data_in[229][4]}} & 8'h53)^

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({8{data_in[229][5]}} & 8'ha6)^
({8{data_in[229][6]}} & 8'h67)^
({8{data_in[229][7]}} & 8'hce)^
({8{data_in[230][0]}} & 8'hed)^
({8{data_in[230][1]}} & 8'hf1)^
({8{data_in[230][2]}} & 8'hc9)^
({8{data_in[230][3]}} & 8'hb9)^
({8{data_in[230][4]}} & 8'h59)^
({8{data_in[230][5]}} & 8'hb2)^
({8{data_in[230][6]}} & 8'h4f)^
({8{data_in[230][7]}} & 8'h9e)^
({8{data_in[231][0]}} & 8'h71)^
({8{data_in[231][1]}} & 8'he2)^
({8{data_in[231][2]}} & 8'hef)^
({8{data_in[231][3]}} & 8'hf5)^
({8{data_in[231][4]}} & 8'hc1)^
({8{data_in[231][5]}} & 8'ha9)^
({8{data_in[231][6]}} & 8'h79)^
({8{data_in[231][7]}} & 8'hf2)^
({8{data_in[232][0]}} & 8'h83)^
({8{data_in[232][1]}} & 8'h2d)^
({8{data_in[232][2]}} & 8'h5a)^
({8{data_in[232][3]}} & 8'hb4)^
({8{data_in[232][4]}} & 8'h43)^
({8{data_in[232][5]}} & 8'h86)^
({8{data_in[232][6]}} & 8'h27)^
({8{data_in[232][7]}} & 8'h4e)^
({8{data_in[233][0]}} & 8'h80)^
({8{data_in[233][1]}} & 8'h2b)^
({8{data_in[233][2]}} & 8'h56)^
({8{data_in[233][3]}} & 8'hac)^
({8{data_in[233][4]}} & 8'h73)^
({8{data_in[233][5]}} & 8'he6)^
({8{data_in[233][6]}} & 8'he7)^
({8{data_in[233][7]}} & 8'he5)^
({8{data_in[234][0]}} & 8'h1c)^
({8{data_in[234][1]}} & 8'h38)^
({8{data_in[234][2]}} & 8'h70)^
({8{data_in[234][3]}} & 8'he0)^
({8{data_in[234][4]}} & 8'heb)^
({8{data_in[234][5]}} & 8'hfd)^
({8{data_in[234][6]}} & 8'hd1)^
({8{data_in[234][7]}} & 8'h89)^
({8{data_in[235][0]}} & 8'h7b)^
({8{data_in[235][1]}} & 8'hf6)^
({8{data_in[235][2]}} & 8'hc7)^
({8{data_in[235][3]}} & 8'ha5)^
({8{data_in[235][4]}} & 8'h61)^
({8{data_in[235][5]}} & 8'hc2)^
({8{data_in[235][6]}} & 8'haf)^
({8{data_in[235][7]}} & 8'h75)^
({8{data_in[236][0]}} & 8'hbe)^
({8{data_in[236][1]}} & 8'h57)^
({8{data_in[236][2]}} & 8'hae)^
({8{data_in[236][3]}} & 8'h77)^
({8{data_in[236][4]}} & 8'hee)^
({8{data_in[236][5]}} & 8'hf7)^
({8{data_in[236][6]}} & 8'hc5)^
({8{data_in[236][7]}} & 8'ha1)^
({8{data_in[237][0]}} & 8'h7c)^
({8{data_in[237][1]}} & 8'hf8)^
({8{data_in[237][2]}} & 8'hdb)^
({8{data_in[237][3]}} & 8'h9d)^
({8{data_in[237][4]}} & 8'h11)^
({8{data_in[237][5]}} & 8'h22)^

```

```

({8{data_in[237][6]}} & 8'h44)^
({8{data_in[237][7]}} & 8'h88)^
({8{data_in[238][0]}} & 8'hfb)^
({8{data_in[238][1]}} & 8'hdd)^
({8{data_in[238][2]}} & 8'h91)^
({8{data_in[238][3]}} & 8'h9)^
({8{data_in[238][4]}} & 8'h12)^
({8{data_in[238][5]}} & 8'h24)^
({8{data_in[238][6]}} & 8'h48)^
({8{data_in[238][7]}} & 8'h90)^
({8{data_in[239][0]}} & 8'h80)^
({8{data_in[239][1]}} & 8'h2b)^
({8{data_in[239][2]}} & 8'h56)^
({8{data_in[239][3]}} & 8'hac)^
({8{data_in[239][4]}} & 8'h73)^
({8{data_in[239][5]}} & 8'he6)^
({8{data_in[239][6]}} & 8'he7)^
({8{data_in[239][7]}} & 8'he5)^
({8{data_in[240][0]}} & 8'h22)^
({8{data_in[240][1]}} & 8'h44)^
({8{data_in[240][2]}} & 8'h88)^
({8{data_in[240][3]}} & 8'h3b)^
({8{data_in[240][4]}} & 8'h76)^
({8{data_in[240][5]}} & 8'hec)^
({8{data_in[240][6]}} & 8'hf3)^
({8{data_in[240][7]}} & 8'hcd)^
({8{data_in[241][0]}} & 8'h69)^
({8{data_in[241][1]}} & 8'hd2)^
({8{data_in[241][2]}} & 8'h8f)^
({8{data_in[241][3]}} & 8'h35)^
({8{data_in[241][4]}} & 8'h6a)^
({8{data_in[241][5]}} & 8'hd4)^
({8{data_in[241][6]}} & 8'h83)^
({8{data_in[241][7]}} & 8'h2d);

```

```

end
endmodule
`endif

```

L. Example IDE TLPs and Test Keys §

This appendix includes several example IDE TLPs, intended to be used to verify correct implementation of IDE. Also included are a set of keys to be used for testing, to simplify IDE validation.

L.1 Example NFM IDE TLP Without Partial Header Encryption §

The TLP shown in § Figure L-1, when processed using the Key and IV shown in § Table L-1, which also includes other relevant inputs and outputs, yields the IDE TLP shown in § Figure L-2.

	+0								+1								+2								+3														
	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0							
Byte 0 →	Fmt			Type					T9	TC			T8	A2	R	TH	TD	EP	Attr			AT		Length															
	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0						
Byte 4 →	Requester ID																Tag								Last DW BE				First DW BE										
	1	0	1	0	1	0	1	1	1	1	0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	0	1	1	1	1	1	1						
Byte 8 →	Address[63:32]																																						
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1							
Byte 12 →	Address[31:2]																												PH										
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0							
Byte 16 →	DATA DW0																																						
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	1						
Byte 20 →	DATA DW1																																						
	0	0	0	0	0	1	0	0	0	0	0	0	0	1	0	1	0	0	0	0	0	1	1	0	0	0	0	0	0	0	1	1	1						
Byte 24 →	DATA DW2																																						
	0	0	0	0	1	0	0	0	0	0	0	0	1	0	0	1	0	0	0	1	0	1	0	0	0	0	0	0	1	0	1	1	1						
Byte 28 →	DATA DW3																																						
	0	0	0	0	1	1	0	0	0	0	0	0	1	1	0	1	0	0	0	1	1	1	0	0	0	0	0	1	1	1	1	1	1						

Figure L-1 Example Memory Write TLP (NFM) §

Table L-1 Inputs and Outputs for Example IDE TLP §

Input/Outputs variables for AES GCM	Values Byte0 [7:0] byte 1 [15:8] Byte N Key, IV, MAC : Follows AES-GCM Specification with Most significant leftmost
<i>Inputs</i>	
Key (K)	8A 95 85 8A 23 1D 41 4A 37 33 F0 E5 7A F6 AE 77 C7 F0 CF BA 75 54 D6 70 83 34 2A 64 2A EF 9D 5D
Initialization Vector (IV)	00 00 00 00 00 00 00 00 00 00 00 01
A	92 00 BC 0C 60 00 00 04 AB CD 00 FF 00 00 00 01 00 00 00 00
P (including PCRC)	00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 88 E2 CE CE

Input/Outputs variables for AES GCM	Values Byte0 [7:0] byte 1 [15:8] Byte N Key, IV, MAC : Follows AES-GCM Specification with Most significant leftmost
<i>Outputs</i>	
C (including encrypted PCRC)	0C F7 B4 8F E6 30 D7 18 CE B9 F4 23 DE EB 68 53 7C DF 22 EE
MAC	53 A7 D7 13 7F 48 1E 7E FB 97 E8 C6

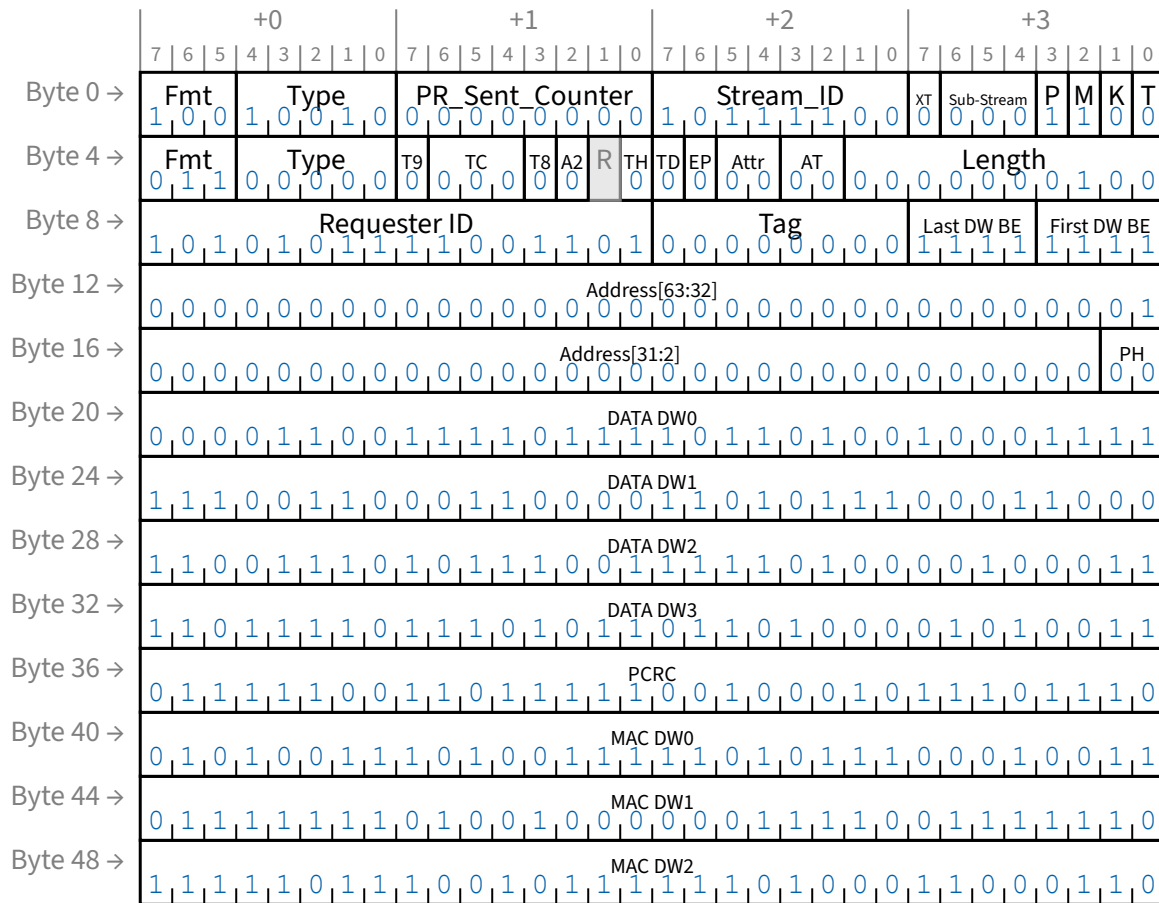


Figure L-2 Example NFM Memory Write IDE TLP §

L.2 Example FM IDE TLP Without Partial Header Encryption §

The TLP shown in § Figure L-3, when processed using the Key and IV shown in § Table L-2, which also includes other relevant inputs and outputs, yields the IDE TLP shown in § Figure L-4.

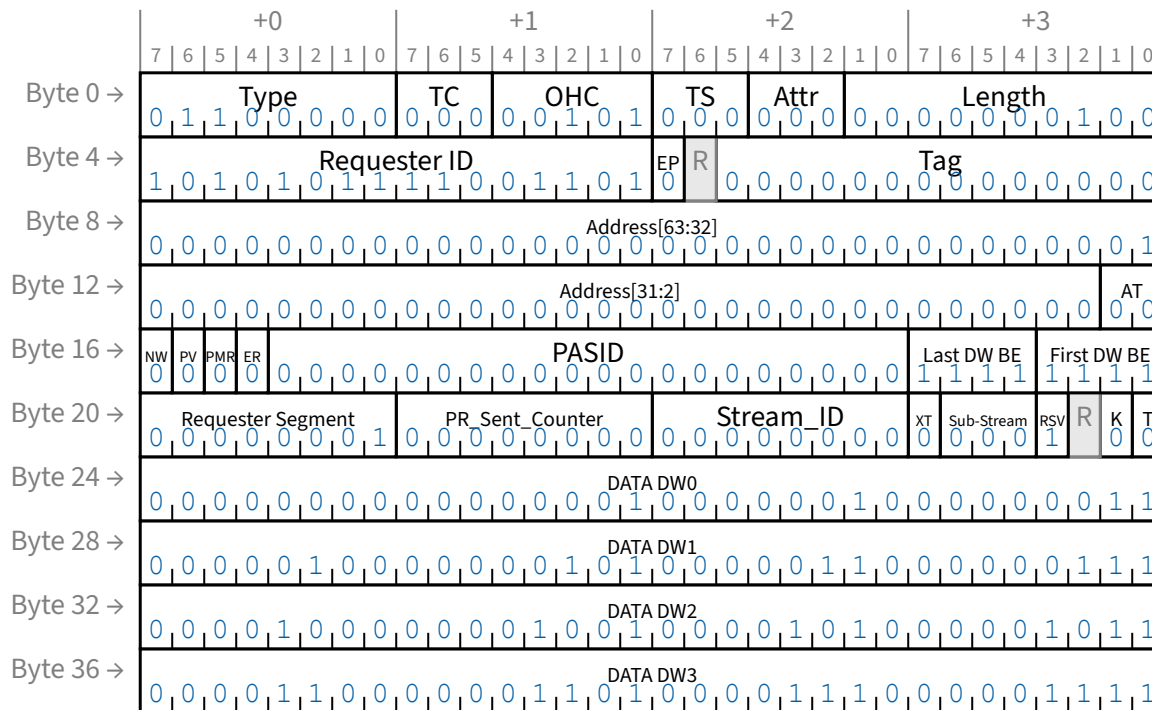


Figure L-3 Example Memory Write TLP (FM) §

Table L-2 Inputs and Outputs for Example IDE TLP (FM) §

Input/Outputs variables for AES GCM	Values Byte0 [7:0] byte 1 [15:8] Byte N Key, IV, MAC : Follows AES-GCM Specification with Most significant leftmost
<i>Inputs</i>	
Key (K)	8A 95 85 8A 23 1D 41 4A 37 33 F0 E5 7A F6 AE 77 C7 F0 CF BA 75 54 D6 70 83 34 2A 64 2A EF 9D 5D
Initialization Vector (IV)	00 00 00 00 00 00 00 00 00 00 00 01
A	60 05 C0 04 AB CD 00 00 00 00 00 01 00 00 00 00 00 00 00 00 FF 01 00 BC 08
P (including PCRC)	00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 88 E2 CE CE
<i>Outputs</i>	
C (including encrypted PCRC)	0C F7 B4 8F E6 30 D7 18 CE B9 F4 23 DE EB 68 53 7C DF 22 EE
MAC	D8 DC 27 8B EF 37 7B 40 D5 1A D6 28

	+0								+1								+2								+3									
	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0		
Byte 0 →	Type								TC		OHC				TS		Attr		Length															
	0	1	1	0	0	0	0	0	0	0	0	0	0	1	0	1	1	1	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	
Byte 4 →	Requester ID																EP	R	Tag															
	1	0	1	0	1	0	1	1	1	1	0	0	1	1	0	1	0		0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Byte 8 →	Address[63:32]																																	
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1		
Byte 12 →	Address[31:2]																										AT							
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0		
Byte 16 →	NW	PV	PMR	ER	PASID														Last DW BE				First DW BE											
	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Byte 20 →	Requester Segment								PR_Sent_Counter								hed	Stream_ID								XT	Sub-Stream		RSV	R	K	T		
	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	1	0	1	1	1	1	1	0	0	0	0	0	1		0	0		
Byte 24 →	DATA DW0																																	
	0	0	0	0	1	1	0	0	1	1	1	1	0	1	1	1	1	0	1	1	0	1	0	0	1	0	0	0	1	1	1	1	1	
Byte 28 →	DATA DW1																																	
	1	1	1	0	0	1	1	0	0	0	1	1	0	0	0	0	1	1	0	1	0	1	1	1	0	0	0	1	1	0	0	0	0	
Byte 32 →	DATA DW2																																	
	1	1	0	0	1	1	1	0	1	0	1	1	1	0	0	1	1	1	1	0	1	0	0	0	0	1	0	0	0	1	1	1	1	
Byte 36 →	DATA DW3																																	
	1	1	0	1	1	1	1	0	1	1	1	0	1	0	1	1	1	0	1	1	0	1	0	0	0	0	1	0	1	0	0	1	1	
Byte 40 →	PCRC																																	
	0	1	1	1	1	1	0	0	1	1	0	1	1	1	1	1	0	0	1	0	0	0	1	0	1	1	1	0	1	1	1	0		
Byte 44 →	MAC DW0																																	
	1	1	0	1	1	0	0	0	1	1	0	1	1	1	0	0	0	0	1	0	0	1	1	1	1	0	0	0	1	0	1	1		
Byte 48 →	MAC DW1																																	
	1	1	1	0	1	1	1	1	0	0	1	1	0	1	1	1	0	1	1	1	1	0	1	1	0	1	0	0	0	0	0	0	0	
Byte 52 →	MAC DW2																																	
	1	1	0	1	0	1	0	1	0	0	0	1	1	0	1	0	1	1	0	1	0	1	1	0	0	0	1	0	1	0	0	0	0	

Figure L-4 Example Memory Write IDE TLP (FM) §

L.3 Example NFM IDE TLP With Partial Header Encryption §

The TLP shown in § Figure L-5, when processed using the Key and IV shown in § Table L-3, which also includes other relevant inputs and outputs, yields the IDE TLP shown in § Figure L-6.

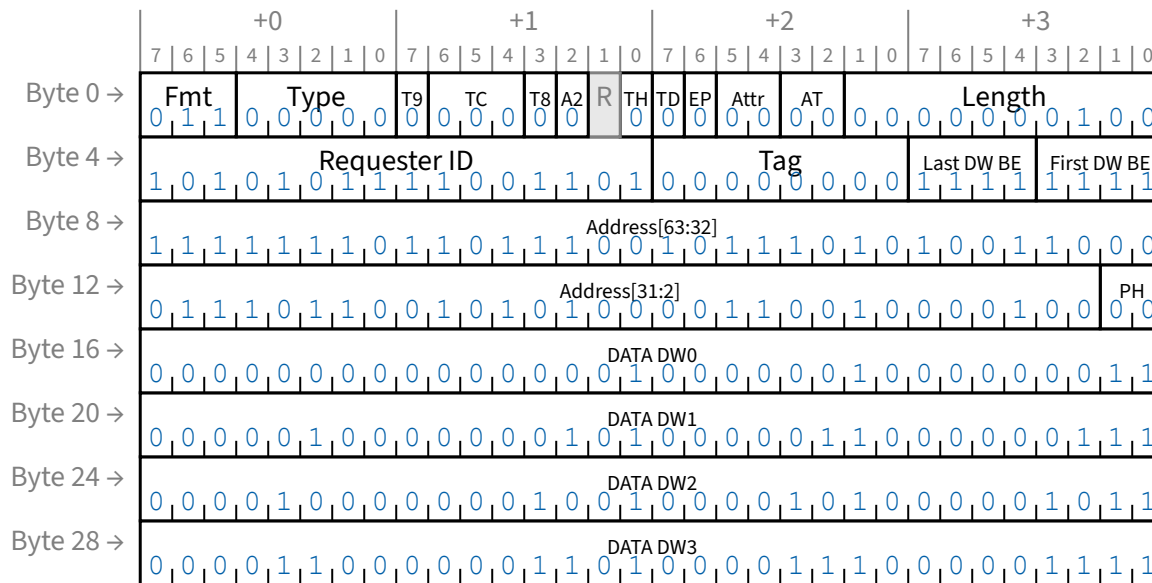


Figure L-5 Example Memory Write TLP with Partial Header Encryption (NFM) §

Table L-3 Inputs and Outputs for Example IDE TLP with Partial Header Encryption (mode 0100b - Address[41:2] Encrypted) (NFM) §

Input/Outputs variables for AES GCM	Values Byte0 [7:0] byte 1 [15:8] Byte N Key, IV, MAC : Follows AES-GCM Specification with Most significant leftmost
<i>Inputs</i>	
Key (K)	8A 95 85 8A 23 1D 41 4A 37 33 F0 E5 7A F6 AE 77 C7 F0 CF BA 75 54 D6 70 83 34 2A 64 2A EF 9D 5D
Initialization Vector (IV)	00 00 00 00 00 00 00 00 00 00 00 01
A	92 00 BC 0C 60 00 00 04 AB CD 00 FE DC B8
P (including Byte Enables, Address[41:2], and PCRC)	FF A6 1D 95 0C 84 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 9C D1 C4 63
<i>Outputs</i>	
C (including encrypted Byte Enables, Address[41:2], and PCRC)	F3 50 AB 19 EE B1 D1 1E C4 B3 FA 2D D4 E1 6E 55 FE 36 E0 2D E0 B8 36 F8 99 8C
MAC	05 F9 2E C3 B3 50 85 45 B1 30 DD A8

	+0								+1								+2								+3									
	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0		
Byte 0 →	Fmt			Type					PR_Sent_Counter								Stream_ID								XT	Sub-Stream				P	M	K	T	
	1	0	0	1	0	0	1	0	0	0	0	0	0	0	0	0	1	0	1	1	1	1	0	0	0	0	0	0	0	1	1	0	0	
Byte 4 →	Fmt			Type					T9	TC	T8	A2	R	TH	TD	EP	Attr	AT	Length															
	0	1	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	
Byte 8 →	Requester ID																Tag								Last DW BE				First DW BE					
	1	0	1	0	1	0	1	1	1	1	1	0	0	1	1	0	1	0	0	0	0	0	0	0	0	0	0	1	1	1	1	0	0	1
Byte 12 →	Address[63:32]																																	
	1	1	1	1	1	1	1	0	1	1	0	1	1	1	0	0	1	1	1	0	0	1	0	1	0	0	0	0	0	0	1	0		
Byte 16 →	Address[31:2]																														PH			
	1	0	1	0	1	1	0	0	0	1	1	0	0	1	1	1	1	0	1	1	1	0	1	0	1	1	0	0	0	1	0	0		
Byte 20 →	DATA DW0																																	
	1	1	0	1	0	0	0	1	0	0	0	1	1	1	1	0	1	1	0	0	0	1	0	0	1	0	1	1	0	0	1	1		
Byte 24 →	DATA DW1																																	
	1	1	1	1	1	0	1	0	0	0	1	0	1	1	0	1	0	1	1	0	1	0	0	1	1	1	0	0	0	0	1	1		
Byte 28 →	DATA DW2																																	
	0	1	1	0	1	1	1	0	0	1	0	1	0	1	0	1	0	1	1	1	1	1	1	0	0	0	1	1	0	1	1	0		
Byte 32 →	DATA DW3																																	
	1	1	1	0	0	0	0	0	0	0	1	0	1	1	0	1	0	1	1	1	0	0	0	0	0	1	0	1	1	1	0	0	0	
Byte 36 →	PCRC																																	
	0	0	1	1	0	1	1	0	1	1	1	1	1	0	0	0	1	0	0	1	1	0	0	1	1	0	0	0	1	1	0	0		
Byte 40 →	MAC DW0																																	
	0	0	0	0	0	1	0	1	1	1	1	1	1	0	0	1	0	0	1	1	1	1	0	1	1	0	0	0	0	1	1	1		
Byte 44 →	MAC DW1																																	
	1	0	1	1	0	0	1	1	0	1	0	1	0	0	0	0	1	0	0	0	1	0	1	0	1	0	0	0	1	0	1	1		
Byte 48 →	MAC DW2																																	
	1	0	1	1	0	0	0	1	0	0	1	1	0	0	0	0	1	1	0	1	1	1	0	1	1	0	1	0	1	0	0	0		

Figure L-6 Example NFM Memory Write IDE TLP with Partial Header Encryption §

L.4 Example FM IDE TLP With Partial Header Encryption §

The TLP shown in § Figure L-7, when processed using the Key and IV shown in § Table L-4, which also includes other relevant inputs and outputs, yields the IDE TLP shown in § Figure L-8.

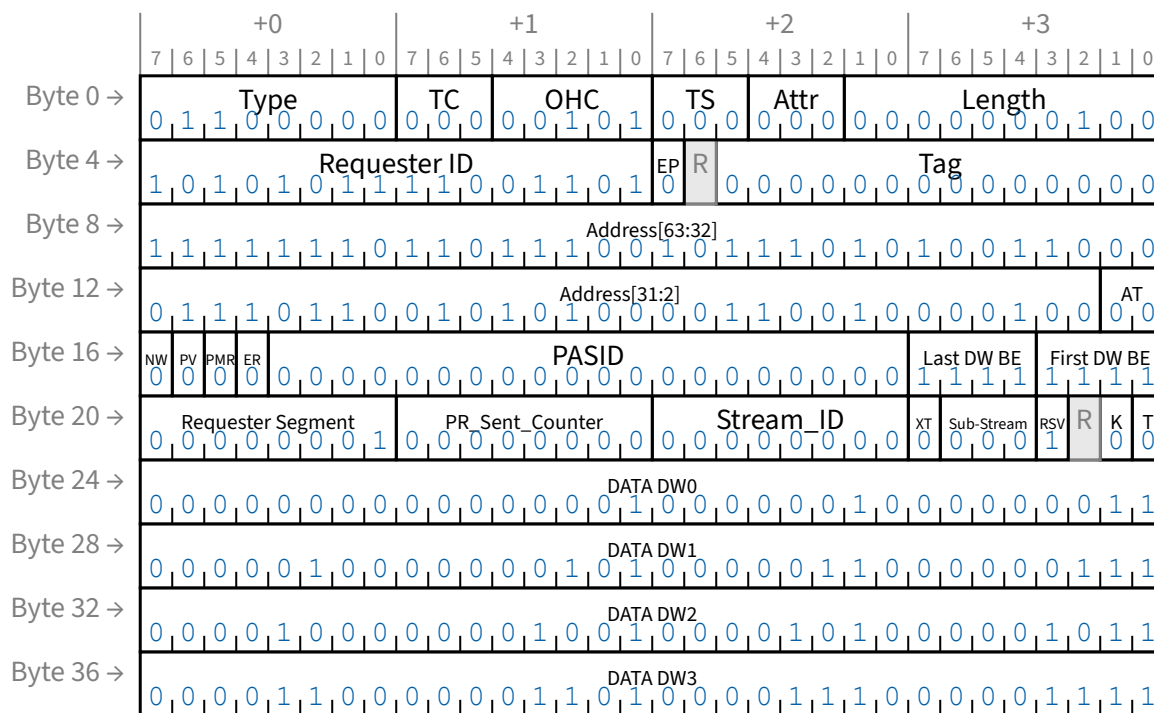


Figure L-7 Example Memory Write TLP with Partial Header Encryption (FM) §

Table L-4 Inputs and Outputs for Example IDE TLP with Partial Header Encryption (mode 0100b - Address[41:2] Encrypted) (FM) §

Input/Outputs variables for AES GCM	Values Byte0 [7:0] byte 1 [15:8] Byte N Key, IV, MAC : Follows AES-GCM Specification with Most significant leftmost
<i>Inputs</i>	
Key (K)	8A 95 85 8A 23 1D 41 4A 37 33 F0 E5 7A F6 AE 77 C7 F0 CF BA 75 54 D6 70 83 34 2A 64 2A EF 9D 5D
Initialization Vector (IV)	00 00 00 00 00 00 00 00 00 00 00 01
A	60 05 C0 04 AB CD 00 00 FE DC B8 00 00 00 01 00 BC 08
P (including Byte Enables, Address[41:2], and PCRC)	FF A6 1D 95 0C 84 00 01 02 03 04 05 06 07 08 09 0A 0B 0C 0D 0E 0F 9C D1 C4 63
<i>Outputs</i>	
C (including encrypted Byte Enables, Address[41:2] and PCRC)	F3 50 AB 19 EE B1 D1 1E C4 B3 FA 2D D4 E1 6E 55 FE 36 E0 2D E0 B8 36 F8 99 8C
MAC	66 6E 87 A3 36 89 ED 1A C5 17 AD 22

	+0								+1								+2								+3									
	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0	7	6	5	4	3	2	1	0		
Byte 0 →	Type								TC				OHC				TS				Attr				Length									
Byte 4 →	Requester ID																EP	R	Tag															
Byte 8 →	Address[63:32]																																	
Byte 12 →	Address[31:2]																														AT			
Byte 16 →	NW	PV	PMR	ER	PASID																Last DW BE				First DW BE									
Byte 20 →	Requester Segment								PR_Sent_Counter								hed	Stream_ID								XT	Sub-Stream				RSV	R	K	T
Byte 24 →	DATA DW0																																	
Byte 28 →	DATA DW1																																	
Byte 32 →	DATA DW2																																	
Byte 36 →	DATA DW3																																	
Byte 40 →	PCRC																																	
Byte 44 →	MAC DW0																																	
Byte 48 →	MAC DW1																																	
Byte 52 →	MAC DW2																																	

Figure L-8 Example Memory Write IDE TLP with Partial Header Encryption (FM) §

L.5 IDE Test Keys §

The keys defined in this section are intended to be used to simplify IDE validation, for example, by making it easier for verification tools to decrypt and MAC-check TLPs. As illustrated in § Table L-5, the defined test keys consist of a constant value in the more significant bytes of the key, and an incrementing value contained in the two least significant bytes of the key, where the test key number corresponds to the value of the incrementing portion.

These keys are intended for use in validation and test environments only - it is essential these keys are not used in production environments. It is recommended that verification tools check for the inadvertent use of test keys in production settings, and provide an appropriate warning to the user. It is also recommended to check for the use of other key values that have been published as test keys, and for trivial keys such as all 0's.

Table L-5 IDE Test Keys §

Key Number	Key Value – Byte0 [7:0] byte 1[15:8].....Byte N Key, IV, MAC: Follows AES-GCM Specification with Most significant leftmost
0	0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00

Key Number	Key Value – Byte0 [7:0] byte 1[15:8].....Byte N Key, IV, MAC: Follows AES-GCM Specification with Most significant leftmost
1	0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 01
2	0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 02
...	0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 0B AD 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 xx xx

